SUBSTRATE ACCUMULATION MODEL

# SAM Volume III

## Computation and Reciprocal History

Part 1: the exact computational record, 28 chapters
Part 2: the exact engine, GEN6 and the Duck as installed

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Part 1 follows the revised manuscript of 20 October 2026 (28 chapters) built from source; Part 2 is the Engine manuscript of 20 October with its continuation to 25 October. Original formatting and historical scientific results retain their recorded scope. The credits describe the development history, not independent scientific endorsement.

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## Reader's contract and source boundary

This expanded edition retains the original twelve-chapter foundation and develops the later completed research in ten additional chapters. It follows the original Volume III design: 10-point Times text, two columns, letter pages, the same margins, Roman chapter identifiers, chapter-relative mathematical numbering and a continuous page sequence.

The original foundation was assembled from the fixed 8 September 2026 SAMA source revision `4c2b5dc1649da5900c90b90d0ce56c223dbb7717`. Its historical v1.2, v0.7, SLCV21R and SLCQ2-RZ identities remain attached to their own stages. The later chapters use local source reports, mathematical contracts and retained derivations through 19 October 2026. The current registry snapshot identifies SLC-GEN5-R1 / SLC-GEN5-CEV1-R1 and its G2-SPIN-CLOCK generation, with subsequent additive source methods identified by their own records. This is read-only authority recovery for manuscript work; no runtime or research campaign is launched by the edition.

Completed results retain their original operands, classifications and comparison sets. Historical failed controls, corrected implementations, stopped campaigns and partial runs keep their recorded status. Exact integer objects, rational expressions and certified enclosures remain distinct mathematical return types. Editorial connections identify how an existing source can be received by another calculation; they are distinguished from newly executed science.

The expansion develops native exact arithmetic and certified logarithms, retained mathematical objects, executable method graphs, learned precision and inverse queries, joint Ising state spaces, the latest layered ground-state solver, certified source integration, paid spin response, source-history reconstruction, matter and tau applications, signed arithmetic transition storage, Mersenne certificates, and prime-event feedback with its frustrated-spin source. The final chapter joins their concrete interfaces to Volumes I and II.

No historical scientific experiment is rerun for this manuscript. The source package includes the retained foundation, exact copied reports and operands used for the additions, source paths and hashes, chapter coverage records, editable TeX, and final typesetting checks. Large native datasets and full run archives retain the canonical locations given by their source reports. The reproducibility register distinguishes copied explanatory sources from those bulk result archives.

The subsequent three-volume research review can start from these source maps and completed interfaces. Selection and execution of new research targets remain a separate next step.

**Edition of 25 October 2026.** At Sean Brady's direction this edition takes the 28-chapter order of the revised manuscript of 20 October 2026 as Part 1, building its six added chapters (exact joint densities at N5000, N2000 ground states, equal observable states and executable histories, exact exchange and symbolic matter, signed arithmetic energy, prime-event computation) from their sources beside the twenty-two chapters of the expanded manuscript, and adds Part 2, the exact engine as installed on 25 October 2026 (the SLC, the one-principle core, the Duck and the collaborative runs), from the Engine manuscript of 20 October with its continuation. The results of the engine's runs of 20–25 October that bear on this volume are received into Chapters XX and XXII as closing subsections, labelled as readings. The reader's contract above governs Part 2 as it governs Part 1.

# Part 1

# The exact computational record

SAM VOLUME III

# N100: Grammar and Exact Work

Chapter I

The first computational question is exact and finite: what object is specified by the admitted source grammar, and how can its complete state sum be evaluated? This chapter keeps the object fixed while following two execution routes. The source census establishes the object; the factorization explains why its formal cardinality need not equal the amount of local enumeration.

The account below follows the chapter source as retained at the fixed 8 September 2026 SAMA revision. Engine names, installed interfaces and campaign status are historical source statements. They do not describe a freshly inspected October deployment. Later source amendments are distinguished where available; no later engine or campaign is inferred.

## I.1 Overview

N100 is an exact frustrated-spin computation whose favorable computational structure comes from its source grammar. The mathematical object is the unchanged 100-spin, 260-edge Hamiltonian with

$$2^{100} = 1{,}267{,}650{,}600{,}228{,}229{,}401{,}496{,}703{,}205{,}376$$

binary assignments. The first complete H14F route returns the full exact density-of-states tensor in 3.828137395 seconds. A later structure-first route detects eight conditional components, executes 165,920 local assignments and convolves them to the coefficientwise-identical tensor in 0.500951441 seconds. The reduced workload is an exact evaluation strategy; it does not reduce the state space, change the Hamiltonian, sample the assignments or request only a ground state.

The 100-row census is itself source-bound. The visible QP lane contains 105 records. Exactly one of the 21 complete five-role packet exclusions—the `p=9, g=0` packet—produces 100 independently counted rows with exact native inventory 16,200 and mean 162. The five excluded-count W9 witnesses remain visible as derived `X1+W8` receipts. The A field remains present as non-row infrastructure, and X1 remains non-row/non-matter. Those are three different boundaries.

The central computational finding is not merely the observed wall time. The source grammar contains exact decomposition certificates, and the compiler can use them while preserving the complete result. Its workload also returns the ledger quantity L162 through an independent arithmetic identity. The N105, N81 and N72 controls show exactly which parts depend on the source grammar and which do not.

## I.2 Opening question and conceptual picture

The governing question is:

> Can a source-derived 100-row grammar expose enough exact structure to compute the complete `2^100` density-of-states tensor without visiting every assignment one by one, while returning exactly the same mathematical output?

The answer is represented by two equivalent exact routes:

- source-derived N100 Hamiltonian
- complete H14F variable-elimination route
- exact retained-role DOS tensor
- same unchanged Hamiltonian
- detect exact conditional components
- compute each local exact tensor
- exact tensor/polynomial convolution
- coefficientwise-identical retained-role DOS tensor

The grammar does not replace the calculation. It tells the exact solver how the complete sum can be factorized.

## I.3 Definitions and cardinality firewall

### I.3.1 Exact frustrated-spin density of states

For binary spins

$$\sigma_i \in \{-1,+1\}, \qquad i = 1,\dots,N,$$

and a declared weighted graph, the exact Hamiltonian has the form

$$H(\sigma) = \sum_{(i,j)\in E} J_{ij}\sigma_i\sigma_j + \sum_i h_i\sigma_i.$$

The density of states is the integer function

$$g(E) = \#\{\sigma \in \{-1,+1\}^N : H(\sigma) = E\}.$$

Completeness requires

$$\sum_E g(E) = 2^N.$$

For N100, `N=100`. Frustration changes the organization of energy and the ground sector; it does not change the formal assignment count.

### I.3.2 Four quantities that must not be merged

- **N100 source rows.** Exact value: 100; Type: independent spins of the declared Hamiltonian.
- **complete configuration space.** Exact value: `2^100`; Type: every formal binary assignment counted by the returned tensor.
- **structure-first work.** Exact value: 165,920; Type: local assignments evaluated before exact convolution.

- **L162 native ledger mean.** Exact value: 162; Type: source-ledger/accounting identity, not a coupling or state count.

The statement "exact `2^100`" refers to the tensor's complete count. It does not say the runner iterated through `2^100` bit strings. Conversely, the 165,920 workload is not a claim that the physical or mathematical problem had only 165,920 states.

#### I.3.3 The row-census exclusions are separately typed

The N100 census is produced after one historical five-row packet is excluded from independent content. Preserve these three boundaries:

- p=9,g=0 packet: five derived W9 witnesses, visible but non-counted
- A-field row: excluded because A is active non-row infrastructure
- X1: response channel, not a physical/source row

Similar arithmetic around the 163-to-162 ledger does not make the three objects equivalent.

#### I.3.4 Source values are not Hamiltonian couplings

The QP rows supply typed identity, topology and role information through a separately declared compiler. Their native numerical values are not inserted as `J_ij` or `h_i` merely because they occur on the same record. The tested Hamiltonian is the source-derived 100-spin, 260-edge graph fixed before the structure-first route.

### I.4 Deriving the 100-row source census

#### I.4.1 Begin with the full visible lane

CR120R begins from 105 visible source rows arranged into 21 complete five-role address packets. The exact source totals are

- source rows 105
- complete five-role packets 21
- address-value sum 2889
- full native inventory 130005/8 = 16250.625

Every source row remains present in the result artifact. The operation is a counted/not-counted overlay, not deletion.

#### I.4.2 Test every one-packet exclusion

Each of the 21 five-role packets is removed in turn from independent-content accounting. A valid candidate must leave 100 rows and close the L162 mean:

$$\frac{M_{native,remaining}}{100} = 162.$$

Exactly one candidate does so:

$$(p,g) = (9,0).$$

The resulting payload is

- excluded typed packet p=9,g=0, five rows
- payload rows 100
- payload native inventory 16200
- payload mean 162=L
- single-packet L162 hits 1

Thus

$$\frac{16{,}200}{100} = 162.$$

#### I.4.3 Verify the derived-parent relation

Each same-role `p=9,g=0` record closes exactly as the sum of its `p=8,g=0` and `p=1,g=0` parents in `M_native`. This is the source-typed W9 route

$$8+1 \to 9$$

expressed on five records. The type controls prevent the scalar 9 from collapsing `W9_CLOSURE_WITNESS`, `C9_CARRIER_ATOM` and `P9_BIGRADE_PARTITION` into one object. Weak-vector and hidden-source p9 support rows are also barred from the selector, and the identity is not extended to g1, g2, composite routes or qA.

#### I.4.4 Reconcile the payload

The derived 100 payload IDs and values reconcile one-for-one with the current `Updated\Particle\Rows.xlsx` extraction recorded by CR120R. This closes the source census used by the N100 computation while retaining the five witnesses as provenance.

### I.5 Constructing the exact N100 problem

#### I.5.1 Freeze the graph and requested output

The declared problem contains:

- 100 independent binary spins;
- 260 source-derived edges;
- induced width 5 under the complete H14F route;
- a retained-role conditional DOS tensor;
- a 16-row scalar closure; and
- exact integer coefficients reconstructed across four CRT primes.

The requested output is the complete tensor, not only the minimum energy or a small collection of moments. The graph, source roles and tensor schema are fixed before either timing result.

#### I.5.2 Why exact routes can be smaller than the state count

The complete sum

$$g(E) = \sum_{\sigma \in \{-1,+1\}^{100}} \mathbf{1}\{H(\sigma) = E\}$$

can be rearranged without approximation. Variable elimination sums internal spins while retaining exact boundary spectra. Disconnected conditional components can be computed independently and convolved. Modular NTT lanes can multiply polynomials exactly, and CRT reconstructs the integer coefficients once the modulus product exceeds the coefficient range.

All such routes remain exact if they reproduce every coefficient and close the complete state count.

### I.6 Complete H14F route from execution to closure

#### I.6.1 Execute the declared Hamiltonian

The first route applies H14F variable elimination to the full source-derived graph. It produces:

- **spins.** Exact result: 100.
- **source edges.** Exact result: 260.
- **represented configurations.** Exact result: `2^100`.
- **occupied energy bins.** Exact result: 177.
- **ground sector.** Exact result: `(-260,256)`.
- **distinct port spectra.** Exact result: 3.
- **singleton root tasks.** Exact result: 2,048.
- **CRT primes.** Exact result: 4.
- **observed runner wall.** Exact result: 3.828137395 s.

The ordered pair `(-260,256)` records the ground energy and its degeneracy. The calculation retains the other 176 occupied bins as well.

#### I.6.2 Close the port marginals

Four boundary bits produce 16 retained port rows. Each row sums to

$$2^{96},$$

because fixing the four retained bits leaves 96 independent spins. Summing all 16 rows gives

$$16 \cdot 2^{96} = 2^4 \cdot 2^{96} = 2^{100}.$$

This is both a marginal check and a full-state closure.

#### I.6.3 Close source symmetries

All 24 source-role permutations reproduce the required tensor relation. The global spin complement also closes. These checks distinguish a complete source-role tensor from an accidental scalar DOS total.

### I.7 Structure-first route from components to the same tensor

#### I.7.1 Detect exact conditional components

The source grammar exposes eight components with sizes

$$15, 15, 15, 15, 15, 10, 10, 5.$$

They are conditional components: any shared retained port variables remain explicit, while the interior variables of each component can be summed locally.

#### I.7.2 Compute local exact tensors

For component `c` and retained boundary assignment `b`, define its exact energy polynomial

$$P_{c,b}(x) = \sum_E g_{c,b}(E) x^E.$$

Exhaustive local work for the eight interiors is

$$W_{100} = 5 \cdot 2^{15} + 2 \cdot 2^{10} + 2^5.$$

Evaluate each term:

$$5 \cdot 2^{15} = 5 \cdot 32{,}768 = 163{,}840,$$

$$2 \cdot 2^{10} = 2{,}048, \qquad 2^5 = 32,$$

so

$$W_{100} = 163{,}840 + 2{,}048 + 32 = 165{,}920.$$

#### I.7.3 Convolve rather than discard cross-component sums

For each retained boundary assignment, multiply the local polynomials:

$$P_b(x) = \prod_{c=1}^{8} P_{c,b}(x).$$

Polynomial multiplication performs the full energy convolution: every choice of one local state from each component contributes to the coefficient at the sum of their energies. Summing or assembling over the retained boundary assignments returns the same retained-role tensor as the complete route.

No state is sampled or dropped. Factorization evaluates the product structure of the full sum.

#### I.7.4 Compare complete outputs

The structure-first wall is 0.500951441 seconds. Relative to the 3.828137395 second H14F route, the observed ratio is approximately

$$\frac{3.828137395}{0.500951441} \approx 7.642.$$

The load-bearing comparison is stronger than timing: the complete tensors have the same coefficient hash and match coefficient by coefficient. The Hamiltonian, state space, energy support and output schema are unchanged.

### I.8 Grammar-to-work identity, derived line by line

The source ledger supplies

$$N100_{native} = 100 \cdot 162 = 16{,}200.$$

The exact component work can be regrouped as

$$165{,}920 = 165{,}888 + 32 = 162 \cdot 1{,}024 + 32.$$

Therefore

$$W_{100} = 162 \cdot 2^{10} + 2^5.$$

Subtract the residual and divide by the ten-bit block:

$$\frac{W_{100} - 2^5}{2^{10}} = \frac{165{,}920 - 32}{1{,}024} = \frac{165{,}888}{1{,}024} = 162.$$

There is a second exact form. Because

$$72^2 = 5{,}184 = 32 \cdot 162,$$

we have

$$W_{100} = 2^5(72^2 + 1) = 32(5{,}184 + 1) = 165{,}920.$$

The residual `2^5` is the binary assignment space of the surviving five-role `p=9,g=1` packet. It is not X1, Theta18 or a physical closure object. The identity connects two independently sourced structures—the row ledger and the exact workload—but the value 16,200 is not inserted into the Hamiltonian and does not mechanically cause the runtime.

### I.9 Source-native controls and deviations

#### I.9.1 N105 restored-packet control

Restore the historical `p=9,g=0` packet. The workload becomes

$$W_{105} = 166{,}912 = 163 \cdot 2^{10}.$$

The difference is

$$\Delta W = 166{,}912 - 165{,}920 = 992 = 2^{10} - 2^5.$$

The restoration also changes the N100/N81 completion-path capacity from 40 to 45 rows. The clean 162 identity is therefore sensitive to the typed packet boundary; it is not an identity forced by any 100-variable benchmark.

#### I.9.2 N81 octahedral control

The N81 source-native control has component sizes

$$[11, 10, 11, 10, 10, 11, 10, 8]$$

and exact work

$$10{,}496 = 2^7(81 + 1).$$

Its octahedral bridge retains 48 cube embeddings in one full-octahedral orbit, the completion path `1-8-9-12`, reflected C4 gluing and 49,152 incidences. The N100/N81 localization remains `40->40` under all nine insertions. These are exact geometry/compiler leads; a causal account of the timing is still a separate research question.

#### I.9.3 N72 dense contrast

N72 remains an independently dense computation surface. It is not edited to look like N100. This control prevents the source-structure finding from being rephrased as a universal rule that nominally smaller N must be faster or that all graphs admit the same component decomposition.

#### I.9.4 Unchanged-parent control

The N100 graph and requested tensor were frozen before the structure-first timing route. The earlier H14F result already supplied the complete tensor. Coefficientwise equality therefore tests an optimization of representation on an unchanged parent rather than a retuned benchmark.

### I.10 Source-native compiler boundary

The complete N100 computation and the later N100-to-F81 semantic compiler are different tasks. G1 audits the latter and records:

- **typed N100 QP source keys.** Coverage: 100/100.
- **Shared72 aggregate connector-incidence eligibility.** Coverage: 72/100.
- **CR211 address rows.** Coverage: 162.
- **CR211 rows with QP elementwise assignment.** Coverage: 0.
- **F81 semantic sites.** Coverage: 81.
- **frozen F81 contact classes.** Coverage: 10.
- **N100 rows with complete elementwise F81 map.** Coverage: 0.

The missing key is

- `QP_CANDIDATE_OR_TEMPLATE_TO_F81_SEMANTIC_SITE_OR_SOURCE_NATIVE_F81_ADDRESS_CLASS`

CR211 explicitly leaves address-to-template incidence null, and the F81 target schema has no QP source key. G1 therefore records `PARTIAL_DOMAIN__ELEMENTWISE_F81_JOIN_OPEN`; the contact lane is `NOT_EXECUTED__SOURCE_NATIVE_ELEMENTWISE_COMPILER_OPEN`. No result or residual selects a map.

This boundary does not retract the exact N100 Hamiltonian calculation. It says that a later semantic join from each source row to an F81 site/address class has not been supplied. Aggregate Shared72 eligibility cannot substitute for 100 elementwise assignments.

### I.11 Established result and current boundaries

The source-bound N100 result is:

1. CR120R fixes a 100-row independent-content census after the unique `p=9,g=0` five-record witness packet is excluded from counting.
2. The A field remains non-row infrastructure, X1 remains non-row/non-matter, and all 105 records remain visible.

3. The unchanged Hamiltonian has 100 spins, 260 edges and a complete `2^100` configuration space.
4. H14F returns the full exact retained-role DOS tensor with 177 occupied bins and ground sector `(-260,256)`.
5. The structure-first route evaluates 165,920 local assignments and exactly convolves them to the same coefficient tensor.
6. Work reduction is not state-space reduction.
7. The exact grammar-to-work identity returns 162 while retaining its `2^5` residual as the p9,g1 binary packet.
8. N105, N81 and N72 preserve the source dependence and scope.
9. The source-native elementwise N100-to-F81 compiler join remains open, and the contact lane was not executed.
10. Historical SLC revision labels in the immutable spine do not replace the then-live `SLCQ2-RZ` source-recorded default identity.

The registered atomic records and focused tests assign no authorized project result classification to this chapter. Exact source verdicts and timings are preserved without manufacturing one or extending the result to arbitrary 100-spin graphs.

### I.12 Connections and forward handoff

- **`SAMA-D000027.`** Connection: Supplies the finite source grammar and row-count distinctions..
- **`SAMA-D000033.`** Connection: Supplies M126/N144/L162 types without merging them with N100..
- **`SAMA-D000035.`** Connection: Makes the grammar, type system, program custody and exact semantic path explicit..
- **`SAMA-D000036.`** Connection: Receives the exact DOS definition, coefficient closure and route families..
- **`SAMA-D000037.`** Connection: Receives the structure-first router and exact fallback policy..
- **`SAMA-P000006.`** Connection: Connects N100 computation back to the particle, Starbreaker and binding parent route..

The immediate handoff to D35 is a language requirement: source identity, operator type, program tape, state and history must be serialized so the grammar certificates can select an exact route without hidden UI or wrapper semantics.

### I.13 Execution architecture and controls

### I.14 Overview

The central Volume III computational finding is not that a $2^{100}$ state space disappeared. It is that the source-derived N100 grammar carries an exact factorization certificate. The unchanged 100-spin Hamiltonian has eight conditional components. Evaluating those components locally and convolving their exact tensors reconstructs the same complete result as the H14F variable-elimination route, coefficient for coefficient.

This separates three quantities that are often confused:

- formal state space = every one of 2^100 assignments counted
- requested exact output = the complete retained-role DOS tensor
- fresh local work = 165,920 component assignments

The source grammar determines the first two. A certified execution route changes only the third. H14F remains the exact fallback when a component, orbit, fiber, separator or rank certificate is absent or fails an installation gate. H14F's fourteen-worker surface is a frozen execution contract where declared, not an informal knob that can be reinterpreted as an arbitrary host setting.

The controls make the result transferable. N105 restores the excluded source packet; N81 supplies a source-native octahedral comparison; N72 remains a dense contrast; mixed partial-join and connected-ring graphs test whether the router can preserve exactness when components are coupled; N96 demonstrates exact cutset branching and fallback. These controls show what the router is: a typed grammar-to-work compiler, not a rule that substitutes the N100 graph for other systems.

Historically, H14F and the structure-first receipts belong to the exact arithmetic/application lineage now preserved in the delegated frozen `SLCV21R` base. Frozen `SLCQ2-RZ` is the sole source-recorded default SLC selection engine. Q2 consumes the byte-identical base services; it does not turn H14F into the present ordinal selector or erase the earlier exact results.

### I.15 Opening question and conceptual picture

The governing question is:

> When a finite grammar defines a complete exact calculation, can the grammar itself certify a smaller execution plan without changing the mathematical state space, Hamiltonian or requested output?

The answer is represented by

- typed source grammar
- declared Hamiltonian and requested tensor
- structural certificate discovery
- admitted exact route
- component convolution
- orbit-sector counting
- separator branching
- rank-aware readout
- or H14F exact fallback
- independent reconstruction of the same tensor

The router never asks which route would make a preferred outcome appear. It asks which exact decomposition is already present in the declared object. If the certificate is missing, the object is not modified to create one.

### I.16 Typed definitions and execution contract

#### I.16.1 Mathematical object versus execution profile

For a fixed Hamiltonian $H$, define the retained-role tensor

$$G(e,r) = \#\{\sigma : H(\sigma) = e,\ R(\sigma) = r\},$$

where $r$ is one of the declared port-role states. Its scalar DOS is

$$g(e) = \sum_r G(e,r).$$

The mathematical object consists of $H$, the assignment domain, the role map $R$, and every requested coefficient $G(e,r)$. H14C and H14F specify how that object is evaluated. They may change elimination order, batching, residue storage, worker persistence and cache policy. They may not change $H$, delete assignments, reinterpret the roles or replace $G$ with a smaller target.

### I.16.2 H14C and H14F

H14C is the completed fourteen-core profile with complete-prime calibration and the accepted retained-capacity contract. H14F adds mixed-width compilation, persistent workers, reduced residue storage and an ephemeral scale-specific projection cache. Both are exact execution profiles.

Where an H14F campaign declares fourteen pinned physical workers, that width is part of the frozen architecture. It cannot silently become "whatever number of host workers is available," because batching, custody and timing receipts are indexed to the declared worker surface. A new worker mapping would be a successor execution profile, not a reinterpretation of the old one.

### I.16.3 Structural certificate

A structural certificate is exact evidence that the requested object can be reconstructed from smaller typed objects. Examples include:

- disconnected components after an explicit conditioning boundary;
- a permutation orbit on which energy depends only on marked population;
- a cutset whose fixed assignments reduce admitted elimination width;
- a fee-exempt fiber whose multiplicity can be lifted exactly; and
- a constrained readout with a known independent rank.

The certificate must contain the reconstruction rule. A low-width-looking plot or favorable runtime is not enough.

### I.16.4 Router and fallback

The routing policy is

- if conditional components close:
- evaluate each component exactly and convolve
- elif uniform orbit sectors close:
- count exact marked-population sectors
- elif a declared cutset lowers admitted width:
- branch exactly on every cutset assignment
- elif the output is a constrained fiber/readout:
- evaluate independent rank and reconstruct the ambient view
- else:
- execute the exact H14F variable-elimination fallback

Every branch returns the complete requested tensor. Failure to certify one branch is a routing outcome, not permission to approximate.

## I.17 The complete H14F N100 result

### I.17.1 Source object

The source-derived N100 object has

- **independent spins.** Declared value: 100.
- **source edges.** Declared value: 260.
- **induced width.** Declared value: 5.
- **retained source packet.** Declared value: $p = 1, g = 0$.
- **role order.** Declared value: plus, minus, anti-plus, anti-minus.
- **complete configurations.** Declared value: $2^{100}$.

The exact configuration count is

$$2^{100} = 1{,}267{,}650{,}600{,}228{,}229{,}401{,}496{,}703{,}205{,}376.$$

H14F treats the complete 100-spin, 260-edge Hamiltonian as one retained-port variable-elimination object. It executes 2,048 singleton root tasks through four CRT primes.

### I.17.2 Retained-role closure

Four binary retained roles give $2^4 = 16$ port rows. Each row sums to $2^{96}$, because fixing the four retained role bits leaves 96 independent binary coordinates. Summing all rows gives

$$\sum_{r=0}^{15} \sum_e G(e,r) = 16 \cdot 2^{96} = 2^{100}.$$

The scalar result has 177 occupied energy bins and ground sector ((-260,256)): energy $-260$ with degeneracy 256. The conditional port rows organize into three spectra according to plus-spin count:

- **0.** Representative state: 0; Ground energy: -260; Ground count: 128; Bins: 174.
- **1.** Representative state: 1; Ground energy: -250; Ground count: 128; Bins: 174.
- **2.** Representative state: 3; Ground energy: -244; Ground count: 256; Bins: 171.
- **3.** Representative state: 7; Ground energy: -250; Ground count: 128; Bins: 174.
- **4.** Representative state: 15; Ground energy: -260; Ground count: 128; Bins: 174.

Thus counts 0/4 share one spectrum, 1/3 share a second and 2 supplies the third. All 24 source-role permutations and the global spin complement close exactly. This is a computationally typed conditional tensor, not an elementwise F81 physical-port assignment.

### I.17.3 Execution receipt

The four prime-kernel walls were approximately

$$0.890328, 0.852483, 0.797375, 0.799622 \text{ s},$$

and the total runner wall was 3.828137395 seconds. The scalar DOS semantic hash is `6e404623175ba60629110fb17f75ca0f0398e690db9de7b7a97766d32d0c2d30` and the retained tensor semantic hash is `33ec2e3f5cd3a1e66eb892730181e89bb1aff71d93b97fbbba4959bb238bd78d`. These hashes become the equality target for the structure-first route.

## I.18 Deriving the N100 structure-first work

### I.18.1 Component census

The unchanged conditional grammar separates into eight exact components of sizes

$$15, 15, 15, 15, 15, 10, 10, 5.$$

The sizes sum to the original independent rank:

$$5 \cdot 15 + 2 \cdot 10 + 5 = 75 + 20 + 5 = 100.$$

For a component of $n_a$ binary variables, direct local enumeration requires $2^{n_a}$ assignments. Therefore the total fresh component work is

$$W_{100} = 5 \cdot 2^{15} + 2 \cdot 2^{10} + 2^5.$$

Evaluating each term,

$$5 \cdot 32{,}768 = 163{,}840,$$

$$2 \cdot 1{,}024 = 2{,}048,$$

and

$$W_{100} = 163{,}840 + 2{,}048 + 32 = 165{,}920.$$

### I.18.2 Why 165,920 does not replace $2^{100}$

Let component $a$ return its exact conditional polynomial tensor $P_a(z, r_a)$. The full tensor is reconstructed by summing compatible role assignments and convolving energies:

$$P_{100}(z, r) = \sum_{r_1 \oplus \cdots \oplus r_8 = r} \prod_{a=1}^{8} P_a(z, r_a).$$

At $z = 1$, each local polynomial counts all local assignments, so

$$P_{100}(1) = \prod_{a=1}^{8} 2^{n_a} = 2^{\sum_a n_a} = 2^{100}.$$

The 165,920 number counts fresh local assignments evaluated by the algorithm. The $2^{100}$ number counts formal global assignments represented by the reconstructed tensor. They answer different questions.

### I.18.3 Same-output equality

The structure-first route completed in 0.500951441 seconds. Relative to the 3.828137395-second H14F route, the observed ratio is

$$\frac{3.828137395}{0.500951441} \approx 7.642.$$

This comparison is meaningful because the retained-role tensors are coefficientwise identical and have the same semantic hash. The route did not change the Hamiltonian, sample states, truncate bins or request only the ground sector.

### I.18.4 Grammar-to-work identities

The work also admits exact source-grammar decompositions:

$$W_{100} = 162 \cdot 2^{10} + 2^5 = 2^5(72^2 + 1).$$

Indeed,

$$162 \cdot 1{,}024 + 32 = 165{,}888 + 32 = 165{,}920,$$

and $72^2 = 5{,}184 = 32 \cdot 162$. Therefore

$$\frac{W_{100} - 2^5}{2^{10}} = 162, \qquad 2^5 \cdot 162 = 72^2.$$

The residual $2^5$ is the formal assignment space of the surviving five-role $p = 9, g = 1$ packet. It is not X1 and not a physical closure object. The identity interprets source grammar and workload; it does not claim that a ledger total was inserted as an energy coefficient.

## I.19 Source-native controls and deviations

### I.19.1 N105 restoration control

N100 excludes the historical $p = 9, g = 0$ witness packet. Restoring it produces N105 and changes the exact work to

$$W_{105} = 166{,}912 = 163 \cdot 2^{10}.$$

The difference is

$$\Delta W = W_{105} - W_{100} = 166{,}912 - 165{,}920 = 992 = 2^{10} - 2^5.$$

This is a source-native perturbation: it restores an identified packet rather than adding an arbitrary edge until timing changes. It also changes the N100/N81 completion-path capacity from 40 to 45 rows.

### I.19.2 N81 octahedral control

The N81 control has component sizes

$$[11, 10, 11, 10, 10, 11, 10, 8]$$

and local work

$$W_{81} = 3 \cdot 2^{11} + 4 \cdot 2^{10} + 2^8$$
$$= 6{,}144 + 4{,}096 + 256$$
$$= 10{,}496 = 2^7(81+1).$$

Its octahedral bridge retains 48 cube embeddings in one full-octahedral orbit, the completion path (1-8-9-12), reflected C4 gluing and 49,152 incidences. This is a geometry/compiler control. Its possible causal relation to runtime remains a research question rather than an installed explanation.

### I.19.3 N72 dense contrast

N72 remains dense. It is not edited to reproduce N100's component structure. This is the correct contrast because it asks whether the router recognizes the actual source object rather than imposing one preferred decomposition on every input.

### I.19.4 Mixed-connectivity controls

The installation tournament adds two controlled deviations:

- a one-edge partial join, width 5, ground sector ((-261,128)); and
- a connected ring, width 5, ground sector ((-268,2)).

Both complete sixteen-row tensors equal an independent original-component bridge contraction coefficient for coefficient. The partial join retains 128 component-orientation choices; the connected cyclic skeleton leaves the two global orientations. These tests show that a changed coupling topology can be handled exactly when its reconstruction rule is explicit.

### I.19.5 Dense N96 fallback

The full N96 graph remains width 24. A frozen five-site cutset creates

$$2^5 = 32$$

exact branches, each admitted to H14F at width 22. This is neither a component convolution nor an approximate shortcut. Every cutset assignment is retained, and H14F performs the exact branch work. The regression ran 4,096 fresh tasks and retained the frozen N96 numerical authority.

## I.20 Failure, correction and routing discipline

The principal wrong route would be to treat N100's favorable graph as the universal computational model. That route is rejected in stages:

- wrong control: substitute the N100 component graph for another grammar
- source identity is lost
- correction: inspect the new object's own component/orbit/cutset/fiber/rank certificates
- choose only an admitted exact route
- retest: reconstruct the complete tensor independently
- compare coefficientwise and replay dense regressions
- fallback: if no certificate closes, retain H14F

The installation gates execute this logic rather than merely state it. They record 12/12 route checks, 18/18 mixed arithmetic checks, 9/9 N18/N54 checks, 16/16 N72 checks and 20/20 N96 checks: 75/75 combined executed checks. Their admission was for a separate owner-authorized promotion decision; the gate campaign itself did not silently rewrite a current pointer.

A second deviation is the source-native N100-to-F81 compiler gap. N100 has 100/100 typed source keys and Shared72 has aggregate connector incidence for 72/100 rows, while CR211 exposes 162 address rows but zero elementwise QP assignments. The missing elementwise join cannot be replaced by aggregate counts or external labels. `G:G1@SAM-RESEARCH` preserves that partial-domain boundary in `SAMA-D000034`; it is not a direct H14F result key for this chapter.

## I.21 Recorded authority and delegated-base status

Historical H14F, structure-first, Exact Write, W9P/Theta, support/Gram, tensor-orbit and bounded application artifacts remain frozen within the `SLCV21R` arithmetic/application lineage. Recorded authority is `SLCQ2-RZ`, the source-recorded default hybrid ordinal selection engine with 70 exact source-visible features. It delegates byte-identical arithmetic/application services to the frozen `SLCV21R` base, retaining the exact H14F fallback and structure-first routing lineage.

`SLCV21R` is noncurrent as a revision, but its declared services remain the delegated base. Q1 is the immutable fallback. Q2 does not install physical calibration, Binding, ATOM3D or RH merely by consuming base arithmetic. H14F therefore remains load-bearing without becoming the current selection engine.

## I.22 Established result, boundary and forward handoff

The complete N100 result counts all $2^{100}$ assignments of the unchanged 100-spin, 260-edge Hamiltonian and returns the full exact retained-role tensor. The structure-first route evaluates 165,920 local assignments and reconstructs that same tensor coefficientwise. The execution-work reduction is established for this source grammar; it is not asserted for every arbitrary 100-spin graph.

The router's general result is procedural and exact: detect the source certificate, preserve the requested object, reconstruct independently, and retain H14F when no reduced route closes. N105, N81, N72, mixed-graph and N96 controls define where each branch applies.

The forward handoff goes to `SAMA-D000041`, where Starbreaker's own grammar is routed without substituting the N100 graph. It also supports the current-Q2 ordinal chapter `SAMA-D000055`, where exact base computation and source-visible selection must remain different types.

SAM VOLUME III

# SAM Language and Exact Program Semantics

Chapter II

A favorable graph decomposition is useful only if the program and its inputs retain their meanings. The formal language places source identity, operator type, state and history into one execution contract. Its real finite kernel is a substantial result in its own right; its explicit exclusions are part of the specification.

The account below follows the chapter source as retained at the fixed 8 September 2026 SAMA revision. Engine names, installed interfaces and campaign status are historical source statements. They do not describe a freshly inspected October deployment. Later source amendments are distinguished where available; no later engine or campaign is inferred.

## II.1 Overview

SAM Language is the semantic boundary between a mathematical grammar and an execution. It owns the parser, resolver, checker, type system, operator definitions, program registry and evaluator. A UI, wrapper or orchestration layer may transport a program, but it cannot acquire authority to recognize a phrase and run hidden scientific logic. Exact program meaning belongs to one declared language path.

That path carries more than a final state. It carries immutable source and program custody, the typed current state, a separately typed accumulated history, deterministic receipts and predecessor hashes. Two programs may end at the same exact state while retaining different route histories. A rejected operation must be atomic, and a tampered receipt must fail custody rather than be silently reinterpreted.

The evidence has three time layers that must remain distinct. SAM Language v0.6 is the preserved formal-simulator campaign whose G01–G16 records test the semantic and custody surfaces in detail. SAM Language v0.7 is the later typed 12-site reflective-tape candidate. Neither is the SLC version or the current SLC release. The live repository authority is `SLCQ2-RZ`; historical language and SLC labels remain source lineage, not current-pointer aliases.

## II.2 Opening question and conceptual picture

The governing question is:

> How does a source grammar become one exact, reproducible computation without allowing interfaces, profiles, mutable tapes or hidden dispatch paths to change its scientific meaning?

The required architecture is

- source text or immutable instruction tape
- parser
- resolver
- checker and type system
- operator/program registry
- evaluator
- typed current state + typed history + custody receipt

The forbidden alternative is semantic branching outside the language:

- UI or wrapper recognizes a phrase
- runs an unrelated calculation
- returns a value as if the language produced it

The first architecture makes source, meaning and result traceable. The second creates two semantic authorities and breaks provenance even if the returned number happens to match.

## II.3 Version and authority firewall

### II.3.1 v0.6: preserved formal-simulator evidence

The v0.6 campaign supplies the focused evidence in this document. It tests one semantic path, formal-profile isolation, the complete 12-site basis, frozen B and X1 operators, ordered-pair schedules, native decompositions, state/history separation, deterministic interfaces, atomic rejection, packaging and 23 wrong controls.

Its source statuses are `PASS` across G01–G16. Those statuses belong to the sealed v0.6 test surface. They are not an authorized SAMA result classification and do not make v0.6 the current language or SLC revision.

### II.3.2 v0.7: typed reflective-tape candidate

The Volume III source spine records v0.7 as a later typed, source-linked reflective SLC C1 tape candidate over twelve sites and 4,096 basis addresses. Its first-class types include

- SLCInstructionTape
- SLCTape12
- SLCTapeInspection

Six declared operations create an immutable zero-origin tape, append B/request/response records, execute it and inspect custody. The executor validates the zero-origin hash chain, dispatches through inherited exact state transitions and returns the same `SLCState12` dossier as the corresponding direct program.

The candidate remains bounded to a straight-line 12-site tape. It has no self-hosted parser, recursive source generation or general SAM-Language-to-Dense-Exact adapter. Tape custody and executed state remain different types.

### II.3.3 Source-recorded SLC: `SLCQ2-RZ`

The immutable Volume III spine contains historical SLC-current wording from its creation time. At the retained source stage, live authority controlled: `SLCQ2-RZ` is the sole frozen/default SLC, with its delegated arithmetic/application base separately typed. SAM Language, the SAM UI and SLC each have their own version lineage.

Accordingly:

- SAM Language v0.6 = preserved formal-simulator evidence
- SAM Language v0.7 = later typed reflective-tape candidate
- SLCQ2-RZ = sole source-recorded default SLC authority

No row in this document promotes one identity into another.

## II.4 Definitions and type boundaries

### II.4.1 Source, tape, state and history

- **source text.** Mutability: fixed for a declared execution; Semantic role: human-readable program origin.
- **instruction tape.** Mutability: immutable append/custody object; Semantic role: ordered typed program records and predecessor chain.
- **current state.** Mutability: transformed by accepted operators; Semantic role: exact endpoint dossier.
- **accumulated history.** Mutability: append-only typed receipt; Semantic role: ordered route, including distinctions erased at the endpoint.
- **inspection.** Mutability: read-only; Semantic role: custody and semantic trace readback.

Endpoint equality means

$$s^{(1)}_{final} = s^{(2)}_{final}.$$

It does not imply

$$h^{(1)} = h^{(2)}.$$

That distinction later supports Complete Exact Write and reciprocal history.

### II.4.2 The twelve-site address domain

Let a basis state be a twelve-bit vector

$$b = (b_0, \ldots, b_{11}), \qquad b_i \in \{0,1\}.$$

Its canonical integer address is

$$a(b) = \sum_{i=0}^{11} b_i 2^i,$$

so

$$0 \le a < 2^{12} = 4{,}096.$$

The finite address surface permits exhaustive basis, arity, range and site-ID checks.

### II.4.3 Operators and profiles

An operator is resolved by identifier through the registry, checked against a typed signature, and executed only after the profile admits it. The formal evidence profile is `FORMAL_CANDIDATE`; normal and research profiles are rejected in G03. This prevents an application mode from injecting semantics into a formal campaign.

### II.4.4 Exact receipt

A receipt binds at least the declared program, predecessor lineage, exact state, history and operator trace. The relevant invariants are byte-stable serialization, deterministic ordering and exact coefficients. Recomputing a hash for a mutated object does not authorize it; predecessor custody is part of the chain.

## II.5 From source custody to one semantic path

### II.5.1 G01 — seal the parent and candidate delta

Before executing semantics, G01 inventories:

- parent files 92
- frozen-source files 150
- candidate executable files 97

Every declared hash matches, no mismatch is reported and the parent remains unchanged. The candidate delta is therefore distinguishable from inherited source. Filename equality or an unsealed working tree would not provide the same custody.

### II.5.2 G02 — traverse the only authority path

G02 checks that parser, checker, evaluator, type system and registry exports resolve to one authority. A representative mixed program emits the trace

- RESOLVE
- `QP_ORDERED_PAIR`
- `SLC_ZERO_REGISTER`
- `SLC_PREPARE_REQUEST`

Operator digits resolve through the registry. No legacy sidecar or source-text dispatch is present. The trace matters because a matching endpoint produced through a hidden path would not be the same semantic result.

### II.5.3 G03 — isolate the formal profile

The same path is then invoked under three profiles. `FORMAL_CANDIDATE` is accepted; normal and research profiles are rejected. Thus the formal evidence does not borrow undeclared application or exploratory behavior.

## II.6 Build and exhaust the formal state surface

### II.6.1 G04 — complete basis construction

G04 enumerates all `2^12=4,096` addresses. It verifies the exact dimension, site-ID order and binary range, then rejects

wrong arity, out-of-range values, wrong register size and unknown/reordered site identifiers.

This establishes a finite formal address space. It does not identify the twelve sites with physical hardware or assign coupling magnitudes.

#### II.6.2 G05 — exact canonical-state invariant

The evaluator returns an exact sparse state. G05 checks:

- canonical sparse-key order;
- integer coefficients;
- the engine contract hash; and
- exact norm.

Reordering terms before hashing, converting coefficients to floats or using an undeclared norm would change the contract even if a display looked similar.

#### II.6.3 G06 — freeze B request and X1 response

G06 installs B prepare/request and X1 response as differently typed operators. For every declared basis column it verifies frozen hashes, matrix shapes, prepare columns, response mapping and involution. B contains a binary flip but is not identical to the flip; request and response cannot be exchanged.

This is the formal premise used by D39. It defines executable operators inside the candidate without assigning a physical B-to-X1 causal weld.

### II.7 Complete reciprocal and decomposition schedules

#### II.7.1 G07 — ordered-pair lift

Twelve sites produce

$$12 \cdot 11 = 132$$

ordered non-self placements. Each placement acts on all 4,096 basis states in both directions:

$$132 \cdot 4{,}096 = 540{,}672$$

forward cases and the same number in reverse. Therefore

$$2 \cdot 132 \cdot 4{,}096 = 1{,}081{,}344.$$

G07 closes all 1,081,344 classicalized schedule cases. Using unordered pairs, one direction or a sample would be a different schedule.

#### II.7.2 G08 — request sign and gauge diagnostics

The four frozen real request candidates `REQ02`, `REQ08`, `REQ09` and `REQ15` are each tested on all 132 placements:

$$4 \cdot 132 = 528.$$

All 528 placement cases pass, leaving four surviving gauges under the installed positive-first-row convention. The diagnostic therefore preserves ambiguity among the four real gauges instead of selecting one from a desired downstream result.

#### II.7.3 G09 — every declared native decomposition

The grammar generates five factor-shape families with exact assignment counts:

- **`1`×`12`.** Assignments: 1.
- **`3`×`4`.** Assignments: 5,775.
- **`4`×`3`.** Assignments: 15,400.
- **`6`×`2`.** Assignments: 10,395.
- **`12`×`1`.** Assignments: 1.

Their sum is

$$1 + 5{,}775 + 15{,}400 + 10{,}395 + 1 = 31{,}572.$$

All 66 rooted-forest parent cases also pass, and the two induction modes agree. The decomposition is native to the declared grammar rather than an external factorization imported after execution.

#### II.7.4 G10 — peel and restore the extra edge

G10 executes 204 exact controls:

- $1 \times 12$ : 132 cases
- $3 \times 4$ : 36 cases
- $4 \times 3$ : 24 cases
- $6 \times 2$ : 12 cases
- total: 204 cases

For each case, the declared extra edge is peeled, the exposed native decomposition is evaluated, the same edge is restored, and the original terminal state returns exactly. Restoring a different edge or accepting graph isomorphism without exact state equality would weaken the control.

#### II.7.5 G11 — frozen classical-limit boundary

G11 checks all eight forward/reverse two-bit rows in the frozen classical limit. Every row matches. The result is deliberately bounded: a complete classical table is not evidence that complex-phase or general quantum semantics have been installed.

### II.8 State, history and interface custody

#### II.8.1 G12 — same endpoint, different route

Chain and star programs produce the same exact coefficients and the same terminal-state hash, but their history hashes differ. In symbols,

$$s_{chain} = s_{star}, \qquad \mathrm{hash}(h_{chain}) \neq \mathrm{hash}(h_{star}).$$

The component partition is not used to erase the route. This is the formal reason an endpoint-returning computation can retain a nontrivial receipt.

#### II.8.2 G13 — deterministic API and CLI receipts

The same declared program is executed repeatedly through API and CLI. Both interfaces are byte-deterministic, their receipts match each other, the CLI exits successfully and no traceback occurs. Interface adapters therefore do not introduce independent semantics or serialization defaults.

#### II.8.3 G14 — atomic rejection and tamper detection

G14 mutates coefficient, exponent, operator argument, predecessor hash and receipt content. Every mutation is detected. Invalid operations reject atomically, leaving no accepted prefix or partially changed state.

This is a semantic property as well as a custody property: malformed input does not acquire meaning merely because its first tokens were valid.

#### II.8.4 G15 — inherited and new replay on a clean wheel

The candidate suite contains

$$154 \text{ inherited} + 29 \text{ new} = 183 \text{ tests}.$$

All 183 pass. The clean wheel contains 37 members, and 13 CLI cases pass with API/CLI parity, no sidecar or UI dependency, no repository-path dependency and no traceback. This retest establishes that the candidate package preserves the parent while carrying the new semantics outside the source checkout.

#### II.8.5 G16 — keep the wrong controls and open boundaries

All 23 declared wrong controls produce their diagnostic outcomes. The release also retains five open domains:

1. complex-phase semantics;
2. measurement and publication semantics;
3. physical connectivity;
4. coupling magnitude and cost; and
5. hardware realization.

A passing formal candidate remains scoped because these boundaries remain in the artifact rather than being erased after the regression run.

### II.9 The v0.7 reflective-tape semantics

The v0.7 candidate makes the instruction sequence itself typed data. A typical execution can be understood as:

1. create a zero-origin `SLCTape12` and record its origin hash;
2. append a typed B/request record whose predecessor is the current tape tip;
3. append a typed X1/response record under the same custody law;
4. seal the straight-line tape;
5. validate the complete predecessor chain;
6. dispatch each record through the inherited exact transition;
7. return `SLCState12` and a separate `SLCTapeInspection`.

If the direct program and tape program contain the same declared operations, their current-state dossiers match. Their custody still arrives through different source objects, so the tape is not replaced by the endpoint.

This reflective layer prepares exact program composition and history without making the language self-hosting. Recursive source generation, a general parser inside the tape and a complete typed-IR-to-Dense-Exact adapter remain open.

### II.10 Source-native N100-to-F81 compiler boundary

G1 asks a different question from G01–G16: whether the exact language and N100 source rows already carry a complete elementwise semantic map to F81. The audit finds:

- 100/100 typed QP source keys
- 72/100 Shared72 aggregate connector-incidence rows
- 162 CR211 address rows
- 0 CR211 QP elementwise assignments
- 81 F81 semantic sites
- 10 F81 contact classes
- 0 N100 rows with a complete elementwise F81 map

The missing relation is

- QP candidate/template
- F81 semantic site or source-native F81 address class.

No outcome or residual selects a map. The contact lane is not executed and the status remains `PARTIAL_DOMAIN__ELEMENTWISE_F81_JOIN_OPEN`.

The boundary shows why formal determinism and semantic completeness are different. The language can parse, type, execute and receipt a program while a particular source-to-target scientific join remains absent.

### II.11 Controls, deviations and promotion discipline

The full chain prevents several distinct deviations:

- **parent source changes under the candidate.** Detecting evidence: G01 hashes and inventories; Preserved correction or boundary: parent remains sealed; candidate delta stays explicit.
- **hidden semantic dispatch.** Detecting evidence: G02 path and trace; Preserved correction or boundary: one parser/checker/evaluator/registry path.
- **profile leakage.** Detecting evidence: G03; Preserved correction or boundary: formal candidate accepts only its declared profile.
- **malformed address gains meaning.** Detecting evidence: G04; Preserved correction or boundary: range, arity, size and ID rejection.
- **request and response collapse.** Detecting evidence: G06; Preserved correction or boundary: frozen hashes, shapes and typed mappings.

- **sampled schedule called complete.** Detecting evidence: G07; Preserved correction or boundary: all 1,081,344 cases executed.
- **preferred gauge selected after outcome.** Detecting evidence: G08; Preserved correction or boundary: all four survivors preserved.
- **endpoint equality erases history.** Detecting evidence: G12; Preserved correction or boundary: equal state hash, unequal history hash.
- **interface changes semantics.** Detecting evidence: G13; Preserved correction or boundary: byte-identical API/CLI receipts.
- **partial mutation survives rejection.** Detecting evidence: G14; Preserved correction or boundary: atomic rollback and tamper detection.
- **source-tree dependency hidden by tests.** Detecting evidence: G15; Preserved correction or boundary: clean-wheel replay.
- **open physical domains disappear after PASS.** Detecting evidence: G16; Preserved correction or boundary: 23 wrong controls and five boundaries retained.
- **aggregate incidence becomes elementwise mapping.** Detecting evidence: G1; Preserved correction or boundary: zero assignments and unexecuted contact lane preserved.

Passing validation is not promotion. v0.6 stays a preserved formal-simulator surface, v0.7 stays a reflective-tape candidate, and SLC Recorded authority is resolved through the live `SLCQ2-RZ` pointer rather than a historical filename.

### II.12 Established result and current boundaries

The source-bound language result is:

1. one declared parser/resolver/checker/type/registry/evaluator path owns exact program semantics;
2. twelve sites give a complete 4,096-address finite basis;
3. B request and X1 response are frozen, differently typed operators;
4. all 1,081,344 ordered-pair schedule cases and all 31,572 declared native decompositions close under their respective tests;
5. current state, immutable source tape and accumulated history remain separate;
6. API and CLI produce deterministic identical receipts;
7. malformed operations reject atomically and declared tampering is detected;
8. 154 inherited plus 29 new tests pass in the clean candidate wheel;
9. 23 wrong controls pass while complex phase, measurement/publication, physical connectivity, coupling/cost and hardware realization stay open;
10. v0.7 supplies a later typed reflective tape but not general Dense-Exact integration; and
11. the N100-to-F81 elementwise semantic join remains open and unexecuted.

The focused G artifacts carry source status `PASS` or the explicit partial G1 status. The registered atomic records assign no authorized project result classification to D35, so this revision does not create one.

### II.13 Connections and forward handoff

- **`SAMA-D000034.`** Connection: Supplies the N100 source grammar and exact structure-first certificates that language must serialize..
- **`SAMA-D000038.`** Connection: Receives immutable programs, signed `Z4^9` operations, separate state/history and exact readback for Complete Exact Write..
- **`SAMA-D000039.`** Connection: Uses the typed B request, X1 response and W9 resolution route..
- **`SAMA-D000042.`** Connection: Receives the separate history receipt for directional event-reel computation..
- **`SAMA-P000006.`** Connection: Uses the language as the computation interface for particle grammar, Starbreaker and binding..

The forward handoff is to Complete Exact Write: compile a declared signed program to its exact group action, apply it without mutating the tape, and retain a reconstructible history even when the endpoint returns.

### II.14 Sources and evidential scope

This chapter is assembled from the following fixed source manuscripts. Source-reported campaign outcomes retain their original scope; no new campaign was executed for this edition.

- Sam language and exact program semantics.

SAM VOLUME III

# Exact State Spaces and Thermodynamics

Chapter III

This chapter moves from one graph to the family of exact methods that may evaluate a declared state-space object. It then follows the resulting density of states into dimensionless thermodynamics. The distinction between an exact numerical object and a physically calibrated energy is maintained throughout.

The account below follows the chapter source as retained at the fixed 8 September 2026 SAMA revision. Engine names, installed interfaces and campaign status are historical source statements. They do not describe a freshly inspected October deployment. Later source amendments are distinguished where available; no later engine or campaign is inferred.

## III.1 Overview

An exact state-space computation answers a stronger question than an optimizer: not merely which assignment has the lowest energy, but how many assignments occupy every possible energy. For a declared binary Hamiltonian this complete answer is the density of states, or DOS. Its coefficients are integers, their sum is the complete formal configuration count, and their arrangement retains the energy organization created by the graph, couplings and fields.

The computation can be exact without visiting every assignment one by one. Gray enumeration, variable elimination, component convolution, modular polynomial arithmetic, orbit sectors and rank-aware readout are different routes to the same requested object. Their right to be called exact comes from reconstruction and equality obligations: coefficientwise agreement, state- count closure, moments, symmetries, port marginals, hashes and independent readers. Runtime is useful evidence about an execution route; it is not the definition of exactness.

This distinction has a second consequence. Exact mathematics and executable custody are separate obligations. SLCX023 retained exact exhaustive and NTT/variable-elimination authorities but failed its frozen same-run custody gates. That failure remains part of the record. SLCX023A corrected the custody mechanism in a separate successor, reproduced the unchanged coefficient authorities and left the failed parent immutable. The chain demonstrates the Courtroom method at its best: retain what failed, identify the type of failure, change only the defective layer, replay the original question and preserve both receipts.

The chapter ends by deriving exact thermodynamic readers from the frozen N96 DOS. Those readers are exact and dimensionless while the physical energy scale remains symbolic. A polynomial can determine partition functions and moments without thereby choosing joules, kelvin, a contact orientation or a physical binding map.

## III.2 Opening question and conceptual picture

The governing question is:

> How can a finite computational system account for every formal assignment, preserve the exact integer answer under different execution routes, and carry that answer into thermodynamics without importing an undeclared physical scale?

The full chain is

- declared typed grammar and Hamiltonian
- exact integer DOS g(E)
- independent route equality and state-count closure
- custody-qualified frozen artifact
- exact polynomial P(x)
- exact partition, moment and response readers
- symbolic-scale physical boundary

Every arrow changes the representation or the question, not the source object. A component factorization may reduce work but may not reduce the state space being counted. CRT may store residues but must reconstruct the original integers. A thermodynamic reader may turn coefficients into rational functions or numerical display values but may not retroactively change the DOS. A physical unit conversion may occur only after a source-authorized scale exists.

## III.3 Typed definitions and domains

### III.3.1 Source grammar, assignment and Hamiltonian

Let $V = \{1, \ldots, N\}$ be the declared independent binary sites, let $E_G \subseteq V \times V$ be the declared interaction edges, and let

$$\sigma = (\sigma_1, \ldots, \sigma_N) \in \{-1, +1\}^N.$$

For integer edge weights $J_{ij}$ and integer local fields $h_i$, the Ising-type Hamiltonian used here is

$$H(\sigma) = \sum_{(i,j)\in E_G} J_{ij}\sigma_i\sigma_j + \sum_{i\in V} h_i\sigma_i.$$

The types are load-bearing:

- $N$ counts independent binary coordinates;
- $E_G$ is topology, not an energy histogram;
- $J_{ij}$ and $h_i$ are declared Hamiltonian coefficients;
- a native grammar-row value does not become $J_{ij}$ merely because it is numerical; and
- a constrained occurrence ledger is not automatically an independent $N$-spin state space.

This is why the N100 compiler keeps source-row selection separate from Hamiltonian coefficients and why later M126/N144/L162 readouts retain their rank and sidecar types.

### III.3.2 Density of states

For each attainable integer energy $e$, define

$$g(e) = \#\{\sigma \in \{-1,+1\}^N : H(\sigma) = e\}.$$

The function $g$ is an exact integer ledger. Summing over all energies partitions the complete configuration space:

$$\sum_e g(e) = \sum_e \sum_\sigma \mathbf{1}[H(\sigma) = e] = \sum_\sigma 1 = 2^N.$$

This derivation is elementary but decisive. Every assignment has exactly one energy, so it contributes once. Frustration changes which energies occur and how assignments are distributed among them. It does not change the cardinality $2^N$ of the formal assignment domain.

A ground-state solver keeps only

$$e_0 = \min_\sigma H(\sigma), \qquad g(e_0),$$

whereas a DOS retains every pair ((e,g(e))). Interior bins, odd moments, response sectors and finite-temperature behavior would be lost if the output were reduced to an optimum.

### III.3.3 Exactness, execution and custody

Three types must not be collapsed:

- **Mathematical exactness.** Question answered: Are the requested coefficients the exact integers?; Required evidence: equality, closure, moments, symmetries, reconstruction.
- **Execution policy.** Question answered: How was the exact object evaluated?; Required evidence: route declaration, worker/batch receipts, performance data.
- **Artifact custody.** Question answered: Did the sealed code, inputs, receipts and outputs retain their declared identities?; Required evidence: hashes, domains, manifests, independent verification.

An exact tensor with a failed receipt is not a successful custody packet. A fast route with no equality proof is not established as exact. A hash-correct artifact does not become the right mathematical object unless the declared calculation also closes.

## III.4 Deriving the exact route family

### III.4.1 Gray exhaustive authority

The direct definition suggests enumerating all $2^N$ assignments. A Gray code orders them so adjacent assignments differ in one spin. If spin $k$ changes sign, only terms incident to $k$ change, giving the incremental update

$$\Delta H_k = -2\sigma_k \left( h_k + \sum_{j:(k,j)\in E_G} J_{kj}\sigma_j \right).$$

Thus one full energy need not be recomputed from scratch at every state. The route is still exponential in the number of visited assignments, but for small $N$ it is transparent and supplies an independent authority against more structured routes. In SLCX023, `GRAY_EXHAUSTIVE` covers the complete address space through $N = 24$.

### III.4.2 Variable elimination

Write a local factor for every edge and field contribution. Eliminating a spin means summing its two values while retaining an exact table over the still-live neighboring boundary. If $B_k$ is that boundary and $z$ marks energy, a schematic elimination step is

$$F_{k+1}(\sigma_{B_k}; z) = \sum_{\sigma_k=\pm 1} z^{\Delta H_k(\sigma_k,\sigma_{B_k})} F_k(\sigma_k, \sigma_{B_k}; z).$$

The coefficients remain integers. Cost follows the largest induced boundary width $w$, roughly through tables of size $2^w$, rather than nominal arity alone. Different elimination orders therefore change work while leaving the Hamiltonian and the final polynomial fixed.

SLCX023 deliberately used two distinct exact versions: a reference NTT/VE route with a frozen min-degree order and a typed NTT/VE route with an independently frozen cell-aware min-fill order. Their boundary tables were retained so equality could be checked below the final scalar output.

### III.4.3 Exact component convolution

Suppose the declared graph splits into independent components $G_1, \ldots, G_m$ after conditioning on an explicit boundary. If component $a$ has DOS polynomial

$$P_a(z) = \sum_e g_a(e) z^e,$$

then additivity of energy gives

$$P_G(z) = \prod_{a=1}^{m} P_a(z),$$

and therefore

$$g_G(e) = \sum_{e_1+\cdots+e_m=e} \prod_{a=1}^{m} g_a(e_a).$$

This is not a sample or truncation. Every combination of component assignments appears once in the convolution. At $z = 1$, closure is inherited:

$$P_G(1) = \prod_a P_a(1) = \prod_a 2^{N_a} = 2^{\sum_a N_a}.$$

The N100 structure-first calculation in `SAMA-D000037` uses this exact identity. The same logic also explains why a reduced work count can still return a complete $2^N$ census.

### III.4.4 NTT lanes and CRT reconstruction

Large integer polynomial products can be evaluated modulo primes. For a suitable prime $p_ell$, an NTT transforms coefficients, multiplies pointwise and applies the inverse transform, all in exact modular arithmetic. One lane returns

$$g(e) \bmod p_\ell.$$

With pairwise-coprime primes $p_1, \ldots, p_L$, CRT returns the unique integer coefficient modulo

$$M = \prod_{\ell=1}^{L} p_\ell.$$

If the declared coefficient bound is less than $M$ (or less than $M/2$ for a signed reconstruction), the original integer is unique. The modular lanes are therefore an exact representation of the coefficient only when the prime product, reconstruction convention and bound all close. A displayed floating-point polynomial is downstream and cannot replace these obligations.

### III.4.5 Orbit sectors and rank-aware readout

When a dense graph is uniform under a declared permutation action, assignments can be grouped by marked population. A sector with $k$ marked sites has multiplicity

$$\binom{N}{k},$$

provided the Hamiltonian is constant on that orbit or the remaining typed coordinates are retained. This replaces assignment traversal with an exact orbit census.

Likewise, an ambient readout with $m$ slots may have only $r < m$ independent coordinates. The exact route evaluates the $2^r$ independent domain and applies the declared readout map. It must not report the ambient $2^m$ ceiling as independent state count. This distinction becomes critical for N144 independent state and L162 constrained occurrence views.

### III.4.6 Route-acceptance checklist

Two routes may differ internally and still compute the same object. Their equivalence is established through a layered checklist:

1. identical occupied-energy support;
2. identical integer coefficient in every bin;
3. $sum_e g(e) = 2^N$;
4. matching exact raw moments $\sum_e e^k g(e)$;
5. declared parity, complement and role-permutation symmetries;
6. matching retained-port marginals and conditional tensors;
7. stable semantic hashes; and
8. an independent reconstruction or reader.

A runtime comparison is interpreted only after these checks establish same-object, same-output work.

## III.5 Deviation chain: SLCX023 to SLCX023A

### III.5.1 The frozen question

SLCX023 asked whether a frozen exact-DOS development ladder could close across six instances of sizes (12,24,36) using independent Gray, reference NTT/VE and typed NTT/VE authorities while also preserving execution and receipt custody. The equality standard was exact: same occupied energies and same integer coefficient in every bin.

The campaign also contained eight deliberately wrong controls and eighteen metamorphic checks. Wrong controls were not noise to be deleted. They were expected-invalid transformations used to demonstrate that the validation surface could distinguish the declared computation from plausible but wrong alternatives.

### III.5.2 Preserved failure

The run completed all six instances, detected all eight wrong controls and passed all eighteen metamorphic checks. Yet its frozen verdict remained

- SLCX023_FAIL__EXACT_DOS_MISMATCH_OR_CUSTODY

because the first coefficient, custody, invariant or execution failure was a stop condition and same-run repair was forbidden. The historical G02 and G11 custody gates failed. This typed outcome matters: the available exact lane structure did not authorize rewriting a failed executable receipt as a pass.

The parent preserved its precommit, executable and execution-ledger hashes and recorded `same_run_repair\=\false`. The failed packet therefore remained reconstructible.

### III.5.3 Correction hypothesis

The appeal did not change the scientific question, Hamiltonians, route definitions or coefficient authorities. It isolated two implementation and custody mechanisms:

- execute the official retest from an isolated retained capsule whose runner, adapter and engine bytes are checked before and after work; and
- distinguish the external lane domain of legacy receipts from the boundary-table schema recorded by corrected receipts.

This is a narrow correction. It predicts that the unchanged scientific outputs will reproduce while the custody gates now close.

### III.5.4 Separate retest and cumulative record

SLCX023A executed as a successor rather than modifying SLCX023. It reproduced all 16 of 16 lane-instance DOS hashes, passed 154 of 154 exactness checks, 64 of 64 moment checks and 18 of 18 metamorphic controls. Twelve of those

metamorphic controls reran transformed solvers and six diagnosed cell repartitioning. All 432 corrected receipts verified with their domain and boundary-table schema separated and with signed links back to the legacy hashes.

The complete chain is therefore

- SLCX023 frozen design
- exact lanes execute
- custody G02/G11 do not close
- parent remains FAIL and immutable
- SLCX023A precommits two custody corrections
- unchanged 16/16 scientific hashes reproduce
- exactness, moment, metamorphic and 432 receipt checks pass
- corrected successor supplements the parent

No hidden $N = 48$ or $N = 60$ instance was opened. The appeal did not install a SAM Language operator or create a physical Ising/gravity claim.

## III.6 Frozen N96 worked example

### III.6.1 Complete configuration count

For $N = 96$,

$$2^{96} = 79{,}228{,}162{,}514{,}264{,}337{,}593{,}543{,}950{,}336.$$

The frozen H14C computation completed four CRT primes and 1,024 root-batch checkpoints. It retained sixteen port rows, each summing to $2^{92}$, hence

$$16 \cdot 2^{92} = 2^4 \cdot 2^{92} = 2^{96}.$$

The scalar DOS occupies 348 energy bins from $-346$ through 348, with ground-state degeneracy 12. State-count, support, parity and raw-moment checks one through four passed. Both active $512 \times 512$ fiber factors remained installed and all four scalar marginals equal the exact N96 result.

The execution completed in 40,307.938817 seconds with the frozen DOS semantic hash `d74fb93facbebf18fa3cfe3ab2d33a0ec013f27b7ab6f326414bd5cdce4a11cb`. The later H14F profile reproduced that DOS hash under a different execution policy. That comparison concerns work routing; it does not replace the N96 Hamiltonian or create a second physical model.

### III.6.2 Shifted exact polynomial

The energy spacing is two, so write

$$E_k = -346 + 2k, \qquad 0 \le k \le 347,$$

and define

$$P_{96}(x) = \sum_{k=0}^{347} g(E_k) x^k.$$

Then

$$P_{96}(1) = \sum_k g(E_k) = 2^{96},$$

with $a_0 = 12$ and $a_{347} = 7$. The degree records the complete occupied energy span after shifting; its coefficients remain the exact degeneracies.

## III.7 From DOS to exact thermodynamics

### III.7.1 Partition function derivation

Let the dimensionless inverse scale be $b = \beta \varepsilon_E$, where $epsilon_E$ remains symbolic, and set

$$x = e^{-2b}.$$

The partition function is

$$Z(b) = \sum_k g(E_k) e^{-bE_k} = \sum_k g(E_k) e^{-b(-346+2k)}.$$

Since $e^{346b} = x^{-173}$ and $e^{-2bk} = x^k$,

$$Z_{96}(x) = x^{-173} P_{96}(x).$$

This factorization separates the exact coefficient ledger from the chosen thermodynamic reader.

### III.7.2 Euler derivatives and moments

Let

$$\Theta = x \frac{d}{dx}.$$

Then

$$\Theta P = \sum_k k g(E_k) x^k, \qquad \Theta^2 P = \sum_k k^2 g(E_k) x^k.$$

The mean shifted index is $\langle k \rangle = \Theta P / P$. Because $E = -346 + 2k$,

$$\langle E \rangle = -346 + 2 \frac{\Theta P}{P},$$

and

$$\mathrm{Var}(E) = 4 \left( \frac{\Theta^2 P}{P} - \left( \frac{\Theta P}{P} \right)^2 \right).$$

Higher raw moments follow by expanding $(-346 + 2k)^m$ and replacing $k^j$ with $\Theta^j P / P$. At $x = 1$, T1 records

$$\langle E \rangle = 0, \quad \langle E^2 \rangle = 1286, \quad \langle E^3 \rangle = 534, \quad \langle E^4 \rangle = 4{,}952{,}792.$$

The nonzero third moment retains an exact asymmetry hidden by the zero mean. The microcanonical mode is $E = 0$ with

$$g(0) = 1{,}761{,}659{,}846{,}545{,}153{,}896{,}312{,}195{,}809.$$

At the positive low-temperature end, the reader approaches $E = -346$, zero variance and entropy $\log 12$. At $x = 1$, it returns mean zero, variance 1286 and entropy $96 \log 2$.

### III.7.3 Conditional response and the open/closed boundary

T2 refines the scalar polynomial into a frozen $4 \times 5 \times 1181$ integer attachment. Every q marginal reconstructs T1. There are twenty response-sector polynomials, ten active and ten inactive, and eight distinct signatures including zero.

The closed trace retains three q signatures and merges $q = 1$ with $q = 3$. The open retained port retains four routing signatures and all five response values. Thus a closed equality cannot be copied backward as an open-contact identity. At $x = 1$, $q = 2$ has exact energy-response covariance 8, while $q = 1$ and $q = 3$ have zero linear covariance but nonzero mutual information. The conditional structure survives beyond a scalar covariance summary.

## III.8 Wrong controls and type firewalls

The useful wrong controls answer specific failure questions:

- **Deterministic count permutation.** What it preserves: $2^{96}$, coefficient multiset; What it changes or violates: moment vector and energy placement.
- **Exact sign reflection.** What it preserves: even moments; What it changes or violates: signs of odd moments.
- **Rational uniform spectrum.** What it preserves: support and total weight; What it changes or violates: integer state-count semantics and moment vector.
- **C4-to-C2 quotient.** What it preserves: some aggregate response structure; What it changes or violates: distinct $q = 0$ and $q = 2$ lanes.
- **Factorized response control.** What it preserves: response totals and scalar marginals; What it changes or violates: actual $q = 1, 2, 3$ joint structure.
- **Open-port/closed-trace comparison.** What it preserves: shared source object; What it changes or violates: routing availability versus final occupancy.

These controls demonstrate why closure alone is necessary but insufficient. Many wrong objects can sum to $2^N$. Exactness belongs to the coefficient ledger, its typed marginals and its independent reconstruction together.

The physical firewall is equally explicit:

- exact integer DOS != physical energy calibration
- symbolic epsilon_E != measured joules
- dimensionless b != temperature in kelvin
- q_A source support != epsilon_E
- A(r) accumulation lift != Ising coupling J_ij
- open routing signature != closed-trace occupancy

BIPM constants may convert units after a physical scale is supplied; they do not choose the scale. The current frozen `SLCQ2-RZ` selection engine delegates byte-identical arithmetic/application work to the frozen `SLCV21R` base. The historical N96, H14C/H14F, Exact Write and thermodynamic artifacts retain their scoped results, but none replaces `SLCQ2-RZ` as the source-recorded default SLC.

## III.9 Established result, boundary and forward handoff

The established mathematical object is the complete exact DOS, not only a ground sector. Multiple exact routes may evaluate it, but all must reconstruct the same integer coefficients and close the same typed marginals. The frozen N96 release accounts for every one of $2^{96}$ assignments, and its exact polynomial supports exact scalar and conditional thermodynamic readers.

The preserved deviation result is equally important: SLCX023 remains the failed custody parent; SLCX023A is the separate correction and retest. The successor repairs executable custody without rewriting the earlier run or changing the scientific tensor.

The forward handoff has three branches:

1. `SAMA-D000037` shows how source grammar becomes an exact work certificate while preserving the full state space and H14F fallback.
2. `SAMA-D000038` develops the exact write transaction and compiled group-law execution surface.
3. A future direct evidence route must register SLCX023/SLCX023A, N96, T1–T6A and current `SLCQ2-RZ` artifacts before those sources can populate a qualified SAMA test crosswalk. Their absence is recorded, not filled with invented keys.

## III.10 Sources and evidential scope

This chapter is assembled from the following fixed source manuscripts. Source-reported campaign outcomes retain their original scope; no new campaign was executed for this edition.

- Exact state spaces and slc computation.

SAM VOLUME III

# Exact joint densities at N5000

Chapter IV

## IV.1 The completed object

We computed the complete joint density of energy, magnetization and four retained spins for six specified complete Ising graphs with 5,000 spins. Every unordered pair interacts, so each graph contains 12,497,500 nonzero pair couplings. The six sources combine three packet constructions with positive or negative uniform fill. All energies, all magnetizations and all sixteen assignments of the retained spins are included.

The primary result contains 1,530,259,992 nonzero records. Each record gives an energy, a magnetization, a retained-spin assignment and an exact integer count of configurations realizing those coordinates. A record is a counted bin, not one individually visited configuration. The counts together account for all $2^{5000}$ spin assignments for each source. Holding the four retained spins fixed leaves $2^{4996}$ assignments in each boundary sector.

The computation also retained a second encoding of the same joint measure. Six sources times sixteen boundary assignments give 96 primary tables; two exponent encodings give 192 logical output rows and 96 matched encoding pairs. All pairs agree on the canonical joint digest, record count and source-derived moments. This is a completed finite enumeration represented by exact generating functions. Its scientific value lies in retaining the full correlation among energy, magnetization and the boundary, so later questions can be answered from the computed object.

Each row includes all sixteen boundary assignments in one encoding. The second encoding reproduces the same row totals. The correction bound measures the total absolute departure from the uniform fill after that fill has been removed; it does not count the complete graph's physical interactions. The small packet/+1 table is the exact uniform ferromagnetic control. Its counts depend only on magnetization and the fixed boundary. [N5-6, N5-7]

## IV.2 What the six sources contain

A packet is a finite graph with fifteen spins arranged in three complete five-spin layers. Perfect matchings connect consecutive layers. One full packet has forty parent edges: thirty within its layers and ten in the two matchings. At N=5000, 333 full packets and one terminal five-spin packet give 13,330 unbridged parent edges. The terminal packet preserves the actual remainder of the construction.

The packet family assigns positive couplings to these parent edges. The signed-packet family changes one edge in every five-spin layer to negative, giving 1,000 negative parent edges and 12,330 positive edges. The signed-packet chain adds 333 gateway bridges with alternating signs: 167 positive and 166 negative. For each family, all pairs absent from the parent receive a uniform fill $J_0 = +1$ or $J_0 = -1$; parent edges retain their signs. Thus the graph is physically complete even though its departure from the uniform fill has a small-component structure.

The retained spins have the ordered labels $\mathscr{P} = (0, 1, 60, 61)$. A boundary mask $b$ from 0 to 15 records their signs: bit $a$ is one when spin $\mathscr{P}_a$ is positive. Retaining the ordered mask makes the output conditional on an explicit physical boundary rather than only on its number of positive spins. Exact duplicate output files can share storage where the source proves equality; their boundary identities remain separately recorded. [N5-2, N5-7]

## IV.3 Separating the collective interaction

A spin takes a value $s_i \in \{-1, +1\}$. Energy is the negative signed sum of pair products, and magnetization is the sum of the spins. With no local fields in these sources,

$$E(s) = -\sum_{i<j} J_{ij} s_i s_j, \qquad M(s) = \sum_i s_i.$$

The joint density $g(E, M, b)$ counts the assignments having those three coordinates. The uniform fill can be handled without materializing every pair during counting. Squaring the total magnetization gives

$$M^2 = N + 2\sum_{i<j} s_i s_j.$$

Writing each pair coupling as $J_{ij} = J_0 + \Delta J_{ij}$ therefore gives the exact identity

$$E = E_c - \frac{J_0}{2}(M^2 - N), \qquad E_c = -\sum_{i<j} \Delta J_{ij} s_i s_j.$$

The dense term is a function of one collective coordinate, $M$. All remaining configuration dependence is in the correction graph. A source edge equal to the fill has zero correction. A source edge opposite to it has correction +2 or -2. This identity changes the representation of the exact Hamiltonian; it does not discard any pair coupling or approximate its contribution.

For counting, let $k = (M+N)/2$ be the number of positive spins and $B = \sum_{i<j} |\Delta J_{ij}|$ the total absolute correction. The shifted correction index $t = (E_c + B)/2$ is an integer. The original coordinates are recovered by

$$M = 2k - N, \qquad E = 2t - B - \frac{J_0}{2}\left[(2k-N)^2 - N\right].$$

The method must retain $k$ jointly with $t$. Summing over magnetization too early would destroy the association needed to restore the dense $M^2$ term. This is why the result is a joint density rather than merely a correction-energy histogram. [N5-2, N5-5]

| Source family | Uniform fill | Primary nonzero records | Correction bound B | Lattice step d |
|---|---|---|---|---|
| Packet | +1 | 79,952 | 0 | 1 |
| Packet | -1 | 461,138,212 | 26,660 | 2 |
| Signed packet | +1 | 63,968,000 | 2,000 | 2 |
| Signed packet | -1 | 461,223,474 | 24,660 | 2 |
| Signed packet chain | +1 | 76,796,480 | 2,332 | 2 |
| Signed packet chain | -1 | 467,053,874 | 24,994 | 2 |
| Total | | **1,530,259,992** | | |

### IV.4 Local tables and exact composition

Within a component, each permitted spin assignment contributes a monomial $x^k y^t$. Its coefficient counts how often those local coordinates occur. Local tables retain gateway spins until their incident bridges are accounted for. Multiplying compatible component polynomials adds positive-spin counts and correction indices, while multiplying their exact integer multiplicities. Summing the gateway assignment joins the compatible cases.

The alternating bridge construction is particularly useful. A bridge whose sign equals the fill has zero correction, so it disappears from the correction graph. The remaining correction components contain at most two complete packets. They are assembled from gateway tables rather than by enumerating all $2^{30}$ assignments of a two-packet component. Repeated components share the same source-bound polynomial, and repeated powers and grouped boundary roots remove repeated arithmetic.

This produces one polynomial $G_b(x,y)$ for each retained boundary. Its coefficient at $(k,t)$ is the desired configuration count after the coordinate decoding above. The correctness argument is constructive: each local assignment appears once in its local table, disjoint component products form every compatible global assignment, and gateway summation closes each interface once. No global configuration is sampled probabilistically. [N5-2, N5-3]

### IV.5 Two encodings and exact GPU arithmetic

The two-variable polynomial is packed into a one-variable exponent using its known coordinate bounds. A common lattice step $d$ divides the correction indices. Put $u = t/d$ and $T = B/d$. The two maps are

$$\phi_K(k,u) = k(T+1)+u, \qquad \phi_T(k,u) = u(N+1)+k.$$

Both are injective on the admitted coordinate rectangle. Division with remainder restores the original coordinates, and output decoding rejects coordinates outside the declared bounds. The two layouts make the same joint counts take different coefficient positions and multiplication layouts. Their agreement tests packing, reconstruction and decoding at the scale of the completed object.

The GPU route multiplies integer polynomials using number-theoretic transforms modulo suitable primes. It then reconstructs the nonnegative integer coefficients by the Chinese remainder theorem. The product of the selected moduli is strictly greater than the full coefficient bound. Each reconstructed coefficient therefore has a unique value in the admitted interval. The stored counts are exact integers; floating-point transforms are not used for them.

N5000 required an enlarged reconstruction buffer. The successor backend admits eighty 64-bit limbs, or 5,120 bits; the N5000 route uses seventy-nine limbs. The serialized count field has 626 bytes, followed with its exact energy and magnetization coordinates to give a 634-byte raw record. The resource planner admits the transform arrays, host output arrays, writer concurrency and disk reserve before the full run. This mattered because exact coefficient growth and output volume remain real costs even when the source factorization is compact. [N5-1, N5-2, N5-4, N5-5]

### IV.6 Validation of the completed density

Each boundary table satisfies conditional magnetization closure. If $r(b)$ is the number of positive retained spins, the count at fixed total positive-spin number $k$ must be

$$\sum_E g(E, 2k-N, b) = \binom{N-4}{k-r(b)}.$$

The writer checks this identity for every admitted $k$. It is stronger than checking only the total $2^{4996}$ because it constrains every magnetization marginal. The energy moments are derived directly from the source couplings and fixed boundary, independently of the polynomial products. To see their content, write the fixed-boundary energy as a constant $c_b$, linear terms $a_v s_v$ on free spins, and pair terms $-J_{uv}s_u s_v$. Under the uniform distribution of the free spins, these distinct spin monomials are orthogonal. With $A = 2^{N-4}$, the exact checks are

$$\sum_{E,M} g = A, \qquad \sum_{E,M} Eg = Ac_b,$$

$$\sum_{E,M} E^2 g = A\left(c_b^2 + \sum_{v\notin\mathscr{P}} a_v^2 + \sum_{\substack{u<v\\ u,v\notin\mathscr{P}}} J_{uv}^2\right).$$

Here $c_b = -\sum_{u<v\in\mathscr{P}} J_{uv}s_u s_v$ and $a_v = -\sum_{p\in\mathscr{P}} J_{vp}s_p$. These source identities constrain the total mass, first energy moment and second energy moment without using the product coefficients as their answer key.

The host output stage writes the exact coordinates and integer multiplicities, checks their widths and support, computes canonical bucket digests, flushes the files, and rereads each complete durable raw file to confirm its hash. Production compares all 96 pairs of independently packed encodings on canonical digest, record count and moments. The preparatory qualification records 75 checks including CPU-equal coefficients at 3,000-, 4,000- and 5,000-bit widths and small source/encoding

cases. The largest packet/-1 boundary pilot passed in both encodings before admission of full production.

A cheap portable replay supplies an additional concrete comparison. Recomputing the packet/+1 control on the CPU reproduced all 32 encoding/boundary rows in 37.768777 seconds, matching the canonical and raw record hashes, moments and counts. That replay verifies the analytically simple control through the public route. It does not replace the full six-source production, whose other cases retain their own exact checks and paired-encoding records. [N5-1, N5-3, N5-5, N5-8]

### IV.7 The computation and its preserved data

Full production completed 192 logical rows and all 96 encoding pairs in 2,824.215 seconds, about 47 minutes. The recorded machine was an RTX PRO6000 Blackwell system with 97,887 MiB of device memory, about 27.2 CPU equivalents, 187,999,997,952 bytes of host RAM and a 4 TB container disk. The largest pilot recorded a process allocation of 83,785 MiB of GPU memory. These are the conditions of one completed execution, not a fitted general runtime law.

Before production, the source-bound compiler encountered the generic 4,096-variable admission ceiling. The completed successor normalized and authenticated the explicit frozen N5000 parent graphs while retaining the generic solver's existing admission policy. The failed preparation remains in provenance. Its relevance is the exact source-binding correction that enabled the finished run; it is not the result of the N5000 campaign.

Lossless zstd compression and byte deduplication reduced the completed files to 89 unique compressed contents. The archive is 609,671,270,157 bytes, compared with 1,045,145,366,546 distinct raw bytes. Compression took 1,260.611 seconds. Each compressed shard was decompressed and checked against its production raw hash. The manifest preserves every logical boundary alias, source, plan, record count, canonical digest and raw hash.

The frozen source-input restoration is a separate completed achievement in the chain. When an earlier backup path was absent, all sixteen original input files were reconstructed byte-identically from the retained source grammar and original construction receipts. The restored parent JSON, coupling bitstreams and manifests match their original SHA256 identities, including their original serialization details. No N5000 bulk output was lost. The final restoration uses the native GEN3 checkpoint route, and earlier restoration attempts remain preserved.

Later custody records complete the data chain. Batches 001 and 002 have Drive size and SHA256 verification. The T500 download records verify batches 003 through 006, with batches 005 and 006 adding 32 shards and 203,896,054,295 bytes in the final stage. All 89 shards and 609,671,270,157 compressed bytes have recorded external custody. The completed final-download record supersedes the early all-batches download hold in the original archive README. These statements cite the retained verification receipts; they do not claim a fresh reread of the full archive during this manuscript revision. [N5-1, N5-6, N5-7, N5-9, N5-10, N5-11, N5-12, N5-13, N5-15]

### IV.8 What the result makes available

Once $g(E,M,b)$ is available, a uniform field can be applied by replacing $E$ with $E-hM$. The partition function for a fixed retained boundary is then the exact finite sum

$$Z_b(\beta,h)=\sum_{E,M} g(E,M,b)\exp\bigl[-\beta(E-hM)\bigr].$$

Boundary fields, partial boundary closure, fixed-magnetization questions and added uniform pair couplings likewise operate on retained coordinates. These operations reuse a completed count rather than enumerate the spins again. The multiplicities remain exact even when an exponential or thermodynamic readout is evaluated numerically with its own precision contract.

The scale increase from N2000 to N5000 is substantive because the larger output retains the same detailed finite observables across 12,497,500 physical pair interactions. The reason it is tractable is also explicit: the collective magnetization removes the uniform complete interaction from the hard local bookkeeping, while repeated bounded correction components control the remaining factorization. The result establishes this exact computation for the six specified sources. It is not a universal exact solver for arbitrary dense frustrated graphs, and the single production timing does not establish an asymptotic runtime exponent. [N5-2, N5-7, N5-16]

### IV.9 N5000 thermal ordering and retained boundary memory

#### IV.9.1 A separately defined thermal family

Later N5000 calculations reuse the exact correction polynomials to determine ordering, fluctuations and boundary response. The thermal question requires an explicit normalization. In the archived unit-fill Hamiltonian, the collective term and sparse correction both have their original integer strengths. Dividing that whole Hamiltonian by $N$ scales both terms. The later fixed-sparse-strength family instead uses

$$H_{N,\kappa}=-\frac{M^2-N}{2N}+\kappa C_N,\qquad \kappa\ge 0,\qquad N=20+30m.$$

Here $C_N=2\sum_{\text{negative edges}} s_i s_j$ for the signed positive-fill sources; the packet control has $C_N=0$. The dense pair coefficient is $1/N$, while the sparse correction has fixed strength $\kappa$. The choice $\kappa=1/N$ recovers whole scaling of the original positive-fill source. Holding $\kappa$ fixed defines the thermal extension used in the transition theorem. This source distinction makes the later physical results interpretable without changing the original integer counts. [N5-14]

Let $F(y)$ be the partition function of one repeated correction component in dimensionless auxiliary field $y$, including its actual sparse Boltzmann factor. The signed-packet motif has $n=15$ spins and the chain motif has $n=30$; a fixed boundary root and a five-spin tail complete each finite source. Gaussian decoupling of the collective term gives the bulk variational pressure

$$\sup_{y\in\mathbb{R}} \Phi_{\beta,\kappa}(y), \qquad \Phi_{\beta,\kappa}(y) = -\frac{y^2}{2\beta} + \frac{1}{n}\log\frac{F(y)}{F(0)}.$$

The inverse temperature $\beta$ is in the declared model units. Define the repeated-component magnetization per spin and zero-field susceptibility by

$$m_0(y) = \frac{1}{n}\partial_y \log F(y), \qquad \chi = m_0'(0).$$

A stationary point satisfies $y = \beta m_0(y)$, and an extremal macroscopic state has magnetization $m_* = y_*/\beta$. The finite root and tail contribute only $O(1)$ to $\log Z$ on each compact field interval. They therefore leave the limiting bulk maximization unchanged while remaining present in the finite conditional weights and boundary-memory observables. The bound $F(y)/F(0) \le e^{n|y|}$ controls the distant Gaussian tails. [N5-14]

### IV.9.2 The first global ordering transition

The exact source factors reduce the ordering question to rational functions of two bounded variables. Put $a = \tanh(2\beta\kappa)$ and $v = \tanh y$, and define

$$P(a,v) = 1 + (-a^3 + 2a^2 - 3a)v^2 + a^2 v^4.$$

Up to positive field-independent constants, the repeated factors are

$$F_{\text{packet}}(y) \propto \cosh^{15} y,$$

$$F_{\text{signed}}(y) \propto \cosh^{15} y (1 - av^2)^3,$$

$$F_{\text{chain}}(y) \propto \cosh^{30} y (1 - av^2)^4 P(a,v).$$

The chain factor has a four-spin path and four negative pairs in its correction core, together with free correction spins. It is not a thirty-spin path. The source-factorization identities were checked against the frozen generating polynomials. The resulting susceptibilities are

$$\chi_{\text{packet}} = 1, \qquad \chi_{\text{signed}} = 1 - \frac{2a}{5}, \qquad \chi_{\text{chain}} = \frac{30 - 14a + 4a^2 - 2a^3}{30}.$$

A local threshold alone would leave open a first-order transition caused by a distant maximum before that threshold. The completed derivation closes that gap by proving the global response inequality $m_0(y) < \chi y$ for every $y > 0$. Its algebraic certificate uses the polynomial

$$S_4(x) = 1 + \frac{x}{3} + \frac{x^2}{5} + \frac{x^3}{7} + \frac{x^4}{9}.$$

For each signed family, the exact rational response satisfies

$$\chi S_4(v^2) - \frac{m_0(y)}{v} = \frac{v^2 Q_f(a,v^2)}{D_f(a,v)},$$

where the denominators are

$$D_{\text{signed}} = 1575(1 - av^2), \qquad D_{\text{chain}} = 4725(1 - av^2)P(a,v).$$

Both are positive on $0 \le a \le 1$ and $0 \le v < 1$. The numerators have bidegrees $(2,4)$ and $(7,6)$ in $(a,x)$, where $x = v^2$. Expanding them in the Bernstein basis on the unit square gives 15 and 56 nonnegative rational coefficients, respectively. All 71 coefficients and the exact reconstruction of the power-basis polynomials are retained. Because each Bernstein basis function is nonnegative on that square, this proves $Q_f \ge 0$ throughout the parameter domain. It is an exact finite certificate of a continuum statement. [N5-14]

The strictly positive remainder in the series for the inverse hyperbolic tangent gives

$$\frac{\operatorname{atanh} v}{v} > S_4(v^2), \qquad 0 < v < 1.$$

Together with $\chi \ge 3/5$, the certificate therefore proves $m_0(y) < \chi y$. Consequently, when $\beta\chi \le 1$, the derivative $\Phi'(y) = m_0(y) - y/\beta$ is strictly negative for $y > 0$. Evenness makes zero the unique global maximizer, including at equality. When $\beta\chi > 1$, the origin has positive curvature, so a nonzero maximizing pair has larger pressure. Thus the unique solution of

$$\beta_c \chi(\beta_c, \kappa) = 1$$

is the **first global ordering transition** for every fixed finite $\kappa \ge 0$ in these specified families. Maximizers satisfy $|y_*| \le \beta$; as $\beta$ decreases to $\beta_c$, compactness and uniqueness of the critical maximizer force $y_* \to 0$. The negative fourth cumulant supplies a nondegenerate square-root onset. The stronger route through global concavity of $m_0$ is false at exact retained counterexamples; it is not needed for this successful first-onset proof. The conclusion concerns the first transition, with no claim to classify every remote stationary point at larger inverse temperature. [N5-14]

| Fixed correction strength $\kappa$ | Signed-packet $\beta_c$ | Chain $\beta_c$ |
|---|---|---|
| 0 | 1.000000000000 | 1.000000000000 |
| 1/8 | 1.122873781777 | 1.136842176652 |
| 1/4 | 1.295422047875 | 1.310642690959 |
| 1/2 | 1.581042157548 | 1.582234699504 |
| 1 | 1.663814710634 | 1.663815941534 |
| 2 | 1.666663067473 | 1.666663067475 |

The exact susceptibility difference is $\chi_{\text{chain}} - \chi_{\text{signed}} = -a(1-a)^2/15$. Hence the chain threshold is strictly higher for finite positive $\kappa$, and both approach $5/3$ as $\kappa \to \infty$. The table reports rounded values from separately retained directed critical brackets. [N5-14]

### IV.9.3 Critical fluctuations and a nonzero boundary response

At the exact critical temperature, let $c_4 < 0$ be the fourth cumulant per spin of the repeated component. The pressure near

zero has a stabilizing quartic term $\Phi(y) = -by^4 + O(y^6)$ with $b = -c_4/24 > 0$. The previously established global uniqueness controls the complement of a neighborhood of zero, and the Gaussian bound controls the distant tail. Rescaling $u = N^{1/4}y$ then gives

$$\sqrt{N}\left\langle \left(\frac{M}{N}\right)^2 \right\rangle \longrightarrow \frac{\sqrt{24/(-c_4)}}{\beta_c^2}\,\frac{\Gamma(3/4)}{\Gamma(1/4)}.$$

For the chain at $\kappa = 1$, the amplitude is 0.546951358570. At the finite map's twelve-decimal rational critical centers, the computed N5000 value of $\langle (M/N)^2 \rangle$ is 0.0077033393, compared with 0.0038595576 at N20000. The exact-critical asymptotic amplitude and the rounded finite input temperatures have distinct definitions and retained numerical receipts. Finite systems remain smooth; their curves are compared with the limiting transition rather than described as finite singularities. [N5-14]

The fixed retained root supplies a second physical consequence. Let $R_b(y)$ be its exact boundary factor including its correction Boltzmann weights. Below and at onset, concentration of the common integral at zero gives

$$\frac{Z_b}{Z_0} \longrightarrow \frac{R_b(0)}{R_0(0)}.$$

The common finite tail and normalization cancel. Applying the declared boundary protocol to these strictly positive limiting weights gives the memory limit directly, without extrapolating numerical data to enormous $N$. At exact chain criticality with $\kappa = 1$, the preparation, between-reading and within-interval differences are respectively

$$-0.430454602216, \qquad 0.000189921648477, \qquad 0.000344266087525.$$

These numbers refer to an explicit observation protocol. A heat-bath generator changes one retained bit at a time with rate $Q_{j,b} = w_j/(w_b + w_j)$ for weights $w_b = Z_b/Z_0$, and its columns sum to zero. Readings retain the least-significant bit, postselect even boundary states, and use intervals $\delta = 1/3$. The preparation difference compares the state-zero start with conditional equilibration inside its initial coarse class. The between-reading difference measures the effect of conditional resets at intermediate readings. The within-interval difference compares those resets with evolution under the coarse generator. Conditional equilibration inside one class is not a globally stationary mixture. The nonzero limiting differences quantify retained boundary response under this declared clock; physical seconds and physical transition rates are separate source assignments. [N5-14]

The finite map evaluates 158 complete directed cases, including N5000 and N20000, correction strengths and temperature offsets, packet controls and four N200 cases. Complete 128-bit Arb integrals retain explicit infinite tails; 84 direct comparisons expand the N200 exact spectrum and sum Gibbs weights at 192 bits. The method therefore checks the new thermal normalization and finite readouts independently of the Gaussian integration. [N5-14]

### IV.9.4 Exact occupied windows and measured probability

A later counting calculation retained a selected thermally occupied N5000 window: 69,904 final records, including 2,056 exact zeros. The window covers positive-spin numbers $2372 \le k \le 2628$, aligned-negative-edge indices $0 \le t_a \le 16$, and all sixteen boundary assignments. Here $t_a = L - d_{neg}$ counts aligned negative edges, with $L$ their total number and $d_{neg}$ the disagreeing-edge count. This $t_a$ is defined for the later window calculation and is not the general shifted correction coordinate used earlier in the chapter. Its original unit-fill labels satisfy

$$M = 2k - N, \qquad E = 4t_a - 2L - \frac{M^2 - N}{2}.$$

The shortcut is mathematical. Reversing the chain's defect variable produces a repeated factor with monomial constant coefficient $\widehat{f}_0 = 32X^6$. If $q_n$ is the coefficient of the reversed defect variable at degree $n$ in its $m$th power, logarithmic differentiation gives

$$q_0 = (32X^6)^m, \qquad n\widehat{f}_0 q_n = \sum_{i=1}^{7} \big((m+1)i - n\big)\widehat{f}_i q_{n-i}.$$

Every polynomial division is exact and checked for zero remainder. Only sixteen recurrence steps are needed for the N5000 window. The remaining pair, free spins and boundary root are then contracted exactly. A separate GPU calculation uses the original defect encoding and modular transforms; all 17,800 intermediate coefficients agree across the N5000 and N20000 windows before final root contraction. Global reversal agrees on all 275,104 final records. Exact zeros are preserved as source-imposed forbidden coordinates. [N5-14]

The N5000 CPU coefficient stage took 0.268194 seconds; the corresponding GPU route took 0.651927 seconds, with another 4.259603 seconds for common root contraction, output and mass evaluation. This short aligned-index regime favors the direct CPU recurrence. It complements the GPU's usefulness for the original much larger full-joint coefficient problem; the measured requests and timing boundaries are explicit. [N5-14]

For a window $W$, probability is evaluated against the **complete** partition function of the thermal family:

$$\Pr(W) = \frac{1}{Z}\sum_{(k,t_a,b)\in W} g(k,t_a,b)\exp\left[\frac{\beta((2k-N)^2 - N)}{2N} - \beta\kappa(4t_a - 2L)\right].$$

| N5000 temperature offset $\beta - \beta_c$ | Captured Gibbs probability | Omitted Gibbs probability |
|---|---|---|
| −0.03 | 53.607483% | 46.392517% |
| 0 | 37.267444% | 62.732556% |
| +0.03 | 17.264479% | 82.735521% |

The exact counts capture substantial probability at and below the transition. Above onset, the ordered magnetization peaks move away from the window centered at zero, reducing

its coverage. Both captured and omitted probabilities are retained as directed intervals. No uncomputed count is silently replaced by zero, and the selected window is not treated as the whole density. The complete N5000 joint construction, the global first-onset proof and the exact occupied window are three substantive completed results with separate output definitions and derivation chains. [N5-14, N5-16]

SAM VOLUME III

# N2000 Exact Joint Densities and Low Energy Physics

Chapter V

Six specified complete Ising graphs with 2,000 spins were solved exactly while retaining energy, magnetization, and four ordered boundary spins. The primary output contains **244,104,392 nonzero integer records**. The retained joint measure accounts for all $2^{2000}$ labelled spin configurations of each graph and supports later field and boundary-response questions without repeating the microscopic enumeration. The subsequent physical readouts determine exact ground energies and degeneracies, analytic residual-entropy limits along a declared size sequence, excitation gaps, field thresholds, and controlled low-temperature heat capacities. These are substantive results about how the specified interactions organize the states. [A1, pp 1-10]

The clearest physical comparison is between negative-fill signed packets and the same packets joined by alternating bridges. The bridges reduce the N2000 ground-state multiplicity by a factor of approximately $2.07536 \times 10^{20}$, lower the ground energy from $-10,840$ to $-10,974$, and reduce the first excitation gap from 10 to 4. On the commensurate sequence $N = 60r$, their pairing of correction components halves the limiting ground-state entropy per spin from $\ln 2/15$ to $\ln 2/30$. The lower ground multiplicity and softer first excitation coexist: they describe different aspects of the exact spectrum. [A1, pp 5-8]

### V.1 The computed object and the six sources

An Ising spin is a binary variable taking the values $-1$ and $+1$. A spin configuration specifies the value at every labelled vertex. The Hamiltonian assigns an energy to each configuration using the pair couplings $J_{ij}$ and site fields $h_i$. In this chapter the sign convention is

$$E(s) = -\sum_{i<j} J_{ij} s_i s_j - \sum_i h_i s_i, \qquad M(s) = \sum_i s_i.$$

All six archived N2000 systems have zero fields and pair couplings $J_{ij} \in \{-1,+1\}$. Every physical pair is coupled, giving $\binom{2000}{2} = 1,999,000$ nonzero interactions. Energies are dimensionless, and the complete interaction has not been divided by $N$. The sources therefore describe finite, unnormalized complete graphs. [A1, p 2, Eqs 1-2]

The retained boundary is the ordered vertex list $P = (0,1,60,61)$. A boundary mask $b \in \{0,\dots,15\}$ records those four spin signs, with a set bit denoting $+1$. The joint density is the integer count

$$g(E,M,b) = \#\{s : E(s) = E,\ M(s) = M,\ s|_P = b\}.$$

Each nonzero record is one supported energy-magnetization-boundary bin and its multiplicity. It is not one separately visited global configuration. Each fixed boundary has total count $2^{1996}$; the 16 boundary sectors sum to $2^{2000}$. Keeping the boundary ordered preserves which retained vertex receives a later perturbation. [A1, p 2, Eqs 3-4]

The construction starts with packets. A full packet contains 15 vertices arranged in three complete five-vertex layers. Perfect matchings join successive layers, giving $3\binom{5}{2} + 2(5) = 40$ parent edges. At N2000 there are 133 full packets and one terminal five-vertex packet, with 5,330 unbridged parent edges in total. The inherited vertex labels are retained explicitly. In the signed family, one edge in every complete layer is negative: there are 400 negative and 4,930 positive parent edges. The chain adds 133 gateway bridges, alternating in sign from $+1$, so 67 bridges are positive and 66 negative. [A1, p 2]

For each of these three parents, all absent pairs receive a uniform fill $J_0 = +1$ or $J_0 = -1$, while existing parent edges retain their signs. This gives six sources. Packet with positive fill is the uniform ferromagnetic complete graph and supplies an analytic control. The other five sources contain frustrated cycles. They share the same spin count and complete physical connectivity, but their correction structures and spectra differ. [A1, pp 2, 4]

The component sizes in the final column describe the implemented packet grouping. The original source retains all 1,999,000 interactions in every case. [A1, p 4, Table I]

### V.2 Why a complete graph admits an exact construction

The dense fill can be carried by a collective observable rather than represented as a large local factor. Magnetization is sufficient for that part because binary spins obey

$$M^2 = N + 2\sum_{i<j} s_i s_j.$$

Write the source as its uniform fill plus the exact correction $\Delta J_{ij} = J_{ij} - J_0$. The complete energy then becomes

$$E = E_c - \frac{J_0}{2}(M^2 - N), \qquad E_c = -\sum_{i<j} \Delta J_{ij} s_i s_j.$$

This is an identity of the full source. Retaining $M$ preserves the association needed to restore the dense contribution after contracting the corrections. A correction-energy marginal alone would lose that association. [A1, p 2, Eqs 5-7]

The correction graph has bounded packet and gateway interfaces. Local integer generating functions count packet assignments according to positive-spin count, correction energy, and the signs still needed by a boundary or a gateway. A full packet has at most $2^{15} = 32,768$ local assignments. The original preparation produced 22 local tables on two A100 devices and compared them with independently arranged CPU Gray-

| Source | Negative physical pairs | Nonzero corrections | Correction bound $B$ | Largest implemented correction component |
|---|---|---|---|---|
| Packet positive fill | 0 | 0 | 0 | 15 |
| Packet negative fill | 1,993,670 | 5,330 | 10,660 | 15 |
| Signed packet positive fill | 400 | 400 | 800 | 15 |
| Signed packet negative fill | 1,994,070 | 4,930 | 9,860 | 15 |
| Signed packet chain positive fill | 466 | 466 | 932 | 30 |
| Signed packet chain negative fill | 1,994,003 | 4,997 | 9,994 | 30 |

code enumeration. The public preparation route uses the CPU enumeration. [A1, pp 2-3]

For a bridge correction $\delta$ joining gateway spins $a$ and $c$, define its nonnegative shifted contribution

$$\tau_\delta(a,c) = \frac{|\delta| - \delta ac}{2}.$$

If $L_{r+1,c}(x,y)$ is the next packet table with gateway sign $c$, a forward contraction has the form

$$F_{r+1}(b,c) = \sum_{a=\pm 1} F_r(b,a) L_{r+1,c}(x,y) y^{\tau_{\delta_r}(a,c)}.$$

The gateway sign is summed only after its interactions have been included. The four externally retained spins remain fixed throughout. When a bridge sign equals the fill, its correction vanishes; in the alternating chain this separates the correction graph into components of at most two full packets. These components are joined from gateway tables, without enumerating all $2^{30}$ assignments of a paired component. Repeated scalar components become polynomial powers, and a length-balanced product tree combines them with the retained component. [A1, p 3, Eqs 10-12]

For storage and polynomial composition, let $B = \sum_{i<j} |\Delta J_{ij}|$, let $k = (M+N)/2$, and shift correction energy to the integer coordinate $t = (E_c + B)/2$. A common lattice step $d$ permits $u = t/d$ and $T = B/d$. The two packed exponents are

$$\phi_K(k,u) = k(T+1) + u, \qquad \phi_T(k,u) = u(N+1) + k.$$

The global coordinate bounds make both maps injective. Quotient and remainder recover the coordinates, with out-of-range values rejected. FLINT polynomial arithmetic, accessed through python-flint, preserves arbitrary-precision integer coefficients throughout. [A1, pp 2-3, Eqs 8-14]

The counting argument follows the construction. A local table counts every permitted local assignment once. Multiplication combines disjoint assignments and adds their observables. Summing gateway signs includes every compatible bridge state once. Restoring the collective term gives the complete physical energy. The resulting coefficient is therefore the multiplicity of one unique $(E,M,b)$. The two exponent orders provide a strong implementation cross-check; they share local source tables and an arithmetic library, so their agreement is not described as agreement between two unrelated mathematical solvers. [A1, p 3]

### V.3 The completed joint output and its validation

The six primary outputs have sharply different support sizes even though they describe the same number of spins and physical pairs. [A1, p 5, Table II]

| Source | Primary nonzero records | Raw GB | Losslessly compressed GB |
|---|---|---|---|
| Packet positive fill | 31,952 | 0.0083 | 0.0042 |
| Packet negative fill | 73,522,212 | 19.0423 | 11.1916 |
| Signed packet positive fill | 10,227,200 | 2.6489 | 1.7201 |
| Signed packet negative fill | 73,556,274 | 19.0511 | 11.1050 |
| Signed packet chain positive fill | 12,265,280 | 3.1767 | 2.1078 |
| Signed packet chain negative fill | 74,501,474 | 19.2959 | 11.2691 |
| Total | 244,104,392 | 63.2231 | 37.3979 |

Here GB means $10^9$ bytes. The exact primary byte totals are 63,223,097,516 raw and 37,397,887,721 compressed. Six cases, 16 boundaries, and two encodings give 192 emitted boundary rows representing 96 distinct case-boundary measures. A boundary row is a whole conditional measure; a record is one supported bin within it. Only one encoding is needed to distribute the measure, while both encodings' scientific receipts are retained. [A1, pp 4-5]

The production took 816.624947 seconds, including row emission and verified durable output. This timing excludes earlier preparation and pilot work and later compression and transfer. Fourteen workers used two FLINT threads each. The recorded CPU time was 16,159.967 seconds, equivalent to 19.79 average CPU cores during the measured interval, under a quota of 27.2. The configured memory limit was 465.66 GiB, and the largest recorded individual worker RSS was 18.69 GiB. Neither is a measured simultaneous fleet peak. RAM-staged rows were copied, fsynced, checksum-read back, and renamed before staging buffers were removed. [A1, pp 4-5]

Correctness was checked at several levels. For a boundary containing $r(b)$ positive spins, counting free-spin assignments gives the exact conditional magnetization identity

$$\sum_E g(E, 2k-N, b) = \binom{N-4}{k-r(b)}.$$

This was checked at every allowed magnetization in every emitted boundary row. The first two energy moments were obtained separately from the couplings and fixed boundary signs. Summing all boundaries yields total count $2^{2000}$, first moment zero, and second moment $1,999,000\,2^{2000}$. Complete canonical row digests compare the full measures under the two encodings. N12 qualification uses independent joint variable elimination, N30 compares an uncompressed component route, CPU and GPU local tables agree, and a deliberately truncated binary is rejected. Source hashes, binary hashes, moment iden-

tities, and coordinate comparisons answer different questions and are retained together. [A1, pp 3-4, Eqs 16-19]

The positive-fill packet control is also available analytically. Its only supported energy at a given $M$ is $-(M^2-N)/2$, and its conditional multiplicity is the corresponding binomial coefficient. Each boundary has 1,997 records, giving 31,952 across all boundaries. A fresh public-runtime computation reproduced all 32 binary hashes for this control, including both encodings, in approximately 3.07 seconds after preparation. This demonstrates recovery of the control's algebra and serialization; the five nonuniform outputs have much larger support. [A1, p 4, Eqs 20-21]

### V.4 Exact ground states from the same source definitions

The paper's physical expansion reconstructed the published source motifs and computed their minimum energy and degeneracy at every magnetization. It used a smaller output than the original full joint measure and summed the retained boundaries. The two outputs have separate computational purposes and separate provenance. [A1, pp 5, 10]

A minimum/count state contains an energy $e$ and an exact multiplicity $c$. Combining alternatives keeps the lower energy and adds multiplicities only when the energies tie. Combining independent assignments adds energies and multiplies counts. Magnetization convolution combines packet states while retaining gateway signs until the bridges are processed. At fixed $M$, the collective fill energy is constant, so minimizing the correction energy and restoring the fill is exact. A final minimum over $M$ returns the global ground energy and sums all tied ground counts. [A1, p 5, Eqs 22-23]

Forward and reversed composition agreed at all 2,001 magnetization sectors for all six N2000 cases. Separate direct complete-pair sums verified every sector minimum and count at N5, N12, and N18. The N2000 ground levels also agree with the separately implemented truncated-spectrum calculation below. These readouts were computed from the source definitions; the revision did not claim that the 63 GB historical joint shards were reread for them. [A1, pp 5, 7-8]

The counts are exact integers. The entropy densities are numerical evaluations of their logarithms. [A1, p 6, Table III]

For positive fill, the complete ferromagnetic background determines the ground ordering. Flipping a nonempty proper subset $S$ from an aligned state changes its energy by

$$\Delta E = 2\sum_{i\in S, j\notin S} J_{ij} \geq 2\{|S|(N-|S|)-2n_-\},$$

where $n_-$ is the total number of negative pairs. At N2000 the three positive-fill cases have $n_- \leq 466$ and $|S|(N-|S|)\geq 1999$, so this change is strictly positive. The two globally inverted aligned states are the complete ground family. [A1, p 5, Eq 24]

For negative fill, the nonzero corrections are positive parent edges with $\Delta J = 2$. Let $m_+$ be their number and let $c(s)$ count correction edges whose endpoint spins differ. The energy becomes

$$E = -\frac{N}{2} - 2m_+ + 4c(s) + \frac{M^2}{2}.$$

Correction components favor internal alignment, while the antiferromagnetic collective term penalizes nonzero magnetization. At N2000 all negative-fill ground states have $M=0$ and lie 20 units above the reference $E_{\text{ref}} = -N/2 - 2m_+$. The terminal packet prevents simultaneous zero cut cost and zero magnetization. Exact contraction resolves that finite-size constraint rather than assigning independent orientations to packets after imposing $M=0$. [A1, p 5, Eq 25]

Changing 400 signed-layer edges shifts the unbridged negative-fill ground energy by 800 while leaving the minimizing multiplicity unchanged at this size. Adding alternating bridges then locks portions of that orientational freedom together. The exact ratio of unbridged to chain ground multiplicity is approximately $2.075362773088\times 10^{20}$. This reduction is a property of the ground states and is not inferred from the number of stored joint bins. [A1, pp 5-6]

### V.5 An analytic residual entropy limit

Residual entropy measures how many configurations remain at the minimum energy. Its density is the logarithm of the ground multiplicity divided by the spin count. A commensurate sequence removes the terminal-packet effect and makes the counting mechanism explicit. Take $N = 60r$ with negative fill. Each unbridged family has $N/15$ ferromagnetic correction components of size 15. Aligning each component gives zero cut cost, and zero magnetization requires half of the components up and half down. Thus

$$\Omega_{0,\text{unbridged}}(N) = \binom{N/15}{N/30}.$$

In the chain, the positive bridges pair adjacent packets, while negative bridges equal the fill and contribute no correction. There are $N/30$ internally aligned size-30 correction components, and half must point each way:

$$\Omega_{0,\text{chain}}(N) = \binom{N/30}{N/60}.$$

These configurations minimize the nonnegative cut and magnetization terms separately, so an excited component cannot supply another state at the same minimum. The ground energies are $-35N/6$ for packet, $-163N/30$ for signed packet, and $-11N/2$ for the chain. [A1, p 6, Eqs 26-30]

The central-binomial asymptotic gives

$$\lim_{r\to\infty}\frac{\ln\Omega_{0,\text{unbridged}}(60r)}{60r} = \frac{\ln 2}{15}, \qquad \lim_{r\to\infty}\frac{\ln\Omega_{0,\text{chain}}(60r)}{60r} = \frac{\ln 2}{30}.$$

The bridges halve the residual-entropy density because they double the number of spins locked into one orientable component. Exact size calculations at N120, N300, N600, N900, N1200, N1800, and N1980 agree with the formulas. N2000 is separate: its terminal packet requires additional cut configurations to attain $M=0$, and its entropy lies above the nearby

| Source | Ground energy $E_0$ | Ground $M$ | Exact ground multiplicity $\Omega_0$ | $\ln \Omega_0/N$ |
|---|---|---|---|---|
| Packet positive fill | -1,999,000 | $\pm$2000 | 2 | 0.000346574 |
| Signed packet positive fill | -1,998,200 | $\pm$2000 | 2 | 0.000346574 |
| Signed packet chain positive fill | -1,998,068 | $\pm$2000 | 2 | 0.000346574 |
| Packet negative fill | -11,640 | 0 | 200771302400067787126123131419286927944400 | 0.047551493 |
| Signed packet negative fill | -10,840 | 0 | 200771302400067787126123131419286927944400 | 0.047551493 |
| Signed packet chain negative fill | -10,974 | 0 | 967403410158179609160 | 0.024160574 |

commensurate trend. The limit is established on the declared $N = 60r$ family rather than estimated from that endpoint. [A1, pp 6-7, Eq 31]

### V.6 Field thresholds and soft excitations

A uniform field couples to the retained magnetization. Let $e_*(M)$ be the exact minimum full energy at fixed $M$. In a field $H$, the ground energy is the lower envelope

$$E_0(H) = \min_M \{e_*(M) - HM\}.$$

The lower convex hull of the integer points $(M, e_*(M))$ gives rational field crossings. The positive-fill cases select $M = +2000$ for every $H > 0$. In negative fill, $M = 0$ persists until the first positive crossing: $H = 3$ for both unbridged families and $H = 17/5$ for the chain. Each first crossing connects $M = 0$ and $M = 10$. Any additional collinear minimizing sectors must be included when computing the degeneracy at the crossing. [A1, p 7, Eq 32]

The first zero-field excitation is a different question. The gap $\Delta_1$ is the energy difference between the ground state and the next supported energy. The ratio $\Omega_1/\Omega_0$ compares the multiplicities at those two levels. [A1, pp 5, 8]

The chain has a softer lowest excitation but a higher initial magnetization threshold. A low-energy excitation need not carry the magnetization required to become favorable in a uniform field. The exact data therefore resolve why the two response measures move in different directions. [A1, pp 5, 7-8]

### V.7 Exact low energy spectra and entropy resolution

The negative-fill energy decomposition also supports an exact bounded spectrum. Local polynomials track positive-spin count and the nonnegative cut count $c$. Their products retain all terms with $c \le 100$. Since cut contributions are nonnegative, a discarded term cannot later return to that range. At N2000 every state satisfying $E \le E_{\text{ref}} + 400$ is represented. As $E_0 = E_{\text{ref}} + 20$, the resulting complete energy window is $[E_0, E_0 + 380]$. It contains 184 supported levels for packet, 185 for signed packet, and 189 for the chain. [A1, p 7, Eq 33]

A separate C++ arbitrary-integer convolution produced these products. At N12 and N18, every supported level below the cutoff agrees with direct complete-graph enumeration for all three negative-fill sources. The N2000 ground levels agree exactly with the separate minimum/count calculation. The spectrum inside the retained window is exact; its construction does not approximate those coefficients. [A1, pp 7-8]

Microcanonical entropy is defined here as $S(E) = \ln g(E)$ with $k_B = 1$. At increasing supported energies, successive entropy slopes are

$$\beta_i = \frac{S(E_{i+1}) - S(E_i)}{E_{i+1} - E_i}.$$

Increasing slopes identify discrete nonconcavity. The analysis also constructs the smallest concave majorant within the window and measures its maximum vertical excess above the entropy points. Counts remain exact; the logarithms were evaluated with 70-digit precision. [A1, p 8, Eq 34]

The entropy curves have appreciable fine-level oscillation. To determine how much is tied to resolution, the analysis sums integer counts into fixed-width bins starting at $E_0$, then takes logarithms and places each value at its bin center. Only complete bins inside the known window are used. [A1, p 8]

The entries are maximum entropy deficits below the finite-window concave envelope. The chain has the largest deficit at individual levels, yet its nonconcavity vanishes at widths 8 and 16 in this window. Both unbridged families retain finite deficits under those coarse grainings. This establishes different shell sensitivity in the three exact spectra. A thermodynamic claim about ensemble inequivalence would additionally require a normalized size sequence and the relevant wider spectrum; the present calculation resolves the finite window and its stated resolutions. [A1, p 8, Table IV]

### V.8 Controlled low temperature heat capacity

The exact window is sufficient for a canonical calculation at low temperature because its omitted contribution can be bounded. Write the excitation energy as $\delta E = E - E_0$. The shifted window partition sum is

$$\widetilde{Z}_{\text{win}}(\beta) = \sum_{E \le E_0 + 380} g(E) e^{-\beta \delta E}.$$

Every N2000 energy is even, so the first omitted excitation is at least 382. The full configuration count is $2^{2000}$, while the retained ground contribution is $\Omega_0$. Therefore

$$\varepsilon_Z = \frac{Z_{\text{tail}}}{\widetilde{Z}_{\text{win}}} \le \frac{2^{2000}}{\Omega_0} e^{-382\beta}.$$

For $0 < T \le 0.25$ in the archived energy units, this bound is below $e^{-190}$ even for the least-degenerate chain. A conservative bound using $\delta E_{\max} \le 1{,}999{,}000 - E_0$ propagates the omitted mass into heat capacity at order $4\beta^2 \delta E_{\max}^2 \varepsilon_Z$; at $T = 0.25$ it is below $e^{-156}$ for every negative-fill case. These bounds control the omitted states. Numerical evaluation precision is handled separately. [A1, p 8, Eqs 35-36]

The expectation and heat capacity are evaluated from the retained counts with 70-digit exponentials and sums:

| Negative-fill source | First gap $\Delta_1$ | $\Omega_1/\Omega_0$ | First positive field crossing |
|---|---|---|---|
| Packet | 14 | 5 | 3 |
| Signed packet | 10 | 2 | 3 |
| Signed packet chain | 4 | 67 | 17/5 |

| Negative-fill source | Exact levels | Width 4 | Width 8 | Width 16 |
|---|---|---|---|---|
| Packet | 7.26627 | 6.27352 | 3.21559 | 1.05748 |
| Signed packet | 3.19872 | 2.76719 | 1.64941 | 0.13406 |
| Signed packet chain | 9.25413 | 0.01281 | 0 | 0 |

$$\langle \delta E \rangle = \frac{\sum \delta E \, g(E) e^{-\beta \delta E}}{\widetilde{Z}_{\text{win}}}.$$

$$\frac{C}{k_B} = \beta^2 \operatorname{Var}(\delta E).$$

| Negative-fill source | Total $C(0.25)/k_B$ |
|---|---|
| Packet | $7.49644 \times 10^{-21}$ |
| Signed packet | $1.35947 \times 10^{-14}$ |
| Signed packet chain | $1.93020 \times 10^{-3}$ |

These are total heat capacities; division by N gives the per-spin quantity. The lowest gaps explain their separation through the leading expression

$$\frac{C}{k_B} \sim \left(\frac{\Delta_1}{T}\right)^2 \frac{\Omega_1}{\Omega_0} e^{-\Delta_1/T}.$$

The chain's gap-4 level with multiplicity ratio 67 creates a much larger response at this temperature than the gap-10 and gap-14 levels of the unbridged sources. The comparison comes from exact low-energy counts with a quantified tail, rather than from output cardinality or a fitted response curve. [A1, p 8, Table V and Eqs 37-38]

### V.9 Reusing the full joint measure

The full joint output preserves more response information than either the ground-state or low-energy reductions. For a uniform field, its fixed-boundary partition function is

$$Z_b(\beta, H) = \sum_{E,M} g(E,M,b) e^{-\beta(E-HM)}, \qquad Z = \sum_b Z_b.$$

It gives magnetization susceptibility $\chi = \beta \operatorname{Var}(M)$ and field-dependent heat capacity $C_H/k_B = \beta^2 \operatorname{Var}(E - HM)$. A specified additional boundary energy $V(b)$ multiplies each $Z_b$ by $e^{-\beta V(b)}$. Because the source already contains its original boundary interactions, $V$ describes a perturbation of those interactions rather than adding their original contribution again. [A1, p 9, Eq 40]

The joint coordinates can also define a deliberate Hamiltonian extension with independently chosen collective and correction strengths:

$$E_{\kappa,\lambda} = \lambda E_c - \frac{\kappa}{2N}(M^2 - N).$$

The correction energy is recoverable as $E_c = E + J_0(M^2 - N)/2$. This permits a normalized collective term while keeping the correction term separately controlled. Dividing the entire archived energy by N would scale both parts and would be a different choice. Any phase comparison must state the new couplings and size sequence. [A1, p 9, Eq 39]

The completed contribution is an exact joint construction for six declared dense systems and a detailed physical reading of their low-energy organization. The graph's dense fill is retained through magnetization; its finite correction interfaces make the counting tractable. The output supports future uniform-field and boundary queries at arbitrary temperature with adequate numerical precision. Arbitrary internal site fields or individual pair changes require further observables unless they are expressible through the retained coordinates. The residual-entropy limits apply to the specified commensurate families, while the N2000 spectra and response values are exact finite-source results. [A1, pp 3, 9-10]

SAM VOLUME III

# Joint State Spaces and Layered Exact Sources

Chapter VI

The N100 construction established an important computational pattern: retain the complete source, then exploit a decomposition that belongs to that source. Later spin work develops that pattern in two directions. A richer integer table retains energy, magnetization and ordered boundaries, allowing new exact responses after the original solve. A separate layered method recognizes collective structure inside dense couplings and returns an exact ground energy, degeneracy and witness. The requested mathematical object determines which of these methods is appropriate.[43, 47]

## VI.1 A joint count is an executable source object

Use the later spin convention

$$E(s) = -\sum_{u<v} J_{uv} s_u s_v - \sum_u h_u s_u, \qquad s_u \in \{-1,1\}. \quad \text{(VI.1)}$$

The sign convention is part of the source identity; the earlier chapters retain their own declared Hamiltonians. Let $M(s) = \sum_u s_u$ and let $b$ be the signs at an ordered list of $p$ retained vertices. The integer object is

$$g(E,M,b) = \#\{s : E(s) = E,\ M(s) = M,\ s|_{\text{ports}} = b\}. \quad \text{(VI.2)}$$

Every entry counts labelled configurations. Summing over $M$ recovers the original boundary-conditioned DOS, and summing over $E, M, b$ gives $2^N$. Ordering the boundary is essential: opposite signs applied to two ports depend on which source vertex receives each sign.[43, 46]

This extra coordinate has a simple independent count identity. If a boundary row has $k_b$ positive signs, its total magnetization marginal satisfies

$$\sum_E g(E, 2k-N, b) = \binom{N-p}{k-k_b}, \quad \text{(VI.3)}$$

with the binomial understood as zero outside its permitted range. The condition checks the full assignment count in each magnetization sector, rather than merely the grand total. The completed joint solver also checks the projection back to its scalar DOS, independent global encodings and small-source direct comparisons. Thus the richer object carries several independently arranged readings of the same source.[45, 46]

The joint table supports an exact response transformation

$$E' = E - hM - \frac{g}{2}(M^2 - N) - \sum_{i=1}^{p} \eta_i b_i. \quad \text{(VI.4)}$$

Here $h$, $g$ and $\eta_i$ are exact rational request parameters. Boundary conditioning selects rows; a fixed-$M$ request selects magnetization sectors. Counts remain arbitrary-precision integers, while changed energies remain rational. Combining equal transformed energies recovers a complete transformed DOS. No microscopic spin enumeration is required for this reading.[43]

One can also find exact ground-state crossings. For each allowed $M$, retain $e_M = \min\{E : g(E,M,b) > 0\}$ after the requested boundary selection. Under a uniform field, the ground energy is the lower envelope of $e_M - hM$. A crossing of two supported sectors occurs at

$$h_* = \frac{e_{M_2} - e_{M_1}}{M_2 - M_1}. \quad \text{(VI.5)}$$

The reader checks all sectors at that exact rational value and returns all coexisting magnetizations and their counts. The implementation uses an exact convex-envelope construction, preserving ties that a rounded field could split.[46]

The retained $N = 120$ signed-packet-chain source with dense fill $-1$ gives a concrete example. At $h = 1/2$ and ordered boundary $[+1, \text{free}, -1, \text{free}]$, the query returns ground energy $-639$, ground magnetization 2 and degeneracy 15, within exactly $2^{118}$ permitted configurations. These are source-coordinate energies and magnetizations. Their physical units are supplied by the calling source map.[46]

## VI.2 Dense completion can expose a sparse correction

For binary signs,

$$\sum_{u<v} s_u s_v = \frac{M^2 - N}{2}. \quad \text{(VI.6)}$$

If $J_0$ is the uniform complete-graph fill, write the actual source as that fill plus its exact correction. Its energy becomes

$$E = E_{\text{corr}} - J_0 \frac{M^2 - N}{2}. \quad \text{(VI.7)}$$

The dense collective part is now a function of a retained coordinate. Local factorization works on the correction graph while carrying $M$. The complete original dense source remains in the final energy.[45]

Subtracting the fill can remove an interaction from the correction graph: an original edge with coupling $J_0$ has zero correction. This can expose independent components and repeated local operators even though every original vertex is coupled to every other vertex. The resulting computational route follows the source identity and the observable being retained; visual graph density alone does not determine its cost.

For disconnected correction components, local energy/magnetization polynomials multiply and their coefficients convolve. Repeated components permit exact polynomial powers and balanced products. Signed single-gateway chains require boundary messages that retain the gateway sign.

The scalar compiler has a further specialization only when the two complete gateway-conditioned energy polynomials agree. It explicitly checks that equality before using it.[43, 45]

That last condition depends on the output. Two gateway rows can agree after magnetization has been summed out and differ as joint polynomials in energy and magnetization. Applying a scalar shortcut to the richer object would erase the new information. The joint compiler therefore retains both signed messages. A field-broken chain serves as a retained control: it rejects the equality specialization and agrees with general variable elimination. This is a useful example of a computational identity carrying its actual domain of applicability into the implementation.[44, 45]

### VI.3 Exact contractions beyond packet families

The shared interface also accepts explicit weighted integer graphs and a declared decomposition. Every vertex and interaction must be accounted for, and retained ports must match their declared block positions. Generic sources without an admitted component or chain decomposition can use joint polynomial variable elimination. Small local tables can be enumerated with native integer Gray-code updates; CUDA local enumeration and CPU FLINT composition supply a separate qualified route for the supported cases.[43, 47]

For ordinary energy-polynomial elimination, the local interaction contributes the nonnegative power $|J| - Js_u s_v$, and a field contributes $|h| - hs_u$. If

$$B = \sum_{u<v} |J_{uv}| + \sum_{u} |h_u|, \tag{VI.8}$$

the resulting polynomial has exponent $E + B$. Subtracting $B$ recovers signed energy. Native integer coefficients preserve counts throughout elimination and convolution. The table schedule and energy-support span are separate resource quantities. Neither is the formal cardinality $2^N$.[47]

The generic finite-factor contraction operation extends the same idea to explicitly listed finite domains and signed integer factors. It sums variables outside a retained set, with no implicit probability normalization. An isolated eliminated variable contributes its domain cardinality. An empty factor collection is the multiplicative identity. These details matter when a source becomes a tensor program: omitted interactions and an empty state space have different algebraic meanings.[47]

### VI.4 What the completed inventory contains

The shared installation retains 120 canonical defaults and 123 source-family records through $N = 120$, with packet and connected frustrated lineages distinguished. Three packet families extend through $N = 1408$. Six complete signed graph constructions at each size through 1408 give 8,448 constructed graphs. Those source definitions are catalogued separately from their solved spectra.[44]

The completed dense magnetization production contains 54 cases: six each at $N = 30, 60, 96, 120, 300, 450, 600, 750, 900$. The full joint energy/magnetization/ordered-boundary production contains 24 cases: six each at $N = 120, 300, 750, 900$. Its compressed joint data total 6,871,786,917 bytes. The installation retains 1,840 native mathematical objects, including the applicable dependencies. These numbers describe different objects and should remain separate in both discovery and exposition.[44]

The installation's 369 shared-route checks and 381 scoped GEN4 checks include actual CPU/CUDA equality, weighted generic elimination, rational responses, crossing ties and portable recovery. A3D41 executes both retained frustrated six-spin phase sources through the shared implementation. Starbreaker recomputes the joint operator of its inherited N100 basis, preserves the original DOS and reads ordered opposite-sign endpoint fields. These adoptions connect the compiler to existing applications through their actual source maps.[44]

For the retained phase chart,

$$z = \frac{s+t}{2} + \mathrm{i}\frac{s-t}{2}, \tag{VI.9}$$

the relevant source relation is $E_{\text{spin}} = 2H_{\text{phase}}$ and $M = 2\sum \Re z$. The binary field therefore supplies an exact phase-alignment response in that model. It does not add material constituents or assign a magnetic unit. The application receives the response using the chart and energy convention it already supplies.[45]

### VI.5 A layered exact ground-state successor

The later dense-source method, H002708, addresses sources whose complete graph defeats small-component or low-width elimination but whose couplings contain a small number of collective signed layers. Its admitted decomposition is

$$J_{uv} = \sum_{k=1}^{K} a_k v_{ku} v_{kv} + S_{uv}, \qquad u < v, \tag{VI.10}$$

where $a_k$ and $S_{uv}$ are integers, $v_{ku} \in \{-1, 1\}$ and $S$ is sparse. With $M_k = \sum_u v_{ku} s_u$, the exact energy is

$$\begin{aligned} E(s) = &-\frac{1}{2}\sum_k a_k (M_k^2 - N) \\ &- \sum_{u<v} S_{uv} s_u s_v - \sum_u h_u s_u. \end{aligned} \tag{VI.11}$$

The implementation may use a floating-point eigensystem to propose the sign vectors. It then checks the displayed integer coupling identity over every pair. The proposal affects admission and work, while the accepted decomposition is an exact identity of the original source.[47, 48, 49]

Frontier elimination of the sparse residual retains the signs still needed by future residual edges and the full vector $(M_1, \ldots, M_K)$. Each state carries four pieces: minimum residual energy, exact number of minimizers, total represented configuration count, and a witness. These serve different purposes. A discarded higher residual energy cannot affect the minimum at fixed frontier and magnetizations, but its configurations still belong in the exhaustive accounting.[49]

For two contributions reaching the same state, the merge rule adds total configuration counts in every case. It keeps the lower energy and its minimizing count; at equal minima it adds their minimizing counts. Joining disconnected residual components uses min-plus convolution over magnetization

vectors, multiplying counts for independent choices. Only after all components have joined does the method add the collective quadratic energy and compare magnetization sectors.[49]

Consequently the retained table evaluates

$$E_0 = \min_{\mathbf{M}} \left[ e_{\mathrm{res}}(\mathbf{M}) - \frac{1}{2} \sum_k a_k (M_k^2 - N) \right], \qquad \text{(VI.12)}$$

while the separate totals satisfy $\sum_{\mathbf{M}} n(\mathbf{M}) = 2^N$. Ground degeneracy counts labelled configurations, including ties between distinct minimizing magnetization vectors. The returned witness is reconstructed and its energy recomputed against the original edge list and fields.[47, 49]

The answer is explicitly a ground-state object: energy, degeneracy, witness and exhaustive-accounting/decomposition certificate. The minimum merge does not retain all excited energies, so it supplies a different object from the full DOS or $g(E,M,b)$. A thermal partition function needs the corresponding full distribution or a separate justified construction. Keeping the return schema explicit lets the application request the object it actually needs.

### VI.6 Latest completed use and the computational boundary

The retained validation admits 52 of 60 planted sources at $N =$ 10 through 16; every admitted source agrees with complete enumeration in energy, degeneracy and witness. Small sources keep the DOS route. An explicitly selected other method is preserved. Automatic selection tries the layered route after variable elimination exceeds its admission. Sources without an admitted decomposition retain a capacity result identifying the relevant limits.[48]

In the recorded Duck1.2.4 Masquerade run, both unchanged raw dense SOLVE tasks complete through this fallback: source A at $N = 360$ in about 6.2 seconds and source B at $N = 180$ in about 4.4 seconds within their workers. Both SELECT tasks fail on an absent mathematical-library objects store. The exact solves and that separate store failure remain distinct outcomes of the same recorded run.[48]

Default limits of six layers, residual frontier sixteen and residual degree eight are implementation admission settings. They describe the retained execution route, with configurable work and memory ceilings. The contribution is the exact decomposition and the state carried by its dynamic program. It extends the original N100 lesson: structural compression can preserve the entire assignment accounting while evaluating the requested source observable with substantially less local work.[47, 48]

SAM VOLUME III

# The Algebra of a Complete Write

Chapter VII

Exact state counting and exact writes answer different questions. A write acts on a state, returns a result and retains the record needed to reconstruct what occurred. Here the translation compiler is derived within its finite group action, while the transaction and history interfaces retain the additional information that an endpoint alone cannot carry.

The account below follows the chapter source as retained at the fixed 8 September 2026 SAMA revision. Engine names, installed interfaces and campaign status are historical source statements. They do not describe a freshly inspected October deployment. Later source amendments are distinguished where available; no later engine or campaign is inferred.

## VII.1 Overview

A complete write is not merely an address increment. It is a coupled typed transaction. An unresolved W8 source participates in a B-activated request; X1 retains the transient response custody; a W9 record carries the resolved target and reciprocal receipt; U coordinates completion; both sites return to W8 while the history remains reconstructible.

The directional address is the finite group

$$\mathbb{Z}_4^9, \qquad |\mathbb{Z}_4^9| = 4^9 = 2^{18} = 262{,}144.$$

Each primitive signed write is a translation in one coordinate. Because translations form a group, any program made solely from those primitives can be compiled to one group element and applied in one bit-sliced pass. This is an exact algebraic compilation, not a heuristic shortcut. Sequential and compiled execution are bit-identical, and applying the inverse translation returns every address.

Endpoint return does not erase event history. Two reciprocal 81-coordinate views split into symmetric and antisymmetric sectors. One common constant direction is quotiented with an explicit anchor, leaving 80 symmetric plus 81 antisymmetric coordinates: a reconstructible rank-161 receipt. The transaction also separates the resolved projection from the conserved orthogonal remainder, supplies exact rational-complex analytic operators, and supports a multiplicative ledger whose write state is recoverable when the prior ledger state is invertible.

The development history is itself part of the result. The first installation corrected a preserved module-loader failure and installed the fast directional primitive. A later Weil fusion remained valid in its scope. H000137 then corrected the broader description: those surfaces did not yet expose the entire paper algebra as one composable transaction. The complete successor added the missing U/X1 lifecycle, common-anchor reconstruction, distinct projection and remainder, analytic operators, composable ledgers and native multi-write composition without editing its parents.

## VII.2 Opening question and conceptual picture

The governing question is:

> What exact algebra lets a reciprocal write change an address, retain both directional views, complete the W8–X1–W9 lifecycle, compose many writes into one action and recover what was written without confusing the final state with its history?

The conceptual transaction is

- W8 source + W8 target
- B activates request-side weld
- X1 holds transient response custody
- signed Z4^9 translation changes directional address
- W9 target record retains axis, orientation, ledger and endpoints
- U completes reciprocal exchange
- both sites return to W8
- W9 receipt and 161-coordinate history remain available

The address after completion may equal a later returned address, but the history need not vanish. State and history are differently typed outputs.

## VII.3 Typed lifecycle and the complete transaction

### VII.3.1 W8 shell

The unresolved shell is

$$W8 = \{0,1\}^8,$$

so it contains

$$|W8| = 2^8 = 256$$

shell states. A W8 site contains a shell-state identifier and a sequence of retained W9 records. Retention does not change the site back into W9; it lets the completed W8 endpoint carry its receipt lineage.

### VII.3.2 Activation weld

Let $H_{W8}(s)$ be the base vector Hamiltonian at shell state $s$, let $\Delta H_{X1}(s;a)$ be the transient response contribution at activation context $a$, and let $B \in \{0,1\}$. The weld is

$$H_{W9}(s;a) = H_{W8}(s) + B\,\Delta H_{X1}(s;a).$$

The two activation values provide an immediate control:

$$B = 0 \quad \Longrightarrow \quad H_{W9} = H_{W8},$$

$$B = 1 \quad \Longrightarrow \quad H_{W9} = H_{W8} + \Delta H_{X1}.$$

This $B$ is the binary activation input. It should not be confused with the analytic difference operator $B_t$ introduced later. X1 is a transient typed contribution and custody flag, not a ninth persistent W8 bit or a matter row.

#### VII.3.3 W9 record and projection

A W9 record can be written schematically as

$$W9 = (s, a, \rho, \ell; q_{before}, q_{after}),$$

where $s$ is the shell state, $a \in \{0, \dots, 8\}$ is the directional axis, $\rho \in \{-1, +1\}$ is orientation, and $\ell$ is the ledger receipt. The projection

$$\pi(W9) = s$$

recovers the W8 shell identity without deleting the other record fields.

#### VII.3.4 U/X1 completion

Activation starts with two W8 sites. The source remains W8 while X1 custody is active and the target holds the transient W9 record. Completion performs

- W8 source / X1 custody / W9 target
- append the same reciprocal receipt to both endpoint histories
- clear X1 custody
- return source and target phases to W8

The terminal equality $W9 = X1 + W8$ is therefore typed lifecycle shorthand, not ordinary material addition. `SAMA-D000039` develops that distinction and its failed literal-half controls in detail.

### VII.4 The signed $\mathbb{Z}_4^9$ address action

#### VII.4.1 Nine base-four coordinates

Let

$$q = (q_0, \dots, q_8), \qquad q_i \in \mathbb{Z}_4.$$

Each base-four coordinate uses two bits. Nine coordinates therefore occupy 18 bits and give

$$4^9 = (2^2)^9 = 2^{18} = 262{,}144$$

addresses. If the low bits form $L \in \{0,1\}^9$ and the high bits form $H \in \{0,1\}^9$, then

$$q_i = L_i + 2H_i.$$

The two nine-bit factors are a packing representation, not independent addresses. Base-four carry couples them.

#### VII.4.2 Primitive signed translation

For axis $i$ and sign $\varepsilon \in \{-1, +1\}$, define

$$T_{i,\varepsilon}(q)_j = \begin{cases} q_i + \varepsilon \pmod 4, & j = i, \\ q_j, & j \neq i. \end{cases}$$

Composing the opposite signed translation restores the changed coordinate,

$$T_{i,-\varepsilon}(T_{i,\varepsilon}(q))_i = q_i + \varepsilon - \varepsilon = q_i \pmod 4,$$

while every coordinate $j \neq i$ is unchanged by both maps. Therefore

$$T_{i,\varepsilon}^{-1} = T_{i,-\varepsilon}.$$

For addition of two packed translations $a, t \in \mathbb{Z}_4^9$, binary low/high vectors obey

$$L_{out} = L_a \oplus L_t,$$

$$H_{out} = H_a \oplus H_t \oplus (L_a \wedge L_t).$$

The last term is the modulo-four carry. Removing it would split the two 512 factors and change the directional action.

#### VII.4.3 Endpoint covariance

For $k \in \mathbb{Z}_4$, define simultaneous endpoint reversal

$$G_k(q)_i = k - q_i \pmod 4.$$

Then

$$G_k T_{i,\varepsilon}(q)_i = k - (q_i + \varepsilon) = (k - q_i) - \varepsilon = T_{i,-\varepsilon} G_k(q)_i,$$

and all other coordinates are unchanged. Hence

$$G_k T_{i,\varepsilon} = T_{i,-\varepsilon} G_k.$$

The implementation exhausts all covariance classes rather than assuming this identity from a sampled address.

### VII.5 Compiling a complete signed-write program

#### VII.5.1 Group-law reduction

Consider a program

$$\mathscr{P} = ((i_1, \varepsilon_1), \dots, (i_m, \varepsilon_m)).$$

Because $\mathbb{Z}_4^9$ is abelian under coordinatewise addition, the program's net translation is

$$\tau(\mathscr{P}) = \sum_{r=1}^{m} \varepsilon_r e_{i_r} \pmod 4.$$

Sequential execution gives

$$T_{i_m,\varepsilon_m} \cdots T_{i_1,\varepsilon_1}(q) = q + \tau(\mathscr{P}) \pmod 4.$$

The compiler therefore reduces the full signed-translation program to the one group element $\tau(\mathscr{P})$, applies one packed addition, and retains the inverse

$$-\tau(\mathscr{P}) \pmod 4.$$

### VII.5.2 Exact racecar result

The benchmark uses 4,096 writes at every one of 262,144 addresses:

$$4{,}096 \cdot 262{,}144 = 1{,}073{,}741{,}824$$

logical primitive transitions. Sequential native execution took 0.989033674 seconds. Compiled execution took 0.000554226 seconds. The observed ratio is

$$\frac{0.989033674}{0.000554226} = 1784.531387.$$

All outputs were bit-identical, and the inverse returned all 262,144 addresses. This measured speedup is reported for the stated workload and benchmark implementation. Exact endpoint compilation applies to programs composed from the declared signed $\mathbb{Z}_4^9$ translations; no uniform runtime factor follows. The compiler does not extend to application dynamics that have not been expressed in this group action.

## VII.6 Reconstructible reciprocal history

### VII.6.1 Two 81-coordinate views

Let the reciprocal histories be

$$L = (L_1, \ldots, L_{81}), \qquad R = (R_1, \ldots, R_{81}).$$

Form symmetric and antisymmetric coordinates

$$S = L + R, \qquad A = L - R.$$

Naively this gives 162 coordinates. A common constant shift occupies a one-dimensional kernel. The interface retains an explicit common anchor $c = S_{81}$, subtracts it from the first 80 symmetric coordinates, and stores

$$H_{nc} = H_{sym}^{80} \oplus H_{anti}^{81}.$$

Thus the quotient has

$$80 + 81 = 161$$

coordinates while the anchor preserves the removed constant class.

### VII.6.2 Reconstruction

Given stored nonconstant symmetric coordinates $\widetilde{S}_i = S_i - c$ for $1 \le i \le 80$, rebuild

$$S = (\widetilde{S}_1 + c, \ldots, \widetilde{S}_{80} + c, c).$$

Then

$$L = \frac{S + A}{2}, \qquad R = \frac{S - A}{2}.$$

Both reciprocal views are recovered exactly. A final endpoint return can therefore coexist with a nontrivial event receipt. The quotient removes only the shared constant redundancy; it does not discard directional information.

### VII.6.3 Lifted dyadic example

For a reciprocal incoming pair ((s,D)) and ((-s,-D)) sharing the same next state $(s_{next}, D_{next})$, define

- left = ( s, D, s_next, D_next, 0, . . . , 0)
- right = (-s,-D, s_next, D_next, 0, . . . , 0).

Then

- symmetric = (0,0,2*s_next,*2D_next,0,. . . ,0)
- antisymmetric = (2*s,*2D,0,0,0,. . . ,0).

The common next state appears in the symmetric sector, while the incoming branch identity remains in the antisymmetric sector. In the p=127 reference chain, all 125 lifted receipts reconstructed their prior and next states.

## VII.7 Completed-mode projection and retained remainder

Let $E$ be an exact energy vector and $C \neq 0$ the completed mode. The orthogonal projection is

$$\Pi_C(E) = C \frac{\langle C, E \rangle}{\langle C, C \rangle}.$$

The resolved response is defined separately as

$$R_{res} = -\Pi_C(E),$$

while the retained orthogonal remainder is

$$E_\perp = E - \Pi_C(E).$$

Orthogonality follows directly:

$$\langle C, E_\perp \rangle = \langle C, E \rangle - \frac{\langle C, E \rangle}{\langle C, C \rangle} \langle C, C \rangle = 0.$$

Therefore the exact norm decomposes as

$$\|E\|^2 = \|\Pi_C(E)\|^2 + \|E_\perp\|^2.$$

Projection is idempotent, and projecting the remainder returns zero. This is why resolved response and orthogonal remainder must remain distinct fields.

### VII.8 Exact analytic operators

Let $h_L, h_R \in \mathbb{Q}(i)^n$ be rational-complex reciprocal views. Define

$$B_t h = h_L - h_R,$$

$$P_+ h = \frac{h_L + h_R}{2}, \qquad P_- h = \frac{h_L - h_R}{2}.$$

The write energy is

$$E_t(h) = \|h_L - h_R\|^2.$$

With Hermitian correlations

$$C_{LR} = \langle h_L, h_R \rangle, \qquad C_{RL} = \overline{C_{LR}},$$

expansion gives

$$E_t(h) = \|h_L\|^2 + \|h_R\|^2 - C_{LR} - C_{RL}.$$

Because $2\|P_+ h\|^2$ and $2\|P_- h\|^2$ separate the symmetric and antisymmetric energies,

$$E_t(h) - \|h_L\|^2 - \|h_R\|^2 = 2\|P_- h\|^2 - 2\|P_+ h\|^2.$$

The executable receipt checks these identities with exact fractions rather than float tolerances. The operator $J$ supplies the declared reciprocal complex structure; $P_+$ and $P_-$ expose its symmetric and antisymmetric views.

### VII.9 Multiplicative ledger and readback

For modulus $M > 1$, prior ledger state $D_{now}$ and write state $s$, define

$$D_{next} = sD_{now} \pmod{M}.$$

If

$$\gcd(D_{now}, M) = 1,$$

then $D_{now}^{-1}$ exists and the written state is recovered exactly:

$$\widehat{s} = D_{next} D_{now}^{-1} \pmod{M} = s.$$

Noninvertibility is a typed outcome and returns no fabricated readback. For a program $s_1, \ldots, s_m$, composition yields

$$D_m = \left( \prod_{r=1}^{m} s_r \right) D_0 \pmod{M}.$$

The separate character map

$$\chi : \mathbb{Z}_4 \to \mu_4 = \{1, i, -1, -i\}$$

uses $\chi(q) = i^q$. The directional write readback is

$$\chi(q_{after}) \overline{\chi(q_{before})} = \chi(q_{after} - q_{before}),$$

which recovers the signed base-four transition in the fourth roots of unity.

### VII.10 Development and deviation chain

#### VII.10.1 Preserved loader failure

The first H000132 execution stopped before algebra evaluation because the Python 3.12 dynamic parent loader had not registered its module before dataclass processing. The unused compiled binary was preserved. A successor changed only loader registration; equations, contract, benchmark task and performance condition remained fixed.

#### VII.10.2 Valid primitive, incomplete total surface

H000132 then installed the 256-state shell, packed $\mathbb{Z}_4^9$ action, covariance, rank-161 quotient, completed-mode receipt and ledger readback. H000136 fused that directional primitive into the Weil adapter and retained prime-spacing, prime-power and W9 history channels.

Those installations were correct in scope, but calling them the complete paper algebra omitted load-bearing layers: the two-site U/X1 lifecycle, explicit activation, common-anchor reconstruction, separate resolved response and remainder, $B_t, E_t, J, P_+, P_-$, composable transactions and native program compilation.

#### VII.10.3 Complete correction and retest

H000137 installed a successor containing the entire surface while leaving H000132 and H000136 unchanged:

- preserved primitive and fusion
- identify missing transaction layers
- freeze complete-algebra contract
- implement one typed interface
- primary execution 19/19
- independent reconstruction 14/14
- sequential/compiled bit identity
- inverse return over all addresses

This is conceptual and implementation correction without retrospective failure inflation: the parents retain their narrower accomplishments, and the successor owns the claim of completeness.

#### VII.10.4 Read-only consumer boundary

The directional Weil adapter may consume exact write state, reciprocal history, prime-spacing and prime-power channels. Consumption does not grant the adapter scheduler, solver or primality authority. Factor assignment and primality assignment remain different result types. An application also cannot omit the dynamics needed to construct its input ledger merely because the ledger readback is exact.

At this retained source stage, `SLCQ2-RZ` is the default selection engine. Complete Exact Write survives as a delegated service in the frozen, noncurrent `SLCV21R` arithmetic/application base. Historical installation pointers describe their own time; they are not current-authority statements.

### VII.11 Established result, boundary and forward handoff

The Complete Exact Write surface executes the complete W8/B/X1/W9/U transaction, the coupled signed $\mathbb{Z}_4^9$ address action, exact inverse and endpoint covariance, reconstructible 161-coordinate history, completed-mode projection, analytic reciprocal operators and multiplicative-ledger readback. Its compiler reduces any declared signed-translation program to one exact group element and reproduces sequential output bit for bit.

The result does not convert unrelated dynamics into translations, assign physical meaning to X1, create scheduler authority, assign primality or alter current Q2 selection. The history surface hands forward to `SAMA-D000042`; the Prime-Ruler application boundary hands forward to `SAMA-D000056`; and the current selection-plane relation hands forward to `SAMA-D000055`.

### VII.12 Sources and evidential scope

This chapter is assembled from the following fixed source manuscripts. Source-reported campaign outcomes retain their original scope; no new campaign was executed for this edition.

- Exact algebra and complete exact write.

SAM VOLUME III

# Lifecycle, Reception and Carrier Return

Chapter VIII

The algebraic action must be received by a typed lifecycle. This chapter follows the source roles without treating a carrier, a matter row, an unresolved site and a completed receipt as interchangeable. Corrections to earlier literal and causal interpretations remain visible alongside the corrected finite constructions.

The account below follows the chapter source as retained at the fixed 8 September 2026 SAMA revision. Engine names, installed interfaces and campaign status are historical source statements. They do not describe a freshly inspected October deployment. Later source amendments are distinguished where available; no later engine or campaign is inferred.

## VIII.1 Overview

The W8–X1–W9 lifecycle describes one reciprocal transition, not three freely interchangeable objects. W8 is the unresolved eight-position substrate state. B/contact prepares the request-side transition. X1 is the one transient reciprocal response channel. W9 is the resolved composite state or terminal receipt of the completed relation,

$$W9 = X1 + W8.$$

The equality is typed. It cannot be read as ordinary inventory addition, as a claim that X1 is matter, or as an equation saying B and X1 are two scalar halves. The exact program surface can require B and X1 as differently typed inputs and return W9 while a physical causal weld remains open. That is not a defect in the lifecycle. It is the reason the lifecycle is useful: syntax, execution, receipt, carrier and physical identification can be developed without being collapsed into one assertion.

The evidence chain is cumulative. CR120V rejected the literal two-half entity formula while retaining a typed outward/return relay. CR120W then preserved a failed attempt to weld the A-field carrier directly to X1. CR120X corrected the route by using Theta18 as a carrier-normalization quantum while keeping X1 as the response type, and CR120Y stress-tested that separation. CR120Z and CR120ZB carried the ordered B-then-X1 grammar into exact and finite-shot quantum sandboxes, but neither installed physical definitions of B or X1.

## VIII.2 Opening question and conceptual picture

The governing question is:

> How can an unresolved substrate state receive contact, return reciprocal information and finish as a resolved write while retaining the identity of the request, response, carrier and receipt?

The compact picture is

- W8 unresolved source/site state
- – B/contact prepares the request –>
- X1 transient reciprocal response
- – typed resolution –>
- W9 resolved composite state and retained receipt

This picture contains two related but nonidentical descriptions.

1. **State description.** W8 and W9 name the unresolved and resolved ends of the relation.
2. **Event description.** B/contact and X1 name the differently typed request-side and response-side participation needed to traverse between those ends.

The lifecycle therefore keeps endpoint state separate from the event that produced it. A terminal W9 receipt records that the typed route completed; it does not turn W9 into a new independent source object. Similarly, X1 can be required by the type system without the installed runtime exposing an independently variable X1 payload.

## VIII.3 Definitions, domains and type boundaries

### VIII.3.1 The four lifecycle objects

- **W8 / `S8_BINARY_SURFACE`.** Type in this document: unresolved binary substrate/site surface; Function: supplies the source-side state to be resolved; Explicit nonidentity: not W9; not an observed matter row.
- **B / `B_CONTACT_OPERATOR`.** Type in this document: non-scalar contact/request operator; Function: prepares or admits the request-side transition; Explicit nonidentity: not X1; not scalar `1/2`.
- **X1 / `X1_AXIS_SELF_CHANNEL`.** Type in this document: axis/response channel with registered scalar value 1; Function: supplies the transient reciprocal response-side occurrence; Explicit nonidentity: not Theta18, matter, a row or two half-operators.
- **W9 / `W9_CLOSURE_WITNESS`.** Type in this document: closure witness and resolved composite; Function: records the completed typed relation `X1+W8`; Explicit nonidentity: not a third primitive object and not scalar-nine `CARRIER9`.

The base identity is

$$\mathrm{RESOLVE}(W8, B, X1) \longrightarrow W9, \qquad W9 = \{X1 + W8\}.$$

The braces emphasize a completed relation. The expression is not a license to remove B, exchange B and X1, or infer that two equal scalar contributions were added.

#### VIII.3.2 Source row, operator, carrier and receipt are different domains

The lifecycle crosses several domains, each with its own counting rule:

- **N100 source grammar.** Counted object: 100 independent source rows/spins; What is not counted there: five derived W9 receipts, X1, non-row A operator.
- **105-row visible QP lane.** Counted object: all source records remain visible; What is not counted there: counted/not-counted overlay does not delete a row.
- **lifecycle runtime.** Counted object: typed operator arguments and one result entity; What is not counted there: scalar values are not an information-capacity count.
- **Theta carrier lane.** Counted object: released carrier unit of size 18; What is not counted there: X1 response identity and ordinary matter lift.
- **history/readback.** Counted object: terminal and ordered event receipts; What is not counted there: an additional material object.

The five W9 records in the N100/RH grammar are terminal records of completed `X1+W8` relations. They remain visible for custody, but they contribute no new physical/source row. Counting them as five more independent objects would change the grammar from 100 to 105 rather than describe its execution.

#### VIII.3.3 Non-row A and the open causal seams

Volume II supplies a narrower installed provenance chain:

- A_OPERATOR: established non-row transform
- B/contact -> X1 axis fee -> W9: supported typed bridge
- A_OPERATOR -> B/contact: open
- direct A_OPERATOR -> X1: open

The A-field carrier, `A_OPERATOR`, its inactive historical row proxy, B, X1 and Theta18 are distinct. Participation in one 162-ledger structure does not fill either open arrow.

#### VIII.3.4 Historical language evidence and current SLC identity

G06 belongs to the preserved SAM Language v0.6 formal-simulator evidence. It supplies frozen B prepare/request and X1 response operators. The later SAM Language v0.7 object is a typed reflective-tape candidate. Neither language version is the SLC described by this retained source revision.

The immutable Volume III technical spine records the SLC authority that was current when that spine was written. At the retained source stage, live authority superseded that earlier pointer wording: `SLCQ2-RZ` is the sole frozen/default SLC, and its byte-identical delegated arithmetic/application base remains separately typed. This chapter uses the historical language and Courtroom artifacts as evidence for the lifecycle; it does not promote their version labels into the current SLC identity.

### VIII.4 Typed lifecycle construction from start to finish

#### VIII.4.1 The formal operator premise

The language-level premise separates preparation from response. In G06, B contains a binary flip but is not exhausted by that flip, while X1 has its own frozen response matrix. The executable checks proceed in four stages:

1. resolve both operators through the single formal registry;
2. verify their frozen hashes and declared input/output shapes;
3. apply B to every declared prepare/request column and X1 to every response column; and
4. apply the involutions and recover the source basis exactly.

The hash, column, mapping and involution checks all pass. Exchanging request and response, reducing B to its binary-flip suboperation, or selecting an unfrozen sign/gauge would test a different operator surface.

#### VIII.4.2 The typed `RESOLVE` gate

CR120A executes

- `S8_BINARY_SURFACE` : BinarySurface, Layer1StructuralCapacity
- – RESOLVE(B_CONTACT_OPERATOR, `X1_AXIS_SELF_CHANNEL`) –>
- `W9_CLOSURE_WITNESS` : ClosureWitness, Layer2Resolution

The runtime does not merely check `8+1=9`. It enforces the operator signature. The exact control sequence is:

- RESOLVE(S8, B, X1) -> W9 PASS
- PROMOTE_VOLUME(W9, D3) -> V27 PASS
- PROMOTE_VOLUME(S8, D3) TYPE_REJECT
- RESOLVE(S8, B) ARITY_REJECT
- RESOLVE(S8, B, B) TYPE_REJECT
- RESOLVE(S8, X1, X1) TYPE_REJECT
- W9 == CARRIER9 AUTHORITY_REJECT

This establishes a typed unresolved-to-resolved gate. Its secondary boundary is equally important: the runtime does not expose an independently perturbable X1 payload or a recoverable W9 preimage, filling or correction certificate.

#### VIII.4.3 What the installed runtime computes—and what it only names

CR120D opens the runtime and follows the calculation step by step:

1. resolve the three named entities;
2. require input types `(BinarySurface,\ContactOperator,\AxisChannel);`
3. check authority;
4. select the registered fixed result entity `W9_CLOSURE_WITNESS;`

5. emit the operator name, inputs, result, typed-plan trace, source trace and warning field.

The X1 registry entry is immutable, has scalar value 1 and has no syntax for preparing `x0`, preparing `x1`, resetting X1, or varying X1 while S8 and B are fixed. The current `RESOLVE` output alphabet is the singleton

$$\{W9\},$$

so the installed fixed-result runtime contains zero bits of input-dependent W9 variation. That is a runtime-contract boundary, not a physical null experiment and not an information-capacity interpretation of the scalar 9.

### VIII.4.4 Structural continuation to V27, F81 and P80

CR120E executes the research-mode structural route

- S8 –B through X1–> W9
- W9 -> V27 -> F81
- `RESERVE_CLOSURE_ADDRESS`(F81, X1) -> P80

The returned `P80_PARTICLE_FACE_CONTENT` has value 80 and authority `STRUCTURAL_ONLY`; normal mode rejects the reserve operation. The supported structural relation is

- P80 structural row inventory
- – X1 non-row closure/contact address
- = F81 completed-face capacity.

This is narrower than a linear quotient theorem. A rank-one quotient would require F81 and P80 vector spaces, a linear reserve map and surjectivity. Those premises are not registered. The source-native search also found neither a punctured $9\times 9$ row map nor a ten-octet grouping. The structural capacity weld therefore remains separate from a row-to-face decoder.

### VIII.4.5 Constructed split, residue and mirror-odd candidate

CR120G asks whether a stateful internal response can be constructed without losing the residue. Its exact rational candidate satisfies

- parent - left - right - kappa = 0
- omit kappa = e2, scalar residual 1
- kappa_B(B0 identity) = e2
- kappa_B(B1 swaps e1/e2) = e1
- H1=[e1] stateful readout = 1
- H2=[e2] stateful readout = 0
- static memoryless control = one fixed value for both histories

The mirror model reconstructs both faces, projects its odd sector away from a predeclared X1 fixture direction and returns rank 80. Transverse and mixed classes are non-null with exact squared norm 5; balanced, X1-only and degenerate classes are null. The wrong projector fails to annihilate X1 and changes the transverse norm from 5 to 4.

This is a validated internal mathematical construction, not an installed SAM route. No registered F81 vector space, X1 vector embedding, inner product, loop split or 80-coordinate decoder is supplied by the language. The model is therefore useful as a discriminating construction while its physical and installed mapping remains open.

## VIII.5 Derived W9 receipts and the source ledger

### VIII.5.1 The 105-visible/100-independent overlay

CR120K preserves all 105 visible QP records and marks exactly five `p=9,g=0` records as derived W9 witnesses. The independent-content census is

- visible records 105
- derived W9 witnesses 5
- independent-content records 100
- baseline M_native 16250.625
- duplicated linear W9 content 50.625 = 405/8
- independent M_native 16200
- independent mean 162

No source record is deleted. The overlay changes only whether the record is counted as independent content.

### VIII.5.2 Worked exact parent examples

Each derived p9 record has p8 and p1 parents of the same role and generation. For example,

$$M_9(\text{QP093A-0019}) = \frac{45}{4},$$

while

$$M_8(\text{QP093A-0016}) + M_1(\text{QP093A-0001}) = 10 + \frac{5}{4} = \frac{45}{4}.$$

Likewise,

$$12 + \frac{3}{2} = \frac{27}{2}, \qquad 1 + \frac{1}{8} = \frac{9}{8}.$$

All five linear residuals are exactly zero. Isolated parent perturbations change the derived candidate while the direct-nine candidate stays fixed, which supplies a dependency discriminator. The records remain separate because downstream `qA`, observed-mass and debit/credit fields can have nonzero residuals even when `M_native` closes linearly.

### VIII.5.3 The CR120T accounting and binding boundary

CR120T carries the ledger proposal through distinct gates rather than treating one exact total as a universal result. The accounting ladder closes:

$$100L = 100 \cdot 162 = 16{,}200,$$

$$100N = 100 \cdot 144 = 14{,}400, \qquad 100M = 100 \cdot 126 = 12{,}600,$$

with

$$L = N + \Theta, \qquad M = N - \Theta, \qquad N = M + \Theta.$$

Scalar arithmetic does not select the source route by itself: p6+p3 can also sum to p9. The typed registry privileges p8+p1 because it supplies the `S8`/X1/W9 roles.

The downstream gates do not promote. The best frozen ID-free, value-blind, shuffle-invariant and conjugate-symmetric F81 rule selects 84 rows with `M_native=14077.125`, not the supplied 81 rows with total 12,600. No three-row exception patch is admitted. The binding successor replaces one feature with the training-frozen `abs(N-Z)/A^(1/3)` term and worsens every reported RMS lane:

- **CR261 20-row holdout.** Baseline RMS MeV: 3.017113068; Candidate RMS MeV: 4.071613571; Change: +1.054500503.
- **CR277 78-row extension.** Baseline RMS MeV: 7.974144225; Candidate RMS MeV: 8.067445504; Change: +0.093301279.
- **CR277 118-row observed.** Baseline RMS MeV: 6.672483502; Candidate RMS MeV: 6.828884165; Change: +0.156400663.

The source verdict is `BOUNDARY_DISCOVERY_SUPPORTED_PROJECTION_OR_BINDING_OPEN`; the binding lane is `NO_FROZEN_BINDING_IMPROVEMENT`. The accounting result remains available without being converted into a binding coefficient.

## VIII.6 The preserved deviation chain

### VIII.6.1 First deviation: the literal two-half entity formula

CR120V recovers a real reciprocal half-slot grammar. The outward and inward accounts are

$$\frac{1}{2}SW_{out} + \frac{1}{2}WRITE_{out} + \frac{1}{2}OUTSIDE = \frac{3}{2},$$

$$\frac{1}{2}INSIDE + \frac{1}{2}WRITE_{in} + \frac{1}{2}SW_{in} = \frac{3}{2},$$

so the complete paired route is

$$\frac{3}{2} + \frac{3}{2} = 3 = D.$$

The failure was assigning this route arithmetic to two entities:

$$B = \frac{1}{2}, \qquad X1 = \frac{1}{2}.$$

B is non-scalar, X1 is an AxisChannel with scalar 1, and the shell has six half-slots rather than two. CR120V therefore rejects the literal entity formula while retaining the typed B-side/X1-side relay as an archival hypothesis. Thirteen of thirteen hashes match and eight of eight wrong controls reject; no physical intervention is executed.

### VIII.6.2 Second deviation: candidate uniqueness is not a weld

CR120W searches 26 partition-one QP occurrences under four frozen gates:

- 26 partition-one occurrences
- 3 legal nonlocal infrastructure rows
- 2 neutral closed carrier-axis shells
- 1 non-additive address role
- 1 shared environmental-axis survivor

The unique survivor is QP093A-0305 `A_FIELD_CARRIER`, with zero native mass, zero arity, `a_kernel_support` route, `carrier_axis`, non-matter ownership and closed carrier support. The candidate ledger keeps twelve additive modes at 81, the mirror/checksum at 81 and closure at 162. Restoring the historical A-field proxy gives the wrong controls 82 and 163.

The attempted executable statement

- `REALIZES_AXIS_CHANNEL`(A_FIELD_CARRIER, `X1_AXIS_SELF_CHANNEL`)

does not close. The result preserves verdict `FAIL_CR120W_SOURCE_OR_CANDIDATE_RECONSTRUCTION`, 25/25 frozen source hashes, 11/12 wrong controls and no same-run repair. The useful discovery is unique typed candidacy; the direct `A_OPERATOR -> X1_AXIS_SELF_CHANNEL` edge remains uninstalled.

### VIII.6.3 Correction: the Theta-normalized dual-depth route

CR120X changes the executable question instead of relabeling the failure. It keeps A, B, X1 and Theta differently typed and asks whether an existing carrier quantum normalizes the retained/native/full ledger:

- A route at d=1 $R^{(d+1)=R}2$=N=144
- B contact 144 -> (126 retained, 18 released)
- Theta normalization 126/18 = 7
- 144/18 = 8 = S8
- 162/18 = 9 = W9
- normalized release 18/18 = 1 = X1 scalar contact
- typed resolution RESOLVE(S8,B,X1) -> W9

Every registered carrier is substituted using native payload and partition scalar. QP093A-0300 `TENSOR_CARRIER`/Theta18 is the sole candidate that reproduces the complete ladder in both comparisons. All 32 CR256 rows remain exact to maximum reported residual `5.000e-08`; only route depth `d=1` gives `R^(d+1)=144=S8*Theta`.

This correction gives Theta18 the carrier-normalization job and X1 the response occurrence. Their ratio is one in this construction, but their types remain different. The campaign records 20/20 source hashes, 11/11 hard gates, 1/1 supportive gate and 20/20 rejected wrong controls. The canonical ledger is 162; restored-row 163 remains the wrong control.

#### VIII.6.4 Retest: model-wide Theta18 stress

CR120Y then removes the possibility that 18 was selected from one attractive ratio. It asks whether the same carrier survives across independent evidence:

- **exact derivation routes.** Result: 6/6 converge on 18.
- **leave-one-family-out folds.** Result: 6/6 retain Theta18.
- **structural output family `{6,9,18,54}`.** Result: Theta18 alone performs the primary-carrier job.
- **A/Theta route depth.** Result: `d=1` only.
- **QP carrier row.** Result: QP093A-0300 only, in both frozen value modes.
- **Starbreaker ledgers.** Result: 96/96 preserve 162-slot and 18/126/18 typing.
- **wrong controls.** Result: 12/12 reject.

Theta18 is retained as the current primary substrate-carrier candidate in the scope of this source result. It remains carrier-only: not X1, matter, rest mass or a fitted mass term. The retest supplements CR120X; neither result rewrites CR120V or CR120W.

### VIII.7 Ordered quantum-sandbox successors

#### VIII.7.1 Exact minimal ordered model

CR120Z supplies a prospective minimal realization of the typed order:

- |00> – B:H_A – X1:CNOT_A->B –> |Phi+>

The resulting Bell state has negativity 0.5, concurrence 1, fidelity 1 and CHSH `2.82842712475`. B alone, X1 alone, X1-before-B and a classical correlated-return state all have zero negativity. In the relay extension, an X1-indexed Bell response on the middle systems heralds a remote pair with the same negativity and CHSH values, while product measurement and unresolved false heralds remain separable. Local marginals stay invariant to maximum trace distance `1.11e-16`.

The noise thresholds reproduce the analytic boundaries: the first grid point with nonzero negativity is `v=0.335`, immediately above `1/3`, and the first frozen-setting CHSH violation is `v=0.71`, immediately above `1/sqrt(2)`. Custody closes 10/10 hashes, 9/9 hard gates and 12/12 wrong controls.

This demonstrates quantum capability of the typed order in the sandbox. H and CNOT are prospective maps, not installed physical definitions of B and X1; W9 is not used as a Bell witness.

#### VIII.7.2 Finite-shot NV boundary

CR120ZB moves the same order discriminator through three inherited NV simulator profiles, finite sampling, readout confusion, SPAM, relaxation, dephasing, drift, linear inversion and PSD reconstruction. Each reconstructed state uses all 15 non-trivial Pauli observables with 4,096 shots per observable. All 2,160 reconstructed states are physical.

- **room-temperature NV.** Forward median negativity: 0.4615; Forward q05: 0.4496; Maximum control q95: 0.0088; Positive gaps: 48/48; Median CHSH: 2.6915; CHSH>2 blocks: 48/48.
- **cryogenic NV.** Forward median negativity: 0.4708; Forward q05: 0.4624; Maximum control q95: 0.0123; Positive gaps: 48/48; Median CHSH: 2.7337; CHSH>2 blocks: 48/48.
- **noisy NV.** Forward median negativity: 0.4426; Forward q05: 0.4126; Maximum control q95: 0.0104; Positive gaps: 48/48; Median CHSH: 2.6256; CHSH>2 blocks: 48/48.

The signal survives and all 12 wrong controls behave correctly, but hard gates pass 8/9. The exact source verdict is `BOUNDARY_X1_ORDERED_QUANTUM_INSTRUMENT_FINITE_SHOT_NV_DISCOVERY__ONE_OR_MORE_FROZEN_GATES_FAILED`. No live backend ran. The result is a simulator-to-device handoff boundary, not a physical B/X1 weld or installed SAM operation.

### VIII.8 What the controls establish

The controls are part of the derivation, not an appendix to it:

- omitting or exchanging B and X1 verifies that the `RESOLVE` signature is typed rather than scalar arithmetic;
- distinguishing `W9_CLOSURE_WITNESS` from scalar-nine `CARRIER9` prevents a value collision from becoming entity identity;
- keeping all 105 records visible while counting 100 prevents source deletion from masquerading as an independent-content result;
- perturbing p8 and p1 parents tests dependency rather than only equality;
- the 163 restored-row lane shows that reinstalling the historical A proxy changes the ledger object;
- the literal-half rejection prevents route slots from being reified as B and X1 entities;
- the CR120W 11/12 control outcome prevents unique candidacy from being promoted through a failed reconstruction;
- leave-one-family-out replay tests whether Theta18 survives removal of any one evidence family;
- reversed-order, product, classical and false-herald quantum controls show that ordinary correlation or label balance is not the ordered signal; and
- the 8/9 hard-gate result in CR120ZB keeps a strong simulated signal and an unpassed device-handoff gate visible at the same time.

### VIII.9 Established result and current boundaries

The source-bound lifecycle can now be stated compactly:

1. W8, B, X1 and W9 have distinct registered types.
2. The formal runtime requires `RESOLVE(S8,B,X1)` and returns W9.
3. W9 is a resolved composite receipt, not a new source row or object.

4. The fixed runtime does not expose independent X1 intervention or an input-dependent W9 outcome alphabet.
5. The literal B-half plus X1-half entity equation is rejected; the typed outward/return relay remains useful.
6. QP093A-0305 is the unique typed A-field-carrier candidate in CR120W, while the direct executable A-to-X1 weld remains open.
7. Theta18, not X1, performs the tested carrier-normalization job across the 7|8|9 ladder and the model-wide stress surface.
8. Exact and finite-shot quantum sandboxes preserve the ordered B-then-X1 discriminator, but no hardware or physical weld is installed.
9. CR120T closes ledger accounting while leaving the F81 projection and binding transfer open.
10. None of these historical test labels changes the live `SLCQ2-RZ` current identity.

No atomic record assigned to this document carries an authorized project result classification. This revision therefore preserves exact source statuses and verdicts without manufacturing one.

### VIII.10 Connections and forward handoff

- **SAMA-D000025.** Connection: Supplies the Volume II connector, substrate-atom and container types and the open A-to-B seam..
- **SAMA-D000034.** Connection: Uses the five derived W9 receipts as a non-counted boundary in the N100 census..
- **SAMA-D000035.** Connection: Owns the parser, type system, operator registry and state/history semantics underlying the formal lifecycle evidence..
- **SAMA-D000040.** Connection: Develops Theta18 as carrier, coupled address and M126-to-N144-to-L162 return..
- **SAMA-D000042.** Connection: Carries the resolved lifecycle into directional history and event-reel computation..
- **SAMA-D000044.** Connection: Uses the typed lifecycle in the cross-scale Nonzero Amplification bridge..
- **SAMA-D000045.** Connection: Uses retained W8/X1/W9 history and Theta activity in relational composition..

The immediate forward handoff is D40: keep X1 as the response channel, move the carrier burden to Theta18, derive its coupled address, and follow the retained state through M126, N144 and L162 without turning a readout ceiling into independent matter.

### VIII.11 Sources and evidential scope

This chapter is assembled from the following fixed source manuscripts. Source-reported campaign outcomes retain their original scope; no new campaign was executed for this edition.

- W8 x1 w9 lifecycle.

SAM VOLUME III

# Directional History and Event Reels

Chapter IX

A history is more than a sequence label attached to a state. Reciprocal endpoints define an event incidence, its Gram and a shared-anchor reel. The elementary positivity of an event is established within that construction; it is not promoted to positivity of every completed arithmetic expression that may later consume the event.

The account below follows the chapter source as retained at the fixed 8 September 2026 SAMA revision. Engine names, installed interfaces and campaign status are historical source statements. They do not describe a freshly inspected October deployment. Later source amendments are distinguished where available; no later engine or campaign is inferred.

## IX.1 Overview

Directional computation keeps two things at once: the state reached by a write and the ordered reciprocal history that produced it. The generic core acts on all 262,144 Theta18 addresses through 18 directional maps, completes the W8/X1/W9 lifecycle and retains 72 exact receipts before any RH, Starbreaker or Mersenne coefficients are applied.

Two reciprocal 81-coordinate histories have an ambient 162-dimensional description. Their symmetric and antisymmetric decomposition contains 80 nonconstant symmetric coordinates, 81 antisymmetric coordinates and one common constant direction. The nonconstant quotient therefore has rank

$$161 = 80 + 81,$$

while the full ambient accounting is

$$162 = 80 + 81 + 1.$$

The retained common anchor reconstructs both sides exactly. Endpoint return does not erase ordered history.

The event-reel construction lifts this local receipt into an append-only global object. For an interaction translation $U_e$, the event incidence is $B_e = I - U_e$ and the unique event Gram is $\Theta_e = B_e^* B_e$. Reciprocal source/target orientations are two references to one intersection event, not two Theta objects. Appending an event adds its positive weighted Gram. A shared-reference N144 AFC cell realizes the W9 pair and the one-Theta lane with equal energy, and an event family with $|\mathscr{E}|$ events has $1 + 143|\mathscr{E}|$ nodes, rank $143|\mathscr{E}|$ and one common scalar kernel.

The installed derivation is cumulative because several stronger shortcuts failed. Generic positive Stieltjes pencils do not force reel contraction; a scalar RII discriminator is insufficient; the local N144 frame alone does not dominate the completed debit; an isolated recurrence lane changes sign; a whole-matrix AFC order fails; individual parity factors fail even when their product closes; and a spherical diagonal Gram order has negative directions. Each failure narrows the selected statement to the exact event Gram, shared-anchor intertwiner and endpoint scalar actually supported by the source.

## IX.2 Opening question and conceptual picture

The governing question is:

> How can every reciprocal write append one exact history event, preserve the common stationary state and reconstruct both directions without either duplicating the event or hard-coding one application's coefficients into the generic history core?

The complete picture is

- local signed Z4^9 write
- W8 source / X1 transition / W9 target
- reciprocal W9 pair W9P
- AFC intersection
- one massless Theta event Gram
- append event to shared-anchor reel
- application adapter reads declared receipt

The source state may return to W8 at completion. The reel nevertheless grows because the receipt records the event, not merely the terminal address.

## IX.3 Generic directional core

### IX.3.1 Exact inventory

The generic core contains

- **Theta18 addresses.** Exact inventory: 262,144.
- **packed factorization.** Exact inventory: $512 \times 512 = 4^9$.
- **directional maps.** Exact inventory: 18.
- **W8/X1/W9 receipts.** Exact inventory: 72.
- **nonconstant history rank.** Exact inventory: 161.
- **nonconstant symmetric coordinates.** Exact inventory: 80.
- **antisymmetric coordinates.** Exact inventory: 81.
- **common constant kernel.** Exact inventory: 1.
- **implementation checks.** Exact inventory: 42/42.

The 18 maps are the nine axes with two signed orientations. Four endpoint reversal classes applied to those 18 directions give the 72 receipt census:

$$9 \cdot 2 \cdot 4 = 72.$$

The core performs exact write, inverse return, direction recovery and history reconstruction before an adapter supplies application-specific weights.

#### IX.3.2 State and path are separate outputs

For reciprocal histories $L, R \in \mathbb{Q}^{81}$, define

$$S = L + R, \qquad A = L - R.$$

One common constant coordinate is redundant in the nonconstant quotient. With anchor $c$, store

$$\widetilde{S}_i = S_i - c, \quad i = 1, \ldots, 80,$$

and all 81 entries of $A$. Reconstruction is

$$S = (\widetilde{S}_1 + c, \ldots, \widetilde{S}_{80} + c, c),$$

$$L = \frac{S + A}{2}, \qquad R = \frac{S - A}{2}.$$

The quotient retains 161 nonconstant coordinates; the anchor carries the one constant direction. A final address alone cannot reconstruct this ordered pair, which is why endpoint-only controls are deliberately insufficient.

#### IX.3.3 Append-only lifecycle

Exact Write completion returns both sites to W8 and appends the same reciprocal receipt to both immutable history tuples. The two endpoint references point to one event object. Appending later events never edits earlier receipts:

- reel_0 = ()
- reel_1 = (theta_1)
- reel_2 = (theta_1, theta_2)
- ...
- reel_n = (theta_1, ..., theta_n)

An extremal or compressed finite reader may summarize this reel. It must not be renamed as the full append-only object.

### IX.4 Event incidence and unique Theta Gram

#### IX.4.1 One translation, two ends, one event

Let $U_e$ be the exact unitary or permutation translation for event $e$. Define the incidence operator

$$B_e = I - U_e.$$

The unique event Gram is

$$\Theta_e = B_e^* B_e \succeq 0.$$

For any state $h$,

$$\langle h, \Theta_e h \rangle = \langle B_e h, B_e h \rangle = \|h - U_e h\|^2 \geq 0.$$

Replacing $e$ by its reciprocal orientation sends $U_e$ to $U_e^{-1}$. The directed incidence changes orientation, but the source/target pair still defines one undirected event energy. On cyclic controls, rowwise incidences sum exactly:

$$(I - U_e)^*(I - U_e) = \sum_{position} b_{e,position}^* b_{e,position}.$$

Thus two W9 directions are the pre-intersection readback of one Theta Gram, not two counted carriers.

#### IX.4.2 Append rule and stationary kernel

With positive event weight $w_e$, the reel Laplacian evolves by

$$L_{n+1} = L_n + w_e \Theta_e.$$

If the underlying event graph is connected on $m$ sites, its incidence Laplacian has rank (m-1), and the constant vector is the kernel:

$$L_n \mathbf{1} = 0.$$

The stationary substrate baseline therefore survives every event. Appending history grows the nonconstant range while preserving one common mode.

### IX.5 N144 AFC frame

#### IX.5.1 F81 contact incidence

The matched F81 contact graph factors as

$$G_{F81} = P_3 \square G_{V27},$$

so its Laplacian is

$$L_{F81} = L_{P3} \otimes I_{27} + I_3 \otimes L_{V27}.$$

The two address-preserving handoffs contribute 27 edges each. Their once- oriented AFC incidence has shape $54 \times 81$, rank 54 and

$$(-B_{AFC})^T (-B_{AFC}) = B_{AFC}^T B_{AFC}.$$

Orientation reversal therefore retains one Gram. Internal incidence has rank 78; completed incidence has rank 80; the only final kernel is $\mathrm{span}\{\mathbf{1}_{81}\}$. Exact LDL reconstruction gives

$$\lambda_2(L_{F81}) = 1.$$

The gap is carried by the completed incidence while AFC remains a zero-mass carrier.

#### IX.5.2 Frozen N144-to-L162 wiring

The shared state is

$$z = (U_{left}^{63}, U_{right}^{63}, \Theta_{shared}^{18}) \in \mathbb{R}^{144}.$$

The same-view embedding duplicates the shared Theta coordinates into the two occurrence lanes:

$$E_{same} z = (U_{left}^{63}, \Theta^{18}; U_{right}^{63}, \Theta^{18}) \in \mathbb{R}^{162}.$$

This map has rank 144. Compose it with the Exact Write quotient $Q_{81}$:

$$A = Q_{81}E_{same}.$$

With the retained anchor $a = \Theta_{17}$, exact composition gives

$$\|Az\|^2 = 2 \sum_{126\ U\ leaves} (z_i - a)^2 + 4 \sum_{17\ nonanchor\ Theta\ leaves} (z_j - a)^2.$$

Thus $A^TA$ is a weighted-star Laplacian with one kernel, 126 weight-two leaves and 17 weight-four leaves. Its spectrum is

$$0^1, 2^{125}, (163 - \sqrt{25417})^1, 4^{16}, (163 + \sqrt{25417})^1.$$

Since $163 - \sqrt{25417} > 2$, the sharp unbalanced lower-frame constant is 2; the reciprocal-balanced constant is 1.

### IX.6 W9P-to-Theta isometry and global reel

#### IX.6.1 Equal-energy local write

Let $G_{144} = A^*A$. For a source event define two Hilbert-valued embeddings:

$$J_e^W h = \mathbf{1}_{144} \otimes h + (e_{U_l 0} - e_{U_r 0}) \otimes \frac{B_e h}{2},$$

$$J_e^\Theta h = \mathbf{1}_{144} \otimes h + e_{\Theta 0} \otimes \frac{B_e h}{2}.$$

The two W9 deviations each occupy a weight-two leaf, so their energy is

$$2\left\|\frac{B_e h}{2}\right\|^2 + 2\left\|\frac{B_e h}{2}\right\|^2 = \|B_e h\|^2.$$

The single Theta deviation occupies a weight-four leaf:

$$4\left\|\frac{B_e h}{2}\right\|^2 = \|B_e h\|^2.$$

Hence

$$(J_e^W)^*(G_{144} \otimes I)J_e^W = (J_e^\Theta)^*(G_{144} \otimes I)J_e^\Theta = B_e^* B_e.$$

The AFC intersection preserves event energy while translating a reciprocal two-lane informational write into one massless history receipt.

#### IX.6.2 Shared-anchor event family

Use one N144 cell per event and identify every cell's quotient-reference coordinate. For event set $\mathscr{E}$, each new cell contributes 143 new nonanchor ranks. Therefore

$$nodes = 1 + 143|\mathscr{E}|,$$

$$rank = 143|\mathscr{E}|,$$

with one common scalar kernel. Stacking events gives

$$(J_{\mathscr{E}}^W)^* G_{reel} J_{\mathscr{E}}^W = (J_{\mathscr{E}}^\Theta)^* G_{reel} J_{\mathscr{E}}^\Theta = \sum_{e \in \mathscr{E}} w_e B_e^* B_e.$$

For the completed RH source, continuous and prime-power events use

$$B_x = I - U_x, \qquad w_\infty(x) = \frac{e^{-x/2}}{1 - e^{-2x}}\,dx,$$

and

$$B_{p,k} = I - U_{k \log p}, \qquad w_{p,k} = \log(p) p^{-k/2}.$$

The completed event operator is the pullback

$$L_D = (J_D^\Theta)^* G_{reel} J_D^\Theta.$$

This is the exact generic-to-RH seam. RH weights enter through the adapter; they are not hard-coded in the generic reel.

### IX.7 Preserved deviation chain

#### IX.7.1 Generic positive-pencil contraction fails

The positive Stieltjes control

$$f(z) = 1 + \frac{1}{1-z} + \frac{1}{2-z} + \frac{1}{3-z}$$

has positive nested sections and decreasing extremal constants, yet

$$\Lambda_2(10) = \frac{5{,}655{,}335}{5{,}212{,}334} > 1.$$

Thus generic positivity does not force adjacent reel contraction. On the actual source, all 2,880 nonleading coefficients across 96 adjacent residuals are positive and yield $0 < \Lambda_N(u) < 1$ for all $u > 0$ on every executed pair. The source-specific signature survives; the generic promotion does not.

#### IX.7.2 Scalar RII control is insufficient

Positive recurrence data $r_N, a_N, \delta_N > 0$ occurs in exact controls whose adjacent residual polynomial has a negative coefficient or nonpositive value. The scalar signature therefore cannot replace the paired event Gram. The correction retains the full incidence operator $B_e^* B_e$.

#### IX.7.3 Local-frame debit domination fails

The W9P-to-Theta isometry is exact, but a rational-rotation control has local event energy $4/5$ against raw matched debit 2. The local N144 frame alone does not supply the completed lower-frame inequality. The source and endpoint conditions remain necessary.

### IX.7.4 Isolated recurrence lane fails

In the two-step recurrence

$$T_N = (u+N^2)^2 T_{N-1} + Q_N R_N,$$

the completed factor

$$R_N = Q_{N+1} - (u+N^2)^2 Q_{N-1}$$

stays positive on the executed source. The isolated lane $D_N = Q_N - (u+N^2)Q_{N-1}$ has negative active coefficients at $D = 0.7, 1$. Completion, not the isolated carried lane, owns the selected sign.

### IX.7.5 Whole-matrix AFC order fails

The determinant and terminal-pivot AFC identities close, but the stronger matrix order

$$D_N^T M_N D_N < N(N+1)H_N$$

has a negative direction already at $D = 0.7, N = 4$. AFC is retained at the exact quotient and terminal scalar level rather than promoted to every matrix direction.

### IX.7.6 Individual parity factors fail

The source terminal scalar factors into a coupled parity product. Each individual factor fails on some exact controls, while 367 controls close the product without both component inequalities. The compensating product is the supported object. A first smoke instrument omitted the odd Fourier lift and was low by $N^2$; its receipt is preserved and the corrected instrument changes only that lift.

### IX.7.7 Spherical diagonal order fails

The mode-local strengthening

$$L_N G_N^o L_N - G_N^{(e,2)} > 0$$

has only 9 positive gaps among 66 tested rows; 57 contain a negative direction. The first witness is $D = 0.7, N = 3$ with minimum eigenvalue approximately $-0.2266605255$. Every corresponding terminal scalar remains positive. The failure therefore rejects the matrix strengthening without discarding the coupled endpoint scalar.

## IX.8 Application-adapter authority boundary

The generic reel owns addresses, writes, receipts, the history quotient, unique event Grams and append-only reconstruction. An adapter may supply a declared source and read the resulting receipt:

- **RH moment adapter.** Permitted use: completed continuous/prime weights and endpoint kernel; Authority not acquired: global RH claim or universal sign by interface alone.
- **Starbreaker consumer.** Permitted use: typed formation-event history; Authority not acquired: scheduler or physical calibration without promotion.
- **Mersenne application.** Permitted use: exact factor-search receipts; Authority not acquired: primality or global scheduling from a factor miss.

A later adapter supplements the reel. It cannot rewrite earlier events or insert its coefficients into the generic core retroactively. Frozen `SLCQ2-RZ` remains the global source-recorded default SLC; the source-recorded RH-local handle is `SLCQ2-RZ-RH-MOMENT-ADAPTER-V1` and remains an application-specific consumer.

## IX.9 Established result and forward handoff

The generic core closes 42/42 checks across its complete address, map, lifecycle and history inventory. The append-only event construction identifies one reciprocal W9 pair with one Theta Gram, embeds that event isometrically in the shared N144 AFC frame and grows the reel by 143 ranks per event while retaining one common kernel.

The failure chain identifies the exact boundary of that result: source- specific completed scalars and full event Grams remain load-bearing; generic positive-pencil, isolated-lane and whole-matrix shortcuts do not replace them. `SAMA-D000043` carries the completed event operator into the separately typed RH route.

## IX.10 Sources and evidential scope

This chapter is assembled from the following fixed source manuscripts. Source-reported campaign outcomes retain their original scope; no new campaign was executed for this edition.

- Directional history and event reel computation.

SAM VOLUME III

# Equal Observations and Executable Histories

Chapter X

## X.1 A common operation separates two histories with the same total

The SLC constructs exact examples in which two objects give the same declared observation and remain distinguishable under a later operation. The retained information is an operand of that operation. The smallest example begins with the ordered histories [1,8] and [8,1]. Both have terminal sum 9. Applying the same connector rule to the first retained entry returns two different rational values. [EQ, pp2–4]

A declared observation is a map that exports selected information from a complete object. Write $\pi$ for that map and $C$ for an admitted response. The object retains its ordered source history in addition to its exported value. For the connector construction, the response rule is the source lift

$$L(p) = p + \frac{p^2}{144}.$$

The first retained entries are 1 and 8. Direct substitution gives

$$L(1) = \frac{145}{144}, \qquad L(8) = \frac{76}{9}, \qquad L(8) - L(1) = \frac{119}{16} = 7.4375.$$

The terminal sum is exactly equal while the common response is exactly unequal. These are rational identities, with no fitted tolerance. An order-insensitive whole-contact control gives

$$L(1+8) - L(1) - L(8) = \frac{1}{9}$$

for either order. The distinction therefore belongs to a specific history-sensitive operation: the lift reads the first retained prefix. The symmetric contact operation reads both operands without their order and agrees. The source's history identifiers name the records; their labels are not numerical operands in the response.

This example states the computational achievement precisely. A system receiving only terminal sum 9 cannot reproduce both required first-prefix responses. It must retain the first prefix, retain the ordered word, or retain another sufficient receipt from which that information can be recovered. The result is about the specified observation and action. It does not claim that the objects agree under an observation vector that already includes the separating response.

## X.2 The exact criterion for a sufficient observation

A fiber is the set of complete objects with one exported observation. A response can be computed from that observation alone exactly when it is constant on every relevant fiber. In symbols, a map $f$ with $C = f \circ \pi$ exists if and only if

$$\pi(x) = \pi(y) \quad \Longrightarrow \quad C(x) = C(y).$$

The proof is short. If $C = f \circ \pi$, equal observations give equal responses by applying $f$. Conversely, if the response is constant on each fiber, define $f$ at an observed value by evaluating $C$ on any object in that fiber. Constancy makes the choice immaterial. [EQ, pp3–4]

The connector pair violates that condition: $f(9)$ would have to equal both $145/144$ and $76/9$. This is the exact obstruction to deleting the prefix before the response is evaluated. For a partial action, admission is also part of the contract. If two objects share an observation but only one admits the action, the observation cannot determine whether the action is legal.

The criterion also governs composition. Equal exported inputs must have the same admitted actions, immediate responses, and exported targets if they are to remain interchangeable in a sequence. A receipt sufficient for one final scalar can fail under a later operation that reads information that scalar discarded. The relevant compression question is consequently which information suffices for the declared action language.

## X.3 Receiver sums and directed currents reconstruct the parents

The receiver construction makes the missing coordinate explicit. Let $L$ and $R$ be nonnegative integer history tensors in a common source-labeled basis. The even carrier and directed current are

$$S = L + R, \qquad J = R - L.$$

They reconstruct the parent tensors through

$$L = \frac{S - J}{2}, \qquad R = \frac{S + J}{2}.$$

For integer tensors the two numerators must be even coefficient by coefficient. The native constructor checks that condition and recovers the supplied inputs. The current may be signed while the stored history multiplicities remain nonnegative; those are different types. [EQ, pp4–5]

Applying the actual current adds $J$ on the left and subtracts it on the right:

$$\Phi_J(L,R) = (L + J, R - J) = (R,L).$$

The sum is conserved, and recomputing the oriented current after the exchange gives $-J$. The inverse transfer therefore uses $-J$. The completed A3D28 roster contains six unequal history-pair types, each in a two-cycle, and four fixed self-pairs. Deleted-current and wrong-current controls reach none of the six unequal targets. Thus the directed coordinate both

distinguishes the orientations and executes the exchange that the sum alone cannot specify.

### X.4 Ordered paths and the reconstructive address

Adjacent receiver exchanges on a triple obey the permutation relations of $S_3$. Write $s_1$ and $s_2$ for exchanging the first two and last two positions. Then $s_1^2 = s_2^2 = \mathrm{id}$, $s_1 s_2 s_1 = s_2 s_1 s_2$, and $(s_1 s_2)^3 = \mathrm{id}$, while $s_1 s_2$ and $s_2 s_1$ have different endpoint actions. The braid words $s_1 s_2 s_1$ and $s_2 s_1 s_2$ reach the same reversed triple through different intermediate triples. Retaining the action word preserves that path distinction after the endpoint permutation has been reported. The completed A3D29 body contains four six-state orbits, 24 states and 48 directed generator edges. [EQ, p5]

The next source construction joins a rooted triple's exterior residue, terminal position, and orientation. An exterior residue records oriented two-direction area: for source vectors a,$b$,$c$ it is $(b-a)\wedge(c-a)$, which changes sign when the directions are exchanged. The source records 48 nonzero two-Write residues and 48 zero one-Write controls. Residue alone recovers roster and orientation but leaves six possible positions. Joining the position coordinate closes that inverse problem.

The completed address has four rosters, six $S_3$ positions, and two orientations, giving $4 \times 6 \times 2 = 48$ outcomes. Every combination occurs once. Decoding reconstructs the source and terminal triples, the two-generator word, action-chain receipts, and signed residue; re-encoding is byte-identical. Removing roster leaves 12 classes of four outcomes, removing position leaves eight classes of six, and removing orientation leaves 24 pairs. These erasure controls show exactly which information each coordinate supplies. [EQ, pp5–6]

For a fixed root, a mixed-radix index $n = 12r + 2p + o$, with $0 \le r < 4$, $0 \le p < 6$ and $0 \le o < 2$, selects the 48 outcomes. The shared source tables remain necessary to reconstruct the full semantic object. This is an exact address into a declared source body, rather than a claim that six bits independently contain every source definition.

### X.5 A whole weave can retain many histories above one invariant

The A3D37 weave carries 48 occurrence-frames per plan: eight cube sites, three rooted slots and multiplicity two. Its roster contains 288 oriented plans, reduced to 144 by the global $C_2$ quotient. The type-resolved weave separates those plans into six classes of 24, while total holonomy has only one class on the released roster. [EQ, pp6–7]

The invariant is exact, and its common value is insufficient to select a single plan. The completed result is the retained six-by-twenty-four history structure. Its load-profile census contains 24 plans with quadratic load 240, 72 with load 192, and 48 with load 144. A later source-weighted action can consume those retained distinctions. The failed total-holonomy singleton selector identifies the information that invariant omits; it does not erase the constructed weave.

### X.6 A receiver functional makes the source difference usable

For the occupied receiver example, the before and after vectors are

$$h_0 = (1,0,0,0,1,1)^{\mathsf{T}}, \qquad h_1 = (0,1,1,1,0,0)^{\mathsf{T}}.$$

Let $Q$ be a real symmetric receiver form and $\varepsilon$ its scale. The quadratic response difference is

$$\Delta H = \varepsilon \,\mathrm{tr}(QD), \qquad D = h_1 h_1^{\mathsf{T}} - h_0 h_0^{\mathsf{T}}.$$

The vectors are orthogonal with squared norm three. Hence $D$ has rank two and nonzero eigenvalues +3 and 3. The transfer therefore specifies one exact linear functional of the receiver form, supported on two source directions. The source records twelve nonzero coefficient weights when the functional is written in independent symmetric-form coordinates. [EQ, pp7–8,13–14]

The complete source difference is already available for further calculation. Assigning a physical $Q$, scale, and energy reference remains a separate source operation. The mathematical functional must not be presented as an already measured nuclear binding energy; its substantive result is the exact form dependence and retained operands.

SAM VOLUME III

# Ordinal Selection and the Delegated Exact Engine

Chapter XI

The latest global selector described by the retained SAMA material is Q2. Its job is to order an admissible catalog using source-visible features, while exact arithmetic remains delegated. This chapter records its frozen comparisons without turning a learned ordering into a physical law or an unrestricted complexity claim.

The account below follows the chapter source as retained at the fixed 8 September 2026 SAMA revision. Engine names, installed interfaces and campaign status are historical source statements. They do not describe a freshly inspected October deployment. Later source amendments are distinguished where available; no later engine or campaign is inferred.

The September SAMA selector chapter cites H000531, H000551 and H000552 for Q2. The accessible August public SAM Research Project live page instead records V21R, and its named Q2 candidate and promotion artifact paths were unavailable during preparation. This chapter therefore preserves the later manuscript account with that retrieval limitation; it does not independently certify a live pointer transition.

**Source-status qualification.** The operational and revision statements in this chapter describe the retained source documents. They are not an instruction to execute, deploy, resume, train, promote or change a service, and they have not been independently verified as present live state for this edition.

## XI.1 Overview

Exact computation can return several admissible receipts for one query. The arithmetic substrate certifies what every receipt is, while the selection plane asks a different question: which admissible receipt has the best exact return quality when the selector is permitted to see only source geometry? `SLCQ2-RZ` is the frozen current answer to that ordinal question.

The target quality is not a learned label invented after execution. For each candidate receipt it is the frozen reciprocal-quotiented lexicographic vector

$$Q = (d_{direction}, r_{inverse}, r_{W8}, r_{D81}),$$

where lower is better and equal vectors remain a complete target tie. Q2 does not read any component used to construct that target. It exposes 70 exact integer features computed from axis, modulus, shell, direction and two typed domain labels. It excludes inverse inputs, inverse directions, returned endpoints, D81 checkpoints, target tiers, custody identities, hashes, execution order and timing.

The data surface contains 5,120 inherited exact receipts, 1,024 canonical queries and 512 reciprocal/replicate-safe groups. Each axis is frozen into 80 TRAIN, 24 HOLDOUT and 24 CONTROL groups. TRAIN supplies 640 queries; HOLDOUT and CONTROL each expose a complete 192-query catalog. Three exact-rational architectures at three predeclared ridge values form nine fits. HOLDOUT fixes the winner before CONTROL is opened:

`Q2-HYBRID_LISTWISE_ADJACENT_ORDINAL_RIDGE-RIDGE-1D1000`.

Against frozen Q1 on the same fresh surfaces, Q2 raises weighted strict-pair accuracy by exactly `63200/594823` on HOLDOUT and `83595/773858` on CONTROL, while recovering every exact best set, creating no false singleton, making no abstention and producing no strict score tie. The installed classification is exactly: **The test result suggests strong contact with the concept.**

The architecture is deliberately layered. Q2 owns the current ordinal selection plane. Frozen `SLCV21R` remains noncurrent as a revision but remains the byte-identical delegated arithmetic and application base. Q1 is an immutable fallback. Neither selection nor promotion assigns physical scale, absorbs ATOM3D, changes Exact Write semantics or authorizes application work.

## XI.2 Opening question and conceptual picture

The governing question is:

> Can exact source-visible event geometry order the complete admissible receipt catalog more faithfully than Q1 without reading any executed value that defines the answer?

The typed route is

- frozen SLCV21R exact query and candidate receipts
- reciprocal/replicate-safe grouping
- target-only exact return-quality vector
- 70-field source-visible feature map
- exact listwise and adjacent-ordinal fitting on TRAIN
- winner lock on complete HOLDOUT
- one comparison on sealed complete CONTROL
- frozen Q2 selection service
- SLCQ2-RZ current pointer

Three separations carry the meaning of the result:

1. the exact substrate determines admissibility and computes receipts;
2. the target oracle evaluates return quality but is hidden from selection;
3. the selector orders candidates from permitted source geometry alone.

Collapsing these surfaces would make the question circular. Keeping them separate lets the result say something precise: the visible structural fields contain enough ordinal information

for the selected exact model to recover the complete best set and nearly all strict pair orderings on two held-out complete catalogs.

## XI.3 Recorded authority and architectural planes

### XI.3.1 Sole source-recorded default revision

`SLCQ2-RZ` has status `FROZEN_COMPLETE_STOPPED_PROMOTED_CURRENT` and is the sole source-recorded default SLC pointer. Older v1.x, V20R and `SLCV21R` revisions retain their artifacts and scoped results; they are not alternative current pointers. The older Volume III technical spine therefore supplies lineage, not present-tense pointer authority.

Q1 remains frozen and immutable as the fallback selector. A fallback is not a second default: it is a preserved predecessor available under the declared service boundary.

### XI.3.2 Selection is not arithmetic

For a query $q$, let $\mathscr{C}(q)$ be the finite set of receipts certified admissible by the exact base. Q2 supplies a score

$$s_q(c) = w^T \phi(q,c), \qquad c \in \mathscr{C}(q),$$

with exact rational coefficient vector $w$ and exact integer feature vector $\phi(q,c) \in \mathbb{Z}^{70}$. It returns all candidates attaining the maximum permitted score, subject to the ambiguity policy described below.

The delegated base still owns DOS construction, exact write, reciprocal return, checkpoints and application execution. Installing a new $w$ changes the ordering of already-admissible candidates; it does not change what the candidate means or how its exact receipt is computed.

### XI.3.3 Four distinct planes

- **exact substrate.** Owner in the installed architecture: frozen `SLCV21R` services; Output: admissible exact receipts and arithmetic.
- **quality target.** Owner in the installed architecture: frozen reciprocal-quotiented oracle; Output: exact lexicographic target vectors.
- **ordinal selector.** Owner in the installed architecture: current `SLCQ2-RZ` Q2 model; Output: score order and complete top-score set.
- **application.** Owner in the installed architecture: separately declared adapter/model; Output: application-specific interpretation or execution.

An ATOM3D model, RH adapter or Mersenne scheduler may call the service only through its own typed contract. None becomes part of Q2 merely by using its ordering.

## XI.4 Target construction and blindness contract

### XI.4.1 Exact target order

The target for candidate $c$ is

$$Q_q(c) = \big(d_{direction}(q,c), r_{inverse}(q,c), r_{W8}(q,c), r_{D81}(q,c)\big).$$

Vectors compare lexicographically, with smaller values preferred. The first coordinate at which two candidates differ decides their strict order. If no coordinate differs, both belong to the same target tier. In particular, a custody identifier never converts an exact target tie into a singleton.

The complete best set is

$$B_q = \{c \in \mathscr{C}(q) : Q_q(c) = \min_{c' \in \mathscr{C}(q)} Q_q(c')\}.$$

Recovery is exact only when the returned set equals $B_q$, not when it merely contains one member of $B_q$.

### XI.4.2 Permitted source fields

Q2 may read only:

- axis
- modulus
- left_shell, right_shell
- left_direction, right_direction
- route_domain
- physical_relation_domain

From these it computes the fixed 70-field map. The two domain labels are constant on this source; they serve as a typed zero-variation interaction check, not as an undeclared ranking signal.

### XI.4.3 Forbidden target and custody fields

The selector may not read:

- inverse input lanes or inverse directions
- returned endpoint lanes
- D81 checkpoint lanes
- quality vector or quality tier
- receipt index or pair identity
- group hash or source hash
- execution sequence
- timing

This prohibition is load-bearing. Inverse, endpoint and checkpoint values construct the oracle and would leak the answer. Custody fields could create memorized ordering without structural meaning. Timings could import hardware and execution-order artifacts into a mathematical selector.

## XI.5 Fresh grouped custody

### XI.5.1 Why the unit is a physical group

Individual receipt splitting is a wrong control because reciprocal directions and replicated views of one physical relation can then appear on opposite sides of a split. A model could learn one view and be scored on its mate. Q2 hashes a fresh namespace together with axis, modulus, unordered shell endpoints

and the two source domain labels. All reciprocal directions and replicates sharing that identity remain in one group.

The frozen census is

- **inherited exact receipts.** Exact count: 5,120.
- **canonical queries.** Exact count: 1,024.
- **reciprocal/replicate-safe groups.** Exact count: 512.
- **reciprocal canonical queries per group.** Exact count: 2.
- **observed group multiplicities.** Exact count: 4, 10, 16.

The nonuniform multiplicities matter. A group contributes according to its actual candidate roster; assuming one uniform multiplicity changes the catalog and its weights.

### XI.5.2 Split derivation

Within each axis, group hashes are sorted and frozen as

$$80\ \text{TRAIN} + 24\ \text{HOLDOUT} + 24\ \text{CONTROL} = 128$$

groups. Across the four axes this gives the 512-group total. The query census is

$$4 \cdot 80 \cdot 2 = 640 \quad \text{TRAIN queries,}$$

$$4 \cdot 24 \cdot 2 = 192$$

queries on each of HOLDOUT and CONTROL.

TRAIN uses the frozen harder tier rule and contains 18,654 group references and 182,688 receipt exposures. HOLDOUT and CONTROL retain every split-local candidate group, yielding complete 192-query catalogs. Selection therefore cannot improve a metric by silently dropping difficult candidates.

## XI.6 Deriving the 70-feature representation

### XI.6.1 Direct and reciprocal alignment

Let the centered source shell endpoints of a query and candidate be exact integers. Q2 forms both alignments:

- direct: query left -> candidate left, query right -> candidate right
- swapped: query left -> candidate right, query right -> candidate left

For each alignment it retains signed residuals before taking absolute or squared summaries. Schematically, with query endpoints ((a,b)) and candidate endpoints ((c,d)),

$$\delta_{direct} = (c-a, d-b), \qquad \delta_{swap} = (d-a, c-b).$$

The map includes the signed components, absolute components, squares, alignment sums and best/other residual summaries. Because both constructions are ordered by a source-visible canonical key before exposure, a whole-lane swap exchanges internal work without changing the final visible vector.

### XI.6.2 Midpoint, span, direction and parity

For endpoints ((a,b)), define signed midpoint numerator and span

$$m = a + b, \qquad \ell = b - a.$$

Using the numerator avoids a floating-point half. Candidate-minus-query relations for $m$ and $\ell$ retain orientation and separation that coarse shell bins erased. Individual direction matches and endpoint parity relations are then appended rather than compressed to a single flag.

### XI.6.3 Two-adic profiles and typed interactions

For a nonzero integer $n$, $v_2(n)$ is the largest $e \geq 0$ for which $2^e \mid n$. Q2 includes four direct/swapped alignment valuations and two span-relation valuations, with the frozen convention for zero. These features expose dyadic shell structure without reading any returned value.

Route-domain and physical-relation-domain equality fields and their declared interactions preserve the type boundary. Fourteen bounded quadratic cross terms add fixed pair interactions; each is clipped at exact magnitude 16,777,215. The clip is part of the pre-execution feature contract and not a post-result adjustment.

### XI.6.4 Why Q2 succeeds where the 20-field map collided

Q1's coarser 20-feature map has strict-pair feature-collision rates 2.3255657565% on HOLDOUT and 2.1898074324% on CONTROL. When two candidates with different target order have the same visible vector, any deterministic linear score must tie them. Q2's signed, unbinned and reciprocal-invariant map has zero strict feature collisions on both surfaces. This removes that representational obstruction before model fitting; it does not guarantee a correct order by itself.

## XI.7 Exact ordinal fitting

### XI.7.1 Three predeclared architectures

The bounded roster contains:

1. `EXPANDED_QUERY_CENTERED_LISTWISE_RIDGE`, which fits query-centered tier grades over the harder TRAIN catalog;
2. `HYBRID_LISTWISE_ADJACENT_ORDINAL_RIDGE`, which appends exact differences for adjacent strict tiers with frozen weight four; and
3. `LEXICOGRAPHIC_CONDITIONAL_MULTI_HEAD_RIDGE`, which learns the first target coordinate globally and later coordinates only inside equality classes of the preceding coordinates.

Each architecture is fitted at exact ridge values

$$\lambda \in \left\{ \frac{1}{1000}, \frac{1}{100}, \frac{1}{10} \right\},$$

giving $3 \times 3 = 9$ fits. There is no floating-point training and no full-scale hyperparameter sweep.

### XI.7.2 Listwise and adjacent observations

For each training query, a target tier grade is centered within its complete candidate list. The listwise design therefore learns relative order without letting different query offsets dominate.

For adjacent strict tiers, choose $c_+$ from the better tier and $c_-$ from the next worse tier. The hybrid supplies the exact difference row

$$\Delta\phi = \phi(q, c_+) - \phi(q, c_-),$$

with a positive ordinal target and frozen weight four. A fitted vector should therefore satisfy

$$w^T \Delta\phi > 0$$

for that adjacent relation. Chaining adjacent relations transports the order through the tier sequence while the listwise rows retain the complete query geometry.

With exact design matrix $X$, target $y$, diagonal frozen observation weights $W$ and ridge $\lambda$, the rational normal system is

$$(X^T W X + \lambda I) w = X^T W y.$$

All inputs and outputs remain `Fraction` values. Exact fitting makes score equality a mathematical event rather than a floating tolerance choice.

### XI.7.3 Winner and tie policy

The three hybrid fits have identical complete-catalog ranking metrics. Only after mathematical metrics and parameter count tie does the frozen key order select the lower regularization identifier. The resulting winner is `Q2-HYBRID_LISTWISE_ADJACENT_ORDINAL_RIDGE-RIDGE-1D1000`.

At service time, a score tie spanning distinct target vectors triggers abstention. A score tie wholly inside one exact target vector returns the complete tied best set. The promoted wrapper names a top-score multiplicity explicitly as `AMBIGUOUS_TOP_SCORE`; it does not choose by receipt ID.

## XI.8 HOLDOUT selection and sealed CONTROL

### XI.8.1 Predeclared material comparator

Before execution, success required all three conditions:

- at least one percentage point of weighted strict-pair gain over frozen Q1 on HOLDOUT;
- positive gain on CONTROL; and
- perfect exact-best-set recovery on both complete surfaces.

HOLDOUT selected and locked the model. CONTROL was opened once afterward for the comparison to the frozen Q1 baseline.

### XI.8.2 Exact results

- **HOLDOUT.** Frozen Q1: `530826/594823`; Q2 winner: `594026/594823`; Exact delta: `63200/594823`; Decimal point gain: +10.625009.
- **CONTROL.** Frozen Q1: `689509/773858`; Q2 winner: `386552/386929`; Exact delta: `83595/773858`; Decimal point gain: +10.802369.

The CONTROL Q2 fraction has denominator half the Q1 display denominator, so the like-denominator comparison is

$$\frac{386552}{386929} = \frac{773104}{773858},$$

and

$$\frac{773104 - 689509}{773858} = \frac{83595}{773858}.$$

Q2 retains 6,376 wrong weighted strict pairs on HOLDOUT and 6,032 on CONTROL. Those residual errors remain in the record. At the service boundary, however, best-tier accuracy and exact-best-set recovery are both $1/1$ on both splits, with zero false singletons, zero abstentions and zero strict score ties.

The authorized classification is:

**The test result suggests strong contact with the concept.**

## XI.9 Deviation, correction and retest chain

### XI.9.1 Candidate-construction failures

Five preserved attempts materially shaped the frozen candidate:

- **001.** Failure: a cross-axis unit fixture mixed an axis-specific assumption into the test; Correction: construct the fixture inside its declared axis/type; Retest consequence: boundary tests reached the intended feature question.
- **002.** Failure: code assumed uniform group multiplicity; Correction: retain the frozen observed roster `{4,10,16}`; Retest consequence: complete catalogs and exposure weights reconstructed exactly.
- **003.** Failure: reciprocal orientation changed the visible feature vector; Correction: canonicalize complete direct/swapped blocks by source-visible keys; Retest consequence: whole-lane swap became invariant.
- **004.** Failure: a serializer could not carry the exact fraction surface; Correction: encode exact numerators/denominators through the declared receipt form; Retest consequence: checkpoint and replay identity remained exact.
- **005.** Failure: the Q1 collision diagnostic read the wrong comparison source; Correction: recompute that diagnostic from the frozen Q1 comparator on Q2 surfaces; Retest consequence: the corrected Q1 collision rates changed; Q2 model selection and CONTROL result did not.

The first four attempts stopped before model selection or CONTROL access. The fifth corrected a predecessor diagnostic only. Keeping these attempts makes clear which faults affected construction and which did not affect the frozen winner.

### XI.9.2 Promotion-wrapper corrections

Promotion also preserved two distinct corrections:

- Attempt 001 expected an `assertions_passed` field, while the independent receipt records `assertion_count=165`. The wrapper was corrected to read the actual schema before installation continued.
- Attempt 002 installed an executable resolution correctly, but the pointer inherited explanatory usage language and stale promotion fields. Pointer metadata was corrected, producing current pointer SHA-256 `751f6e6a6dbe01cfdc335cf303824887148c9eb657ded691264e37c6f61fa318`, and the promotion surface was revalidated.

Neither correction retrained Q2, reopened Q1 or mined CONTROL.

### XI.9.3 Wrong controls retained as lessons

The following alternatives would answer a different question and are not the installed controls:

- receipt-level random splitting, because it leaks reciprocal or replicated physical relations;
- incomplete HOLDOUT or CONTROL candidate rosters, because they can hide misranking and false singleton behavior;
- target-visible fields, because they make ordinal prediction circular;
- custody-ID tie breaking, because it converts exact ambiguity into an arbitrary order;
- floating score tolerances, because exact equality would depend on runtime;
- opening CONTROL during model choice, because CONTROL would cease to be the postselection surface.

## XI.10 Freeze and promotion process

The candidate freeze passed 8/8 release checks, 14/14 reconstruction checks, 165/165 independent assertions and, after the diagnostic correction, 8/8 release checks again. The promotion wrapper then passed:

- **pre-install.** Result: 22/22.
- **post-install.** Result: 12/12.
- **independent promotion validation.** Result: 20/20.
- **final release.** Result: 6/6.

Promotion changed the live selection pointer through H000551; H000552 records the pointer-metadata correction. It preserved the candidate, Q1 and delegated base artifacts rather than rewriting them.

The promotion performed no retraining, CONTROL mining, physical calibration, ATOM3D specialization, remote transfer, prime work, publication or deployment. These are not missing steps in Q2. They are separate campaigns requiring their own contracts and authority.

## XI.11 Established result and boundary

This document establishes the following typed statement:

> On the frozen Q2 grouped catalogs, the 70-feature exact hybrid listwise-plus-adjacent ordinal selector materially improves strict-pair ordering over frozen Q1 and recovers every complete exact best set. Its promotion makes `SLCQ2-RZ` the sole source-recorded default SLC selection revision, while exact arithmetic remains delegated to frozen `SLCV21R` and Q1 remains the immutable fallback.

It does not establish a physical energy scale, a new DOS, a change to Exact Write, a global optimizer for every application, native GPU credit, ATOM3D specialization, an RH conclusion or a Mersenne status. The next chapter uses Q2 only as a typed application dependency and keeps its Mersenne scheduler and certificates separate.

## XI.12 Sources and evidential scope

This chapter is assembled from the following fixed source manuscripts. Source-reported campaign outcomes retain their original scope; no new campaign was executed for this edition.

- Slcq2 rz ordinal selection engine.

SAM VOLUME III

# Starbreaker: Formation on Its Own Grammar

Chapter XII

Starbreaker is the first major formation application in this volume. It receives its own ledgers, events, stages and occurrence views. The useful transfer from exact computation is a routing discipline, not substitution of the N100 graph. Both successful reconstructions and failed physical or numerical controls belong in the account.

The account below follows the chapter source as retained at the fixed 8 September 2026 SAMA revision. Engine names, installed interfaces and campaign status are historical source statements. They do not describe a freshly inspected October deployment. Later source amendments are distinguished where available; no later engine or campaign is inferred.

## XII.1 Overview

Starbreaker is a formation application with its own native grammar. It retains 2,976 ledgers, 482,112 typed atoms, seed/collapse/final stages, two event orders, two D18 pair maps, same and reversed occurrence views, contact incidence, record origin and constrained M126/N144/N144+d/L162 readouts. N100 contributes structure-first routing discipline; its 100-spin graph is not substituted for this application.

The exact state ladder begins with M126 retention. Fee-exempt Theta18 provides an independent 18-coordinate lift, so $2^{126} \cdot 2^{18} = 2^{144}$. L162 is an ambient occurrence view whose valid image still contains $2^{144}$ states, not $2^{162}$ independent matter configurations. Active mismatch coordinates travel in an N144+d sidecar rather than being promoted to another object or capacity.

The structure-first replay detects those native quotient, fiber and readout certificates. It reproduces the completed parent byte for byte, passes 78/78 checks and never invokes dense H14F fallback. The W9 history refinement adds a rank-161 target-local order-history receipt, but the mode is emit-only: it may record cross-stage memory and may not alter scheduler grouping or allocation without a separate tested promotion.

The Courtroom evidence supplies a complete methodology chain. CR210 reconstructs a scoped external-collapse directional-contact surface. CR211 freezes a typed 162-slot particle-ledger interface while leaving address-to-template incidence null. CR005c identifies why stage ordering cannot be promoted to physical escape time. CR005d builds a dimensionless continuous flow and carrier-QNM source bridge. CR005e stops on a numerical trace-residue gate; CR005f freezes one exact trace-free correction; CR005g runs the unchanged morphology design and records the boundary that no stable Starbreaker/CCSN morphology match cleared the gates. CR005h constructs a complete per-carrier ledger but fails three controls; CR005i corrects only those controls, independently replays all 96 scenarios and 53,568 carriers, adopts the byte-identical ledger and leaves CR005h immutable.

## XII.2 Opening question and conceptual picture

The governing question is:

> How does Starbreaker carry a source-native formation history from typed ledgers through exact state accounting, directional contact, continuous source construction and controlled application readouts without replacing its grammar, erasing failed tests or promoting dimensionless receipts into physical observables?

The application picture is

- Starbreaker ledgers and typed atoms
- seed / collapse / final formation stages
- lexicographic and reverse event orders
- exact M126 retention
- fee-exempt Theta18 lift to N144
- optional active mismatch sidecar N144+d
- constrained L162 occurrence views
- emit-only W9 rank-161 history
- dimensionless source/application bridges
- physical calibration joins remain explicit

The route is layered. A state count is not a waveform. A source quadrupole is not strain. A carrier with positive source influence is not the substance transported by a gravitational wave. Keeping those types separate is what allows the exact and exploratory work to accumulate safely.

## XII.3 Native objects, counts and type boundaries

### XII.3.1 Ledger and atom census

Each of 2,976 ledgers carries one 162-slot typed surface, giving

$$2{,}976 \cdot 162 = 482{,}112$$

structured atoms. The slots are not assumed occupied merely because they exist. Unary, pair, triad, carrier, support, scalar and rejected classes remain distinct, and categorical relations remain unresolved until their own geometry stage.

Starbreaker has three stages and two event orders. Therefore the number of stage/order snapshots is

$$2{,}976 \cdot 3 \cdot 2 = 17{,}856.$$

Executing both D18 pair-map controls doubles the runtime rows:

$$17{,}856 \cdot 2 = 35{,}712.$$

These identities are direct state-accounting checks. A dropped ledger, stage, order or pair map changes the total immediately.

### XII.3.2 Persistent and transient types

The formation engine preserves the following firewall:

- ***O*.** Role: persistent matter identity; Nonidentity: not a substrate site phase.
- **W8.** Role: unresolved/returned substrate-site phase; Nonidentity: not a retained history rank.
- **W9.** Role: direct target phase during active contact; Nonidentity: not an independent source row.
- **X1.** Role: one transient H2 mirror bit; Nonidentity: not matter and ledger contribution zero.
- **U.** Role: coordinated reciprocal record exchange; Nonidentity: not a new material object.
- **M126.** Role: retained matter/output state; Nonidentity: not universal occurrence transport.
- **Theta18.** Role: fee-exempt directional carrier/fiber; Nonidentity: not retained mass lift.
- **N144+d.** Role: active defect-sidecar custody; Nonidentity: not extra independent capacity.
- **L162.** Role: constrained occurrence readout; Nonidentity: not $2^{162}$ independent physical states.

Completion returns both contacted sites to W8 and clears X1 while retaining updated records. "State returns; history does not" means the endpoint type can repeat while its reciprocal receipt remains nonzero.

### XII.3.3 Stage and control surfaces

The two event orders are lexicographic and reverse-lexicographic. The retained pair-map controls are `D18_P1_SHIFT_14` and `D18_M1_SHIFT_08`. Same and reversed views test occurrence reconstruction from opposite readout orientations.

These are application fixtures. A pair map or contact-incidence field does not become a physical matter mirror, q orientation, Theta partition or elementwise pairing unless a separate source selects it.

## XII.4 Construction interfaces

### XII.4.1 Particle-ledger interface

`CR:CR211@15` freezes one prototype 162-slot ledger and 321 distinct QP templates. Its structural equalities include

$$18+126+18=162,$$

$$9\cdot 18=162,$$

and six copies of the 27-address H27 rank census. The CR210d boundary retains the typed values 81/144/126/162.

The most important output is negative in a precise sense: address-to-template incidence is explicitly null. No slot is claimed to contain a particle merely because the register has that slot, and no elementwise QP-to-F81 assignment is manufactured. The interface supplies typed storage and structural authority; physical content, mass, binding and Starbreaker dynamical incidence remain open.

### XII.4.2 Continuous-flow and carrier-QNM bridge

`CR:CR005d@21` replaces the earlier affine stage probe with a zero-fit continuous structural flow. The three stored states are joined by exponential radial collapse, a bounce impulse at dimensionless phase $t=1$, and constant radial plus shortest-branch angular release. The maximum reconstruction error at the stored states is $2.191\times10^{-14}$.

For unit carrier positions $x_c(t)$, a trace-free quadrupole has the form

$$Q_{ab}(t)=\sum_c\left[x_{c,a}(t)x_{c,b}(t)-\frac{\delta_{ab}}{3}|x_c(t)|^2\right].$$

The source surface uses $d^2Q_{ab}/dt^2$; a third derivative supplies a power proxy. The carrier-QNM response convolves the source with the sealed dimensionless kernel

$$g(u)=\frac{e^{-(4/45)u}\sin((3/8)u)}{3/8},$$

so $\omega_R M=3/8$ and $\omega_I M=4/45$ are fixed rather than fitted.

The run tracks 1,993 dual carriers. Of these, 1,541 satisfy escape before closure on the continuous flow:

$$\frac{1{,}541}{1{,}993}=0.7732062217762168.$$

The median closure-minus-escape margin is 0.04020605301428759. Across the construction there are 10,515,538 transition roots and zero root failures. All 14/14 construction gates and 10/10 controls pass; all 96 scenarios and 482,112 atoms are present. Maximum relative trace error is $2.5997906629060057\times10^{-15}$.

The bridge remains dimensionless. Unit carrier weights are not matter masses, phase time is not seconds, and the QNM response is not gravitational-wave strain, luminosity, frequency in hertz or a detector forecast.

## XII.5 Exact formation-state derivation

### XII.5.1 M126 retention

The structured application begins with two exact N72 custody packets. Each retains

$$2^{72}=4{,}722{,}366{,}482{,}869{,}645{,}213{,}696$$

binary states as a $64\times10$ conditional tensor. Their declared matter join is rank 126 and accounts for

$$2^{126}=85070591730234615\cdot10^{21}+86584365185794205 2864$$

M126 configurations across 4,096 two-sided sectors. The scalar background has 127 oriented marked sectors and 64 occupied energy bins from $-7875$ through 63.

M126 retains record origin without promotion in 468 stage/order snapshots and requires promotion in 17,388. Those counts sum to the complete snapshot census:

$$468 + 17{,}388 = 17{,}856.$$

The split demonstrates why M126 is a retention/output state, not a universal occurrence-transport state.

#### XII.5.2 Fee-exempt Theta18 lift

Theta18 contributes 18 independent directional coordinates, so

$$2^{126} \cdot 2^{18} = 2^{144}$$

and

$$2^{144} = 22300745198530623141535 \cdot 10^{21} + 718272648361505980416.$$

The lift is fee-exempt: no Theta multiplicity is folded into X1 and no retained mass is attached to the carrier fiber.

#### XII.5.3 N144+d and L162 readout

For a same-view mismatch mask, define

$$d_t = \text{popcount}(m_t), \qquad 0 \le d_t \le 18.$$

The application route is

- d = 0 -> exact N144 independent union state
- 0 < d < 18 -> N144 plus active d-coordinate sidecar
- d = 18 -> L162 constrained occurrence readout

The L162 ambient ceiling is $2^{162}$, but each valid same or reversed image contains $2^{144}$ states. Their ratio is

$$\frac{2^{162}}{2^{144}} = 2^{18} = 262{,}144.$$

Thus L162 supplies 18 constrained occurrence coordinates over the N144 image; it does not add 18 independent matter spins.

#### XII.5.4 Byte-identical structure-first replay

The v1.0 replay inspects Starbreaker's own certificates and chooses the M126 quotient, N144 fiber lift, N144+d sidecar and L162 constrained readout. Dense H14F fallback is unnecessary. The exact route census is

- **N144.** Rows: 1,262.
- **N144+d.** Rows: 17,922.
- **total non-extreme.** Rows: 19,184.

The replay retains all 2,976 ledgers, 20,415,176 onset events per order and 35,712 runtime rows. It passes 78/78 checks in 5.251708 seconds, and fresh runtime output is byte-identical to the completed parent. The method changes; the Starbreaker grammar and output do not.

#### XII.5.5 Emit-only W9 history

The target-local W9 history operator is present across discovery/validation, both pair maps and both endpoint gauges. Every branch has cumulative and cross-stage rank 161. Seed equals its stage-local control; collapse and final retain nonzero cross-stage memory.

`W9HistoryMode` is emit-only. It may return the typed history sidecar but has no authority to change phase-pool grouping, N144 allocation, N144+d fallback or L162 routing. A scheduling use would require a separate test and promotion.

### XII.6 Focused application evidence

#### XII.6.1 External-collapse directional-contact replay

`CR:CR210@15` regenerated all 1,218 three-dimensional simulations, rebuilt the no-fit external contact from fresh outputs and passed all 58 Starbreaker regression tests. All 28 precommitted source/output files reproduced their SHA-256 digests.

Both remnant ladders remain strictly ordered across $p = 0,1,2,3$. All 42 individual ladders rise at every step, all 42 inward $p = 3$ allocations exceed their same-total uniform controls, and both aligned allocations are maximal among all 24 radial permutations.

The source verdict is a scoped qualitative directional-contact pass. It does not identify $p$ with physical compactness, identify a remnant with a black hole, test faster formation time, solve hydrodynamics or relativity, calibrate A or promote the packing law physically.

#### XII.6.2 Inventory-corrected pixel-bridge retest

`CR:CR005g@21` executes the morphology question frozen in CR005e and corrected in CR005f. Ten public three-dimensional weak-CCSN matter-strain waveforms are compared with all 96 Starbreaker post-bounce unit-carrier quadrupole sources across 13 observer directions. Each source becomes the same normalized $48 \times 64$ time-frequency power grid.

For normalized pixel distributions $P, Q$, the primary statistic is

$$\mathscr{A}(P,Q) = \sum_{pixel} \sqrt{P_{pixel} Q_{pixel}}.$$

The best-template search is replayed inside 47 time-shift nulls and the locked reversal controls. All 12/12 construction gates and 11/11 controls pass. The door results are

- **discovery.** Observed median: 0.150857; Shift q90: 0.155669; Shift q95: 0.155861; Clears gate: no.
- **validation.** Observed median: 0.068025; Shift q90: 0.094694; Shift q95: 0.096175; Clears gate: no.

- **final.** Observed median: 0.076474; Shift q90: 0.073062; Shift q95: 0.073193; Clears gate: not q95/all.

The largest individual affinity is 0.407647 for `s9.0.swbj15.horo.3d.gw.dat` against Starbreaker `REF_0272` in view `face_yz_m`. Because the precommitted stable-door gates do not close, the source result is

- `BOUNDARY_PIXEL_SURFACE_CONSTRUCTED__NO_STABLE_STARBREAKER_CCSN_MORPHOLOGY_MATCH`

This is normalized morphology only, not amplitude or mechanism identity.

#### XII.6.3 Independent carrier-ledger replay and freeze

`CR:CR005i@21` independently replays all 96 scenarios and all 53,568 carrier rows under the corrected controls. The parent authority matches exactly. The maximum discrete-linearity observed/bound ratio is $5.253916 \times 10^{-3}$, and the maximum integrated-power ordering difference is $1.616856 \times 10^{-9}$, below the frozen $10^{-6}$ limit. Static quadrupole residual and derivative are zero. Isotropic cancellation residual is $1.370646 \times 10^{-16}$.

The byte-identical CR005h carrier ledger is adopted and frozen. Seven slots are strong source-contributor candidates—16, 13, 14, 15, 4, 2 and 3—and slot 17 is directional. Positive leave-one-out influence means reinforcement of the internal source proxy. It does not mean that a carrier is transported GW material.

### XII.7 Preserved deviation and correction chains

#### XII.7.1 Preserved CR005e numerical stop

`CR:CR005e@SAM-RESEARCH` freezes the external morphology design before execution: ten Princeton three-dimensional weak-CCSN waveforms, 96 Starbreaker scenarios, 252 carriers per scenario, 13 directions, 2,049 samples, three trimmed derivative-edge samples, a $48 \times 64$ pixel grid, Hellinger affinity, same-door matching, 47 shifts and reversal controls.

The first Starbreaker scenario stopped because independently finite-differenced diagonal tensor components retained relative trace residue

$$9.43617769285563 \times 10^{-9},$$

above the frozen $10^{-10}$ gate. No match, null distribution, door statistic or scientific verdict opened. CR005e remains immutable.

#### XII.7.2 Trace-free projection correction

`CR:CR005f@SAM-RESEARCH` freezes one permitted correction. Trace freedom requires

$$S_{xx} + S_{yy} + S_{zz} = 0.$$

After the two locked finite differences, the successor sets

$$S_{zz} := -(S_{xx} + S_{yy})$$

before observer projection. This changes no source, door, sample count, grid, view, statistic, null, evidence ladder or physical boundary. It is an exact algebraic projection with zero fitted parameters, not a relaxed threshold. CR005f is the correction precommit; the executed retest is CR005g.

#### XII.7.3 Per-carrier ledger control failure

`CR:CR005h@21` constructs a complete 53,568-row per-carrier attribution ledger. Every occurrence receives a unit-occurrence trace-free quadrupole, individual power, exact leave-one-out aggregate-power change, bind/escape class and frozen identity. Positive

$$\Delta P = P_{aggregate} - P_{without\ carrier}$$

marks reinforcement; negative $\Delta P$ marks coherent cancellation.

The candidate content was complete, but exactly three construction controls failed, so the campaign verdict remains

- `FAIL_PER_CARRIER_GW_LEDGER_CONSTRUCTION`

The ledger is preserved rather than deleted or tuned to pass.

#### XII.7.4 Control-appeal correction

CR005i precommits three implementation corrections:

1. G02 compares the sealed CR005d verdict against its exact string instead of the wrong shortened prefix.
2. G05 evaluates third-difference linearity inside a frozen roundoff envelope and retains an independent integrated-power guard, rather than demanding literal floating equality after derivative amplification.
3. G08 first checks static constancy and isotropic cancellation at the quadrupole level, then bounds derivative roundoff, rather than demanding a literal post-third-difference zero.

The envelope is fixed before the appeal runner:

$$512\,\varepsilon_{mach}\,\|D^3\|_\infty \max_t \sum_i |Q_i(t)|.$$

Candidate scores, thresholds, histories, flow and every ledger row remain unchanged. The correction step and the independent replay in ğ5.3 are separate links. CR005h remains the immutable failed parent after adoption.

The two cumulative chains are

- CR005e failure -> CR005f correction precommit -> CR005g executed retest
- CR005h control failure -> CR005i bounded correction -> CR005i independent replay/freeze

## XII.8 Observability and physical boundary

### XII.8.1 Escape and closure observability boundary

`CR:CR005c@21` begins with the frozen three-stage history grammar:

- 101 -> contact, open, returned contact
- 010 -> open, transient contact, open
- 100 -> contact, open, open
- 110 -> contact, contact, open

There are 8,340 history-101 relations and 59,191,245 ending-zero relations. For every 101 relation, a piecewise-affine Cartesian probe finds one outward crossing on seed-to-collapse and one inward crossing on collapse-to-final. Median coordinates are

$$\lambda_{release} = 0.566282338,$$

$$\lambda_{return} = 1.132859211,$$

with open interval 0.617486748.

These values establish local near-far-near geometry. They do not establish physical timing because $\lambda$ is a stage-normalized affine coordinate and the order $\lambda_{release} < \lambda_{return}$ is already encoded by the 101 bits. The frozen record has no physical solver clock, independent Home/horizon seal or independently defined release surface. A pair's own inward recrossing cannot also be the external closure used to test its escape.

The open physical joins are therefore explicit:

- phase-to-seconds calibration;
- numerical carrier mass/energy weights;
- lossless T3 open-q to T4 closed-energy mapping;
- independently selected physical mirror and release surface;
- complete-event feedback acceptance;
- scheduler use of W9 history; and
- full RH-aware event semantics.

T6B carries exact dimensionless thermodynamic response through all 2,976 ledgers while preserving the carrier inventory and complete Theta18 address multiset. It does not assign physical energy, temperature, duration, rate, fusion threshold, numerical A or a physical mirror. Global SLC authority remains frozen `SLCQ2-RZ`; bounded Starbreaker services retain their own typed application lineage beneath that source-recorded default selection surface.

## XII.9 Moving-information E/M1, SO1 and matter joins

### XII.9.1 E/M1 returning two-body relation

The isolated SB1-E/M1 successor compiles all 17,922 active Theta18 packets into `384/384` elliptic two-body relations. Ordered electric and magnetic roles carry the same exact C4 quarter turn as the Complete Write source: the state returns while the relation retains the signed forward/reverse history needed by later interference and receiver stages. **The test result suggests strong contact with the concept.**

### XII.9.2 SO1 coherent spin-orbit wave

SB1-SO1 composes orbit, E-spin and M-spin signed waves before taking the Gram across 2,048 synchronous source cases. All 2,048 pass. Component self Grams remain planar rank two; coherent totals contain

- 1,920 positive rank-two waves
- 128 exact cancellations
- 0 rank-one cases.

Each cross family is nonzero in 1,024 cases. The interference census is 896 constructive, 1,024 destructive and 128 completely cancelling. The nonzero signed wave has period two while its shell-constant Gram has period one. Unit, primary and independent checks pass `11/11`, `13/13` and `18/18`. **The test result suggests strong contact with the concept.**

### XII.9.3 Live ATOM3D reception and re-emission

J1 executes the live Starbreaker source, E/M1 emissions and ATOM3D receivers in one causal chain of 19,958 custody ledgers. J2 then uses the retained signed off-fibre history to emit 1,536 ordered generation-two Theta18 packets and extends custody to 20,345 rows. The same source compiled across all nine TDI3 relations yields an exact two-mode matter innovation positive for every real `rho`. The receiver construction and controls are developed in `SAMA-D000058`.

E/M1, SO1, J1 and J2 are isolated/non-promoted classical exact-information successors. SO1 assigns no physical scale or numerical gravitational-wave strain, and none changes a current pointer.

## XII.10 Established result and forward handoff

Starbreaker's exact formation object retains its native ledgers, atoms, stages, orders, controls and readouts. Its structured execution closes the M126, N144, N144+d and L162 accounting; its structure-first replay is byte-identical and avoids dense fallback; and its W9 history remains an emit-only rank-161 receipt.

The application evidence is cumulative rather than selectively cleaned. The external-collapse, typed-ledger, continuous-flow, pixel-morphology and per-carrier surfaces each retain their declared boundary. Both failed-control chains preserve the parent, freeze a narrow correction and replay the original question without result-driven replacement.

The next application layer must supply the specific physical joins rather than rename current dimensionless objects. No exact Starbreaker count or source-contributor ledger by itself supplies strain, luminosity, seconds, hertz, detector response or transported GW substance.

## XII.11 Sources and evidential scope

This chapter is assembled from the following fixed source manuscripts. Source-reported campaign outcomes retain their original scope; no new campaign was executed for this edition.

- Starbreaker formation engine.

SAM VOLUME III

# Source Histories, Directional Reception and Inverse Reconstruction

Chapter XIII

## XIII.1 From an event address to an executable history

The directional core developed earlier supplies exact phase addresses and reversible Writes. Its applications must additionally state what leaves the source, which channels arrive together, what the receiver records, and which unknowns its inverse is asked to recover. The completed source-history work turns these questions into explicit computational experiments. A returning address, a recovered program and a recovered source preparation are distinct outputs. The examples below calculate each from its declared observations rather than reading the answer from source identifiers.

The applied chain begins with Starbreaker's actual generation-two packets and the A3D41 contact receiver. It then varies supplied initial information, simultaneous sources and spatial observations. Finally, it follows complete accepted-reading histories through declared dynamics. These applications use the exact inverse and retained arithmetic interfaces developed elsewhere in this volume, while keeping their own source maps and executable questions central. Sean Brady supplied the research direction; the technical maps, derivations and independent reconstructions are attributed to Codex in the cited completed records.

The classifications remain those of each executed comparison. Full two-channel reception and the source-history reconstructions achieve *The test result suggests strong contact with the concept.* Mixed and partially observed cases retain their source-defined ambiguities and classifications. The current dynamical constructions use explicit source kinetics and attempt-time units; their energy and physical propagation maps remain attached to their own recipients.

## XIII.2 Starbreaker's packets retain motion above a returning phase

The completed J4 packet roster supplies 18,318 incoming histories. Its three selected relations are the original edges 002, 006 and 084, each with forward and reverse rays. For incoming address $q_{\mathrm{in}}$, the declared propagation law is

$$q_{\mathrm{out}}(e,d,k) = q_{\mathrm{in}} + h_e + k v_{e,d} \pmod 4.$$

These are nine-coordinate phase records. The source has 3053 incoming addresses per relation/direction family, and four shells per incoming history. The applied R4 calculation expands the frozen translations into signed unit Writes, executes the retained translation and propagation, and then performs an explicit inverse-translation reconstruction. Formation and production of the original J4 response remain the supplying stages. [1]

All 73,272 shell phase states match the frozen source. Every incoming address is recovered, and every one of the 18,318 paths returns to its starting address after reconstruction while retaining nonzero accumulated winding. The calculation executes 366,360 signed native unit Writes. It reproduces 52,116 unique raw output addresses and 21,156 raw address collisions. Relation, ray direction and incoming ancestry distinguish the colliding histories.

This is an immediately useful example of state and history being separate computational objects. The returning phase is the current state; the executed signed lift and winding are retained path information. Raw endpoint equality therefore leaves meaningful distinctions in the saved source computation. The unit-Write decomposition specifies the adapter's intermediate microsteps. Physical spin and orbital motion require their own source assignment rather than being inferred from that decomposition.

Every packet has an authenticated native calculation receipt, with zero failed or incomplete calls. The recorded 277.202 seconds includes preparation, current-runtime execution, phase attachment and receipt writing. There is no matched earlier-engine timing comparison. The achieved result is exact packet propagation and inverse recovery on the actual roster, with its full phase and signed-history output available to later receivers. For that result, *The test result suggests strong contact with the concept.*

## XIII.3 One event emits two complementary observations

The common-reception construction starts from two existing native readouts of one A3D41 event: signed phase $h$ and axis occupancy $|h|$. It forms their typed pair at the source, then sends both through the same J4 response $T$:

$$(h,|h|) \longmapsto \mathrm{diag}(T,T)(h,|h|).$$

The phase advances once. A unit Write rotates its signed component through the four-phase cycle, while occupancy follows the two-phase axis quotient. The occupancy channel is computed before reception; taking the absolute value of the received signed signal gives another observation. [2]

For the three relations, write $A = X_{002}$ and $B = X_{006}$, with the exact template identity $X_{084} = -B$. The two received components have the form

$$S = z_0 A + (z_1 - z_2)B, \qquad U = a_0 A + (a_1 - a_2)B,$$

where $z_j$ carries signed phase and $a_j$ its axis occupancy. A nonzero complex minor certifies independence of $A, B$ at each of the 128 declared response settings. A Write on relation zero is therefore distinguishable from a Write on either other relation. For relations one and two, equal signed increments require opposite phases, hence equal parity. Equal occupancy increments require opposite parities. Both conditions cannot hold simultaneously. [3]

This local calculation explains the completed inverse. Beginning with the declared initial phase and directions, the decoder compares each observed change with every remaining legal next Write, keeps all matches and advances the phase. Its input excludes the actual order, source address, barrier and source hashes. A separate implementation enumerates law-generated words using Gaussian-integer phases and reconstructs the contact profile through the full incidence Gram.

Both channels and their ordered observations recover all 98,304 declared three-Write histories, their orders, phase paths, contact profiles and native barriers. Signed reception alone recovers 73,728 orders and 77,824 barriers; occupancy alone recovers 49,152 orders and 65,536 barriers. Both endpoints without intermediate records recover no unique orders. Their 6144 barrier recoveries occur where all six orders already share a barrier. These are controlled comparisons on the same source families.

The motivating pair makes the missing information tangible. At phase $(0,2,0)$, positive Writes and response setting one, orders 012 and 021 have the same entire signed history. Their contact profiles are respectively $(2,4,6,2)$ and $(2,4,4,2)$, with barriers four and two. Their occupancy records differ and the two-channel inverse recovers them. The construction is an exact discrete receiver; moving emission, stationary-site encounters and measured propagation/readout are the identified physical extension maps.

### XIII.4 Recovering a program can require less than recovering its source

The information-bridge series enlarges the application from three distinct Writes to all six signed letters, with repeats, reversals and changing directions. It includes all sixty-four phase triples, every word of length one through three and four response settings. The source executes 16,512 histories and 46,464 Writes, producing 66,048 received histories across those settings. Word length, legal alphabet and calibration remain supplied. [4]

Per response setting, full observations with initial phase and directions recover all 16,512 programs, source histories and barriers. With directions alone, the complete program is still recovered in every case, but only 16,288 initial source histories and barriers are unique. With initial phase alone, 10,944 programs are recovered. With neither initial phase nor directions, 8192 programs are unique. For the two direction-supplied full-program comparisons, *The test result suggests strong contact with the concept.* For the partially supplied comparisons and endpoint-only tests, *The test result suggests the concept is possible.*

A single negative Write on edge 002 illustrates the separation. Starting from $(0,t,t)$ for any of four $t$ values gives the same observed event. Its program is unique while the barrier can be zero or two. The executable answer should consequently contain a unique Write and a retained source/barrier candidate set. Returning only one aggregate success flag would erase this difference.

The complete length-one records contain every transition of the sixty-four-state graph. At each response setting, 384 transitions reduce to 360 distinct keys of direction, before-observation and after-observation. Every key has one signed Write label. Concatenating valid transitions therefore gives a local word decoder for any finite walk under those same supplied-direction and full-sampling conditions. The executed word census remains lengths one through three; the local composition statement follows from the complete transition table. Missed events, physical superposition and another alphabet require their own observation contract.

### XIII.5 Superposition and spatial sampling expose different inverse fibers

Two sources writing into the same receiver supply another applied question. Both initial states and directions are known. The series compares all nine source-choice pairs in each family, giving 589,824 mixed received cases. Native vector-current computation supplies the two-lane sum; its separate recovered-parent and odd-current fields are excluded from the mixed observation. [4]

The mixed signed-plus-occupancy signal uniquely recovers 137,088 of 147,456 source-choice pairs per response setting, or 92.96875%, with maximum answer-set size three. Signed mixing alone recovers 87,616. Duplicating the mixed two-lane signal produces exactly the same inverse partition. Separately observed contributions recover every pair. For the mixed-source result, *The test result suggests the concept is possible.* For the separated-source reference, *The test result suggests strong contact with the concept.* The additional information comes from a genuinely distinct source contribution, rather than another copy of the same scalar output.

The same computational principle appears in a spatial inverse that directly uses the declared radial accumulation law $a(r) = M(<r)/r$. The source roster contains 141 distributions of six units among six shells, with each shell amount zero, one or two. Six receivers between successive shells recover all distributions. Multiplying each reading by its radius gives enclosed amounts; differences return the shell populations. Reading only after shells one, three and six leaves fifteen classes and only two unique distributions. The outer reading is always $12/13$, so one outer reading or six copies of it retains one class of all 141 sources. For the full inverse, *The test result suggests strong contact with the concept.* For this fixed-total outer-only recovery, *The test falsifies the concept.*

For the six directions $\pm x, \pm y, \pm z$, total $M$, signed dipole $D$ and trace-free diagonal quadrupole $Q$ also recover every source. Opposite-pair totals obey $P_k = (Q_k + M)/3$, and the directed populations are $(P_k \pm D_k)/2$. Omitting signed quadrupole or dipole leaves larger inverse fibers. The powers $(M^2, \sum D^2, \sum Q^2)$ give only nine classes and one unique distribution. They even merge two geometrically distinct opposite-pair multisets: $(0,0),(1,2),(1,2)$ and $(0,1),(0,1),(2,2)$ share powers $(36,2,54)$. Full signed moments preserve the distinction. This is a completed angular-information application of the exact receiver, with no temperature-spectrum or cosmological parameter assignment added.

### XIII.6 From source factors to declared temporal dynamics

A static source density does not by itself select an attempted-event clock. The shared dynamics applications make that kinetic completion explicit. For positive source weights $w_i$, bit-attempt rate $\nu_b$ and departure factor $c_i$, a neighbouring flip $i \to j$ has rate

$$Q_{ji} = \nu_b c_i \frac{w_j}{w_i + w_j}, \qquad \pi_i \propto w_i / c_i.$$

The diagonal is minus the outgoing column sum. Ordered readout masks act on the unnormalized accepted substate; normalization occurs only when its full accepted-history probability is known. The endpoint without intervening masks is retained separately. [5][6]

Three shared operations supply the generator, ordered readout and projected memory. SB uses its exact sixteen-boundary-state source density at inverse temperature $1/100$, unit boundary attempts, unit departure clocks and all-minus preparation. Three equally spaced observations require coordinate zero to remain negative. Full history acceptance is 0.6382671065; the two-state memoryless approximation gives 0.6322980041. Their interval difference excludes zero, and the tested partition is certified nonlumpable.

For a declared coarse map $C$, lift $R$ with $CR = I$ and hidden remainder $r_0$, projection has the exact form

$$\begin{aligned} A_c &= CQR, \quad D = (I - RC)Q|_{\ker C}, \\ B &= (I - RC)QR, \\ M(t) &= CQe^{tD}B, \quad I(t) = CQe^{tD}r_0, \\ \dot{v} &= A_c v + I(t) + \int_0^t M(t-u)v(u)\,du. \end{aligned}$$

The full generator is $Q$, with $D$ acting on the retained hidden complement. The installed operation returns both sampled memory kernel and initial-memory forcing. The applied conclusion is specific: the visible bit alone omits information needed by the next accepted observation. This sixteen-state boundary model is an explicit kinetic source, rather than a newly solved trajectory of every microscopic N100 spin.

### XIII.7 The completed Starbreaker extension through five thousand sites

The larger SB study uses frozen exact polynomial factors to calculate the boundary weights. For boundary $b$,

$$P_b(x,y) = R_b(x,y) \prod_j F_j(x,y)^{a_j}.$$

The variable $x$ counts up-spins, while $y$ carries the losslessly compressed correction index. At inverse temperature $1/N$, substituting the exact exponential correction and multiplying Arb polynomials gives coefficients $A_b(k)$. The partition is a complete finite sum over $k = 0, \ldots, N$, with the original quadratic magnetization factor. Both original encodings retain their source and plan identities. Common exponent shifts are restored in exported absolute logarithms and cancel only in the specified boundary ratios. [7]

This source-factor route avoids requiring absent historical bulk shards. Eighteen qualification cases compare the factor contraction with expanded exact polynomial summation at $N = 12$ and retained certified partitions at $N = 100, 500$. The new anchors $N = 2000, 3000, 4000, 5000$ produce twenty-four additional cases; the combined study contains 409 sizes and 2454 cases. No magnetization sector is cut. Every new full-minus-memoryless accepted-history interval is strictly positive, and every tested two-state partition remains certified nonlumpable.

At $N = 5000$, positive-fill full acceptance is about 64.1593–64.1757%, while negative fill gives 63.2314–63.2392%. The memory contributions, in percentage points, are

| Source family | $+1$ fill | $-1$ fill |
|---|---|---|
| packet | 0.933746 | 0.017661 |
| signed packet | 0.919123 | 0.009812 |
| signed chain | 0.918002 | 0.009810 |

All six acceptance curves and memory differences decline from the prior $N = 1800$ anchor. Empirical slopes over $N = 2000$–$5000$ range approximately $-0.464$ to $-0.474$ for positive fill and $-1.0003$ to $-1.0007$ for negative fill. These are measured finite-range slopes; an asymptotic law remains the source-defined next derivation. For persistent source-dependent boundary memory through the completed roster, *The test result suggests strong contact with the concept.*

The application sequence now gives Volume III several finished objects: exact returning packets with retained winding, a two-channel local inverse, source and program inverse fibers, spatial distribution inverses, complete ordered accepted-state evolution, and source-factor dynamics that retain quantified memory. Each makes an executable question concrete. Their useful reuse depends on carrying the particular observations, source preparation and clock with the result, so another consumer can apply the completed computation without changing what it answers.

SAM VOLUME III

# The Native Mathematical Runtime: Exact Values, Certified Functions and Retained Composition

Chapter XIV

The early exact-state and language chapters establish a grammar whose accepted operations preserve their mathematical meanings. The later native runtime turns that foundation into a shared research system. An exact source can become a certified logarithm, a complete spectral factorization can enter a new matter calculation, and a returned matrix can become an operand of another registered operation. The computation retains the source, basis, error account and derivation needed to interpret each result. This chapter develops that working mathematical architecture through the completed logarithm, capacity and retained-input constructions.[87][88]

The historical language candidates and Q2 selector retain their original dates and roles. The source snapshot used here selects SLC-GEN5-R1 with CE SLC-GEN5-CEV1-R1 and generation G2-SPIN-CLOCK. Older GEN2 and GEN3 operation prefixes remain supported calls, rather than separate current global engines. The current revision records shared precision from 64 through 4096 bits, general complex matrices through dimension 2048 and supported Hermitian logarithms through 512. These bounds are admission capacities. Individual thermal, polynomial and augmented-response algorithms retain their own smaller envelopes.[87][89]

## XIV.1 Three mathematical forms of a returned value

An exact rational, an exact finite logarithmic expression and a certified interval answer different questions. Rational arithmetic can return $7/3-4/3=1$ without approximation. A logarithmic calculation can retain $\log 2$ as an exact expression even though a finite decimal representation is unavailable. A numerical consumer can then enclose that expression between exact rational endpoints at a declared working precision. The interval is the returned numerical object; its width records the remaining certified uncertainty.

The native scalar contract accepts integers and exact rational or decimal strings. Interval endpoints are exact lower and upper values. Binary floating-point operands are not silently promoted into scientific exactness. Administrative host timing has a separately typed finite representation and does not supply mathematical source data. This separation lets the same engine manage deadlines while preserving the exact scalar semantics of its calculations.[88]

A GEN5 logarithmic expression is a rational-weighted sum

$$\mathscr{L} = c + \sum_j q_j \log r_j, \qquad r_j > 0, \quad q_j \in \mathbb{Q}.$$

Arguments can remain factored powers. An exponent such as $10^{18}$ need not be expanded into the corresponding enormous integer before its logarithm is represented. Canonical factor cancellation gives

$$\log 12 - \log 3 - 2\log 2 = 0$$

as an exact normalized result. Dimensioned inputs provide a compatible positive reference, so logarithms act on dimensionless ratios with their source meaning attached.[88]

Signed combinations preserve cancellation before numerical exponentiation. Equal positive and negative contributions can disappear exactly; small remaining differences use stable source operations such as `expm1`. Increasing precision then improves the arithmetic enclosure of the surviving expression. It does not remove a declared source radius, alter a physical unit or infer a missing reference.

This distinction appears directly in the completed precision controls. The normalized expression $\log 4 - 2\log 2$ cancels exactly at 64 bits. The positive quantity $\log(1+2^{-400})$ is also certified at 64 bits through the stable logarithm-one-plus path. The apparent number of decimal zeros in an input is consequently an incomplete guide to the arithmetic work. Algebraic structure, cancellation and the requested observable determine the useful route.[90]

## XIV.2 A source scalar and a spectral profile are different logarithms

The certified GEN3 source-log construction makes the target explicit. Its registered mean/cut source admits an ordered rational vector $a_0,\dots,a_{s-1}$. Put $M=\sum_i a_i$. The source quantity is

$$Q = \frac{M^2}{s} + \sum_{r=1}^{s-1} \frac{(\sum_{i<r} a_i - rM/s)^2}{r(s-r)}.$$

The native record's `mean` field retains the signed sum $M$. Its name is not an instruction to replace $M$ by $M/s$.[91]

The same registered kernel has exact eigenvalue rule

$$\lambda_0 = 1, \qquad \lambda_k = \frac{1}{k(k+1)} \quad (k \ge 1).$$

Projection onto its polynomial modes gives weights $w_k = A_k^2/\|p_k\|^2$. The scalar target and spectral target are

$$\log(1+Q), \qquad \sum_k w_k \log(1+\tau\lambda_k).$$

They are separately typed readouts, and neither is a log determinant. For the source $(1,-1)$ and $\tau=1$, the complete spectral expression is $2\log(3/2)$, while the scalar expression is $\log 2$. The difference is mathematical content, not numerical disagreement.[91]

After modes through $m$ have been retained, the exact residual mass is

$$R_m = \sum_i a_i^2 - \sum_{k=0}^{m} w_k.$$

It bounds remaining source energy by $R_m\lambda_{m+1}$. Full degree has zero tail; a negative residual is rejected. Adaptive refinement adds missing modes instead of recomputing the retained projections. New values of $\tau$ reuse the same source modes. A direct source-cut route separately evaluates complete $Q$ without requesting spectral projections.

This eigenvalue rule belongs to the admitted mean/cut kernel. It is not imposed on an arbitrary exchange matrix because both happen to possess spectra. The source contract retains basis, dimensionless units, reference and ordered history. The logarithm consequently acts on a specified source quantity, while other operators use their own registered spectral definitions.[91]

Numerical log evaluation uses rational range reduction and an explicit-tail rational atanh series. A tolerance-met return, a valid budget-limited enclosure and an exact symbolic expression remain distinct result states. Exhausting a term or mode budget retains the last valid certificate. An overlapping interval does not become an exact equality or ordering. These behaviors give a later consumer enough information to choose whether it needs more arithmetic, more source modes or merely the existing expression.[91]

### XIV.3 Operator functions retain support and order

A matrix logarithm requires its operator domain. The installed supported Hermitian path handles positive support with an exact zero kernel, retaining the projector and zero-extending the supported logarithm. An uncertain kernel remains unresolved. A negative-support input is not reinterpreted as a positive operator. The older modular call returns $-\log$, while ordinary GEN5 operator logarithms return $+\log$; the convention is part of the operation identity.[92]

The completed logarithm expansion found a useful algorithmic change. The previous sixteen-state method returned unresolved spectra for a fully coupled positive repeated-mode source and repeated noncoordinate kernel blocks, although analytical projectors gave their exact spectra. The successor certifies a complete positive spectral window without separating every repeated eigenvalue.

A positive Gershgorin lower bound can supply that window directly. Otherwise directed LDL pivots establish positivity, and a validated inverse row-norm bound $U$ gives

$$\lambda_{\min} \geq U^{-1}.$$

The existing series then uses the complete spectral window and its tail. Certifying strict positivity before symbolic rank elimination also avoids unnecessary kernel work. Repeated modes retain their complete multiplicity without requiring a separately resolved eigenbasis.[93]

The source-defined gain was visible before the larger capacity change. An initial 64-state attempt spent more than 159 seconds in symbolic rank work and was interrupted. The completed positive-window route took 1.306 seconds for its dense 64-state call. The matched 128-state candidate took 5.939 seconds. These timings belong to the declared nine-family, 256-bit, one-thread comparison, including its native return boundary. They identify the practical value of a better certificate rather than a general dimension-independent speed claim.[93]

The qualification retains noncoordinate rank-one support, repeated kernel blocks, small positive modes, finite term exhaustion and irreducible source widths. Its meaningful failed and unresolved cases remain beside the successful certificates. Independent rational scalar-log/projector references check every emitted matrix entry. The report's classification is: *The test result suggests strong contact with the concept.*[93]

Ordered operator response uses the same distinction between a full operator and a scalar summary. For

$$A(t) = A_0 + tA_1 + t^2A_2 + \cdots,$$

an ordered log jet retains all admitted matrix words. A Fréchet direction need not commute with $A_0$. The calculation differentiates the actual operator function rather than substituting a scalar derivative of a selected eigenvalue. Formal derivative order remains separate from a finite-displacement Taylor remainder, which needs its declared source control.[88]

### XIV.4 Capacity is qualified at a specified workload

The capacity extension increased the shared range from 64–1024 to 64–4096 bits, general matrices from 512 to 2048, and supported logarithms from 128 to 512. One native limits its definition governs admission, discovery and ordinary refinement. Thermal compatibility retains its matrix-128/log-16 contract; GEN5 thermal jets must fit their complete augmented operator within dimension 512. A large base matrix and a high derivative order cannot each consume the full bound independently.[89]

The completed dense matrix fixture uses $A = I + J$ in dimension 2048 and a source vector with zero sum, so $Ab = b$. Every matrix entry is nonzero, and all 4096 returned real/imaginary components were checked. The supported logarithm fixture uses dimension $n = 512$ and

$$A = 2I + \frac{J}{8n}, \qquad P = \frac{J}{n}, \quad P^2 = P.$$

Its exact operator identity is

$$\log A = (\log 2)I + \log(17/16)P.$$

Every matrix entry and the full trace enclosed the independent reference. The large dimensions were qualified at 128 bits; 4096-bit scalar and small-matrix work was qualified separately. Their combination is an admission capacity, not a measurement of every possible large, high-precision workload.[89]

The same extension reused a completed $N = 10^{16}$ Gibbs source for order-two boundary response at 4096 bits with zero repeated original Gibbs integrations. Its original pressure uncertainty remained in the output. More arithmetic headroom improved the downstream numerical work without recasting the retained source as newly integrated at that precision. This gives capacity a useful scientific interpretation: it expands which subsequent calculations a completed source can support.

The measured dense 2048 product took 26.303637 seconds, and the dense 512 supported-log case took 46.675142 seconds at their stated settings. Complete dense output preserves every entry, so storage and exact-certificate work grow with matrix dimension squared and precision. The runtime's compact spectral routes address that output problem when a complete factor representation answers the research question more efficiently.[89]

### XIV.5 A completed result becomes a native operand

Retained mathematics makes reuse part of the execution semantics. An object carries a kind, complete source binding, mathematical value, provenance and dependencies. Discovery can list a read-only collection without loading all coefficients. Selecting a reference requires an exact source match. Adopting that reference brings its dependency closure into the ordinary session store; the new calculation can then consume it recursively inside a matrix, polynomial or symbolic payload.[94]

The eleven retained-result operations separate discovery, selection, retrieval, publication, execution/reuse, moments, factor comparison, symbolic composition, dependencies, export and import. Publishing a caller-supplied exact operand identifies its source; it is distinct from obtaining a newly executed mathematical answer. Export carries the chosen roots and their derivation closure. Import authenticates the supplied closure without repeating arithmetic, while preserving its original evidence.

Execution reuse is keyed by the complete scientific payload and exact implementation identity, apart from its immutable output name. A different source, operand, parameter or implementation is a new mathematical request. Conversely, a repeated consumer can receive the same completed calculation under a separately retained consumer lineage. This prevents a file name or friendly label from becoming the mathematical equivalence rule.[94]

The completed collection contains 1441 typed objects. It includes exchange levels $L = 2,3,4,8,10,12,13,14$, 846 distinct larger-level integer polynomials, all 165 retained Li7 T/K pairs and 32 symbolic quartic classes. Missing levels $5,6,7,9,11$ remain absent from that seed collection. New research can publish further admitted objects through the same interface without treating an index gap as a previously solved spectrum.[95]

The associated L14 composition demonstrates the mathematical consequence. Each local quartic family closes 268,435,456 states over 47 partitions, represented by 90,772 factor/weight terms. The batch composed all 32 families using the retained exchange and local operators. It performed zero new exchange solves and zero new quartic solves. A complete local-pair/collective composition took 1.235260 seconds; the 32-family composition calls took 36.009 seconds.[95]

Those are symbolic spectra retaining $J, \mu, g$ and the optional collective $\nu$. Equal all-to-all unordered-pair coupling is explicit; a physical energy scale is a separate source assignment. Each algebraic root's multiplicity is its factor multiplicity times Schur-weight multiplicity, with coincident specialized energies added. Compact references avoid expanded polynomial-degree limits and repeated coefficient materialization.

Exact moments and factor comparison become inexpensive receiving questions on that retained object. The L14 normalized moment sequence begins $1, 7/2, 203/8, 1393/8$; the L12/L14 comparison finds 23 shared factors while retaining each level's multiplicities. Fresh-process recovery returned the L14 composition with composition and moment arithmetic disabled. The acquired mathematical object, rather than a report about it, was available to the new consumer.[95]

### XIV.6 From a spectrum to a complete receiving calculation

The six cooperating GEN5 kernel families share exact expressions, operator functions, thermal response, histories/protocols, source/receiver composition and retained refinement. They use one matrix wire and ordinary source-bound references. A finite partition calculation therefore receives all declared degeneracies,

$$Z = \sum_r g_r \exp\left(-\beta E_r + \sum_a f_a Q_{ra}\right),$$

while a noncommuting block source receives its full Hermitian exponent and direction. A retained complete Gibbs source supplies its absolute $\log Z = N$ pressure and boundary response without replacing normalization by boundary ratios.[88]

Complete spectral factors can likewise supply a partition sum after real-root isolation and multiplicity reconstruction. Exact preparation can survive numerical refinement while only arithmetic precision changes. These paths connect the dense exact-state foundations to modern finite thermal and operator calculations. The source remains the same mathematical object through each change of representation.

The architecture also distinguishes a numerical bound from a physical interpretation. A supported log, entropy, source action or complete thermal derivative answers its registered question. It receives a physical energy, clock or matter meaning only through the corresponding source and reference map. This is how Volume III carries the constructive mathematical results of the first two volumes forward: it preserves their actual operands and makes subsequent operations composable, rather than substituting one recurring number or matrix dimension for a complete source definition.[88][94]

SAM VOLUME III

# Executable Methods, Adaptive Precision and Learned Questions

Chapter XV

The next computational development concerns how a problem selects its calculation. A retained spectrum can support several equivalent methods; a numerical goal can admit several working precisions; and a hidden-source question can be answered through different permitted probes. The completed SLC work makes these choices explicit mathematical objects. A method is an executable checked graph. A precision policy proposes work and then faces the actual certificate. An investigative query is selected from its measured observation partition and retained for later use. These constructions extend the earlier formal language and ordinal-selection chapters through actual native execution.[96][90][97]

## XV.1 A method library contains executable definitions

The installed GEN5 mathematical library initially supplied 34 parameterized recipes. Its six operations discover, admit, compile, execute, fit and select methods. A recipe declares exact parameters, applicability assumptions, source namespace, operation dependencies, checks and a returned observable. Compilation binds it to a source instance; execution follows the admitted graph and retains each returned dependency. Merely admitting a definition does not certify every possible instance.[96][98]

The recipe language permits up to 32 parameters, 64 steps and 128 checks. An input can be an exact rational, interval, matrix, polynomial, logarithmic expression or retained object. Step outputs can feed later steps through explicit field references. Forward references, cycles and undeclared placeholders are rejected. There is no dynamic source-code evaluation inside a recipe. The mathematical content consists of installed native operations with their existing source, basis and precision contracts.[98]

Applicability is executable. A matrix can be required to be square, two parameters to have the same shape, or a scalar to be positive. A failed assumption returns an inapplicable instance before mathematical steps execute. A failed later check retains the completed dependencies and the explicit failure. This gives the caller a concrete account of which part of a proposed route was actually applicable.

Checks also preserve their mathematical distinctions. Complete exact equality compares canonical exact values. Interval containment checks that a target lies in an enclosure. Successful native admission records the operation's contract; it is different evidence from an independently checked algebraic identity. Every returned step has explicit coverage. A successful instance therefore returns a selected mathematical output together with the checks that give that output its stated role.[98]

The catalogue covers exact log cancellation and ratios, operator determinants and responses, classical and quantum information, complex Grams and four-point Fourier action, finite spin partition functions and cumulants, complete factor spectra and failure/reset protocols. A custom symbolic recipe also executed the independent coefficient identity

$$(x-1)(x-2-g) = x^2 - (3+g)x + (2+g).$$

Its legacy mathematical stage used the same retained-result path developed in the preceding chapter.[96]

## XV.2 Source indexing and implemented mathematics have different scopes

The library's pinned source index covers 372 OpenAI result families and 719 manuscripts. That index preserves family/manuscript relationships, author attribution, source availability, formalization metadata and licenses. It is broader than the executable recipe set. Five donor families received focused source review; four supplied concrete finite transfers.[96]

Those transfers are specific. Family 107 supplies a tensor Gram identity and native matrix compositions. Family 130 supplies an exact four-point Fourier transform over $\mathbb{Q}(i)$ with its full orthogonality identity. Family 221 supplies finite spin partition, normalized-weight and directional-cumulant calculations. Family 234 supplies the normalized finite pressure

$$N^{-1}(\log Z - N\log 2)$$

under the exact state-count condition $\dim = 2^N$. Their original assumptions remain attached to their source definitions. The finite implementations do not inherit every asymptotic or thermodynamic statement from their donor papers.

Family 109 remains reviewed source material for integer multiplication and precision refinement; its all-length multiplication algorithm is not an installed recipe in this result. The index also does not mean all 719 manuscripts were downloaded or every formalization was executed. The report explicitly records local Lean execution as not run. These distinctions locate the completed computational transfer precisely while keeping the larger source collection available for future method development.[96]

Namespace transfer is equally explicit. A `SAM_land` method can enter a STANDARD problem through a declared method-only mapping naming the donor and recipient interpretation. The compiler inspects the complete adopted dependency closure, including retained parameters and recipe defaults. The mapping transfers the specified method; it does not silently rename the donor's physical law or source assumptions. A learned selector crossing the same namespaces needs its corresponding explicit model mapping.[98]

## XV.3 Learning selects equivalent executed routes

The native method learner receives pairs of successful execution traces for one exact problem. Their selected outputs must

be canonically equal. Problem identity includes task type, normalized inputs, source binding and precision. Training, development and reserved splits retain whole problems, so the same source instance cannot contribute examples to both fitting and reserved evaluation.[98]

The default target is native operation count. A measured elapsed-time target is also available, with its execution conditions retained. Cost ties do not manufacture winner labels. Features describe mathematical structure: dimensions, nonzero and negative counts, rational bit lengths, operation types, assumptions and retained-input counts. They do not contain the mathematical answer. Exact class-balanced CART fits the method comparison and uses development data to select its bounded tree depth.

Selection compiles feasible candidates, ranks their ordered pairs and preserves rational score ties. It returns compiled methods; their execution remains a subsequent native operation on the new source. Imported models are advisory. A model package preserves its native record, feature schema, training dependencies and objective, while a newly selected result still undergoes the current recipe checks.[98]

The actual 54-preparation frustrated-spin example supplies a concrete lesson. Its normalized source rows form a $54 \times 18$ batch, and its common receiver is an $18 \times 34$ map. All 1836 returned receiver intervals contain the independently retained exact endpoints, and the complete output matches the earlier native CPU result at 1024 bits. The original Gibbs integrations were not repeated.[96]

Two recipes calculate that same observable. The direct route uses one product; the alternate reconstructs the receiver through two adjoints and then multiplies, using three operations. The lesson separates six training, twelve development and thirty-six reserved preparations. The learned selector chooses the direct route for the complete batch. That full batch is a new batch shape assembled from the original preparations; the thirty-six-preparation subbatch is the reserved source split. The result is a learned execution choice with a measured cost difference, rather than retrieval of a previously labeled answer.[96]

The later spin/clock installation adds five executable compositions: source projection, hidden residual map, residual payment, generalized hidden-direction projectors and fifth-coordinate lower response. It brings the recorded method count to 39 and native operation count to 254 in the G2-SPIN-CLOCK generation. The earlier 34-method/248-operation report remains its historical installation snapshot. Both describe completed stages of the same retained architecture.[53]

## XV.4 Adaptive precision proposes work; the certificate accepts it

Precision learning has a related but distinct target. A case names an installed numerical operation or compiled method and an executable width or sign goal. The controller proposes a starting precision, executes native trials and checks the returned complete enclosure. Failure can trigger higher precision. The learned choice never replaces the mathematical acceptance test.[90][99]

For selected intervals $I_j = [l_j, u_j]$, an additional declared output radius $e$ and target width $w$, acceptance is

$$\max_j (u_j - l_j + 2e) \le w.$$

A matrix path checks every real and imaginary component. Source radii already present in the native return stay present; the separate additional radius is paid before acceptance. Positive, negative, nonzero and exact-zero goals operate on the selected real interval under their own native criteria.[90]

The default ladder starts at the installed minimum and doubles to the declared maximum. Explicit starts override advice. An exact successful case can remember its cheapest observed successful precision or elapsed cost. The word observed matters: an untested lower precision is not reported as a measured optimum. Models accelerate unfamiliar cases, while checked case memory can reuse what a particular calculation already established.[99]

The completed lesson trained on ten one-log cases and four development cases. Their executions supplied forty witnessed rows, and development selected a depth-one native tree. Six reserved cases used two-log signed expressions with coefficient scales and accuracy goals absent from training. The baseline required eighteen numerical trials; the learned policy required six, reaching the baseline's first successful ladder point in every case. Fresh stores kept the comparison free of preexisting target results. The same imported model repeated eighteen-to-six on the qualified pod.[90]

The primary matched-work result is a 67 percent reduction in numerical trials. Recorded local execution-loop totals were 338.723 milliseconds and 130.001 milliseconds, excluding planning, transfer and cold return. The report retains one exceptional development witness; certification still determines every future acceptance. Its classification is: *The test result suggests strong contact with the concept.*[90]

Several controls show what precision learning must distinguish. A declared source radius of $10^{-6}$ against a $2^{-80}$ width goal returns source-refinement-required with zero arithmetic trials. A retained interval $[1,2]$ cannot be narrowed to a point by increasing arithmetic precision; its observed width plateau requests source or truncation review. A goal beyond the policy ceiling returns its unresolved ceiling status. An initially unresolved nearly singular exact solve, by contrast, succeeds after escalation to 256 bits.

Imported execution traces do not become recipient-local training witnesses. The recipient must execute the case locally to obtain its runtime-owned marker. This prevents a supplied or rehashed provenance string from fabricating measured labels. Mathematical reuse and evidence transfer remain useful, but they retain the distinction between possessing a completed result and witnessing a new local training outcome.[99]

## XV.5 Exact inverse reconstruction preserves the requested target

The earlier native inverse release provides another form of computational intelligence: choose an observation that resolves the property actually requested. Its source relation retains every compatible ordered history. A receiver frame or

difference may leave several states compatible, and the planner evaluates permitted probes using exact weighted information. Applied observations refine the original relation instead of replacing its reconstruction target.[100]

In the declared unknown-phase fixture, the original observation leaves sixteen histories. For initial-state reconstruction, the selected probes reduce the relation

$$16 \longrightarrow 4 \longrightarrow 1.$$

For the original action ratio, history counts follow the same sequence while ratio alternatives reduce $8 \to 2 \to 1$. The original barrier is known after the first probe, even though four histories remain. Its next plan returns no gain. Knowing the requested property and knowing the whole source history are different stopping conditions.[100]

Exact information uses the declared probability measure. Binary, uneven and three-way tests retain rational weights and logarithmic terms such as $\log 3$. These information quantities remain distinct from source-action ratios. Source restrictions and unavailable probes also remain explicit; an unavailable observation does not remove a compatible source from the relation.

The completed qualification passes 55,917 assertions, covering 384 original source members, 2304 probe extensions, equal-observation fibers, variant receivers, short ordered histories and guards. Appending a probe preserves the original target even when the extended history's barrier changes. A resumed session retains its full sealed decision plan, alternative predictions and all maximizing ties. The result's classification is: *The test result suggests strong contact with the concept.*[100]

This inverse machinery is a completed finite method under its declared source and observation contract. It supplies a useful basis for learned investigative queries: the learner can compare the partitions generated by permitted actions, count their costs and retain a method whose information matches the original question.

## XV.6 A learned query changes how information is gathered

The unfamiliar-query lesson starts from sixteen hidden phase contexts and two primitive Writes, $A$ and $B$. Each query reports one selected value: a signed receiver difference, its sign, or one of the native logarithmic history channels. Full source frames remain in receipts and are not policy observations. The objective orders correct resolution first, readout count second and Write count third.[97]

The first attempt is the optimal adaptive tree over fourteen primitive queries. The learner then enumerates words of lengths two through four, with the same seven readouts and primitive continuations where needed. Its 196 generated methods plus the incumbent give 197 candidates. The selected `AB:DELTA` query performs two Writes before returning one combined receiver difference. `BA:DELTA` is the other exact best tie.

The selected partition resolves all sixteen training contexts. Readouts fall from 28 to 16 while Writes rise from 28 to 32. After freezing, the same method transfers to three withheld direction bindings: all 48 fresh cases are correct, with 48 rather than 84 readouts and 96 rather than 84 Writes. New method searches are zero. The improvement changes how information is collected; it does not make every cost decrease.[97]

The broader clock-history lesson acquires four additional variation queries and grows its library from six to ten methods. Training resolves 128 cases with 168 rather than 220 readouts; development uses 68 rather than 80. In final longer-history transfer, however, a primitive variation query already resolves every case in one report. The learner selects that simpler option and records no advantage over the primitive policy. The successful method remains useful without assigning a gain to a source where the gain was absent.[101]

Fresh known-source recovery restores the acquired decision machine and reconstructs 32 histories with 43 readouts, with zero new model rows, policy planning or method searches. A separate declared-question bundle answers 64 questions with 32 probes instead of 40 sequential probes. The joint history target supplies the narrower state, ratio and barrier answers under the same source contract. Both results retain exact question/transcript bindings and their corrected completion records.[101][102]

The division of contributions is explicit. Codex supplied the task mapping, legal query grammar and generic finite search; native source executions supplied every candidate's observation partition and exact cost. These experiments learn and reuse a method. They do not measure an autonomous change of the search algorithm or assign physical seconds to computational resets and Writes.[97]

## XV.7 Persistent histories and language retain their own boundaries

The modern persistent-history kernel retains occurrence identities and every directed event, rather than only a final total. Range queries and edits preserve $U, D, V$, net change, maxima and all ties. Point edits provide explicit replacement incident edges, and the earlier tree root remains usable. Gaps remain gaps; the consumer can request complete positive segments rather than silently joining unrelated runs.[88]

Its logarithmic algorithmic work has a specific meaning. Range and edit paths use $O(\log N)$ tree work, with exact integer arithmetic, output and export counted separately. Repeated finite protocols use binary powering with $O(\log n)$ ordered matrix compositions at fixed state dimension and derivative order. Those protocols retain success, failure, reset, record and recipient channels, whose complete transition closes stochastically. A survival result keeps its accepted mass and declared source clock. The logarithmic work count is not a claim that every matrix or source integral has logarithmic arithmetic complexity.[88]

Native language is another typed interface to this computational system. The later compiled operations parse sourced concepts, sentence frames, argument fragments and declared arithmetic. For basic arithmetic, the language kernel builds the operand graph and the existing exact operation evaluates it; a conversational wrapper does not become a replacement calcu-

lator. Source meanings retain their document identity and page, while runtime status comes from current authority.[103]

Its admitted capacities and historical lessons have their own scope. The installed code contains indexed-library and streamed-training operations, typed frames and conversation state. Those capacities are not counts of current learned data. The live conversation authority records that chat, its sustained training programme, saved models and designated data were owner-withdrawn; the services are stopped and guarded against restart. Shared native operation code remains installed. This edition presents the mathematical and language contracts as retained computational work and preserves that current removal status.[104]

Across methods, precision and questions, the completed pattern is consistent. Source identity and exact mathematical outputs make alternate routes comparable. Witnessed costs make a particular route choice learnable. A retained graph, model or decision plan makes that choice reusable. Actual execution and its source-defined certificate still determine the returned result. Volume III thus adds a constructive account of method acquisition to the exact-state foundation while retaining every important distinction among a formal language, a learned selector, a numerical certificate and a physical source interpretation.

SAM VOLUME III

# Exact Exchange and Symbolic Matter

Chapter XVI

## XVI.1 A local source becomes an interacting exact object

The exchange program constructed a continuous route from retained Li7 source operators to finite interacting chains and then returned their spectra to symbolic matter calculations. Its completed outputs include nineteen arrangement classes, a fully matched finite polynomial Bethe benchmark, a complete length-thirteen exchange spectrum accounting for 67,108,864 states, and composition with thirty-two local-matter families in 2.011662 seconds of recorded assembly. The assembly reused completed exchange factors and local quartics. [EX, pp2–10]

The local exchange space $V$ has four states. Chain length $L$ counts copies of this four-state factor; it is distinct from isotope mass number, Ising spin count and Starbreaker layer count. $T$ is the retained local return operator, $K$ the signed source-role interaction, and $Q(g) = T + gK$ their symbolic local family. On the chain, $P$ exchanges two factors, and the periodic exchange Hamiltonian is $H_{\mathrm{ex}} = \sum_{j=1}^{L} P_{j,j+1}$. Independent parameters $J$, $\mu$, $g$ and later $\nu$ remain symbolic.

The signed interaction uses inherited source incidence vectors $d$ and their diagonal component matrices $D^a$:

$$K = \sum_{a=1}^{3} D^a T D^a.$$

All thirty-five assignments of three proton roles and four neutron roles produce nineteen exact $K$ spectra and nineteen exact $T + K$ spectra. The classes are the sixteen reversal pairs and three palindromic singletons. Equal species counts or equal counts of unlike neighbors do not determine the result. The source-role operator reads additional arrangement information. Species erasure sends $K$ to zero, and sign erasure changes its spectrum. [EX, p2]

The full coupling calculation constructs $35 \times 48 = 1{,}680$ four-state component polynomials $\det(qI - T - gK)$. Five exact coupling evaluations determine each degree-at-most-four coupling dependence, and every coefficient array agrees with an independent determinant expansion over all twenty-four permutations. Comparing complete 192-state characteristic polynomials gives equality only at $g = 0$ between distinct arrangement classes. The nineteen whole-spectrum classes therefore remain distinct for every nonzero coupling. This concerns equality of entire spectra; it does not exclude individual level crossings. [EX, p3]

## XVI.2 The canonical quartic retains the actual source matrix

In the canonical component, $T$ has diagonal entries 84 and off-diagonal entries 8 on the path 2–1–3–4. $K$ has diagonal $(672, 2016, 672, 672)$, with path entries 96,32,32. After ordering along that path and subtracting $(84 + 672g)I$, define

$$a = 8 + 96g, \qquad b = 8 + 32g, \qquad h = 1344g, \qquad x = q - (84 + 672g).$$

The centered matrix is tridiagonal with diagonal $(h, 0, 0, 0)$ and off-diagonal $(a, b, b)$. Its characteristic polynomial is

$$p(x; g) = x^4 - hx^3 - (a^2 + 2b^2)x^2 + 2hb^2x + a^2b^2.$$

This follows from successive tridiagonal minors: $p_0 = 1$, $p_1 = x - h$, $p_2 = xp_1 - a^2$, $p_3 = xp_2 - b^2p_1$, $p_4 = xp_3 - b^2p_2$. At $g = 0$, the uncentered eigenvalues are $80 \pm 4\sqrt{5}$ and $88 \pm 4\sqrt{5}$. In the original 192-state generated Li7 instance, each has total multiplicity forty-eight. The quartic is derived from the retained matrix rather than fitted to sampled eigenvalues. [EX, pp2–3,12]

The 1,680 component occurrences yield 165 distinct ordered operator pairs, thirty-two symbolic quartics and 162 uniform-field compatibility classes. These count different properties. An ordered pair keeps the actual matrices in an aligned basis; a quartic keeps their eigenvalue multiset; a compatibility class determines which placed operators preserve the fixed transfer family. Equal local spectra do not by themselves establish that differently placed operators are compatible. [EX, pp5–6]

## XVI.3 The source extension supplies the exchange law

Sean Brady's explicit extension lifts the completed receiver-current swap to two separately owned four-state factors:

$$P|i, j\rangle = |j, i\rangle.$$

The local return operator is not silently replaced by a permutation. $T$ and $K$ retain their source roles; $P$ is the newly specified joint action. It preserves the sum of one-hot contents, reverses their directed difference, and squares to the identity. The rational exchange family is

$$R(u, v) = (u - v)I + \eta P, \qquad \eta \neq 0.$$

The Yang–Baxter consistency condition reduces to a scalar coefficient identity. For this family the residual coefficient is $\eta^2[(u - w) - (u - v) - (v - w)]$, which vanishes identically. Native exact word coefficients on the sixty-four-state three-factor space and an independent permutation replay check the construction. The $R$-matrix belongs to established integrable-system mathematics; the project result is the specified source extension, its exact execution, and the resulting reusable source-to-matter chain. [EX, pp3–4]

## XVI.4 A complete finite polynomial Bethe benchmark

A transfer matrix is obtained by letting an auxiliary four-state factor interact successively with the chain and tracing out that

auxiliary factor. With auxiliary label $a$, the local operators and ordered product are

$$L_{aj}(u) = uI + \eta P_{aj}, \qquad M_a(u) = L_{aL}(u)\cdots L_{a1}(u),$$

$$\tau_L(u) = \operatorname{tr}_a M_a(u), \qquad [\tau_L(u), \tau_L(v)] = 0.$$

At the regular point its logarithmic derivative gives

$$H_{\mathrm{ex}} = \eta\, \tau_L(0)^{-1}\tau_L'(0) = \sum_{j=1}^{L} P_{j,j+1}, \qquad L+1 \equiv 1,$$

with no identity offset in the retained gauge. At $L = 2$ the two periodic bonds give $H_2 = 2P_{12}$. This convention fixes the spectral numbers used in every subsequent matter calculation. [EX, p4]

The finite benchmark constructs sixteen exact nested polynomial tuples: two at $L = 2$, four at $L = 3$ and ten at $L = 4$. Each is checked against the simultaneous eigenspaces of every transfer coefficient, including the full GL4 multiplet multiplicities. All 336 states are accounted for, and five hundred exact transfer-coefficient commutators vanish.

| Chain length | Exchange eigenvalue and multiplicity | Total states |
|---|---|---|
| 2 | 2:6; 2:10 | 16 |
| 3 | 3:4; 0:40; 3:20 | 64 |
| 4 | 4:1; 2:50; 0:60; 2:110; 4:35 | 256 |

Polynomial divisibility also handles singular Bethe-root presentations. At $L = 4$, the tuple $Q_1 = z(z+1)$, $Q_2 = Q_3 = 1$ gives the regular collected eigenvalue $\Lambda(z) = 4z^4 + 4z^3 - 2z - 1$ and energy $\Lambda'(0)/\Lambda(0) = 2$. Dividing individual terms at the singular roots would obscure that exact result. The benchmark instead collects the numerator, checks exact divisibility, and compares the resulting polynomial with the actual joint eigenspaces. The complete finite result is $L = 2, 3, 4$; the large-chain spectrum below uses a separately established representation route. [EX, pp4,11]

## XVI.5 Exact representation structure closes length thirteen

The full tensor space has dimension $4^L$, but the exchange operator lies in the permutation algebra. Schur–Weyl decomposition separates a local-matter representation $U_\lambda$ from a permutation representation $S_\lambda$ for each partition $\lambda$ of $L$ with at most four rows. Exchange acts on $S_\lambda$, and multiplying its eigenvalue multiplicity by $\dim U_\lambda$ recovers the full tensor account. The exact dimension closure is

$$\sum_{\substack{\lambda \vdash L \\ \ell(\lambda) \le 4}} \dim U_\lambda \, \dim S_\lambda = 4^L.$$

Momentum splits the permutation blocks further. Exact modular characteristic polynomials and bounded Chinese remainder reconstruction avoid uncontrolled rational coefficient growth. At $L = 10$, momentum reduces the largest block from 768 to 77. A separately recorded fifty-four-sector comparison takes 117.611 seconds for the rational characteristic-polynomial stage and 0.962 seconds for its modular counterpart. These are times for that stage and comparison. [EX, pp4–5]

The raw-coordinate route retains an invertible basis matrix $B$ and the retained action $H_B$ without first normalizing the basis. The pencil $H_B - xB$ has determinant $(-1)^n \det(B) \det(xI - B^{-1}H_B)$, so its exact normalization and coordinate maps recover the same spectrum. At $L = 13$, opposite momenta are paired over the real cyclotomic field $\mathbb{Q}(\zeta_{13} + \zeta_{13}^{-1})$, of degree six. A 7,920-dimensional rational primitive-parity problem becomes a largest 1,320-dimensional matrix over that field. Supported GPU work constructs exact projector/action residues and columns; CPU arithmetic performs certified lifting and spectral computation. [EX, pp5,13]

The completed L13 record closes all thirty-nine partitions and accounts for $4^{13} = 67{,}108{,}864$ states. Its 713.1002004-second clock is the resumed hybrid completion, importing thirty-two finished partitions and completing the remainder. It is not a fresh whole-L13 timing. The completed factors and multiplicities are the result used by the subsequent matter calculation. The attached article's L14 run was only launched at its cutoff and is not included in that completed count. [EX, pp5,10; H001563]

## XVI.6 One retained exchange solution supports thirty two matter families

A uniform sum of local operators commutes with every site permutation. For a local $Q$, the identity $P(Q \otimes I) = (I \otimes Q)P$ gives $[P, Q \otimes I + I \otimes Q] = 0$. Consequently the homogeneous Hamiltonian

$$H = JH_{\mathrm{ex}} + \mu \sum_{j=1}^{L} Q(g)^{(j)}$$

uses the same exchange decomposition. If $q_i(g)$ are the four local roots and $n = (n_0, n_1, n_2, n_3)$ is a Schur occupation with $\sum_i n_i = L$, then its combined energy is

$$E_{\lambda,e,n} = Je + \mu \sum_{i=0}^{3} n_i q_i(g).$$

The multiplicity is the exchange multiplicity $m_\lambda(e)$ times the exact Schur-weight multiplicity $K_{\lambda,n}$. Coincident energies add their multiplicities. This formula explains the reuse: a new local family needs the retained exchange factors, local roots, occupations, source maps and multiplicities; it does not require the exchange operator to be solved again. [EX, p6]

Both rows perform zero new exchange solves and zero new local-quartic solves. The L13 shared template retains 133 distinct exchange factors. Its assembly clock measures the composition stage, separately from obtaining those factors. This is the main computational significance: a completed many-body spectrum remains an exact operand for an entire family of further matter calculations. [EX, pp6–7; H001565]

The all-length one-site criterion also identifies when non-identical local sources may be mixed. For $L \ge 2$ in the homo-

| Retained exchange input | Partitions | Local families | States per family | Factor and weight pairs | Assembly seconds |
|---|---|---|---|---|---|
| L12 | 34 | 32 | 16,777,216 | 54,633 | 0.842111 |
| L13 | 39 | 32 | 67,108,864 | 35,184 | 2.011662 |

geneous untwisted family, a sum $\sum_j A_j^{(j)}$ commutes with every transfer matrix exactly when $A_j - \mathrm{tr}(A_j)I/4$ is independent of site. Uniform traceless content is sufficient; scalar offsets may differ and remain in the final energy. The retained example $K_{036} - K_{012} = 336I$ has that property. The necessity follows from commutation with the cyclic shift at $\tau_L(0)$, then partial tracing its shift defect. This is an all-length algebraic statement with its declared transfer-family assumptions. [EX, p6]

### XVI.7 Fusion and a conserved current connect composite channels

The pair projectors $E_\pm = (I \pm P_{12})/2$ select ten symmetric and six antisymmetric local states. The exact fused/fundamental $R$-polynomials act on forty- and twenty-four-dimensional spaces. All fifteen bivariate coefficient matrices in the mixed Yang–Baxter comparison vanish in each channel, checked in the faithful regular representation of $S_4$. Uniform matter actions lift into the same fused bases and have eigenvalues $q_i + q_j$, with $i \le j$ in the symmetric channel and $i < j$ in the antisymmetric channel. [EX, pp6–7]

On three factors the oriented exchange current is $C = \sum_j [P_{j,j+1}, P_{j+1,j+2}]$. It commutes with $H_{\mathrm{ex}}$ and obeys

$$C^2 = 3H_{\mathrm{ex}}^2 - 27I, \qquad \det(xI - C) = x^{24}(x^2 + 27)^{20}.$$

The current's diagonal pair-parity blocks vanish; both cross maps have rank twenty. It therefore transports the two twenty-dimensional mixed-symmetry sectors between the symmetric-pair and antisymmetric-pair carriers while preserving their exchange energy. The fully symmetric twenty-dimensional sector and fully antisymmetric four-dimensional sector form its kernel. These are exact operator identities and ranks, with no physical time assigned to $C$. [EX, pp7–8,13; H001565]

### XVI.8 The collective interaction preserves the exact family

The completed compatibility analysis also supplies a genuine two-object interaction. Equal coupling of every unordered pair gives

$$Y_Q = \sum_{i<j} Q^{(i)} Q^{(j)} = \frac{1}{2}\left[Q_{\mathrm{tot}}^2 - (Q^2)_{\mathrm{tot}}\right].$$

Expanding the square proves the identity, and both uniform sums commute with every site permutation. Thus $JH_{\mathrm{ex}} + \mu Q_{\mathrm{tot}} + \nu Y_Q$ preserves the entire transfer family at every length. On a Schur occupation, define $s = \sum_i n_i q_i$ and $r = \sum_i n_i q_i^2$. Its energy is

$$E_{\lambda,e,n} = Je + \mu s + \frac{\nu}{2}(s^2 - r),$$

with the inherited multiplicities. Existing exchange and weight records therefore already support this collective two-object term. [EX, p8]

The nearest-neighbor density family spanned by $T \otimes T$, $T \otimes K + K \otimes T$ and $K \otimes K$ has a different completed answer. For all 165 retained pairs, its strict three-site commutator outputs have rank three. For $L \ge 5$ no nonzero combination in that specified span commutes with the retained $H_{\mathrm{ex}}$. This delimits that proposed geometry; the collective construction is an explicit compatible successor. Spatial coefficients, nuclear calibration and all-length polynomial Bethe completeness remain separate questions. None is required to state the completed finite spectra or the all-length compatibility identities already derived.

SAM VOLUME III

# Exact Application Reuse in Matter and Learned Reconstruction

Chapter XVII

## XVII.1 A method becomes useful through its source contract

The same exact computational interface can serve different scientific objects while keeping their questions distinct. A Li-6 construction asks which source arrangements minimize a declared action and how a receiver distinguishes retained ties. An ATOM3D observer asks which learned expressions return genuinely different observations on a task roster. A tau reconstruction asks whether native source distinctions help identify useful measured objects and training methods. The completed applications below supply actual operands, source maps and outputs for each question.

Their common feature is source-bound reuse. A retained expression, inverse table, source factor or method memory can be applied again without rebuilding the calculation that produced it. The meaning of the reused object travels with its source inventory, observation roster and task. Equal array sizes or matching labels do not establish a cross-domain identification. The applications are therefore connected here through their executed mathematical interfaces and source-defined results, with no new scientific computation performed for this manuscript.

## XVII.2 Li-6 grammar retains the information in circulation

The isotope-specific Li-6 construction contains three proton and three neutron objects on six named sites. Its nine oriented relations give site-incidence rank five and cycle rank four. Together those coordinates recover all nine edge currents. A source action based only on endpoint currents would omit the circulation distinction used by the completed grammar. The source retains six one-body addresses, one hidden-nine support, road, color owner and weak vector, while distinguishing partition, depth, route, source role and conjugation. [8]

There are five motif covers. A cover assigns every relation once, either through nine pair motifs or one three-owner motif and six remaining pair motifs. The full 321-row catalog supplies a typed source; candidate constituent projection selects the declared pair and triad rows. Unused scalar or rejected-control records remain in the catalog without becoming constituents of this isotope. The reference masks retain the separated source obtained by removing its two depth joins.

The completed calculation selects two covers. Without hidden-loop response, each favored three-owner circulation cover has sixty-four minimizing histories and action $91/2$. The hidden-nine depth-loop response selects the same sixteen histories in both covers at observed action

$$S_{\min} = \frac{737}{16} = 46.0625.$$

The pair-only cover instead has minimum $1385/16$ with hidden response. Two other triad covers have $363/4$. The selected native-channel minimum is $969/16$ with the same two covers and sixteen histories. All ties are retained. Across six center placements and 128 response settings, the selected family is unchanged.

This source shows two distinct reductions: circulation selects the motif covers, and the hidden-loop response refines their histories. The endpoint-only triad comparison favors the pair-only cover. Replacing it by the retained circulation changes the assembly preference on the same constituent inventory. An explicit same-site-current/different-circulation witness records the information responsible for the change.

The factorized computation retains 245,760 minimum cases and 119 distinct minimizing sets across its masks, channels, direction maps, receiver variants, covers, placements and response settings. The hardware calculation evaluates 5,563,227 primitive response values, agreeing exactly across H14F direct forms, Ryzen independent expansions and actual Radeon 780M integer contractions. Two source-feature batches avoid materializing a large Cartesian product while their inverse maps still cover 2,097,152 state/reference rows. For the achieved grammar assembly and source-action selection, *The test result suggests strong contact with the concept.* The action 46.0625 is in candidate source units; its measured binding-energy assignment has a separate physical recipient.

## XVII.3 Active probes resolve a family whose source action is tied

The original sixteen histories, two covers and six placements give 192 retained Li-6 configurations with the same observed action. The signed-readout application asks how an observer can distinguish them using new native quarter Writes. It evaluates the response to an intervention instead of selecting among tied constructions through a hidden identifier. [9]

At known history 14336, a single Write on edge 002 or 006 already distinguishes the twelve cover/placement alternatives. With initial history unknown, all eighteen scalar action probes resolve ninety-six pairs. Their remaining ambiguity is simultaneous global-current reversal. The exact search over 153 probe pairs finds eight tied pairs with the same ninety-six-class distinction; no single probe has more than twenty-four outputs on the roster.

The selected decoder uses two independent response experiments. Each starts from the same restored initial configuration: one positive Write on the source edge of index two, and one negative Write on the source edge of index five. The measured changes are accompanied by the signed N01 site-current polynomial $\operatorname{Re}\alpha + \operatorname{Im}\alpha$, whose value is $+1$ or $-1$. The two action changes distinguish the reversal pairs; the signed bit supplies orientation. The native finite inverse recovers state, cover and

placement in all 192 cases. The experiments are independent restored-state probes, rather than two consecutive changes to an already perturbed source.

The source computation executes 288 transitions, 3264 construction evaluations and 3456 signed response differences with their retained arithmetic graphs. Independent rational checks match every difference. All ninety-six inverse pairs have the predicted reversal association, and fresh graph recovery applies the usable decoder without replacing its source. For the completed response-based distinction, *The test result suggests strong contact with the concept.*

One difference is only 43 inside response values of order $10^{12}$. Retained exact signed arithmetic computes it while reusing 560 prior graph nodes. This is a concrete application in which preserving exact operands matters to the scientific receiver: a small cover distinction remains available above a large source baseline. It does not itself assign the experimental resolution, magnetic coefficient or MeV conversion. The decoder identifies the admitted finite source family at response setting one, with every cover and placement still available.

## XVII.4 The observer chooses distinct reports without retraining memory

The next ATOM3D application keeps learned memory fixed and changes only method selection. Its predecessor contains thirty-six atomic methods and ten clock methods. Several compatible expressions can produce the same vector of reports over a declared task's source histories. Reporting all of them spends observations without distinguishing another candidate. The native distinct selector computes that equivalence in the installed exact arithmetic. [10]

For compatible methods $e, f$, the task-specific decision is

$$D(e,f) = \sum_s \big(q_e(s) - q_f(s)\big)^2.$$

Every difference, square and sum is native exact signed-log computation. Zero identifies equal report vectors on the declared roster. A signed sum of differences would be insufficient because its terms can cancel. In calibration, twelve compatible methods have eight distinct report vectors. Seven of sixty-six pairs are equal and fifty-nine differ; twenty-two unequal pairs have signed differences summing to zero. The squared test preserves all twenty-two distinctions.

The quotient retains method identities and fiber multiplicities. Its exact residual method-identity information is $(5/6)\log 2$. The selector follows the original compatible-method priority, takes the first representative of each exact report class and fills up to four distinct slots. All expressions and ties remain in memory. Their equivalence is bound to the task, source roster and frozen method memory, rather than declared as a global identity between expressions.

The previous 224-history reuse comparison keeps all histories correctly recovered while reducing scalar reports from 516 to 508. Fresh confirmation declares all 256 legal length-eight words on one A3D29 orbit, disjoint from 416 predecessor paths. Both selectors recover $256/256$ histories; the distinct selector uses 625 reports instead of 633. Saved policies replay all confirmation cases exactly on the opposite host, with no new source-model generation, ranking or search.

The cost comparison retains its ordering: unresolved histories first, scalar reports second, executed query nodes third. Fewer reports can require additional arithmetic. Query nodes increase from 1899 to 2163 on fresh confirmation, while scalar bytes decrease from 2749 to 2715. Selector preparation is separately measured; recovery reuses its saved policy. Thus the reported result is improved selection of existing learned methods, not newly trained expressions or a blanket reduction of every computational cost. *The test result suggests strong contact with the concept.*

The completed tests remain available while this research line is owner-paused. Its eight services are stopped and the verified local checkpoint is retained. The pending final T500 checkpoint-copy status is carried by its live authority. That operational boundary does not alter the completed native equivalence and paired-selector outcomes.

## XVII.5 An exact symmetry accelerates full memory in a matter source

Application reuse can also improve execution without reducing the represented state. The two retained A3D41 six-spin phase graphs have sixty-four states, zero fields, inverse temperature $1/4$, unit attempts and unit departure clocks. Global spin reversal sends state $s$ to $63-s$ and leaves every pair energy and flip gap unchanged. The actual generator therefore commutes with that reversal. [11][12]

For paired states set $x_i = p_i + p_{63-i}$ and $y_i = p_i - p_{63-i}$. The transformed generator has two independent thirty-two-coordinate propagation blocks. This is an invertible representation of all sixty-four coordinates. The symmetric block is a probability generator; the antisymmetric block carries signed response. A visible-bit mask couples the blocks:

$$x^+ = (x+Sy)/2, \qquad y^+ = (Sx+y)/2,$$

where diagonal $S$ records the accepted bit of each representative. Repeating propagation and these masks reconstructs the original unnormalized accepted substate for arbitrary initial preparation and every declared finite ordered reading sequence.

This exact representation retains the preparation dependence exposed by the full source. In one graph, two initial states with the same visible bit have three-read acceptance approximately 0.47596 and 0.95360; in the other, corresponding values are 0.40943 and 0.99147. Formal refinement of the two visible classes reaches all sixty-four classes in both sources. The parity method accelerates those full histories while retaining their hidden-state information.

The pilot's propagation medians improve by factors 2.004 and 2.176 on one i9 core. Including the separately measured common generator cost gives factors 1.343 and 1.265. Both endpoint and accepted-substate operators are checked on all sixty-four basis preparations. The source states, timing scopes and complete retained response remain explicit. *The test result suggests strong contact with the concept.* The adapter is a qualified research candidate rather than an unrecorded global

operation installation.

### XVII.6 Tau learns correspondence and training utility separately

The tau pilot asks a different applied question: can native SAM distinctions organize measured constituent observations and help choose useful reconstruction methods? The source vocabulary includes measured constituents, pair/three-owner roles, amount versus directional contribution, signed receiver values, native Write barriers and relative grammar alternatives. These are declared source mappings into measured data. The pilot acquires finite empirical correspondences over that vocabulary, while actual matched training supplies a separate method-utility record. [13]

Correspondence acquisition uses 8192 training observations. For each task, exact native conditional information selects among eight role recipes, then evaluates all fifteen pairs of six native readout families jointly. Each candidate report has four training-median bits, and constituent count supplies one-, two- and three-or-more-object contexts. Exact ties and conditional target probabilities remain in the acquired memory.

For transverse-residual positivity, the selected representative is energy near-jet, paired with signed assembly and relative grammar. Identification selects momentum near-jet with endpoint and signed assembly. Six-class decay selects momentum near-jet with signed assembly and relative grammar. Their conditional bundle information is approximately 0.329848087, 0.181193140 and 0.347745237 nats, with respectively 875, 818 and 881 saved cells: 2574 source-bound correspondence cells in total. Tied role alternatives remain available. The positivity question concerns signs of transverse residual components, a different target from the final trained angular-error metric.

The saved probability tables can predict new known cells after restart without another neural fit or information-planning calculation. On eligible new observations, positivity accuracy is 30.10% versus the same-row count-only 27.53%; identification is 80.36% versus 72.18%; decay is 54.51% versus 53.58%. The known/eligible counts are $2256/2485$, $4792/4985$ and $2260/2485$. The remaining 229, 193 and 225 eligible observations are explicitly unknown cells. A source-bound memory can therefore supply useful predictions and an exact account of where its finite acquisition ends.

### XVII.7 Matched fitting selects methods the information ranking did not

Thirteen methods, three seeds and twenty epochs produce thirty-nine fits and 780 epochs on 20,000 training rows and 5000 validation rows. Checkpoints and choices freeze before 5000 new confirmation rows are opened. Every fit uses the corrected residual process, training-only component scales, an anchored momentum head, fixed hidden widths, zero-start added inputs and matched seed/epoch orders. Shuffled controls break only the added native-feature-to-row association. [13]

Validation selects an established momentum-Q3 direction method with transverse assembly/response, an established charge-mixed identification method and the corrected energy-Q3 decay base. Their new confirmation means are 0.010031877 radians, 0.982346871 AUROC and 0.692733319 macro-F1. The selected direction readout reduces its measured-jet starting error from 0.010917399 radians by 8.11% on this new sample.

The best learned-bundle alternatives are also fixed by validation. Their new direction error is 0.010072426 radians, 0.40% above the selected reference; their identification AUROC is 0.982390874, a gain of 0.000044002; their decay macro-F1 is 0.692669915, a change of $-0.000063404$. The declared learned-bundle improvement questions retain their separate classifications. For direction, *The test falsifies the concept.* For identification, *The test result suggests the concept is possible.* For mean decay improvement, *The test falsifies the concept.* For the pilot's acquisition-and-reuse question, *The test result suggests the concept is possible.*

These outcomes locate the distinction between information and training utility. A native report can carry conditional information about a declared task without being the validation-selected input for its trained reconstruction loss. The completed method memory consequently records actual pre-confirmation fitting results alongside correspondence memory. It returns the selected three training plans after restart with zero new fits and zero native ranking calls. Confirmation outcomes do not enter those method rankings.

Eight of nine matched association controls improve when actual native-to-row pairing is retained. Fifteen class-15 decay examples are missed by both selected decay readouts in every seed. The latter is a precise retained target, alongside unacquired cells and the mismatch between a sign-category information question and angular reconstruction utility. Every candidate, control, role tie and task classification stays in the result rather than being compressed into the pilot's aggregate outcome.

### XVII.8 The executable application survives its training session

The tau result includes 117 saved validation-prediction matrices, each replayed exactly before confirmation scoring. Its native calculation also retains 117 actual metric trajectories with separate worsening, improving, total-variation and net accounts. Epoch clocks remain separate from constituent admission, native Writes and physical time. Three additional native graphs calculate exact three-seed validation means and pairwise method differences; their rankings reproduce the frozen method choices. The complete application authenticates 246 managed receipts with no failed or incomplete calls. [13]

This provides two reusable objects with different sources: conditional probability cells learned from the acquired observation table, and method choices learned from the sealed matched fitting record. Recovering those memories repeats neither their correspondence ranking nor the sealed fits. Applying them to newly encoded observations retains the actual native source-map computation. The inference task receives their retained vocabulary, eligible-cell definitions, role ties and task-specific metric. The next learning question can use these objects while identifying exactly which new source distinctions

or feedback it adds.

The broader full-data training remains owner-paused, and TAU-SLC-PILOT3 stays installed. Current storage authority records that selected historical tau payloads and traces were removed; historical receipts retain the outcomes of those calculations without establishing present recoverability from deleted local paths. The completed semantic/method pilot and its exact current statement remain the authority for the application described here. [14]

Together these applications give the computational volume a source-level account of useful reuse. Circulation and signed probes preserve distinctions inside a tied matter construction. Native report equivalence improves observation selection while retaining the learned methods. Exact parity preserves an entire measured history with lower propagation cost. Tau keeps correspondence evidence separate from optimization feedback. Their shared contribution is executable specificity: an identified source enters an identified method and returns an actual result whose ties, remaining unknowns and subsequent reuse are still available.

SAM VOLUME III

# Certified Source Integration and Retained Response

Chapter XVIII

An exact finite table becomes more useful when another calculation can receive it with its source, clock and dependencies intact. The completed spin work supplies two detailed examples. One transfers certified integration to original signed-Write source paths and then executes complete carrier trajectories. The other compiles retained Gibbs-source matrices into response methods and improves the receiver by calculating a new source observable. Both examples develop computation as a chain of mathematical objects, with the receiving operation kept explicit.[50, 53, 56]

## XVIII.1 The source path precedes the numerical integral

For the original signed W1/W4/W7 source, execution first returns the actual wrapped phase state $(n_a, n_b, n_p)$, orientation and declared path role. The path is then

$$a(\theta) = (n_a - n_p)\pi/2 + k_a\sigma\theta, \tag{XVIII.1}$$

$$b(\theta) = (n_b - n_p)\pi/2 + k_b\sigma\theta, \tag{XVIII.2}$$

with $0 \le \theta \le \pi/2$. The role coefficients $(k_a, k_b)$ are $(1,0)$, $(0,1)$ and $(-1,-1)$. Its original density is $E = 6 - 2\cos a + 2\cos b$, with role derivative

$$g = 2\sin a, \quad -2\sin b, \quad -2\sin a + 2\sin b, \tag{XVIII.3}$$

respectively. The elapsed-source integral is

$$I = \int_0^{\pi/2} \frac{12\sqrt{E^2 + g^2}}{E^2 + 36}\,\mathrm{d}\theta. \tag{XVIII.4}$$

The initial state, signed role and orientation determine the integrand. An aggregate endpoint or total action is not substituted for that source history.[51]

The successful transfer evaluates this same integral in two coordinates. Angular quadrature is paired with the half-angle chart $t = \tan(\theta/2)$:

$$\cos\theta = \frac{1 - t^2}{1 + t^2}, \quad \sin\theta = \frac{2t}{1 + t^2}, \quad \mathrm{d}\theta = \frac{2\mathrm{d}t}{1 + t^2}. \tag{XVIII.5}$$

Both charts run at 384 bits. Initial quarter-turn sines and cosines are exact integers. Certified analytic branch handling remains active for the square roots. The two resulting enclosures must be finite, positive, overlap and have radius below $10^{-60}$. The full half-angle enclosure is passed onward; no midpoint or intersected point replaces it. A constant-density branch is also checked against $\pi/2$.[51, 54]

The clock receives this operand through native products and solves, preserving the original calibration and wait:

$$\Delta t = \text{ratio}\left[\frac{RI}{c(1 - A_{\text{los}})} + \text{waiting gap}\right]. \tag{XVIII.6}$$

Internal carrier time, elapsed source time, propagation length and the later clock readout remain separate quantities. This distinction connects directly to Volume I: a source-derived duration enters a specified receiver and does not acquire every other clock meaning merely by being positive.[51, 53]

## XVIII.2 Five original trajectories received the refined operands

The original frozen H02–H06 experiment completed with zero of five full trajectories and independently reproduced all five failures. The separate integration successor refines only two incoming source-duration enclosures in each job, with each new enclosure required to lie inside its original one. Original coefficients, waiting gaps, events, initial states and mathematical receiver remain fixed. The receiver stays at 192 bits with logarithm tolerance $10^{-35}$ and a 512-term bound.[50, 51]

All five successor cases complete, each with its full independent reproduction. They execute 4,781 original DAG operations and pass 481,979 declared predicate/native-status checks. The independent reconstruction contributes 80,940 checks across the five completed comparisons. It reconstructs the signed source paths with separate high-precision angular quadrature and evolves the unchanged full 273/546-dimensional carrier equations before comparing native outputs.[50, 52]

The comparison covers all six carrier segments and four paid/image events: complete states, source operators, clocks, memory, outgoing channels, driver and energy accounts. The result therefore concerns the full composed trajectory. The refinement is a numerical successor with its own receipts, alongside the preserved frozen failure record. For this integration transfer and its complete compositions, *The test result suggests strong contact with the concept.*[50]

The trajectory structure makes the result informative. H02 and H03 have the same source endpoint in different orders. Their active-duration sums are approximately 0.3156476595 and 0.3181630929, while their final receiver times at common internal endpoint 0.35 are approximately 1.5928865558 and 1.5928814590. The order of those two source durations and the order of the two final-clock readings concern different observables. Their final native intervals are disjoint in 81 of 273 current coordinates and 162 of 546 first-return coordinates.[50]

H04 returns to its original wrapped source address with cancelling signed phase action. Both active durations remain equal and positive. Its receiver retains the intervening evolution: initial and final intervals are disjoint in 85 of 273 coordinates, with paid work and balancing driver retained. H05 and H06 retain their declared waits separately. These are executable examples of the history distinction introduced in the earlier chapters: equal source endpoints or cancelling signed

action need not identify the receiver state.[50]

### XVIII.3 A dependency graph can cross session boundaries

The original large jobs are streamed by ordered step. Later cases open a fresh native session every hundred original steps, receiving prior operands through authenticated outputs and checkpoint dependencies. This changes memory ownership while retaining each operation and edge in the mathematical program. The completed custody comparison reconstructs all 4,781 executed inputs and checks 6,328 dependency edges in the segmented cases, including 1,858 crossing session boundaries.[50, 51]

The resulting execution is useful beyond this campaign. A retained node must carry the source binding, requested operation, implementation identity, exact operand/enclosure and upstream references. A report containing an old result reference is evidence about that result; adopting the complete calculation requires the referenced dependency closure. These are two distinct reuse actions, and the installed research library records which representation is available.[54]

The recorded segmented cases peak at roughly 1.9–2.1 GiB RSS, compared with about 5.9 GiB for the single-session H02 run. All five native cases together use 2,760.488 CPU seconds. Their summed wall times are 3,071.815 seconds; the source does not identify that sum with concurrent makespan. The computational contribution is the measured, dependency-preserving composition with bounded live memory.[50]

The subsequent installation makes the original-source integral available as a normal native operation, keeps the historical compiler selectable, and exposes thirteen research bundles with 27,816 operand paths. Qualification reproduces the ten retained angular/half-angle integral pairs and executes a fresh 26-operation source-clock DAG. It does not repeat the five full trajectories. The frozen zero-of-five and successor five-of-five records remain separate library inputs.[53, 54]

### XVIII.4 Source matrices become executable methods

The Gibbs-source response uses a different physical and computational source: the reversible single-site heatbath with $A = N(I-P)$, original Gibbs inner product and $N/\xi$ site-attempt clock. Its matrix recipes retain that source binding. They do not inherit the signed-Write clock merely because both programs use spin-derived mathematics.[55]

Let $b_4 = (d, e, u_d, u_e)$, with Gram $G_4$, Dirichlet matrix $D_4$ and hidden Gram $W_2$ for $(w_d, w_e)$. The source-projection method forms

$$H = G_4^{-1} D_4, \qquad T_4 = D_4 G_4^{-1} D_4 + E W_2 E^*. \tag{XVIII.7}$$

Here $E$ is the source's matching hidden-coordinate embedding. The hidden-residual method receives the source-shifted form $L$ and evaluates

$$Z = L^{-1} E W_2, \quad B = Z^* G_4 Z, \quad Q = B + W_2. \tag{XVIII.8}$$

The retained and complement terms are computed with their full mixed entries. Generalized hidden directions are taken relative to the actual $W_2$ metric, rather than an arbitrary Euclidean coordinate norm.[55]

For the completed five-coordinate basis $b_5 = (d, e, u_d, u_e, w_d)$, let $g = G_5[:, 0]$. The lower-response recipe solves

$$(\xi G_5 + D_5) a = g, \qquad L_5 = g^* a. \tag{XVIII.9}$$

Its normalized response is $\xi L_5 / G_5[0,0]$. The matrices are certified native values with rational interval endpoints. Their source names and ordering remain explicit in the binding, so coordinate relabelling cannot silently permute hidden positions.[55]

These five installed recipes cover source projection, hidden residual maps, residual payment, generalized projectors and fifth-coordinate lower response. Qualification executes a complete retained chain, compares its intervals with completed calculations and imports the exported chain into a second fresh consumer. It evaluates no new source integral. Computational composition and source enrichment are separate steps.[53, 55]

The installation is dated: its report records then-proposed $J_{21}$ and $F_{11}$ source contractions. Later completed source work supplies them through its own reports and operands. Treating an earlier method contract as a statement of the latest scientific frontier would conflate a reproducible historical interface with subsequent source calculation. The latest edge result below uses the completed $G_5, D_5, T_5$ and its actual $F_{11}$.[56, 57]

### XVIII.5 A new edge observable enriches the receiver

Write $k = D_5[:, 4]$ and retain the actual fifth-source residual

$$c = G_5^{-1} k, \quad r = A w_d - b_5 c, \tag{XVIII.10}$$

$$\beta_r = F_{11} - k^* c, \quad h = T_5[:, 4] - D_5 c. \tag{XVIII.11}$$

The new target is an upper enclosure of $d_r = \langle r, Ar \rangle$, to be received by a separate sixth-coordinate lower response. It is calculated from original-source edge information.[57]

For the seven core features $\phi = (d, e, Cd, Ce, C^2 d, C^2 e, C^3 d)$, the kernel evaluates the full symmetric edge Gram

$$\mathscr{E}_{ij}(y) = \frac{1}{2} \sum_x w_y(x) \sum_{\ell=1}^{20} \rho_\ell(x) \Delta_\ell \phi_i \Delta_\ell \phi_j. \tag{XVIII.12}$$

All $2^{20}$ labelled core states and all twenty core flips enter. Post-flip source fields retain the exact finite-$N$ shifts. Signed mixed products remain in the matrix; they are not replaced by isolated diagonal envelopes before contraction. One family kernel serves its three original preparations through their own coefficient maps.[57]

The source adapter reuses retained normalizers, Chebyshev moments, nodes and full-tail bounds, adding the numerical payments required by this new observable. Its analytic envelope is $40 A_i A_j$ times the source bound, where the seven feature amplitudes are $(2, 2, 4, 12, 160, 480, 6400)$. The largest multiplier is 1,638,400,000. A more demanding edge observable therefore receives its own interpolation, roundoff and tail accounting. The old source precision is not automatically assigned to it.[57]

Full physical-field and background contributions are then received. If $r = r_{\mathrm{core}} + v$, the completed comparison supplies

bounds $\|v\|_\pi^2 \le \delta_\pi^2$ and $\langle r_{\mathrm{core}}, C r_{\mathrm{core}}\rangle_\pi \le I_{\pi,+}$. With $C \le 20I$, the source certificate is

$$d_r \le \left(\sqrt{I_{\pi,+}} + \sqrt{20\delta_\pi^2}\right)^2 + \frac{N-20}{2}(\beta_{\mathrm{ph}}/N)^2 L_r^2. \tag{XVIII.13}$$

The square retains a paid core/complement mixed contribution; the last term retains the remaining background inventory. This is a full-source upper enclosure of the specified scalar.[57]

### XVIII.6 The latest result and its separate endpoint comparisons

H002689 completes all 54 preparations and 1,998 original histories, using 144 new positive nodes across eighteen source families. The receiving upper certificate changes from approximately 28.4187–28.4264 to 9.7473–9.7579. Its 65.6713–65.7013 percent reduction compares these cap upper endpoints. It does not identify the certificate cap with exact $d_r$.[56]

For $K = \xi G_5 + D_5$, the receiving extension is

$$L_6 = L_5 + \frac{(h^* K^{-1} g)^2}{\xi\beta_r + d_{r,+} - h^* K^{-1} h}. \tag{XVIII.14}$$

Every one of the 1,998 Schur denominators is strictly positive and every native gain is positive. The 108 independently arranged full six-coordinate controls overlap the Schur calculation. All original histories retain their $N/\xi$ clock.[56, 57]

There are two distinct endpoint comparisons. All 1,998 new lower endpoints improve the frozen C1 plus explicitly included C2 native WORD frontier. In 1,989 cases, the new lower endpoint also exceeds the upper endpoint of the earlier local-cap variational-lower evaluation. That earlier interval encloses a calculated lower estimate; it is not an upper bound on the full physical response. The comparison therefore demonstrates a separated gain in the lower estimate.[56]

The source classification is *The test result suggests strong contact with the concept.* The full mixed two-residual Dirichlet block and the next squared-action norm $\|Ar\|^2$ retain their separate source questions. The completed scalar does not fill either operand by inference. The full mixed block can reuse retained core edge data, but requires joint physical/background and signed cross-term payments establishing its matrix ordering.[56, 58]

### XVIII.7 What becomes reusable across the volumes

The two examples expose a common computational interface. A source supplies actual ordered operands. A numerical method returns a certified value or exact finite object. Native composition carries its dependencies into the receiver. The final readout retains the clock and observable to which its bounds apply. Reuse can occur at each level: local source factors, saved joint tables, previously calculated matrices, identical readout requests, or whole authenticated method chains.

Volume I receives calibrated source durations, field histories and ordered records. Volume II receives exact phase-source responses, hidden-state retention and complete physical receiving bounds. Volume III supplies the corresponding source objects, algebra, execution graph and retained computations. The integration successor demonstrates a successful method transfer with the scientific source fixed; the edge result demonstrates an improved receiver after a specifically identified new source calculation. Their different forms of progress now have concrete places in one computational account.

SAM VOLUME III

# Signed Arithmetic Energy and Conditional Continuation

Chapter XIX

## XIX.1 Retaining the original full energy makes history computable

The RH application computes exact signed Möbius source energies while preserving their original mean, centered prefixes and metric. That retained state supports spectral readout, signed scale transport, explanation of excursions and finite future certificates. The attached research paper reports 12,836 complete forcing prefixes, exact conditional four-admission bounds at demanding states and a proved conditional implication from a continuing source estimate to the Riemann Hypothesis. These are distinct results with distinct scopes. [RH, pp4–10]

For a positive dyadic scale $s$ and a stop $0 \le t \le s$, define $a_i = \mu(s+i)$ for $i < t$ and $a_i = 0$ afterward. The Möbius function is zero on integers with a repeated prime factor and is $(-1)^k$ on a product of $k$ distinct primes. Put $M_a = \sum_i a_i$ and define the centered prefix

$$q_a(r) = \sum_{i<r} a_i - \frac{rM_a}{s}.$$

The full source energy is

$$Q_s(a) = \frac{M_a^2}{s} + \sum_{r=1}^{s-1} \frac{q_a(r)^2}{r(s-r)}.$$

Its positive-definite kernel $K_s$ has inverse $\Pi + L_s$, where $\Pi$ projects onto constant vectors and $L_s$ is the path Laplacian with edge weights $r(s-r)$. The constant eigenvalue of $K_s$ is one; its other eigenvalues are $1/[k(k+1)]$, $1 \le k < s$. For normalized eigenmodes $\phi_k$, define $z_k = \sum_i a_i \phi_k(i)$. The full energy then has the exact spectral representation

$$Q_s(a) = z_0^2 + \sum_{k=1}^{s-1} \frac{z_k^2}{k(k+1)}.$$

Expanding in these polynomial modes preserves the original energy as a weighted sum of squared source coefficients. [RH, pp4–5]

A truncated mode sum $F_m$ through $m = \min(s-1, \lceil \sqrt{s} \rceil)$ has uniform omitted energy at most one. Put $N_a = \sum_i a_i^2 \le s$. When $m < s-1$, orthonormality gives the explicit enclosure

$$0 \le Q_s(a) - F_m \le \frac{N_a - \sum_{k=0}^{m} z_k^2}{(m+1)(m+2)} \le 1.$$

When every mode is retained the tail is zero. This is a proved geometric enclosure, rather than an empirical statement inferred from the observed prefixes.

The first nonconstant mode explicitly consumes history. With integer-normalized $p_1(i) = s-1-2i$ and signed prefixes $P_r = \sum_{i<r} a_i$, summation by parts gives

$$A_1 = (s-2t+1)M_a + 2\sum_{r=1}^{t-1} P_r.$$

At $(s,t) = (8192, 6994)$, $M_a = -10$ and the signed prefix area is 149,308, giving $A_1 = -240{,}666$. Its mode contributes approximately 0.158034139, while the mean contributes 0.01220703125. The endpoint mean alone cannot reproduce that prefix-area contribution. [RH, p5]

## XIX.2 Signed grouping retains the cancellations in the source

The divisor calculation groups actual signed factor-pair terms into product bands while retaining all interband crosses. At (8192,6994), the sum of band self energies is approximately 28,252.560385784 and the signed cross sum is 28,252.184906209; their joint retained energy is 0.375479575. At (8192,1670), the corresponding numbers are 3,340.554542529, 3,339.542843749 and 1.011698779. [RH, pp5–6]

The large cancellation is part of the exact quantity. Summing only positive band self energies would answer a different question. The returned records retain the original source, stop and metric so that regrouping preserves the same joint energy. A dual calculation also returns the exact maximizing finite test vector and a sum-of-squares attainment certificate, making the demanding direction a readable operand rather than the outcome of a random direction search. [RH, p5]

## XIX.3 Signed scale transport has an explicit arithmetic forcing

At the next dyadic scale, let $\mathbf{1}$ denote an indicator and define the stopped fine source and parity forcing by

$$z_j = \mu(2s+j)\mathbf{1}_{j<2t},$$

$$b_i = \left[\mathbf{1}_{s+i \text{ even}}\mu(s+i) + \mu\big(2(s+i)+1\big)\right]\mathbf{1}_{i<t}.$$

Pair the fine source into $v_i = z_{2i} + z_{2i+1}$ and let $d_i = z_{2i} - z_{2i+1}$. The Möbius parity relation gives $v = -a + b$. Both even and odd fine cuts are recovered from the coarse centered cuts and the within-pair differences:

$$q_z(2r) = q_v(r) = -q_a(r) + q_b(r),$$

$$q_z(2i+1) = \frac{q_v(i) + q_v(i+1) + d_i}{2}.$$

The excess

$$\mathscr{E}(s,t) = \frac{1}{2}Q_s(b) - \langle a,b\rangle_s - \frac{1}{2}Q_s(a) = \frac{1}{2}Q_s(v) - Q_s(a)$$

is a signed quantity. The exact full-energy transport is

$$Q(2s,2t) = Q(s,t) + \mathscr{E}(s,t) + \mathscr{R}(s,t), \qquad \mathscr{R}(s,t) \le 3.$$

For an odd fine stop, the additional admission term has upper bound two. The source supplies inherited energy, forcing self energy, signed cross interaction and geometric remainder as parts of one reconstruction. The bound is one-sided; it does not replace negative terms by their magnitudes. [RH, p6]

At the next admission, write $\alpha$ for the parent coefficient, $\beta$ for its forcing coefficient, $p = (K_s a)_j$, $q = (K_s b)_j$ and $k = (K_s)_{jj}$. Exact expansion yields

$$\Delta\mathscr{E} = (\beta-\alpha)q - (\beta+\alpha)p + \left(\frac{\beta^2}{2} - \alpha\beta - \frac{\alpha^2}{2}\right)k.$$

The weighted pairings $p$ and $q$ determine how the new signed coefficient interacts with the state already present. Signs alone are insufficient. If $\beta = \alpha$ throughout a consecutive interval, $v$ is unchanged and the excess telescopes as

$$\mathscr{E}(t) - \mathscr{E}(t_0) = Q_s(a_{t_0}) - Q_s(a_t) \le Q_s(a_{t_0}).$$

This is a source-conditioned run identity valid for any length of such a run. [RH, pp6–7]

### XIX.4 Complete finite scans locate the excursions

Every prefix in those four listed ranges satisfies the tested ceiling $\mathscr{E} \le 1$. The proposed half ceiling has thirteen exact counterexamples. The separate 15,364-prefix signed-divisor gain campaign measures another quantity and is not added to this census. [RH, pp7–8]

At scale 4096, forcing contributes approximately +0.245027 from the stress interval's start to the peak, while weakening inherited energy contributes +0.005809. The excess recrosses one half at stop 1077 and falls below the interval's starting excess at 1130. At the scale-8192 peak admission, $\alpha = \beta = 1$ leaves $v$ unchanged; the excess rises because the inherited parent energy decreases. The broader buildup is still forcing-led. Retaining the increments distinguishes the cause of a final peak admission from the cause of its preceding accumulated rise. [RH, pp7–8]

### XIX.5 The current weighted state certifies a finite continuation

The prospective certificate starts with the actual current source and permits every future sign compatible with known squarefree support. A squareful coefficient is zero; an unrevealed squarefree coefficient may be either +1 or 1. The method enumerates all such branches through at most four admissions, updates the exact weighted state on each branch, and returns the largest attained excess and terminal drift interval. It does not receive future nonzero Möbius signs. Relaxing their dependencies includes every actual continuation in the admitted family. [RH, pp7–9; H001428]

All three states certify remaining below one over that horizon. The recovery state also certifies remaining below one half. The recrossing state has a current value below one half but does not certify that tighter horizon bound. The successful operation is a rigorous finite family bound from the weighted state; it is not a forecast of which future branch occurs. The companion frozen sign-pattern predictor failed its proposed overshoot bounds on separate evaluation windows. Those failures delimit the predictor, while the branch certificates retain their different exact basis. [RH, pp7–9]

### XIX.6 A continuing estimate has a proved arithmetic consequence

The analytic target is a uniform upper bound on the excess. If, for every $\eta > 0$, a constant $C_\eta$ independent of dyadic scale and stop satisfies

$$\mathscr{E}(s,t) \le C_\eta s^\eta,$$

then the original full energy is uniformly subpower and the Riemann Hypothesis follows. This is a proved conditional implication. The finite scans and four-step certificates do not supply its uniform hypothesis. [RH, pp9–12]

To see the chain, let $H_j = \max_t Q(2^j,t)$. Signed scale transport and the odd-stop allowance give

$$H_{j+1} \le H_j + C_\eta 2^{j\eta} + 5.$$

Summing that recurrence makes $H_K$ subpower in $2^K$. The mean term of the original energy bounds every stopped dyadic Möbius sum by $\sqrt{sQ}$. Dyadic decomposition then gives the Mertens bound of order $x^{1/2+\varepsilon}$ for every $\varepsilon > 0$. Partial summation extends the reciprocal zeta Dirichlet series to the half-plane $\Re w > 1/2$, and the zeta reflection relation completes the conditional RH conclusion. The source estimate, the transport implication and the finite evidence remain distinguishable at every step.

The paper also proves a flexible sufficient transfer threshold: a smaller-source energy exponent may approach two from below at a specified logarithmically decreasing saving and still imply the required subpower estimate. At exactly exponent two, that recurrence permits power-law energy. The strict saving belongs to the stated analytic hypothesis. The computational result is that the source-preserving system has made that hypothesis an explicit question about retained signed operands, rather than changing the energy whenever a stricter empirical ceiling fails. [RH, pp9–10,12]

| Scale | Complete prefixes | Maximum excess | Maximizing stop | Exceedances of one half |
|---|---|---|---|---|
| 32 | 33 | 0.063413987429 | 1 | 0 |
| 512 | 513 | 0.210217371429 | 481 | 0 |
| 4096 | 4097 | 0.514400059116 | 1071 | 5 |
| 8192 | 8193 | 0.520439137454 | 7798 | 8 |
| Total | 12,836 | | | 13 |

| Starting state at scale 4096 | Current excess | Upper over next four admissions | Terminal drift interval | Branches |
|---|---|---|---|---|
| Peak $t$=1071 | 0.514400059 | 0.589424759 | [0.069410267,+0.075024700] | 32 |
| Recrossing $t$=1077 | 0.493799231 | 0.600499614 | [0.096696737,+0.106700383] | 64 |
| Recovery $t$=1130 | 0.261231122 | 0.343086070 | [0.072035796,+0.081854948] | 64 |

SAM VOLUME III

# Reciprocal History and the Completed Arithmetic Interface

Chapter XX

Reciprocal History extends the state/history distinction into a mathematical interface with completed arithmetic. This chapter records the finite constructions, the failed direct identification and the corrected source route. It does not supply a proof of the Riemann hypothesis. The universal joins named below remain open in the retained source.

The account below follows the chapter source as retained at the fixed 8 September 2026 SAMA revision. Engine names, installed interfaces and campaign status are historical source statements. They do not describe a freshly inspected October deployment. Later source amendments are distinguished where available; no later engine or campaign is inferred.

**Source-status qualification.** The operational and revision statements in this chapter describe the retained source documents. They are not an instruction to execute, deploy, resume, train, promote or change a service, and they have not been independently verified as present live state for this edition.

## XX.1 Overview

Reciprocal History begins with paths, not scalar state counts. On the unchanged N100 graph, every undirected edge has two orientations. The resulting 520 directed-edge states support a nonbacktracking transfer operator that records ordered continuation, reversal and primitive-cycle data. Pairing 10,700 oriented primitives with their reverses produces 5,350 reciprocal classes with exact carrier-weighted norms and power laws. This is not the N100 DOS renamed: the DOS counts assignments by energy, while RH retains ordered history on the same source grammar.

The stage chain contains a decisive preserved failure. P2 direct asked whether the history object itself supplied the ordinary-prime Euler coefficient. Its first coefficient was 100 rather than 2, so that direct identification is falsified at its exact scope. P2A and P2B are successors, not retrospective relabels. P2A reconstructs ordinary primes through the radix-12 first-singularity tower $n = \mathrm{ord}_q(12)$. P2B supplies coefficient one from the N100 role/depth grading, while N95 and N105 controls return index zero. No nontrivial-zero ordinate chooses the tower.

The accumulation field supplies the real coordinate that turns multiplication into translation:

$$q = \frac{A_{total} - A_0}{1 - A_0} = \frac{r_c}{r}, \qquad u = -\log q = \log \frac{r}{r_c}.$$

Scaling $r \mapsto nr$ becomes $u \mapsto u + \log n$, and the half-density becomes $n^{-1/2}$. On this coordinate the operator $L = -d^2/du^2 + 1/4$ has endpoint null modes $e^{\pm u/2}$, giving the domain on which the completed prime-power translations act.

P3 retains the completed arithmetic

$$\xi(s) = \tfrac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s) = \xi(1-s)$$

and the paired completed Weil form. Its continuous energy, finite centered von-Mangoldt displacement sum and floor term form one object; splitting them into individually divergent pieces changes the calculation.

Four authority surfaces remain distinct: the public RH claim, the internal universal-nonnegativity assignment, optional finite F81 spectral research, and operational SLC/Starbreaker integration. The earlier retained operational adapter is `SLCQ2-RZ-RH-MOMENT-ADAPTER-V1`. The later source amendment below records Q3RHV2 as the RH-local successor and preserves Q2 as a noncurrent exact oracle. Neither adapter merges those questions.

## XX.2 Opening question and conceptual picture

The governing question is:

> How does a finite source grammar carry reciprocal path information into an ordinary-prime tower, a real logarithmic translation coordinate and the completed arithmetic form while preserving the exact failure and authority boundaries encountered along the way?

The chain is

- N100 source graph
- 520 directed nonbacktracking states
- primitive histories and reciprocal classes
- P2 direct obstruction retained
- P2A radix-12 prime tower + P2B coefficient-one grading
- accumulation-derived u = log(r/r_c)
- pi/Gamma completion and xi reciprocity
- completed Weil quadratic form
- separately typed public, internal, spectral and operational surfaces

No successor is allowed to erase the failed direct identification. Each new arrow supplies a different missing structure.

## XX.3 The reciprocal-history object

### XX.3.1 Directed-edge state space

N100 has 260 undirected source edges. Orient each edge both ways. The directed state census is therefore

$$2 \cdot 260 = 520.$$

Write a directed edge as $e = (u, v)$. The nonbacktracking transfer $B$ has entries

$$B_{(u,v),(v,w)} = \begin{cases} 1, & w \neq u, \\ 0, & w = u. \end{cases}$$

Thus a step may continue through $v$ but may not immediately undo the edge just traversed. Powers $B^n$ count ordered nonbacktracking histories of length $n$, and traces $\mathrm{tr}(B^n)$ retain closed directed history. Primitive-cycle and determinant receipts separate repeated powers from primitive orbit data.

### XX.3.2 Reciprocal pairing

Reversal sends

$$(v_0, v_1, \ldots, v_n) \longmapsto (v_n, v_{n-1}, \ldots, v_0).$$

P1A3/P1B retain 10,700 oriented primitives. Pairing each history with its reverse gives

$$\frac{10{,}700}{2} = 5{,}350$$

reciprocal classes. The paired object retains the common contribution and the directional difference, with integer carrier-weighted norms and exact power laws.

### XX.3.3 Why RH is not DOS

The N100 DOS is

$$g(E) = \#\{\sigma : H(\sigma) = E\}.$$

The reciprocal-history transfer is indexed by directed edges and ordered continuations. Two histories may end at the same state and contribute to the same DOS sector while remaining distinct under reversal, primitive length or ordered path. Conversely, a DOS coefficient contains no data sufficient to reconstruct an oriented primitive. The two objects share source provenance; they do not share type.

## XX.4 Executed stage chain and deviation

### XX.4.1 P0 through P1B

The foundation is cumulative:

- P0 builds the exact 520-state nonbacktracking transfer and retains traces, reversal, primitives and determinant receipts.
- P1A compiles tensor and Euler types.
- P1A2 keeps W9 as five derived non-counted receipts.
- P1A3/P1B pair the 10,700 oriented primitives into 5,350 reciprocal classes with native norms.

The five W9 receipts remain outputs of completion. They do not increase the native source-row census.

### XX.4.2 Preserved P2 direct failure

P2 direct tested the simplest possible bridge: identify the history-prime Euler coefficient directly with the ordinary-prime coefficient. The first history coefficient is 100, while the target ordinary-prime coefficient is 2. Because

$$100 \neq 2,$$

the direct identity is falsified at that scope. The mismatch is structural, not a tolerance or serialization fault.

For this exact direct-identification test, the registered classification is: **The test falsifies the concept.**

The correct record is

- P2 direct question
- first exact coefficient 100
- required direct coefficient 2
- direct identification falsified
- obstruction retained

Neither P2A nor P2B changes these two numbers.

### XX.4.3 P2A successor: radix-12 first singularity

For an ordinary prime $q \nmid 12$, 12 is a unit modulo $q$. Its powers form a finite subgroup of $(\mathbb{Z}/q\mathbb{Z})^\times$, so there is a least positive integer

$$n = \mathrm{ord}_q(12)$$

with

$$12^n \equiv 1 \pmod{q}.$$

Minimality places $q$ in the first cyclotomic singularity:

$$q \mid \Phi_n(12)$$

subject to the exact prime-power/order bookkeeping of the tower. P2A uses this first block order to reconstruct ordinary-prime placement. It adds a radix-12 arithmetic route that P2 direct lacked.

### XX.4.4 P2B successor: source grading

P2B uses the N100 role/depth grading

$$C_0|C_1|C_2 = 20|24|5,$$

$$H_0|H_1|H_2 = 8|7|0$$

to supply the compiled family index 1. The source-native N95 and N105 controls return index 0. This distinguishes the selected N100 grading from nearby row- count perturbations without consulting any nontrivial-zero ordinate.

The successor logic is

- P2A: ordinary prime q -> first radix-12 order ord_q(12)
- P2B: N100 role/depth grading -> coefficient one
- controls: N95 and N105 -> zero

## XX.5 Accumulation to logarithmic translation

### XX.5.1 Field coordinate

The accumulation relation is

$$q = \frac{A_{total} - A_0}{1 - A_0} = \frac{r_c}{r}.$$

Taking the negative logarithm gives

$$u = -\log q = -\log \frac{r_c}{r} = \log \frac{r}{r_c}.$$

If the radial ratio is multiplied by $n > 0$,

$$r \mapsto nr \quad \Longrightarrow \quad u \mapsto \log \frac{nr}{r_c} = u + \log n.$$

Multiplicative arithmetic has become additive translation.

### XX.5.2 Half-density

The symmetric translation difference acts at $\pm t/2$. For $t = \log n$, the decaying half-density factor is

$$e^{-t/2} = e^{-\log(n)/2} = n^{-1/2}.$$

This derives the prime-power weight from the logarithmic displacement rather than inserting it as a separate numerical convention.

### XX.5.3 Endpoint operator

Define

$$L = -\frac{d^2}{du^2} + \frac{1}{4}.$$

For $f_\pm(u) = e^{\pm u/2}$,

$$f_\pm'' = \frac{1}{4} f_\pm,$$

so

$$L f_\pm = -\frac{1}{4} f_\pm + \frac{1}{4} f_\pm = 0.$$

The two endpoint null modes generate the two-moment boundary. The declared test domain is

$$\mathscr{D}_0 = L(C_c^\infty(\mathbb{R})),$$

which keeps compact support and endpoint annihilation visible.

Accumulation supplies this coordinate route. It is not identified with the N100 DOS, a local P5A symbol or a numerical energy coefficient.

## XX.6 Completed arithmetic

### XX.6.1 Lambda and xi

Start from

$$\Lambda(s) = \pi^{-s/2} \Gamma(s/2) \zeta(s).$$

The completed entire function is

$$\xi(s) = \frac{1}{2} s(s-1) \Lambda(s).$$

The $\pi$/Gamma factor supplies the archimedean completion, while $s(s-1)$ handles the pole bookkeeping. The resulting reciprocity is

$$\xi(s) = \xi(1-s).$$

Prime powers enter through the von Mangoldt weight $\Lambda(n)$, not only through primes at exponent one.

### XX.6.2 Translation energy

Let

$$(T_a h)(u) = h(u + a).$$

Define the reciprocal difference

$$B_t h = T_{t/2} h - T_{-t/2} h$$

and its nonnegative route energy

$$E_t(h) = \|B_t h\|^2.$$

The construction is direction-sensitive before the norm and reciprocal after pairing.

### XX.6.3 Completed Weil form

The compiled form is

$$\begin{aligned} Q_\xi(h) = &\int_0^\infty \frac{e^{-t/2}}{1 - e^{-2t}} E_t(h)\, dt \\ &+ \sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}} \left[ E_{\log n}(h) - 2\|h\|^2 \right] \\ &- \kappa_0 \|h\|^2. \end{aligned}$$

where

$$\kappa_0 = \gamma - \log(3A_0) + \frac{\pi}{2} + \log 2.$$

For compactly supported $h$, $T_{t/2} h$ and $T_{-t/2} h$ have disjoint support once $t$ exceeds the support diameter. Then

$$E_t(h) = 2\|h\|^2,$$

so the centered summand vanishes. The prime-power sum is therefore finite on each compactly supported input.

The continuous route-energy integral, centered arithmetic sum and floor subtraction are one completed object. Separating them into divergent pieces would change the domain and value of the form.

## XX.7 Spectral and operational successors

### XX.7.1 P4 finite F81 route

P4 constructs a finite exact F81 positive/self-adjoint 211-mode quotient and a paired 422-dimensional determinant. This is an optional finite spectral route. It is not the completed P3 object and does not replace the source- native completed form.

### XX.7.2 P5A failed scale-extension route

P5A tested a uniform determinant scale extension and was falsified at its declared scope. The failure rejects that optional bridge. It does not rewrite P3 arithmetic, the P2 direct obstruction, or the separately retained public and internal surfaces.

### XX.7.3 Source-recorded RH native engine and retained moment oracle

In this retained source, the RH-local SLC handle is frozen `Q3RHV2`. Its authenticated facade executes Q3/V6 Complete Write, constructs caller-supplied RH rosters and binds exact threshold events and certificates into immutable CE/CAS records. The Q2 moment adapter remains a frozen noncurrent exact oracle; Q3 ranking is optional prioritization only and never an RH premise.

The base roster closes 1,726/1,726 pairs. A fresh $N = 192$ roster closes 573/573. A fixed q7 depth has one failure; the source-visible rule

$$q = \lceil \log_2 N \rceil = 8$$

repairs that case. The adapter is operational evidence. It neither changes global `SLCQ2-RZ` nor silently closes universal RH joins.

## XX.8 Native mode-6 relational contact continuation

### XX.8.1 Preserved failures redirect the source comparison

Exact signed Abel/PNT2 reconstruction falsifies the exact-data absolute PNT2 majorant route. The isolated pole–Gamma–Lebesgue deterministic frame also has negative Schur complements and negative lower/full LDL pivots at `D=10,20,40,80`, falsifying that block-alone positive-frame concept. These are useful exact failures: they preserve signed arithmetic as load-bearing and redirect the program from an absolute or isolated positive frame to a relational source comparison.

### XX.8.2 Relational Schur law and intrinsic innovation

For positive reels, the exact native law is

$$b\left(\sum_i G_i\right) \geq \sum_i b(G_i),$$

with equality controlled by common minimizers and strictness by kernel intersection. A literal `Z4^2` Complete Write witness has separate floors zero and combined floor one. Write retains unitary custody; amplification occurs in the composed positive reel afterward.

Endpoint conditioning annihilates the completed pole. At normalized smooth contact, the debit-parallel connection

$$\Gamma_A = \frac{1}{2} A_\Gamma^{-1} A'_\Gamma$$

removes arbitrary transport and identifies the intrinsic current

$$I_A = q^*(C' - \Gamma_A^* C - C\Gamma_A)q = H\mu' = J.$$

Each admitted prime-power event separates into an exact linear debit and a triple-zero relational remainder, reducing the threshold cell to

$$z^*(K'_{bg} + R'_k)z > 4w.$$

The relational law and its validation are classified: **The test result suggests strong contact with the concept.**

### XX.8.3 Particle spin, orbit wave and returning history

The N100 four-role reciprocal tensor aggregates to `2^99 I_2` across 177 occupied bins and is strict on 174. That tensor result suggests strong contact. Its unaugmented two-mode H000724 innovation has a negative least current at all four sampled supports, so positivity at all four for that unaugmented route is separately falsified.

The source-native orbit-wave successor carries the same exact C4 quarter turn:

$$P_{C4}T = TJ, \qquad T^T L_{C4} T = I_2, \qquad J_\Theta^* G_\Theta J_\Theta = L_{C4}.$$

Here `P_C4^4=I` while cumulative wave action has positive derivative. The state returns, but history does not; information does not sit still. Primary, precision, independent and release checks pass `84/84`, `12/12`, `141/141` and `151/151`, and the result suggests strong contact.

The active continuation is the exact weight and placement of accumulated dimensionless Theta/wave action in the completed current. Universal mode 6, product-cone, signed-tail, Schur/atlas and whole-domain joins remain open.

## XX.9 Four authority surfaces

- **Public.** Object: documented SAM RH claim and public repository; Current boundary: external validation remains pending.
- **Internal.** Object: $Q_\xi(h) \geq 0$ for every $h \in \mathscr{D}_0$; Current boundary: universal derivation remains active; not an extra gate on the public surface.
- **Spectral.** Object: finite F81 determinant and related exact identifications; Current boundary: optional research; P5A uniform bridge falsified.

- **Operational.** Object: Q3RHV2 plus source-native Starbreaker/Theta continuations; Current boundary: exact event/custody engine installed; universal joins remain open.

The owner-designated internal status is

- `SAM_RH_CLOSURE_PENDING_EXTERNAL_VALIDATION`

The public repository, internal universal-sign program, finite spectral research and application engineering may inform one another, but none acquires authority merely by sharing vocabulary.

### XX.10 Wrong controls and preserved boundaries

The route retains several decisive wrong controls:

1. Rename N100 DOS as RH: rejected because ordered directed history is absent.
2. Identify the P2 history coefficient directly with the ordinary-prime Euler coefficient: falsified by 100 versus 2.
3. Use N95 or N105 grading as the selected coefficient-one family: both return zero.
4. Let a nontrivial-zero ordinate select the prime tower: excluded; the tower is derived from $\mathrm{ord}_q(12)$.
5. Split continuous, arithmetic and floor terms before completion: rejected because the separate pieces are not the compiled object.
6. Promote the P5A uniform determinant bridge into a required closure: its fixed route is falsified and remains optional.
7. Use a fixed RH adapter depth when the source-visible rule changes with $N$: the q7 miss is preserved and q8 is the declared correction at $N = 192$.

These controls preserve rather than weaken the positive completed arithmetic already in custody.

### XX.11 Established result and forward handoff

The established RH route begins with an exact 520-state nonbacktracking history operator, pairs 10,700 oriented primitives into 5,350 reciprocal classes, retains the P2 direct failure, supplies the P2A/P2B successor prime route, derives the logarithmic translation coordinate from accumulation and builds the completed $\xi$/Weil form.

The current operational adapter is separately typed, and the four authority surfaces remain explicit. `SAMA-D000056` uses prime coordinates for an exact Mersenne factor-search application; that application does not enter the RH completed-form chain retroactively.

### XX.12 Sources and evidential scope

This chapter is assembled from the following fixed source manuscripts. Source-reported campaign outcomes retain their original scope; no new campaign was executed for this edition.

- Reciprocal history and the rh route.

### XX.13 Results received, 22–24 October 2026: the original source's next transitions

The original fixed-sharp odd Möbius source of this chapter ($\varepsilon = 1/2$, $s = 3/2$, $k = \pi/\log 2$; retained cut $Y = 4096/3$, 1365 coordinates, 217 odd-prime cuts) was moved to its next changed source accounts on the installed exact engine, in the verification form: everything given, two host implementations written separately from the rules, predictions written and hashed first (ironclad H002768–H002775).[105] The complete price $P(Y) = u^{\mathsf{T}}(H_{\mathrm{adm}} - C_{\mathrm{Self}})u - C_{\mathrm{scalar}}$ built twice by separate code agrees to $8 \times 10^{-80}$ and gives the retained signed target $-2.223195245316$ to every digit; natively at 512 bits (1673 calls, 881 checks) the whole price at both cuts by bilinear forms of exact integer masks closes the six-account ledger as a zero enclosure. At the first transition, $(4096/3, 1365) \to (1367, 1367)$, the direct difference $\Delta P = -7.3001 \times 10^{-4}$ equals the ledger's sum to $5 \times 10^{-80}$; the storage of H002629 releases $+4.2713 \times 10^{-7}$ and pays the crossing rise of $+2.4238 \times 10^{-7}$ with the frozen-chart identity $\Delta P_\chi - \Delta F_\chi = 2a\Delta\ell$ closing; the exterior term, which has no storage recipient, carries the price. The adaptive receiver was reproduced the same way ($P(Y_0) = -0.4348853154523694$, the program's other published target) and at this transition rises by $+2.715 \times 10^{-6}$, structurally (frozen-receiver $+2.494 \times 10^{-6}$): the sign of the exterior change belongs to the receiver's weights, the admitted energy rising at the five smallest cuts and falling at 154 of the other 212.

Seven further accounts, $1371 \to 1389$, were computed unattended for both receivers by a two-cut form (fourteen PASSes, 44,066 native calls, fourteen chain identities zero enclosures, so the ledger is seamless from $4096/3$ to 1389; 266 of 266 host rows inside the native enclosures). The sharp price falls and the adaptive rises at every account; six of seven sharp changes lie within 0.6% of $-8.66 \times 10^{-4}$ regardless of the admitted count. The sharp storage identity $1 + \bar{\ell}/m$ is exact at all seven steps, with its $\Delta\ell$ sign flip at $1383 \to 1385$ as declared; the adaptive storage obeys a different law. Three addenda then located the sign: the sharp change is $dP/dY \times \Delta Y$ cut by cut with new coordinates entering at zero kernel, so $P(Y)$ is continuous and the ledger samples it; the window sum $\sum_{n>Y/2} R_n$ by sector and owner is NEGATIVE_PP $-57{,}331$ plus BALANCED $+26{,}222$ against an unrestricted sum of $+785$, so "the fall is the admission rule's choice"; and under a weight-free window table the adaptive and sharp receivers read the same calculus with opposite signs, the adaptive weights ($\le 1.005$) seeing $+190$ and the sharp weights (to 4.58) seeing $-31{,}179$, with $S_{\mathrm{sharp}} = \sum_p w_p S_1(p) - \mathrm{Self}$ exactly. The next account was predicted before computing: $\Delta P_{\mathrm{sharp}}(1389 \to 1391) = -8.6305 \times 10^{-4}$, $\Delta P_{\mathrm{adaptive}} = +6.449 \times 10^{-6}$. Of the lead's pre-registered predictions across these entries, roughly a third failed in a clause each and are recorded with the results. No complete-price constant, uniform arithmetic exponent or all-scale bound is assigned by any of this; the recorded next step is the predicted account, and the program was parked by the owner on 25 October 2026 until Volumes I–III were brought to the present edition.

**Received 25 October 2026: the window calculus verified (H002784).** The addenda of 24 October had read the sweep's constant fall per account as calculus: the change at an account is $\mathrm{d}P/\mathrm{d}Y \times \Delta Y$ with $\mathrm{d}P/\mathrm{d}Y = Y^{-s-1}S(Y) - \mathrm{d}C/\mathrm{d}Y$ and $S(Y) = \sum_{Y/2<n\le Y} R_n$ the window sum of the pair census by sector and owner. A Duck window in the verification form (run `rh_window_1`, four peers, lead outside) reproduced that census with six separately written codes: every sector sum, $S_{\mathrm{sharp}}(1389) = -31\,178.547\ldots$ to thirty digits, the standing changes $1389 \to 1391$ ($-8.63047 \times 10^{-4}$ sharp, $+6.449 \times 10^{-6}$ adaptive) to $10^{-32}$ of the host, each witnessed natively with a failing control. Three results stand beside the ledger. Every census sign decision is exact integer arithmetic: a pair $(m,n)$ is negative iff $2^{4j+1}m^2 < n^2 < 2^{4j+3}m^2$, and no decision can be unresolved since $(n/m)^2 = 2^{\mathrm{odd}}$ is impossible for odd $m,n$ (the lead's own text had called the pair $(67,379)$ negative; $379^2 < 32 \cdot 67^2$ by seven, so it is positive, and all four peers caught it). The owner profile above the pure prime-pair threshold is a closed form, $O_p(Y) = 2w_p \sum_q \min(\cos \pi \log_2(q/p), 0) \cdot \#\{h\}$, a function of $p/Y$ with peaks at $p/Y = 0.380, 0.0955, 0.0240$ at every $Y$ tried from 1389 to 60 000; the "fixed owner profile" of the sweep was a 1.6 % change in $Y$. And the admitted sharp window sum, negative at every account of the sweep, changes sign between $Y = 57\,000$ and 60 000 in three codes — the sharp weights saturate near 4.66 and the prime-pair half's weighted ratio to the small-owner mirror falls about 0.30 per unit $\ln Y$ through one. That last statement is about the window sum alone: the curvature term $\mathrm{d}C/\mathrm{d}Y$ was not computed there, the arithmetic was floating point at a crossing of two halves of order $5 \times 10^7$, and the rules were applied literally far outside the support they were written for. Whether they are meant to reach that far is the owner's decision, not the ledger's; the overnight accounts $1391 \to 1429$ test the ledger, not this sign. Nothing here is a claim.

SAM VOLUME III

# Signed Arithmetic Sources and Retained Transition Storage

Chapter XXI

## XXI.1 From the arithmetic interface to an executable source

The reciprocal-history construction in Chapter XX retains the path grammar, prime tower and completed arithmetic form. The later research makes its signed-source calculations more explicit and less expensive. It supplies exact interval-prefix evaluation, common-grid transport, an ordered Gram forest and a source-specific transition storage. Each construction retains the original coefficients, their quadratures and the terms that have already been paid.

This chapter is a Codex editorial synthesis of completed mathematical and computational records. Sean Brady is originator and conceptual director; OpenAI ChatGPT and Codex are AI research collaborators and co-authors. Source rearrangements, native recipients and derived certificates retain their original Codex attribution. The numerical results belong to their recorded runtime and source configurations. Their use here starts no new arithmetic campaign.

## XXI.2 Exact range queries preserve the whole signed recipient

One completed Type II construction uses

$$u(n)=\begin{cases}\mu(n), & n \text{ odd},\\ 0, & n \text{ even},\end{cases}\qquad \kappa=\pi/\log 2.$$

At scale $a$, put $U=V=\lfloor a^{1/4}\rfloor$ and $Z=a^{7/8}$. For fixed source index $m$ and odd $k$, the admissible $c$ values in the original near-lattice sum lie in one exact interval:

$$c\le V,\qquad c>\max\{Z/m,V/k,a/(mk)\},\quad c<2a/(mk).$$

The strict product boundaries remain in the integer admission. Define three signed prefixes

$$P(t)=\sum_{c\le t}u(c)c^{i\kappa},\qquad S(t)=\sum_{c\le t}u(c)/c,\qquad \Theta(t)=\sum_{c\le t}u(c)\theta_{V,c}.$$

The lattice part becomes one difference of $P$ values, with its original outside phase. The other two tables retain the continuous and boundary compensation. The complete recipient remains

$$F=E_{\text{near}}-\frac{VS(V)}{2}\{A(a/V)+(a/V)T(a/V)\}.$$

The same signed sum contains the lattice, correction and mean. Shared-prime pairs, source zeros and small-prefix mean terms survive the rearrangement.[15][16]

For rational $a=A/D$, all admission endpoints can be chosen by integer arithmetic. With $n_Z=\lfloor a^{7/8}\rfloor$, the interval is

$$L=1+\max\{\lfloor n_Z/m\rfloor,\lfloor V/k\rfloor,\lfloor A/(Dmk)\rfloor\},\qquad H=\min\{V,\lfloor(2A-1)/(Dmk)\rfloor\}.$$

An empty interval contributes exactly zero. Even and squareful source zeros can be skipped, and strict $ck>V$ excludes $k=1$. There are $O(a^{3/4})$ source $m$ indices and $O(a^{1/8})$ odd $k$ indices. The resulting $O(a^{7/8})$ query count describes computational work. The size of the actual signed amplitude remains its separate arithmetic question.

The completed production includes 22 distinct fresh pod windows from 4,194,301.875 to 40,960,000.125, with a preserved duplicate execution and a corrective replacement window. At $a=16{,}777{,}216.125$, the directed complete value is approximately $-137.312426025$; at 16,777,216.375 it is $-137.3124361512$. The fractional pair retains the moving boundary at $V=64$. Every GPU discovery value receives a complete independent 256-bit native evaluation. For the exact prefix construction and completed source panel, *The test result suggests strong contact with the concept.*[15]

## XXI.3 A common grid separates arithmetic from geometric payments

The common-grid continuation groups fine odd coordinates $n=4p+1,4p+3$. Its zeroth and first moments are

$$d_0(p)=\{u(4p+1)+u(4p+3)\}/4,\qquad d_1(p)=\{u(4p+1)+3u(4p+3)\}/16.$$

Both fine and coarse responses use the same moving $A$ through $8A$ coordinate grid. The retained kernel is

$$W_A(p,t)=\int_A^{2A}w(p/a)w(t/a)\,\frac{da}{a^3}.$$

With $q=1/\sqrt{2}$, the leading recipient is

$$Q_{00}(A)=\sum_{p\ne t}[d_0(p)d_0(t)-q\,u(p)u(t)]W_A(p,t)+\frac{1}{8}\sum_p u(4p+1)u(4p+3)W_A(p,p).$$

Its first line contains the four actual progression-pair products jointly opposed by the coarse product. The second line retains the distance-two within-block correlation. Neither is replaced by an independence model.[17]

The complete source identity is

$$E(4A)-qE(A)=Q_{00,\text{cent}}(A)+C_{\text{actual}}(A)+e_{\text{off}}(A)+\varepsilon_4(A),\qquad Q_{00,\text{cent}}(A)=Q_{00}(A)-c_{\text{repay}}/A,$$

where

$$c_{\mathrm{repay}} = (q-\tfrac{1}{4})\frac{6\kappa^2}{\pi^2(1+4\kappa^2)}.$$

Geometry, first moments and literal self transport have explicit payments for every real $A \geq 16$:

$$\begin{aligned} |C_{\mathrm{actual}}| &\leq 1.969606/A, \\ |e_{\mathrm{off}}| &\leq 5.532052A^{-2} + 1.068576A^{-3}, \\ |\varepsilon_4| &\leq 2.733795A^{-3/2}. \end{aligned}$$

The displayed constants are outward-rounded source budgets. Their existence leaves a precise uniform arithmetic recipient: one fixed $C_{00}$ controlling the centered joint covariance on the specified infinite mesh. That constant remains unassigned.

All 3422 completed block recipients recover their full defect. The leading and first-moment centers each have 3378 negative and 44 positive values; the complete defect retains all 78 original positive cases. At $A = 124$, a positive leading center of approximately $5.72178 \times 10^{-7}$ is repaid by the actual negative diagonal remainder, giving complete defect $-9.00791 \times 10^{-6}$. At $A = 19530$, both the leading center and diagonal remainder are positive, while the moment correction is negative; the complete defect is $1.13847 \times 10^{-9}$. These are different signed source arrangements inside the same exact identity. [17]

The follow-up explicitly pays the known scale-dependent terms for every $q = 4^{-1+\varepsilon}$, $0 < \varepsilon < 1$. Its sufficient source target is a joint $O(A^{-1})$ bound after the exact diagonal repayment. The known terms have formulas for all $q$; the unknown source-correlation coefficient remains a single named arithmetic assignment. For these completed uniform transport and source reductions, *The test result suggests strong contact with the concept.*[18]

### XXI.4 A determinant retains the transverse source direction

The retained determinant branch uses the full Mertens source $M(x) = \sum_{n \leq x} \mu(n)$ and

$$E(N) = \int_1^N M(x)^2 \, \frac{dx}{x^2}.$$

Its signed-support construction gives two cumulative source columns $U, V$ with $M = U + V$. Relations between rough products retain the actual parity signs. One source copy in the dual covector remains fixed at the actual Möbius sequence. The resulting two-coordinate Gram measures a complete source response, including its transverse direction. For a block $k \to K$, set $w = k^{-2} - K^{-2}$. [38][42]

Let that Gram be $H$, its covector $a$, and define

$$\begin{aligned} B &= (1,1)H(1,1)^T, \quad e = a_1 + a_2, \\ c &= (1,1)H(0,1)^T, \quad d = H_{22}, \quad q = a_2. \end{aligned}$$

Exact two-dimensional elimination gives

$$a^T H^{-1} a = \frac{e^2}{B} + \frac{(Bq - ec)^2}{B(Bd - c^2)}.$$

Thus its factor-four certificate is exactly

$$(4wB - e^2)(Bd - c^2) \geq (Bq - ec)^2.$$

The first term is the response along the actual combined source; the second pays transverse mismatch. Both the determinant and the signed covector enter the quantitative estimate.

At the retained block $241 \to 249$, collective cross-base relations reduce normalized squared dual norm from 502.841935 to 3.427634, below four. At $377 \to 386$, the rough triple $385 = 5 \cdot 7 \cdot 11$ is decisive: retaining it as an independent amplitude gives 19.391695830996, while its actual relation $\mu(385) = -\mu(77)$ gives 0.773021470913. The endpoint corrections of individual free coordinates remain in the complete signed product readout. The constant-size Gram is retained alongside the growing source-query and product ledger.[41][42]

The completed exact representations, transport identities and finite certificates have classification *The test result suggests strong contact with the concept.* The uniform route retains *The test result suggests the concept is possible.* Its all-scale determinant estimate remains the named source assignment.[38]

### XXI.5 The auxiliary inverse receives its whole complement

The later complementary-resolvent construction evaluates the entire auxiliary inverse through an exact four-channel source metric. Both exterior tails remain. Its finite source response is the original Galerkin response plus a positive-semidefinite correction of rank at most four; the underlying complementary operator remains infinite. Twelve native recipients carry the whole source response.

For the retained rank-40 packet at pole $33/2$ and parameter $1/3$, the whole $h_0$ response is 2.50093808544, against Galerkin 2.49293778112, with correction 0.00800030431736. The rank-40 half-pole calculation returns its whole positive metric and corrected source response, while its subsequent inverse-heavy Woodbury Schur is `INVERTIBILITY_UNRESOLVED`. The successful primitives and unresolved successor are both kept. [39]

Original signed source forms retain $Q$, physical norm and $H^1$ norm with every mixed entry. A positive auxiliary metric does not change the type of the original signed form. In exact kernel/pivot coordinates $T = [X, E]$, its next receiving block is

$$T^* Q T = \begin{pmatrix} Q_{KK} & Q_{KE} \\ Q_{EK} & Q_{EE} \end{pmatrix}.$$

When the actual kernel block is positive definite, the remaining full-source sign is carried by $Q_{EE} - Q_{EK} Q_{KK}^{-1} Q_{KE}$. The completed kernel forms and the proposed mixed Schur have separate status. For the completed whole-inverse and signed-source recipients, *The test result suggests strong contact with the concept.* [39][40]

### XXI.6 An ordered Gram forest keeps both quadratures

A second computational line acts directly on the original weighted source. Put $s = 1 + \varepsilon$, $c_n = u(n) n^{ik}$, $k = \pi / \log 2$,

and

$$L_Y(n)=\begin{cases}[n^{-s}-\min(Y,2n)^{-s}]/s, & n<Y,\\ 0, & n\ge Y.\end{cases}$$

For each group having the same already-processed smooth factor, retain its ordered aggregate

$$C=\sum_n c_n,\qquad W=\sum_n c_nL_Y(n),$$
$$\mathscr{E}=\sum_{m,n}c_m\overline{c_n}L_Y(\max(m,n)).$$

For consecutive ordered groups $A,B$, the exact combine is

$$C_{AB}=C_A+C_B,\qquad W_{AB}=W_A+W_B,$$
$$\mathscr{E}_{AB}=\mathscr{E}_A+\mathscr{E}_B+2\Re(C_A\overline{W_B}).$$

The cross term is part of the full energy. An AVL forest stores this associative ordered receiver, with genuine deletions and sorted-median construction of outgoing groups. Each physical node retains its original integer address, coefficient, phase and kernel. [19][20]

Preindexed distinct-prime events move a node from group $h$ to $hp$. Only affected roots change. Complete outgoing self and surviving boundary energy remain separate. If $S$ is the number of nonzero coordinates and $A_{\mathrm{inc}}$ their distinct-prime incidences, aggregate/control work is

$$O((S+A_{\mathrm{inc}})\log S+\pi(N)).$$

Source generation, coefficient admission, precision-dependent arithmetic and retention have their own costs.

At $N=8192$, the original source has 3321 nonzero nodes and 6363 prime incidences. Every incidence moves once. The observed maximum tree height is twelve; after the final cut every group is a singleton. Matched complete-cut arithmetic takes 0.177855 seconds, compared with 9.930477 seconds for the complete-group baseline, a measured ratio 55.83. The baseline already omits costly full-group serialization, so this is an arithmetic comparison. Fourteen retained cases at 512 bits agree across 764 complete cuts; the full campaign retains 102,044 checks and 25 cold-exact native results. *The test result suggests strong contact with the concept.* [19]

### XXI.7 Paid subaccounts locate the cross-prime task

The tree also exposes an exact source split. For a complete prime cut, let $S_p$ be self energy, $B_p$ the surviving boundary energy, and

$$X_p=M_{p^-}-B_p-S_p.$$

The literal squarefree source cocycle gives

$$X_p=-2\cos(k\log p)D_p+R_p,\qquad D_p=\sum_{\substack{n<Y\ \text{odd squarefree}\\ p\mid n}}L_Y(n).$$

The matched prime-ratio term $D_p$ has an exact all-scale Euler-product payment. At $\varepsilon=1/2$, an elementary upper is $16a/(9\sqrt{3})$, $a=1-2^{-s}$. The original off-diagonal overlap $R_p$ is retained after that payment.

The independent inequality

$$x-\log(1+x)\ge\frac{x^2}{2\max(1,1+x)},\qquad x>-1,$$

supplies a source-level quadratic credit. With

$$Q_p=\frac{(S_p+X_p)^2}{2\max(B_p,M_{p^-})},$$

the completed sharp and adaptive receivers have their explicit self/logarithmic credits. They act on the actual overlap, with no intermediate absolute-value replacement.[20]

The same work retains a useful failed route. At scales $Y_j=(4/3)2^j$, the terminal prime block has coherent positive overlap. Its positive forcing diverges for fixed $0<\varepsilon<1$, under the established prime census used in the source derivation. For independent all-real summable prime-block payments in that specified form, *The test falsifies the concept.* Earlier signed sectors must participate in the compensation. At $Y=16384/3$, the finite adaptive terminal contribution is $+0.005434532$, earlier signed forcing is $-0.245163459$, and the whole receiver is $-0.239728928$. The failure and the useful complete signed receiver are both retained.[19]

### XXI.8 The complete crossing has a signed rank-one payment

The latest original-source calculation fixes $\varepsilon=1/2$, $Y=4096/3$, $N=1365$, and core $K=\{3,15,33\}$. It retains all 1362 exterior coordinates, source one and every zero, all owner/self sectors and both transported quadratures. The completed divisor quotient preserves the literal source and reduces the core to direction $\nu=(1,-1,-1)$. Put

$$m=-\nu^TR_K\nu>1/16,\qquad \chi=\nu^TC,\qquad \ell=\chi y.$$

For anchor $a$ and the full exterior vector $y$, the exact square identity is

$$-m|a|^2+2\Re(\overline{a}\ell)=-m|a-\ell/m|^2+|\ell|^2/m.$$

The untouched exterior term $y^*N_{\mathrm{out}}y$ stays in the full source account. The positive coefficient payment is the complete rank-one factor $\chi^*\chi/m$; it needs no dense exterior-square expansion.[21][22]

At the actual source $a=-1$, $y=\mu_{\mathrm{exterior}}$, the native values are

$$m=0.0759247800948322793\ldots,$$
$$\ell=0.0578748989347488838\ldots,$$
$$2a\ell=-0.115749797869497768\ldots,$$
$$\ell^2/m=0.0441160833462246285\ldots.$$

The completed fixed-margin payment $16\ell^2$ is $0.0535920628273183598\ldots$. The contracted unsigned envelope is $0.0969190987930023511\ldots$, and the full three-row unsigned envelope is $0.292100805586072730\ldots$. The reductions 17.6817, 54.4815 and 84.8970 percent compare separate baselines. They are distinct losses removed by the actual metric, source signs and retained row combination. [21]

The result-selected continuation evaluates the actual signed shell. For saturated core and frozen coefficients,

$$\ell' = Y^{-s} J_\chi, \qquad J_\chi = \sum_{Y/2<n<Y} \kappa_n \mu(n),$$

where prime denotes logarithmic-scale differentiation. The full 683-coordinate annulus gives

$$J_\chi = -5.01612235373511102\ldots.$$

Native composition retains the full mixed Gram of $(\ell, J_\chi)$, including their negative correlation. Consequently the positive-payment derivative is $-0.000151581702305060533\ldots$, while the negative-square derivative is $+0.000350438333171710846\ldots$. Their sum is the actual crossing derivative $+0.000198856630866650313\ldots$.

Writing

$$F_\chi = m(a-\ell/m)^2 \geq 0, \qquad T_{\text{core+cross}} = P_\chi - F_\chi,$$

with $P_\chi = \ell^2/m$, supplies the exact frozen-chart storage identity $\Delta T = \Delta P_\chi - \Delta F_\chi$. At this cut storage release pays the increasing crossing even while the positive payment falls. All 1255 successful external native outputs receive independent reconstruction, totaling 544,057 checks. For these tested source and transition relationships, *The test result suggests strong contact with the concept.* [21]

### XXI.9 Complete transition recipients and the retained boundary

The completed finite price work keeps a compensated reference floor, once-paid divisor and cycle alternatives, and the full self/partition account. Its complete 139-member finite cut-three reconstruction has negative net price. A corrected source-kernel ceiling is

$$K_Y \preceq K_\infty/(1-2^{-s}).$$

The earlier unscaled ceiling is retained as a corrected route. Mixed positive-semidefinite allocation and pointwise cycle price have different mathematical meanings; an exact negative-PSD control has positive pointwise price. No allocation spends the same coefficient credit twice.[23]

The remaining recipient is concrete: combine exterior energy and scalar credit with the crossing storage, actual moving margins, owner weights, donor/spent masks, removals and reference changes. The latest source derivative holds in its frozen chart; the complete moving-source account retains $m'$, coefficient changes and both quadratures. A complete fixed all-scale price $K$, or a uniform signed-source exponent closing the selected parent target, remains unconstructed in these records. The achieved contribution is the exact source evaluator, retained signed arithmetic and paid transition structure now available to that assignment.

SAM VOLUME III

# Prime Coordinates and Exact Mersenne Factors

Chapter XXII

The prime application asks a narrower question than primality: can a declared structure expose exact factor opportunities? The geometry is an information-coordinate language. Modular certificates decide factors, and a finite failure to find one remains only that. Later bounded application amendments are recorded under their own provenance.

The account below follows the chapter source as retained at the fixed 8 September 2026 SAMA revision. Engine names, installed interfaces and campaign status are historical source statements. They do not describe a freshly inspected October deployment. Later source amendments are distinguished where available; no later engine or campaign is inferred.

**Source-status qualification.** The operational and revision statements in this chapter describe the retained source documents. They are not an instruction to execute, deploy, resume, train, promote or change a service, and they have not been independently verified as present live state for this edition.

### XXII.1 Overview

The Mersenne application begins from an owner-defined change of coordinate: prime marks are informational radix coordinates on a closed two-dimensional sphere, not equally spaced marks on an ordinary integer ruler. Exact prime integers remain indispensable. They generate the marks, define operator composition and carry every modular certificate. The conceptual change is in the role assigned to a prime, not in its arithmetic identity.

The rational stereographic chart

$$x = \tan(\chi/2), \qquad S(x) = \left(\frac{2x}{1+x^2}, \frac{1-x^2}{1+x^2}\right)$$

turns multiplication into an exact sphere action $T_n(x) = nx$, inversion into $J(x) = 1/x$, and logarithmic displacement into $du = dx/x = d\chi/\sin\chi$. Prime-only marks $p^- \to p$ form reversible adjacent cells. On each cell the program places three reciprocal voices:

$$\alpha = 2+\sqrt{3}, \qquad \beta_p = \frac{p+\sqrt{3}}{p-\sqrt{3}}, \qquad \gamma_p = \beta_p \beta_{p^-}^{-1}.$$

Repeated squaring generates dyadic pulses. A contact at depths $i, j$ has shallow depth $s = \min(i, j)$ and delay $d = |i-j|$. Transport into the shallow frame produces an exact echo $E$ satisfying

$$E^{2^s} = 1 \pmod{q}, \qquad \mathrm{ord}_q(E) \mid 2^s.$$

The executed MP-S0–MP-S4B lineage compiles those objects into ECHO16 and exposes the genuine Mersenne factor coordinate

$$k = b2^e, \qquad q = 2kp+1.$$

A prime $q$ is assigned as an exact factor of $M_p = 2^p - 1$ only through the order certificate $\mathrm{ord}_q(2) = p$. MP-S5's sealed structural $k$-order returns 103 factored objects against 100 for equal-budget ascending counting, an installed strong-contact result in the application-scheduler scope. MP-S6 assigns 1,282 exact factors among 5,390 exponents. MP-S7 and MP-S8 preserve a 1,858-row roster after official-factor removal and complete 9,595-coordinate shell misses. Those misses remain `PRIMALITY_UNASSIGNED`.

The clean-sheet route is a successor to two preserved failures. M4's 13,818 RH prime-sidecar ranking and M4A's 7,708 candidate-local Theta18/W9 ranking both fail their fixed direct-successor questions. Their classifications remain **The test falsifies the concept.** The clean-sheet compiler does not rename those outcomes.

The owner has resumed only the bounded CEV1/MP successor and its initial training. All other production, farm, frontier and publication work remains paused. This chapter records the installed CEV1 and MPV2 successors without launching arithmetic, remote work, promotion or distribution.

### XXII.2 Opening question and conceptual picture

The governing question is:

> Can the prime-coordinate, reciprocal-history and dyadic structures be compiled into an exact ordering of genuine Mersenne factor coordinates, while keeping factor assignment, PRP status and Mersenne-primality certification strictly distinct?

The complete route is

- prime integer p and predecessor pˆ-
- prime-only reversible sphere cell
- alpha, beta_p and gamma_p reciprocal voices
- dyadic pulse depths and delayed echo
- structural k = b*2ˆe coordinate
- prime q = 2*k*p + 1
- exact modular order certificate
- factor assigned OR finite miss remains PRIMALITY_UNASSIGNED

This is an application compiler, not a universal replacement for the SLC substrate, Q2 selector, Reciprocal History, RH authority or Lucas–Lehmer certificate lane.

### XXII.3 Informational radix with exact arithmetic custody

#### XXII.3.1 The owner-defined role

The owner interpretation is:

Prime numbers are informational radix coordinates used to express orbit on a closed two-dimensional sphere.

The formalization retains two layers:

- **arithmetic encoding/certificate.** Meaning: $p$ and $q$ are exact certified prime integers; modular powers and divisibility are exact.
- **SAM object role.** Meaning: each prime emits one indecomposable mark and multiplicative sphere action.

Uniform integer ordinal is not the ruler. Composite integers still appear inside arithmetic values such as $n$, $k$ and (q-1); they simply do not emit prime-ruler marks of their own.

#### XXII.3.2 Orbit and harmony are typed terms

`Orbit` denotes the discrete generated family

$$z, z^2, z^4, \ldots, z^{2^n}$$

on the exact information-state surface. It is not a claim of physical motion around a sphere. `Harmony` denotes exact equality, reciprocal identity or modular closure between separately generated voices. It is not an acoustic analogy substituted for calculation.

### XXII.4 Closed sphere and prime-only ruler

#### XXII.4.1 Rational sphere chart

Set $x = \tan(\chi/2)$. The point

$$S(x) = (X(x), Z(x)) = \left(\frac{2x}{1+x^2}, \frac{1-x^2}{1+x^2}\right)$$

lies on the unit circle section of the closed two-dimensional sphere because

$$X(x)^2 + Z(x)^2 = \frac{4x^2 + (1-x^2)^2}{(1+x^2)^2} = \frac{1+2x^2+x^4}{(1+x^2)^2} = 1.$$

The two limits $x = 0$ and $x = \infty$ close the chart at reciprocal poles.

#### XXII.4.2 Multiplicative actions and reversal

Define

$$T_n(x) = nx.$$

Then operator composition is exactly integer multiplication:

$$(T_m \circ T_n)(x) = m(nx) = (mn)x = T_{mn}(x).$$

With $J(x) = x^{-1}$,

$$(J \circ T_n \circ J)(x) = J\left(\frac{n}{x}\right) = \frac{x}{n} = T_{1/n}(x).$$

Thus reversal conjugates expansion by $n$ to contraction by $1/n$.

Differentiating $x = \tan(\chi/2)$ gives

$$\frac{dx}{x} = \frac{\frac{1}{2}\sec^2(\chi/2)}{\tan(\chi/2)} d\chi = \frac{d\chi}{2\sin(\chi/2)\cos(\chi/2)} = \frac{d\chi}{\sin\chi}.$$

Therefore

$$du = \frac{dx}{x} = \frac{d\chi}{\sin\chi}.$$

The logarithmic coordinate $u = \log x$ converts $T_n$ into translation by $\log n$, joining the sphere chart to the logarithmic translation language of Reciprocal History without merging their application authorities.

#### XXII.4.3 Marks and adjacent cells

MP-S0B emits one object for each prime $p$. A mark retains:

- the exact integer $p$ and reciprocal point $1/p$;
- indecomposable action $T_p$;
- the two exact sphere points $S(p)$ and $S(1/p)$; and
- the adjacent arithmetic ports (p-1) and (p+1).

If $p^-$ is the preceding prime, the directed adjacency $p^- \to p$ defines one reversible cell with scale

$$\rho_p = \frac{p}{p^-}, \qquad \rho_p^{-1} = \frac{p^-}{p}, \qquad \Delta u_p = \log\frac{p}{p^-}.$$

Its exact spherical chord is computed from the two rational sphere points. MP-S0A/S0B install 172 prime-only marks and 171 adjacent-prime cells. No composite between consecutive marks becomes an extra ruler object.

### XXII.5 Three reciprocal voices

#### XXII.5.1 Native Lucas–Lehmer voice

Let

$$\alpha = 2 + \sqrt{3}, \qquad \alpha^{-1} = 2 - \sqrt{3}, \qquad \alpha\alpha^{-1} = 1.$$

Modulo $M_p = 2^p - 1$, define

$$L_i = \alpha^{2^i} = a_i + b_i\sqrt{3}.$$

The reciprocal trace is

$$s_i = L_i + L_i^{-1} = 2a_i.$$

Because $L_{i+1} = L_i^2$,

$$s_{i+1} = L_i^2 + L_i^{-2} = (L_i + L_i^{-1})^2 - 2 = s_i^2 - 2 \pmod{M_p}.$$

At $i = 0$,

$$s_0 = \alpha + \alpha^{-1} = 4,$$

so the common reciprocal mode is exactly the Lucas–Lehmer recurrence. MP-S0 reconstructs this mode for all 171 prime exponents through 1024.

### XXII.5.2 Candidate prime-mark voice

The Cayley lift

$$\beta_p = \frac{p+\sqrt{3}}{p-\sqrt{3}}$$

has reciprocal conjugate

$$\beta_p^{-1} = \frac{p-\sqrt{3}}{p+\sqrt{3}}.$$

Its norm is one, so its dyadic family

$$R_j = \beta_p^{2^j}$$

lives on the same reciprocal quadratic-unit surface as the $\alpha$ family. The prime integer selects the voice; the modular and quadratic arithmetic retains exact custody.

### XXII.5.3 Incoming transition beat

The transition from the previous prime cell is

$$\gamma_p = \beta_p \beta_{p^-}^{-1}.$$

This object records change between consecutive prime voices rather than a second copy of the current mark. Its inverse reverses the cell:

$$\gamma_p^{-1} = \beta_{p^-} \beta_p^{-1}.$$

The three families answer different questions: $\alpha$ carries the native Mersenne recurrence, $\beta_p$ carries the current prime mark and $\gamma_p$ carries the adjacent-prime transition.

## XXII.6 Beats within beats

### XXII.6.1 Dyadic depth and delay

Repeated squaring of any voice produces pulses at depths

$$z^{2^0}, z^{2^1}, \ldots, z^{2^n}.$$

For a contact between depth $i$ in one voice and depth $j$ in another, set

$$s = \min(i,j), \qquad d = |i-j|.$$

If, for example, $j = i+d$, then

$$z^{2^j} = \left(z^{2^i}\right)^{2^d}.$$

The exponent $2^d$ transports the deeper pulse into the shallower frame. The residual relation is the base echo $E$. Every retained MP-S2 and MP-S3 factor component satisfies

$$E^{2^s} = 1 \pmod{q}, \qquad \operatorname{ord}_q(E) \mid 2^s.$$

The echo is therefore an exact finite-order modular object with a dyadic closure depth, not a visual correlation.

### XXII.6.2 Nested compiler

Factor the Mersenne multiplier coordinate as

$$k = b2^e,$$

where $b$ is odd and $e = v_2(k)$. The structural base $b$ and dyadic scale $2^e$ echo the prime-cell and pulse decomposition. The complete compiled route is

- adjacent-prime cell
- alpha/beta/gamma reciprocal relation
- shallow depth s and echo delay d
- base echo closure
- odd structural base b and dyadic scale 2^e
- k = b*2^e
- q = 2*k*p + 1

This is the exact content of `beats within beats`: nested coordinate relations are compiled into a candidate factor address that the modular lane can certify or reject.

## XXII.7 Exact Mersenne factor certificate

### XXII.7.1 Deriving the candidate form

Let $p$ be an odd prime and suppose prime $q$ divides

$$M_p = 2^p - 1.$$

Then

$$2^p \equiv 1 \pmod{q}.$$

The multiplicative order $r = \operatorname{ord}_q(2)$ divides $p$. For a nontrivial Mersenne factor, $r \neq 1$; since $p$ is prime, $r = p$. Fermat's theorem also gives $r \mid q-1$, hence

$$p \mid q-1.$$

Because $q$ is odd, (q-1) is even, so for some positive integer $k$,

$$q-1 = 2kp, \qquad q = 2kp+1.$$

Conversely, if the candidate $q = 2kp+1$ is prime and the exact order check returns

$$\operatorname{ord}_q(2) = p,$$

then $2^p \equiv 1 \pmod{q}$ and therefore $q \mid M_p$. The implementation may certify the modular closure with `pow(2,p,q)=1` together with the declared prime/order conditions. It never needs to materialize the enormous integer $2^p - 1$.

#### XXII.7.2 Result-type boundary

- **prime $q$, candidate form, exact order $p$.** Authorized type: exact factor of $M_p$.
- **finite $k$- or shell search finds no factor.** Authorized type: `PRIMALITY_UNASSIGNED`.
- **PRP result with its certificate.** Authorized type: production PRP/certificate status only.
- **exact Lucas–Lehmer terminal.** Authorized type: Mersenne-primality baseline/confirmation status.

A finite miss never crosses rows in this table. It is not a prime, probable prime or verified Mersenne-prime assignment.

### XXII.8 Executed discovery-to-compiler chain

#### XXII.8.1 Object installation: MP-S0 through MP-S0B

MP-S0 installs the $\alpha$ reciprocal common mode on 171 prime exponents through 1024. MP-S0A installs the exact sphere actions. MP-S0B emits 172 prime-only marks and 171 reversible cells. These stages define the object language before evaluating factor-search advantage.

#### XXII.8.2 Harmony and transition discovery: MP-S1 through MP-S3

- **MP-S1.** Exact observation: self-harmony exposes factors for 52/158 composites and 0/13 exact positives; Classification: **The test result suggests the concept is possible.**.
- **MP-S2.** Exact observation: $\alpha/\beta_p$ harmony adds five factors beyond MP-S1; all 37 contacts are phase-shifted echoes; Classification: **The test result suggests the concept is possible.**.
- **MP-S3.** Exact observation: $\alpha/\gamma_p$ adds four held-out factors; all four are delayed echoes; Classification: **The test result suggests the concept is possible.**.

The zero counts on exact-positive controls remain part of the record. The factor contacts motivated compilation because they closed under an exact echo identity, not because every composite was detected.

#### XXII.8.3 Compilation and blind-range anatomy: MP-S4A/MP-S4B

MP-S4A compiles the three voices into candidate-local deterministic ECHO16 with checkpoints. MP-S4B applies it to a blind range and factors 50/136 composites and 0/1 exact prime. Inspecting the exact factor anatomy exposes $q = 2kp+1$, turning the voice/echo structure into the genuine factor-coordinate interface. MP-S4B's classification remains **The test result suggests the concept is possible.** ECHO16 is a language and local compiler at this point, not yet an established global ordering.

#### XXII.8.4 Structural scheduler: MP-S5

MP-S5 seals a structural ordering of $k = b2^e$ before comparison. At equal budget it factors 103 objects versus 100 under ascending $k$ counting and reaches shared contacts earlier. Its authorized classification is:

**The test result suggests strong contact with the concept.**

The result concerns this Mersenne factor-scheduling application. It does not promote the structural ruler to the global Q2, Starbreaker or RH scheduler.

#### XXII.8.5 Frontier deployment: MP-S6 through MP-S8

MP-S6 deploys the structural ruler on 5,390 prime exponents:

$$5{,}390 = 1{,}282 + 4{,}108.$$

It assigns 1,282 exact factors without constructing $M_p$; 4,108 rows remain campaign survivors. The classification is **The test result suggests the concept is possible.**

MP-S7 removes 2,250 survivors by independently verified official factors:

$$4{,}108 - 2{,}250 = 1{,}858.$$

MP-S8 executes every one of 9,595 first-singularity-shell coordinates for each of those 1,858 rows without assigning a factor. The 1,858-row roster is the immutable structural freeze, and every row remains `PRIMALITY_UNASSIGNED` at that boundary.

A later dated status triage yields a 1,204-row operational distribution derivative after removing 647 distinct verified PRP composites, routing two reliable-but-unverified PRP results to verification, removing two verified Lucas–Lehmer composites and holding three active assignments. That derivative does not rewrite the 1,858-row MP-S8 result.

### XXII.9 Preserved direct-ranking failures and wrong controls

#### XXII.9.1 M4 direct prime-sidecar ranking

M4 joined 13,818 ordinary-prime spacing sidecars to the completed Mersenne curriculum. It compared pointwise, era-balanced pairwise, smooth-minimax and parent-blend routes across six held-out eras and 19 exact positives. Every non-parent route had a negative worst-era change, so M3E remained selected.

The exact direct-ranking concept is classified:

**The test falsifies the concept.**

#### XXII.9.2 M4A candidate-local Theta18/W9 ranking

An unsealed successor first treated Theta18 as an 18-neighbor prime window. That was the wrong object. The corrected source-visible bridge for each q30 survivor was

- static U63 + dynamic U63
- theta18_cross_hex
- reciprocal left/right F81 views
- W9HistoryMode
- nine forward + nine reverse Z4 maps

M4A then evaluated all 7,708 q30 survivors with exact reciprocal, directional and reconstruction checks. Mechanical validation passed 22/22. Its coupled channel used

$$q(p)=\frac{1}{p}, \qquad A_{lift}(p)=\frac{1-A_0}{p},$$
$$A_{total}(p)=A_0+A_{lift}(p), \qquad u(p)=\log p.$$

the weight $\log(p)/\sqrt{p}$ and the declared pi-bearing floor constant. The fixed candidate-local channel did not stably replace M3E. Its exact-scope classification is:

**The test falsifies the concept.**

Its receipt is typed `CANDIDATE_LOCAL_COMPLETED_CHANNEL_RECEIPT_NOT_FULL_Q_XI`. It neither evaluates the full completed RH form nor changes RH universal-nonnegativity authority.

### XXII.9.3 Successor relationship

The clean-sheet route asks a different executable question. Instead of using an RH-derived scalar to rank Mersenne positives directly, it orders genuine trial-factor coordinates and retains exact modular factor decisions. This successor does not revise M4 or M4A. It preserves them as the deviation chain that redirected the application toward an operationally exact target.

Wrong controls for the clean-sheet route include:

- declaring a finite factor miss to be a primality result;
- comparing unequal search budgets in MP-S5;
- allowing official factor lookups to enter a blind factor-search surface;
- letting a receipt or exponent identity break a structural-score tie;
- treating the sphere as a physical trajectory rather than the declared exact information-state chart;
- calling the factor scheduler a global SLC/RH scheduler; and
- replacing Lucas–Lehmer certification with the structural filter.

## XXII.10 Bounded CEV1/MP resumption and exact custody

H000725 resumes only the bounded CEV1/MP successor and its initial training. The remaining pause has the following literal operational consequences:

- no new prime benchmark, factor farm, production process or supplemental exploratory run is started;
- Farm V1.1, V9/V13 and the balanced factor pass remain inactive;
- the active remote M143064041 power-9 run is not stopped, altered, queried for new credit or incorporated into this documentation;
- H14F remains a fourteen-physical-worker contract; no 30-worker result exists;
- the 31/127 execution-arity observations remain unserialized diagnostics, while the formal binary-versus-adaptive-31/127-versus-fixed-L162 rotated comparison remains unexecuted; and
- no public prime output, pointer, assignment, promotion or distribution is changed here.

### XXII.10.1 Native CEV1 MP domain

Frozen non-promoted `SLCQ3-RZ-CEV1` compiles `p`, `2^p-1`, `p[ln(2)]`, D18/D81/D162 and exact `q=2kp+1` coordinates into a source-visible 200-feature integer model distinct from the Q3 reciprocal weights. Its first checkpoint trains on 16 discovery runs and evaluates three disjoint held-out runs:

- 25,786 strict correct comparisons
- 190 ties
- 1,281 reversed comparisons
- 27,257 held-out comparisons total.

Base, independent and complete-source checks pass `8/8`, `22/22` and `12/12`; producer-independent reconstruction passes `19/19`. MP52 is barred from training. **The test result suggests strong contact with the concept.**

### XXII.10.2 MPV2 certified frontier and provenance firewall

Isolated MPV2 separates discovery from certification and carries exact state through a chunked CAS certificate DAG. Component-local valuations, combined `F`, exact cyclotomic closure, every frontier edge, zero-yield work and H000723/H000724 wave learning from independently certified intermediate growth survive 252/252 integrated and 59/59 producer-independent checks. Exact closure alone supplies `ACTIVE -> EXHAUSTED_NO_MORE_CERTIFIED_MASS`; an ordinary completed bounded wave remains `ACTIVE`.

Conventional Prime95-compatible arithmetic, residues and checkpoints and PRP-derived scheduling examples remain useful under their exact provenance. They do not become native CEV1/Q3 execution or a primality premise by origin. Native custody begins only after independently certified factor growth.

Topology-derived endpoint observation is installed, but the authenticated remote-worker, CPU-replayed 780M and physical-T500 execution-custody join is not. Native deployment therefore remains fail-closed. No real-hardware benchmark or continuous MPV2 training process is running. **The test result suggests strong contact with the concept.**

For $p=143064041$, $k=320000$ would form

$$q=2(320000)(143064041)+1=91560986240001,$$

a 47-bit candidate coordinate. The active run did not use that gate. The calculation records a possible address only; it is not an executed factor result, and a miss at any finite gate would remain primality-unassigned.

The paused frontier-prime objective below ten minutes remains unfinished in the Lucas–Lehmer certificate lane. This statement preserves the live boundary without weakening any completed MP-S0–MP-S8 result.

### XXII.11 Established result and forward boundary

This chapter establishes a continuous conceptual-to-technical chain:

1. exact prime integers emit informational sphere marks;
2. adjacent marks carry reciprocal $\alpha$, $\beta_p$ and $\gamma_p$ voices;
3. dyadic voice contacts yield finite-order delayed echoes;
4. the echo anatomy factors $k = b2^e$ and forms $q = 2kp+1$;
5. exact order $p$ assigns a genuine factor of $M_p$;
6. the structural ruler supplies an application-specific search order; and
7. finite misses remain `PRIMALITY_UNASSIGNED`.

The installed application results and their exact classifications are retained stage by stage. No further computational step follows from this document while the owner pause is active. Any successor requires fresh owner authority and a new campaign that preserves frozen artifacts, exact status types and the untouched active-remote boundary.

### XXII.12 Sources and evidential scope

This chapter is assembled from the following fixed source manuscripts. Source-reported campaign outcomes retain their original scope; no new campaign was executed for this edition.

- Prime ruler and mersenne application.

### XXII.13 Results received, 20–21 October 2026: the FEOP local operator and the bound

The FEOP transfer operator (classes S, X, Z, T; a $4\times 4$ split-level matrix $A$ and a no-split matrix $B$ normalised by $q = p^{-2}$; schedule $h = j\cdot e$ with $e = v_p(Q)$ per offset) was re-derived four times and verified by two code bases with predictions first at $p = 11, 13, 17, 19, 23$, at $p = 5$ with $L = 4$, and on joint moduli with 11 and 13 (ironclad H002759, H002761).[106] Its spectrum is $(B-I)(B-qI) = 0$ and $(A-I)(A-qI)^2 = 0$ with a Jordan block; the remainder is $N\cdot R = g/(p+1)$ from $L \geq g-1$; the joint recurrence has roots $\prod_{p\in S} p^{-2}$; the stationary $(g+2)$ form and the four-state form are reconciled by an explicit map $\Phi$. The operator fails only where two offsets share a class, under whole-cluster death, when $p \nmid Q$, or under a wrong per-offset schedule; the children at a split are Z 1, T 1, dead $2(p-1)$, settled $p^2-2p$. For the bound BOUND_C02, which remains a candidate, the runs gave a depth-uniform two-sided bracket with native receipts for its two steps on small cases, an exact Selberg remainder in its frame (polynomials $4T^3+19T^2+12T+1$ at $z = 7$ and $14T^3+34T^2+15T+1$ at $z = 11$), the finding that the measured joint discrepancy is neither additive across primes nor linear in $T$, and a fibering helper that carries native coverage past the 64-row source-word limit with partition receipts. The operator is the stated route to the bound's deeper coverage; no run has addressed the bound since 20 October 2026.

SAM VOLUME III

# Reusable Mersenne Certificates and Arithmetic Compiler Boundaries

Chapter XXIII

## XXIII.1 A certificate is a reusable mathematical object

Chapter XXII develops the prime-coordinate and exact factor interface. The later computational work improves how exact evidence is constructed, shared, selected and consumed. Its progress resides in explicit certificate dependencies and source-bound arithmetic. A factor interval, a recursive primality certificate, a probable-prime run and a learned cost observation retain their own result types.

This chapter is Codex editorial synthesis of completed records in the Sean Brady/OpenAI ChatGPT/Codex research collaboration. New support reductions, arithmetic plans and exact implementation changes retain the technical attribution of their original campaigns. Historical installed branches are named at their executed versions; the current sole global runtime has its separately maintained selector.

## XXIII.2 Certified support and shared premises

The native recursive route retains a known certified factor product $F$ of $N-1$. Its sufficient closure condition is

$$F \mid N-1, \qquad F^2 > N.$$

The certificate also retains the required exact modular witnesses and recursively certified prime factors. A checked child can participate in many parents without being re-created each time. Complete canonical contents identify a reusable subtree; changed contents require validation.

This gives different forms of useful reuse. Reference validation consumes evidence already checked in the current session or a verified restored checkpoint. Batch validation shares child subtrees and repeated powers across its admitted targets. Fresh construction searches and certifies missing support. The installed native service records which of those operations occurred, together with actual arithmetic and reuse counters. [24][25]

For M1279, first native certificate validation takes 0.103482 seconds, including translation of the historical reachable 61-node inventory into the newer schema. Warm reference validation has median 0.875 milliseconds. The original construction took 356.585453839 seconds. A distinct bounded fresh attempt took 0.212750 seconds, retained $F = 524286$, and returned `KEEP_LOOKING` at the order-142 component boundary. These results identify where the work was saved: validated evidence makes a strong reusable input, while fresh missing factor support remains an actual construction task. [24]

The certified progress channel addresses $F$, keeping unknown $N$ as an exact integer. Multiplying certified support by $c$ changes its exact logarithmic margin by

$$\Delta\log(F^2/N) = 2\log c.$$

Those increments have unequal mathematical gain and measured cost. The retained history distinguishes a zero progress increment from zero elapsed work. The proposed fixed-sphere or golden-cadence mapping in the B300 handoff remains a proposed scheduler recipient; the exact certified-support margin is already an available source operand.[26]

## XXIII.3 Exact bulk elimination retains its frontier

For Mersenne exponent $p$, candidate factors have the original coordinate $q = 2kp+1$. The native finite batch computes the exact modular condition for each admitted proper-factor lane. Its `uint64` coordinates use `uint128` modular products, with overflow rejection before dispatch. An emitted factor is independently replayed. A completed no-factor interval records its last checked coordinate and returns `KEEP_LOOKING`.

Across 128 finite intervals containing 1,048,576 declared positions, the installed batch performs 1,020,043 actual modular checks and four exact eliminations. Every first factor and frontier agrees with arbitrary-integer replay. Equal-work reference arithmetic takes 2.859704 seconds, compared with 0.185043 seconds for the installed cold batch. Reusing the identical packet takes 0.003852 seconds and adds zero modular checks. The reuse result consumes authenticated coverage; the cold result executes the actual new lanes.[24]

The staged p2–p292 workload combines composite-exponent divisibility, 44 retained exact evidence inputs and the native factor kernel. It returns 284 exact terminals in 0.018891 seconds; seventeen rows enter screening and seven remain unresolved. A checkpoint contains complete component, certificate, reference and coverage state. Fresh restore reconstructs that state through the source-bound replay before admitting its references. The installed controls also reject altered and resealed evidence.[24][25]

The MP52 finite trial illustrates the terminal boundary at large exponent. For $p = 136279841$, the trained head executes 1,048,576 exact checks in 0.151517382 seconds. It finds no factor, preserves the complete frontier and sets next $k = 1{,}048{,}577$. The finite construction budget $p \le 4096$ prevents an automatic certificate-construction dispatch at that target. The completed result is the exact screening interval and its recoverable continuation. [27]

## XXIII.4 Learning work order without changing terminal mathematics

The installed farm pilot learns one concrete decision: certification first or bulk elimination first. It measures both orders on p2–p127, three repetitions per order, with earlier checkpoints retained. Exponents enter without historical answers, target certificates or successful factor coordinates. The minimum-median cost rule selects `BULK_FIRST` before later rows execute.

Both later-row arms recover the same p127 checkpoint. On the 165 p128–p292 rows, three repetitions per arm return the same 156 exact terminals and nine `KEEP_LOOKING` rows. Median project execution after recovery falls from 2.077462737 to 0.962855588 seconds, a 2.157606 ratio. Component builds fall from 71 to 28, certified-$F$ history appends from 224 to 94, and trial divisions from 66,934 to 25,604. New modular checks increase from 45,079 to 57,523. More cheap exact elimination avoids expensive component construction; the terminal mathematics remains the same.[28]

Recovery plus project medians are 4.042923356 and 2.751570008 seconds, retaining that additional cost. Fresh installed adoption reproduces the outcomes. Repeating the recovered completed packet adds zero checks, components or certificates. For route-order optimization, *The test result suggests strong contact with the concept.* [28]

The larger CSV experiment supplies a useful contrast. Its completed deep pass covers 15,064 batches, with 96,409 training and 24,096 evaluation rows drawn from 120,505 sampled exponents. An opening shallow interval supplied no factor and no action-order learning signal; the successor used two deeper finite intervals. The chronological `RH_HISTORY` adapter fitted empirical signed rewards using exact retained arithmetic. Those coefficients describe farm costs, rather than the Möbius source of the RH chapter.

In the completed evaluation, `LOW_FIRST` and `MP_MEAN` each test 3,148,317,882 lanes for 120 terminals. `RH_HISTORY` tests 3,149,431,994 lanes for the same 120 terminals. The completed training corpus and recovery remain valuable retained objects; this evaluation supplies no added lane-reduction result. [29][30]

### XXIII.5 Inverse support and recursive certificate reduction

The B300 campaign asks an exact finite inverse question: which subsets of M1279's fifteen supplied root factors still meet $F \mid N-1$ and $F^2 > N$? Native custody reconstructs all 32,768 subset candidates. Exactly one is sufficient: mask 32767, containing all fifteen factors. The root has no removable row under that subset relation.

The result-led successor applies sufficient-support reduction recursively to child certificates with a declared bit-work proxy. Unique modular powers fall from 277 to 163 and dependency edges from 177 to 95. All eighteen large powers remain. Six native executions validate their resulting certificates as `MERSENNE_PRIME_VERIFIED`. A concurrent-farm CPU comparison changes from 222.115 to 183.862 milliseconds, while the unchanged-root comparison itself shows timing variation. [31]

The combined controller investigates 28 supported source/action combinations over 150 proof-learning cycles. An acquired reduced artifact changes which reuse action is eligible; a certified source terminal removes that exponent from later factor work. This is an executed cost-learning and evidence-reuse loop.

The final quiet-pod comparison retains a useful regression: reduced M1279 validation takes 343.239230 milliseconds, original validation 331.883348, and the controller's subsequently selected original route 349.868179. The mathematical support reduction survives, while measured context changes the preferred execution. The remote program remains `PAUSED_BY_OWNER`; its handoff records the unavailable final receipt transfer separately from the observed outputs.[26]

Resident GPU history has its own achieved consumer. At the retained 52-batch snapshot, the farm has executed 53,925,413,920 GPU lanes and holds 220 GiB of populated history in five buffers. A separate readout consumes final residues to select factor candidates for native checking. Midpoint residues and squared bases are retained for later investigation. Source cursors, terminals and reconstruction recipes are durable; evictable GPU buffers retain their volatile role. For the executed sufficient-proof reduction and source-linked history readout, *The test result suggests strong contact with the concept.*[31]

### XXIII.6 Shared powers and source-bound arithmetic plans

The local Prime95 research adapter follows the remaining large premises inside the reduced proof. Its 24 source-bound trials combine three support policies and three arithmetic plans, including original baselines. All eight retained certificate rows are admitted by native validation. M607 drops from 117 to 75 unique powers; M1279 from 277 to 163, with its original 177 dependency edges reduced to 95.

For the reduced M1279 certificate, a shared squaring ladder performs 5430 squarings instead of 25,197 in the adapter's independent binary implementation. Multiply count remains 14,612, and all eighteen powers with modulus size at least 1024 bits remain. The savings come from sharing arithmetic histories inside required premises.

Single-run arithmetic observations are 102.229 milliseconds for binary, 49.400 for the shared ladder and 73.714 for the prefix DAG. These exclude the separate exact-power checks and native validation. Fresh checkpoint recovery validates all eight rows. The retained prototype keeps its separate independent validation; native consumption of the shared-power plan is the named next interface. Its two offline Prime95 controls remain external backend results. For this qualified local method composition, *The test result suggests strong contact with the concept.*[32]

### XXIII.7 Exact skipping, measured acceleration and a retained near-tie

The later sparse CPU iterator improves an already-sieved factor-search baseline. A fixed 4620-period wheel generates permitted residues; remaining sieve roots clear a compact bitset, and set-bit iteration visits actual survivors. The $q = r$ exception is restored explicitly. Both paths perform the same modular powers with the same arithmetic. The saving concerns preparation and traversal of rejected coordinates.

Four fresh 1,048,576-coordinate confirmation windows give summed median compiled times 51.538 milliseconds baseline and 46.076 sparse: 1.1186 throughput ratio, or 10.60 percent less elapsed time. Independent reconstruction matches

the complete ordered survivor lists over sixteen windows and 7,340,032 coordinates. One unfavorable initial window is retained. For this finite implementation improvement, *The test result suggests strong contact with the concept.*[33]

A distinct GPU experiment learns sieve size, retaining the same factorer and exact candidate classes. Three held-out paired runs at 80–81 bits reduce median four-class time from 23.606 to 23.083 seconds, with 1.02266 throughput ratio. All seventeen launches pass 104 known-factor startup tests. The selected 512-Mib sieve is a configuration result on that workload; the measured full-band extension remains separate.[34]

The exact sieve-transition successor derives, for invertible class stride,

$$t_{\text{new}} = (t_{\text{old}} - B) \bmod r.$$

Moving a class start by $4620B$ therefore reuses its first-cleared-bit offset. Full reconstruction agrees with the entire 364,544-byte packed state over at least 480 transitions. For exact transition reuse, *The test result suggests strong contact with the concept.*

The matched runtime comparison nearly ties: three 79–80-bit pairs have medians 11.561 and 11.540 seconds; the held-out pair is 23.145 and 23.140. Later pairs reverse the initial advantage. The record establishes no repeatable material whole-runtime gain for that recurrence. Its checked transition remains an experimental implementation, while the optimized stock route retains its working-default role.[35]

### XXIII.8 Jacobian compilers retain their source domain

The same computational discipline clarifies the Jacobian program already developed in Chapter XXVI. An exact endpoint compiler, ordered event receipt, path-jet discriminator and Newton-signature compiler are different mathematical outputs. Each completed identity has a source map and an admitted chart. A successful target reconstruction does not identify those outputs with one another.

The current Jacobian program retains JC1 and the HF1–HF6TH results and their corrections. Its source-manifest entries distinguish original executables, independent validations, historical local grammars and current routing. The clean restart remains unexecuted. Its next source question derives invariants naturally from the polynomial fibers and regular-value set before selecting any later SAM compiler. [36][37]

This provides a precise connection with the prime and RH advances. Mersenne reuse consumes exact certificate premises; RH trees consume the original signed coefficient/kernel account; Jacobian comparison requires an explicit compiler preserving its own source invariants. A future matter recipient has four distinct joins: payload compilation, preservation through reciprocal delivery, an identified matter consumer, and a controlled load-bearing output change. No retained JC1/HF field has completed that chain. The completed polynomial/history results remain Volume III inputs under their established classifications. [36]

### XXIII.9 What the arithmetic advances establish

The most effective completed changes exploit mathematical structure already present in the source: shared certified subtrees, sufficient factor support, exact rejected-coordinate avoidance, persistent coverage and retained signed aggregates. Cost learning chooses among declared exact actions. Its measurement can prefer extra cheap checks, change preference after a regression, or retain a checked method whose whole-runtime benefit has not appeared.

Each of these outcomes contributes to the computational object. Exact terminals and open frontiers, arithmetic counts and elapsed costs, source history and volatile working states remain recoverable under their actual scope. Those distinctions make the resulting certificate and consumer useful to later Volume III work.

SAM VOLUME III

# Prime Event Computation and Microscopic Reliability

Chapter XXIV

The prime-event program constructed a seven-spin stochastic decision gate inside an event-created sieve and improved its reliability by changing specified microscopic interactions. The original gate produced an exact prime/composite decision sequence through one billion for one of 64 seeds per accumulation profile. A successor selected from the source algebra before its trajectories were generated produced exact sequences for 47 of the same 64 seeds per profile. With that source then frozen, 13 of 16 fresh seeds also produced exact sequences through one billion. The matched total fell from 569 to 40 errors across the two paired profiles, a 92.9701 percent reduction. These are completed experiments with retained source, trajectory, scoring and archive records. [A2, Sections II–III; PE2]

The chapter follows the construction that made this improvement intelligible. An accepted event writes future return channels. Two banks turn the channel history into gate inputs. Six hidden spins provide controllable pair and triple terms for an output spin. Full finite-state dynamics carries the hidden configuration from one reading to the next. Local replay identified the input sector responsible for the baseline's first composite errors; elimination of the hidden spins explained that sector; a balance equation selected the revised couplings. The resulting observations can then be compared with numerical survival calculations and stated resource budgets.

### XXIV.1 The event created sieve

A candidate is an integer n, processed sequentially from 2 to a declared horizon N. A return channel is a stored period established by an earlier accepted candidate. If p is accepted, its channel returns at 2p, 3p and subsequent positive multiples within the horizon. Channels enter two banks alternately in creation order, beginning with bank 0. A bank reports whether at least one of its channels returns at the current candidate. A logical input r_i equals +1 when bank i returns and -1 otherwise.

The ideal decision accepts exactly when neither bank returns. This recurrence generates the primes by induction. Initially 2 is accepted. If all accepted candidates below n are precisely the earlier primes, then a prime n has no earlier prime divisor and receives no return. A composite n has an earlier prime divisor, whose installed channel returns at n. Thus the ideal rule accepts every prime and rejects every composite. The arithmetic resides in integer periods and their future multiples; the seven-spin system implements the decision from the two bank inputs. An independent sieve supplies reference labels for scoring the generated sequence. [A2, Section IIA; PE4, Section 3]

A stochastic decision changes the history. A falsely accepted composite installs its own channel, and an omitted prime fails to install one. Either changes the parity of the number of accepted events and can therefore alter the bank assignment of later channels. Hidden microscopic state also carries between candidates. An exact trajectory means no incorrect acceptance or rejection anywhere through N, not merely a final event count that happens to match. At N=10^9, an exact trajectory emits all 50,847,534 primes and no composites.

### XXIV.2 The microscopic gate and its effective margins

The gate has an output spin e and six hidden spins h_j, each taking values -1 or +1. The two bank inputs are clamped during a candidate. An accepting reading has e=-1. The source is specified at dimensionless inverse temperature beta=1 by pair couplings to the hidden spins and explicit hidden fields. Let x=(e,r_1,r_2). Its energy is

$$E(e,h \mid r_1,r_2) = -\sum_{j=1}^{6} h_j[(Wx)_j + \eta_j].$$

Two hidden spins form pair modules of strength u, with rows (u,u,0) and (u,0,u). Four more form a triple module of strength t, with rows (-t,-t,t), (-t,t,-t), (t,-t,-t), (t,t,t). The hidden fields are eta=(0,0,t,t,t,t). Every microscopic interaction is a pair interaction or a declared one-spin field. The effective three-spin term appears after the hidden variables are summed. [A2, Section IIB; PE4, Sections 4–5]

Because the hidden spins are conditionally independent at fixed e and inputs, their partition factors multiply. The conditional free energy is therefore

$$F(e \mid r_1,r_2) = -\sum_j \log\{2\cosh[(Wx)_j+\eta_j]\} = C(r_1,r_2) + eH(r_1,r_2),$$

where C contains terms independent of the output. The output field and its two coefficients are

$$H = -a(r_1+r_2) + br_1r_2, \qquad a(u) = \tfrac{1}{2}\log\cosh(2u),$$

$$b(t) = \tfrac{1}{2}\log\left[\frac{\cosh(2t)^4}{\cosh(4t)}\right].$$

The equilibrium probability of an accepting output is 1/(1+exp(-2H)). With no returns, H=2a+b and the gate favors acceptance. With one returning bank, the two pair contributions cancel and H=-b. With both banks returning, H=b-2a. The positive magnitudes b and 2a-b are the two composite-blocking margins. They identify how strongly the source favors rejection in the two composite input sectors.

For the baseline u=4 and t=2, the coefficients are a=3.653426466 and b=2.960949986. The one-bank margin is

2.960949986; the two-bank margin is 4.345902946. The baseline consequently gives substantially weaker static protection to the one-bank sector. This is a source calculation, not an explanation inferred only from aggregate error counts.

### XXIV.3 Carrying microscopic state through the decision

The experiment retains all 2^7=128 free-spin configurations. A heat-bath generator Q_0 assigns a transition rate 1/(1+exp[E(z)-E(s)]) when z differs from s by one spin flip, with diagonal entries chosen so each generator column sums to zero. The attempt-rate multiplier is Gamma=512. A candidate's total exposure tau_k is split among r confirming readings, giving the transition matrix T=exp(Q_0 Gamma tau_k/r). Original production uses r=3.

Every reading samples the complete microscopic state. Its output must equal -1, and all three readings must accept before the candidate is emitted. If D projects onto states with e=-1 and v_s is the incoming state vector, the acceptance probability is

$$p_{\mathrm{accept}}(s) = \mathbf{1}^T (DT)^r v_s.$$

The final sampled state initializes the next candidate whether the candidate was accepted or rejected. All prescribed random readings are consumed even after an earlier reading rejects. This preserves both state history and random-stream alignment. Cubing the equilibrium acceptance probability would discard finite relaxation and correlations between readings; it is not the production calculation. [A2, Section IIC]

Two accumulation profiles use a common 35-phase duration sequence. The uniform profile has accumulation 1/4, and the rough profile varies its accumulation over the phases. Exposure is proportional to duration times the square root of one minus accumulation, with normalization matching total exposure per cycle. These are supplied dimensionless cadence profiles. Each seed uses shared random streams across profiles, so the profiles are paired observations of that seed.

### XXIV.4 Replay located a correctable microscopic weakness

Baseline seed 1000063 was exact through one billion under both profiles. The mechanism study compared each other seed's first incorrect decision with that seed at the same candidate. Because neither path had yet erred, the return histories and bank inputs were identical. Across 126 local comparisons, replacing the failing seed's three draws with the successful seed's draws corrected all 126 decisions while retaining the failing incoming state. Replacing the incoming state while keeping the failing draws left all 126 decisions incorrect. In 32 comparisons the incoming states already matched. The realized random triple was the local discriminator in these comparisons. [A2, Section IIIA; PE4, Section 9]

Of those first errors, 120 were composite emissions and six were missed primes. Every first composite emission had exactly one bank returning. This links the replay directly to the source field: the pair terms cancel in precisely that sector, leaving b as the blocking margin. The replay covered 230 saved chunks, 1.15 billion decisions and 264 local windows, checking replayed event bits and checksums. It resolved the local first-error mechanisms; it did not generate entire altered futures after each hypothetical intervention.

One missed-prime example makes the meaning of microscopic replay explicit. At prime 3,510,937, the failing seed and successful seed both enter state 103. The failing readings are 56,56,101; the successful readings are 56,56,56. The third failing draw lies in state 101's narrow rejecting interval, about 3.77Œ10^-11 wide. This is an identified numerical event in a retained dynamics, rather than an unexplained exception assigned to a special seed property.

### XXIV.5 The algebraically chosen successor

At fixed a>0, the smaller of the two blocking margins obeys min(b,2a-b)<=a, with equality precisely when b=a. The successor therefore kept the pair-module strength u=4 and raised the four triple rows and their hidden fields together until b(t)=a. This choice was fixed before successor trajectories were examined.

Let C=cosh8 and z=cosh(2t)^2. The balance equation becomes z^2=C(2z-1). Its physical positive-strength root gives

$$t_* = \tfrac{1}{2}\operatorname{arcosh}\sqrt{C+\sqrt{C^2-C}} = 2.3464897298984759\ldots.$$

The resulting a=b=3.6534264659876117 in the archived binary64 source output. Pair strength, temperature, cadence, three-reading rule, scheduler and matched random streams remain fixed. The triple strength and associated hidden fields increase by 17.32 percent. This establishes a maximum of the weaker static margin at fixed pair coefficient. The subsequent experiment determines the effect on full trajectories. [PE3]

The finite-state calculation shows both sides of the change. The one-bank three-reading false-emission probability falls from about 1.91Œ10^-8 to about 3.0Œ10^-10, approximately a factor of 63. The two-bank probability rises from 4.74Œ10^-12 to about 3.01Œ10^-10, approximately a factor of 64. The no-return sector also becomes more reliable. Equalizing the blocking margins transfers protection from the stronger composite sector to the weaker one; the frequency of those sectors matters to sequence survival. [A2, Table I]

### XXIV.6 Completed billion index results

All 64 seeds 1000000–1000063 were evaluated under both accumulation profiles using frozen baseline records for the matched comparison. Every successor trajectory completed through one billion. The final observed results are:

Per matched profile, 62 seeds improved, one was unchanged, and one worsened. All balanced matched seeds were exact through 10^8; the worst had two errors at 10^9. The complete matched error count fell from 569 to 40 across profiles. Balanced uniform and rough event sequences agreed for every matched seed pair. The profiles remain paired because they share streams; they do not turn 64 seeds into 128 independent seeds. [PE2]

The fresh continuation fixed the successor source and used all 16 seeds 10000000–10000015 with disjoint base and auxiliary random-seed ranges. There was no source retuning or

| Source and profile | Seeds | Composite emissions | Missed primes | Exact trajectories |
|---|---|---|---|---|
| Baseline uniform | 64 | 271 | 13 | 1 |
| Baseline rough | 64 | 272 | 13 | 1 |
| Balanced uniform | 64 | 17 | 3 | 47 |
| Balanced rough | 64 | 17 | 3 | 47 |
| Fresh balanced uniform | 16 | 2 | 1 | 13 |
| Fresh balanced rough | 16 | 2 | 1 | 13 |

seed screening. Thirteen of 16 were exact through one billion per profile; each of the other three had one error. The errors were composite emissions at 82,866,449 and 203,033,716 and a missed prime at 997,044,317. Fifteen of 16 were exact through 10ˆ8. These fresh streams test the frozen source beyond the matched roster; there is no corresponding fresh baseline ensemble. [A2, Section IIIC; PE2]

The significance is a completed progression from mechanism to source design to observed improvement. It does not rest on a selected exceptionally long successful seed: the matched roster and fresh roster are reported in full, with per-seed outcomes retained. The experiment measures stochastic sequence reliability in a constructed sieve. Its exact finite realizations do not issue arbitrary-input primality certificates or establish an unbounded guarantee; the prime-event sources do not establish a relation to FEOP.

### XXIV.7 How retained history changes a later input

A baseline profile pair exposes a concrete feedback path. Rough seed 1000053 accepts an extra composite at 312,596,611. That changes event-count parity. When prime 483,920,363 is later accepted, its channel enters a different bank than in the uniform partner. At 967,840,726=2Œ483,920,363, its return meets the period-2 return. The partners now present one-bank and two-bank inputs and make different decisions. The earlier erroneous period does not divide the later candidate; its influence passes through channel creation order and bank assignment. [A2, Section IIID; PE4, Section 10]

This example establishes a history-mediated change in a later gate input and decision. The average effect of post-error feedback on the ensemble error count is a separate quantity. Zero-error survival ends at the first error, before post-error amplification can occur. That distinction permits a clean survival calculation along the ideal input history without pretending that all later erroneous histories are identical.

### XXIV.8 Statistical and state aware explanations of the reliability scale

For a Poisson reference with mean error count lambda, the zero-error probability is exp(-lambda). The baseline uniform mean is 284/64=4.4375, giving 0.011825, compared with observed 1/64=0.015625. The matched balanced mean is 20/64=0.3125, giving 0.731616, compared with 47/64=0.734375. The balanced histogram has 47 zero-error, 14 one-error and three two-error trajectories. Fitted Poisson expectations for these counts are approximately 46.82,14.63 and 2.29. The sample Fano factors, variance divided by mean, are 0.9721 for uniform baseline, 0.9619 for rough baseline and 1.0032 for matched balanced. [A2, Section IVA and Table III]

These fitted comparisons describe the archived counts. A separate calculation in the attached expanded analysis predicts survival without fitting those counts. Along an error-free history, the ideal sieve determines each input. The microscopic correct-decision operator is A_n=(DT_n)ˆr for primes and T_nˆr-A_n for composites. Starting from the no-return Gibbs vector, the product of these substochastic operators gives the model's zero-error probability: incorrect paths are removed at their first wrong decision.

The analysis does not multiply one billion matrices directly. It calculates the minimum and maximum wrong-decision probabilities over all 128 incoming states for each input and cadence phase, then counts how often each phase/input combination occurs on the ideal history. Every conditional distribution given survival lies within those extrema. Products of their complementary probabilities bound survival without assuming independent successive decisions. Numerical implementation uses logarithms of 1-epsilon, with binary64 propagators.

The ideal-history census contains 50,847,534 no-return candidates, 129,835,417 bank-0-only candidates, 77,222,775 bank-1-only candidates and 742,094,273 both-bank candidates, totaling 999,999,999 decisions. The resulting uniform-profile numerical survival envelopes are [0.0158014,0.0158362] for baseline and [0.717107,0.717210] for balanced. The latter predicts about 45.90 exact cases out of 64 and 11.47 out of 16, compatible with the observed 47 and 13. This is a model-based explanation of the recorded reliability scale. The quoted envelopes cover incoming-state variation in the numerical model; they are not directed-interval certificates of exact real matrix exponentials. [A2, Section IVB–C and Table IV]

The census also explains the cost of weakening the two-bank margin: both-bank candidates are the largest sector. Their balanced contribution is about 0.224 of total hazard 0.332. A useful trajectory objective therefore weights local error risks by their actual opportunities, rather than maximizing the weakest static margin alone.

### XXIV.9 Resource comparisons make the design conclusion precise

Source strength must be defined before alternatives are compared. Summing absolute couplings gives B_J=4u+12t. Including the four hidden fields gives B_Jeta=4u+16t. Baseline budgets are (40,48); balanced budgets are (44.157877,53.543836), increases of 10.39 and 11.55 percent. The 17.32 percent figure refers specifically to t. These are dimensionless source-strength conventions; no device energy consumption was measured.

The attached analysis compares uniform strengthening of the original u:t=2:1 ratio with balancing under these stated bud-

gets. It also tests four or five confirming readings while dividing the same candidate exposure among them. The following are numerical predictions for the uniform profile, separate from the observed matched and fresh ensembles. [A2, Section V and Table V]

| Control | Confirming readings | Predicted zero error probability through 10^9 |
|---|---|---|
| Original baseline | 3 | 0.015801–0.015836 |
| Original balanced | 3 | 0.717107–0.717210 |
| Uniformly scaled baseline matched in coupling budget | 3 | 0.708104–0.708408 |
| Uniformly scaled baseline matched in coupling plus hidden field budget | 3 | 0.769661–0.769936 |
| Uniformly scaled baseline matched in a+b | 3 | 0.592883–0.593218 |
| Balanced source at original coupling plus hidden field budget | 3 | 0.005023–0.005027 |
| Baseline with unchanged total exposure | 4 | 0.773634–0.773842 |
| Baseline with unchanged total exposure | 5 | 0.720757–0.735827 |

Balancing retains a small advantage over uniform strengthening at matched coupling-only budget. Uniform strengthening performs better when hidden fields are counted in the strength budget. Four-reading confirmation is competitive at the original source strengths and unchanged exposure, but changes sampling and measurement costs. Five readings suppress false emissions further while increasing missed-prime opportunities; more confirmation does not monotonically improve the combined fidelity objective.

These comparisons place the observed intervention among explicit design alternatives. The original intervention achieved a large measured improvement with pair strength and cadence fixed. It does not establish balancing as a universal optimum at every resource convention. The controls identify which resources and error opportunities a subsequent optimization must price.

### XXIV.10 Validation and what the retained records support

The balanced source passed 1,080 source checks including all 280 profile/phase/input transition matrices against a separately implemented direct Padé exponential. After column normalization, the maximum entry discrepancy was 1.5821Œ10^-14; the initial unnormalized comparison and its column-mass drift are retained. Short-run qualification contributed 64,574 checks, including scalar CDF sampling, independent modulo-based return reconstruction and final-state/count/checksum agreement. [PE2–PE3]

The matched campaign checked all 25,600 chunk hashes and chains and scored all 127,999,999,872 candidate decisions against the independent sieve. It checked 1,572,864 retained microscopic prefix samples against their output decisions. The fresh continuation checked 6,400 chunks, scored 31,999,999,968 decisions and checked 393,216 prefix samples. Together these phases contain 159,999,999,840 scored decisions across paired profiles, a work count rather than a count of distinct integers or independent statistical replications. Whole-sequence scoring is extensive; complete independent replay of every microscopic transition is not claimed.

Native source construction, qualification, matched execution and fresh continuation retain their result, export and checkpoint receipts. The archived source and matched data comprise 26,220 files and 25,821,384,418 bytes; the fresh archive contains 6,535 files and 6,350,494,109 bytes. Final hash receipts record successful T500 custody. A standalone clean replay of baseline seed 1000063 through one billion matches its final microscopic state, event count and checksum while emitting exactly the reference prime sequence. [A2, Section VI; PE2]

The achieved result is a source-linked reliability method: construct a finite microscopic gate, retain its dynamics and event history, localize a failure, choose a coupling change from the algebra, and measure the complete successor ensemble. The finite successes, local derivations and resource comparisons remain distinct and jointly usable. Their scope is the specified stochastic source, digital recurrence, cadence and observed or numerically analyzed ensembles. Physical gate calibration, an average causal measurement of post-error feedback, and new matched ensembles for the alternative resource designs remain separate tasks.

SAM VOLUME III

# Prime Event Continuation and Retained Clocks

Chapter XXV

The prime-event construction was continued far beyond its matched billion-index reliability study. The retained local CPU trajectory completed 1,039,515,000,000 decisions with zero false events and zero missed primes before its intentional stop. A separate three-GPU continuation of the same seed reached eight trillion decisions with three false emissions and no missed primes. These are completed trajectories with their own source, checkpoint and scoring records. Their lengths and error counts describe those realized paths; the matched and fresh-seed ensemble comparisons remain the separate reliability evidence developed in the preceding chapter.[81, 82, 83, 84]

This chapter preserves the later developments of the expanded manuscript: measured transfer from exact graph responses, lower-leakage gates, a finite first-error bound, long raw execution, and exact clock-sensitive response. The event index, microscopic preparation and internal activity clock each retain information that later operations can use. Complete numerical and source scopes are stated with the result they concern.

## XXV.1 Source support and driven history before the feedback successor

The selective compiler preceding the seven-spin gate enumerated all 67 four-boundary sign subgroups. A single subgroup projection retains an interaction family closed under XOR: two retained disjoint pairs also permit their four-spin product, while overlapping retained pairs permit the remaining pair. Unequal nonnegative copies of gauge-transformed stars can remove unwanted companions. The completed search examined 6,560 nonempty copy vectors with multiplicities zero, one or two; its derivation extended the relevant minima to arbitrary nonnegative integer multiplicities in that dictionary. Independent local modules then permit the two pair strengths and the triple orientation to be selected separately. Character cancellation fixes the interaction support at every temperature. Matching its amplitude is a separate calculation at the selected temperature.[63]

For inverse temperature $\beta$, summing one hidden pair module gives $2\cosh[\beta u(e+r)]$. The four-hidden-spin triple module has partition factors $16\cosh(4\beta t)$ and $16\cosh(2\beta t)^4$ in its two visible-parity sectors. Thus the effective coefficients before setting $\beta = 1$ are

$$a(\beta,u) = \frac{\log\cosh(2\beta u)}{2\beta}, \qquad b(\beta,t) = \frac{4\log\cosh(2\beta t) - \log\cosh(4\beta t)}{2\beta}.$$

The original scale-two source has $b = 2.9609499856382886$, one-bank gap $2b = 5.9218999712765772$ and equilibrium false-output probability about 0.00267293946 per reading. The finite three-reading experiment retains the full state rather than treating those readings as independent equilibrium samples.[64]

The earlier driven experiment used externally specified return periods. For periods $5/7$, pair strength two and $\Gamma = 8$, balanced correctness was 71.737824% with uniform accumulation, 71.236764% with an exposure-matched rough profile, and 71.525318% when those same rough values were reversed. At fast relaxation the ordering difference fell below numerical comparison resolution. These calculations establish that equal total exposure does not determine a source's response to an ordered input sequence. The clamping and accumulation profiles were supplied drives; the experiment did not assign their work or backreaction.[66]

A retained feedback reconstruction shows the distinct role of event creation order. In one rough-profile path, an extra event at 312596611 changed event-count parity before prime 483920363, sending that prime's channel to a different bank. At $967840726 = 2 \times 483920363$, period-two and prime-period returns occupied one bank in the uniform path and separate banks in the rough path. One path emitted falsely while the other suppressed. The earlier extra event did not divide the later candidate; its effect passed through event order and bank assignment. The mechanism study replayed 230 selected five-million chunks and 264 detailed windows. The example establishes that feedback path, alongside the ensemble and first-error evidence in the preceding chapter.[68]

## XXV.2 A measured bridge from exact graph responses

The spin/prime dataset made the common interface explicit. Its completed first build joined 6,000 spin cases at sizes 1–1000, 91,620 boundary ground-response records, 183,240 certified thermal readouts and 183,272 interaction-coefficient records, together with the two historical gate variants and their observed outcomes. It preserved graph identity, port ordering, temperature, source pointers and aliases. The 1,056 seed/horizon records include repeated measurements; the database does not turn them into 1,056 independent runs. Full joint density tables were not expanded simply to make the join.[70]

Every indexed spin graph had zero local fields. All 91,620 odd-degree coefficient intervals contained zero, as required by global inversion. A direct odd triple could therefore not be extracted from an unchanged zero-field four-port source. The completed connection instead pins one port. For the original even four-port expansion, setting a boundary spin to $p$ yields

$$\beta F(e, r_1, r_2 \mid p) = \text{input terms} + e(d + ur_1 + vr_2 + wr_1r_2).$$

The four-body term becomes the visible triple $w$; the pair joining output to the pinned port becomes the output bias $d$. A declared local output field can cancel that bias at the specified

temperature. This is a concrete boundary condition and source modification. It explains how odd gate terms arise while the original zero-field inversion theorem remains intact.[65][70]

For the $N = 1000$ positive-fill packet at $\beta = 4/1000$, the selected pinned role and sign convention give dimensionless coefficients

$$d = u = v = -1.907980462097066,$$
$$w = +1.730671594798363.$$

After the specified bias cancellation, the triple-to-pair ratio is $b/a = 0.9070698726632545$. Transferring that ratio into the existing six-hidden-spin compiler at gate temperature $\beta = 1$, and fixing the mean pair coefficient to 3.6534264659876117, constructs a new 128-state model. Its finite-rate transition calculation does not inherit the dynamics or relaxation time of the original 1000-spin graph.[65]

The measured comparison covered both profiles, 35 phases and all 128 incoming states. Worst uniform one-bank false-emission probabilities were $1.91437 \times 10^{-8}$ for the original gate, $5.10148 \times 10^{-9}$ for the new $N = 62$ positive packet ratio, $2.60902 \times 10^{-9}$ for $N = 300$, and $2.31833 \times 10^{-9}$ for $N = 1000$. The last gives an 8.26-fold improvement over the historical gate. The already balanced design remained better at $3.04423 \times 10^{-10}$. The negative-fill $N = 1000$ packet supplied a contrasting ratio of approximately $8.66 \times 10^{-7}$ and one-bank leakage 0.124998814. Boundary response is therefore a useful design feature, but its sign and scale must actually be retained.[65]

Keeping the pinned output bias provides an instructive control. The unadjusted $N = 1000$ design suppresses one-bank false emissions to about $7.0 \times 10^{-19}$, but its worst no-bank missed-emission probability is $2.67 \times 10^{-6}$. Cancelling the bias brings the latter to $1.79 \times 10^{-9}$ while giving one-bank error $2.32 \times 10^{-9}$. Three-confirmation readout treats emission and suppression asymmetrically, so even a favorable smallest equilibrium margin is not a complete dynamic objective.

Entropy is part of this transfer. The retained thermal response reconstructs the pinned field through

$$\frac{\beta}{2}\Delta F = \frac{\beta}{2}\Delta\langle E\rangle - \frac{1}{2}\Delta S.$$

For the selected no-return packet field the actual decomposition is

$$3.6386520569 = 4.0074318192 - 0.3687797623.$$

Ground energies alone would miss the entropy contribution that changes the gate bias. The exact joint and boundary machinery of Chapter VI supplies the source identity, normalization and thermal information needed to make this connection reproducible.[65]

The source search examined 11,268 source/temperature pairs with 192 role/gauge choices, both with and without bias cancellation: 4,326,912 candidate evaluations. Twenty-four selected candidates received 96 Arb truth-table checks, and six compiled gate designs received finite-rate evaluation. No new closed-loop stream was generated by this connection experiment. Positive packet and frustrated signed-chain ratios at $N = 1000$ differ by only about $5.95 \times 10^{-7}$; their conditional errors are correspondingly almost equal. The completed result identifies boundary-response structure as the effective design input. It does not identify frustration itself as the cause of the measured improvement.[65]

### XXV.3 Uniform leakage improvement and the first-error theorem

A separate continuation increased the balanced source strength while retaining the same readout. At $u = 5$, the analytic balance gives $t = 2.8465622403941495$ and $a = b = 4.653426410750604$. Temperature remains one, attempt exposure remains $\Gamma = 512$, and there are still three readings. Under balance, the no-bank field is $3a$ and every blocked input field is $-a$. Independent equilibrium readings suggest an error scale near $e^{-6a}$; this explains the design direction, while the actual result uses the complete finite-rate sampler.[71][72]

All $35 \times 4 \times 128 = 17,920$ phase/bank/state inputs strictly improved. The worst blocked-input emission probability fell from $3.04422969 \times 10^{-10}$ to $8.0581104 \times 10^{-13}$; worst no-bank suppression fell from $9.06033914 \times 10^{-10}$ to $2.250089 \times 10^{-12}$. The minimum improvement across individual inputs exceeded 377.78. These are probabilities of the encoded CDF sampler. Its binary64 transition tables were independently compared with matrix exponentials, but exact finite-table counting has its own precise meaning.[71]

Let $L = 2^{53}$ and let $C_s(j)$ be a stored row CDF. The number of uniform-grid draws selecting state $j$ is

$$W_{sj} = \lceil LC_s(j)\rceil - \lceil LC_s(j-1)\rceil,$$

with endpoints clipped to the draw range. Each row sums to $L$. Set $V_0(s) = 1$ and

$$V_{k+1}(s) = \sum_{j:e(j)=-1} W_{sj}V_k(j).$$

Then the exact three-reading emission probability is $V_3(s)/L^3$, with denominator $2^{159}$. This counts retained-state paths through all three readings. It establishes exactness for the encoded sampler under independent uniform-grid draws, rather than asserting that rounded tables equal exact real exponentials.[72]

The conditional table also supports a useful finite first-error theorem. Suppose accepted history is exactly the primes before the next candidate, and fresh draw triples are independent uniform-grid draws conditional on the entire past. Let $c$ bound blocked-input emission over all phases and incoming states, and $d$ bound no-bank suppression. The induction above assigns a nonzero mask to every composite and mask zero to every prime, regardless of the microscopic preparation. Disjointly decomposing by the position of the first error gives

$$\Pr(\text{an error through } N) \le [N - 1 - \pi(N)]c + \pi(N)d,$$

with the bound capped at one. Dependence of incoming states across candidates is allowed. The hypothesis concerns new conditional draws; a fixed PCG64 seed is a realized deterministic algorithmic path. After an error, accepted periods can be wrong, so this proof is a first-error bound and does not bound every subsequent error in a corrupted feedback history.[72]

### XXV.4 Raw execution, exact continuation and resource evidence

The lower-leakage source was actually run without correction, learner, factoring veto or replacement of event history. The initial paired CPU pilot used seed 9368001 and an extra draw seed 10368001. Both original and $u5$ sources completed 100 million decisions with zero false emissions and zero missed primes, producing 5,761,455 events. Median complete rates were 8.540 and 8.487 million decisions per second respectively, including truth observation, compression and durable checkpoints. Six timing trials, three per source, executed 600 million decisions in total, reusing the same 100-million prefix for each source; they measure timing and reproducibility, not six independent accuracy exposures.[73]

An indexed-history integration then reached 200 million unique decisions for the $u5$ seed, with 11,078,937 accepted events and zero errors. Resident-reference, continuous-indexed and stopped-at-85-million/restored paths agreed at every checked chunk and endpoint. Six hundred million executed decisions across those paths represent the same 200-million unique prefix. The bounded 8 MiB decoded cache reduced resident memory and authenticated old history on demand. Payload was durably written before checkpoint publication. An uncommitted journal was deliberately quarantined and regenerated; this establishes the tested orphan-recovery case, rather than an unstated arbitrary-power-failure theorem. The complete checkpoint preserves actual source state, RNG and history index.[74]

A later RTX3090 implementation qualified the raw 128-state source, rather than applying the earlier eight-state GPU controller's checked-repair rule. Reusable RAM buffers and exact disjoint PCG64 subsequences gave 14.43–14.70 million decisions per second for the whole 100-million prefix, including CPU startup, initial GPU parity checking, observation, compression and durable writes. Later 90-million windows reached 17.60–17.89 million per second. The matched CPU reference, all checked chunk states and draw-stream endpoints agreed; stopping at 35 million and restoring reproduced uninterrupted execution. Repeated GPU qualification paths remain one unique 100-million source prefix. This was a qualified backend and bounded performance result, not a silently started continuous production run.[75]

The installed local resource successor preserved seed 9370001 at migration, both PCG64 states and all 23,336 prior history records. Its scientific configuration remained byte-equivalent while the execution cache and checkpoint index changed. Adjacent production windows showed a 10.3% complete-rate gain, from 9.772 to 10.776 million decisions per second; this observation includes overlap and contention rather than isolating a kernel change. The later 908.480-billion rollout observation was subsequently superseded by the completed local endpoint.[76][77]

The local CPU trajectory was intentionally stopped on 3 October 2026 at 13:15:20 UTC after 1,039,515,000,000 decisions, through candidate 1,039,515,000,001. It retained 39,036,981,351 events in 207,903 five-million chunks, with zero false events and zero missed primes. Its final 500-million chunk-loop window ran at 7.152970 million decisions per second; current throughput is zero because this is a stopped result. Final checks authenticated the frozen source, all 207 index blocks and positions/ranks of all indexed records, plus the final journal and complete terminal state/RNG agreement. That check did not replay the trillion-decision path or rehash every earlier journal. STOP remained present and continuation disabled.[81][82][83]

The same seed's separately continued three-GPU execution stopped at exactly 8,000,000,000,000 decisions on 3 October at 16:49:18 UTC, with 279,010,070,814 actual events, three false emissions and no missed primes. The false positions were 2,274,828,165,298, 2,379,059,006,438 and 2,687,315,963,345. Each error reproduced in a retained one-billion CPU replay with preceding history, microscopic state and both RNG streams. Every false emission fed back unchanged. The independent segmented sieve only scored decisions. The final hundred-billion interval averaged 155.410939 million decisions per second; this dated production result uses a later Blackwell configuration and does not replace the earlier RTX3090 pilot measurement. Its 1,600,000 committed records, final state and archive-chain custody were retained. Three GPUs separated speculative proposals, bank-mask construction from authenticated actual history, and ordered event extraction/scoring; CPUs retained ordered state consumption, exact random streams and the sole durable writer. Local and pod executions overlap on the same seed, so their prefixes are not summed as independent accuracy exposure. Both endpoints are intentional stops, not current running counters.[84][85][86]

### XXV.5 Native clocks retain information beyond accepted indices

The clock continuation returns to the balanced seven-spin source with its retained kinetic policy, $\beta = 1$, $\Gamma = 512$, $\eta = 3/8$, and actual source-time observation schedule. It does not treat the $u5$ production horizon as a new clock experiment. For the actual state $S_t$ and bank $b$, let $\lambda_b$ be the sum of all seven flip rates and $\kappa_b$ the output-spin flip rate. Two native activity coordinates are

$$\tau_{\mathrm{all}}(t) = \int_0^t \lambda_b(S_s)\,ds, \qquad \tau_{\mathrm{out}}(t) = \int_0^t \kappa_b(S_s)\,ds.$$

The compensated flip counts are martingales under the source process. Their usual expectation identity applies to the declared unconditioned or appropriately stopped horizon; it does not automatically hold after selection on a future accepted history.[62][78]

For any positive activity $a$, time change gives the column generator $Q^a = Q\,\mathrm{diag}(1/a)$ and speed measure proportional to $a\pi$, where $\pi$ is stationary for $Q$. Dividing by the output hazard makes the output flip rate exactly one from every microscopic state. Therefore, if $C_e$ groups only the visible output,

$$C_e Q^{\mathrm{out}} = H C_e, \qquad H = \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix}.$$

This is an exact two-state output quotient in output-activity time, for every bank and hidden preparation. Its apparent

simplicity does not remove the source's memory. The actual readings occur at random stopping coordinates $\tau_{\mathrm{out}}(t_j)$ inherited from source-time waits. Substituting equal clock waits changes the protocol. The all-spin activity clock does not even have this output-only quotient, because $\kappa/\lambda$ varies within each output sector.[78]

The full accepted clock law uses the tilted kernel

$$F_b(z,t) = \exp[(Q_b - z\,\mathrm{diag}\,\kappa_b)t].$$

The masked product $\mathbf{1}^\top DF_3DF_2DF_1\rho$ is the clock-weighted acceptance mass. Divide by its value at $z=0$ to obtain the conditional Laplace transform. Normalization and all three masks are part of the observable.

At bank zero, initial states 24 and 102 both show output $-1$ and have acceptance mass 0.999999999093968. Their conditional transforms at $z=1$ are nevertheless 0.9999998550562574 and 0.2252263286074421. Directed mean-clock bounds put the first mean below 0.000141473371361 and the second above 1.49064947744, a separation of at least 1.490508004067729 activity units. Bank-one Laplace values also differ, but its displayed mean intervals overlap and do not order those means. These are declared hidden preparations, not equal-frequency production populations.[62]

A finite raw-feedback experiment ran 32 fresh seeds through candidate 2048: 65,504 decisions, 23,256,750 actual flips and 54,138 output flips. Each seed accepted the same 309 events without errors, yet all 31 comparison seeds had different internal clock coordinates from the first. Doubling every rate while halving every laboratory wait reproduced states, readings, bank history and both clocks bitwise. The achieved clock is therefore a source-path quantity invariant under the common time-unit change. Its units are activity; a physical lapse or calibration in seconds was not supplied by this construction.[62]

### XXV.6 An accurate event quotient can omit clock memory

The next completed comparison retained output and hidden bits 5 and 6, giving eight coarse coordinates. Let $C$ group these blocks and $R$ lift each block using its declared conditional Gibbs weights. Then $CR = I_8$, $P = RC$ is a projection, and the omitted residual has dimension 120. The stored rate model is bound as exact binary rationals with diagonals rebuilt exactly; conditional weights are normalized within blocks. This does not silently assert that the stored Gibbs weights are stationary for the rounded generator.[79][80]

For a masked, possibly tilted interval operator $T$, decompose the full state as $x = Ru + v$. The exact update is

$$u' = CTRu + CTv,$$
$$v' = (I - P')T(Ru + v).$$

The signed injection $CTv$ is the source information lost by replacing the residual with zero. This is the same structural issue treated for spin preparation and selected readout in Chapter XVIII, now evaluated on the actual prime gate and clock.

For three readings, set $T_i = DF_b(z,d_i)$ using the actual source-time waits $d_i$, and write

$$G_{\mathrm{full}} = CT_3T_2T_1,$$
$$G_8 = CT_3PT_2PT_1P.$$

The exact difference is $G_{\mathrm{full}} - G_8 = \Delta_0 + \Delta_1 + \Delta_2$, with

$$\Delta_0 = CT_3PT_2PT_1(I-P),$$
$$\Delta_1 = CT_3PT_2(I-P)T_1,$$
$$\Delta_2 = CT_3(I-P)T_2T_1.$$

These are respectively the preparation replacement, first-reading reset and second-reading reset. Keeping the full first interval removes only $\Delta_0$. Adding all three signed operators restores the evaluated full response for every preparation; it is not an autonomous eight-state production algorithm.[80]

For the retained bank-zero protocol, the ordinary eight-coordinate endpoint discrepancy was at most $6.175\times10^{-20}$ in $L^1$, while the clock-weighted discrepancy was about 0.774772. States 102 and 120 share all retained coordinates, yet their conditional Laplace responses differ by 0.7747725033. Any initializer using only those coordinates must therefore miss at least one response by 0.3873862517. Keeping the exact first microscopic interval reduced the complete clock-weighted endpoint discrepancy to at most $2.636\times10^{-11}$ uniformly over all probability preparations. Event-only agreement had hidden a substantial preparation response.[79]

For bank one, the post-first-interval unnormalized discrepancy was at most $1.575\times10^{-14}$, but acceptance was only about $3.014\times10^{-10}$. Its conditioned bound is consequently $5.222\times10^{-5}$. Retaining this denominator prevents a rare accepted branch from appearing accurate merely because its total mass is small. Joint clock closure would require both a generator intertwining and a block-constant clock hazard; the latter fails in all eight blocks for all four banks. The exact two-state activity-clock output quotient remains valid because its random stopping map still carries that hidden information.[79][80]

### XXV.7 What has been established, and the next source-bound join

The completed connection is substantial: an exact thermal response supplies interaction features; a pairwise microscopic gate realizes them; its finite-rate probabilities guide a source change; raw feedback creates recurrence history; and complete masked operators expose timing information that accepted indices alone cannot recover. The large-source transfer has not yet received a matched closed-loop prime-stream comparison, and the full three-term clock correction has not yet become a smaller autonomous production representation.[65][71][79]

Two focused assignments follow directly. One is to select new boundary-derived designs against the actual three-reading objective across every bank sector, then compare the selected gates in matched raw feedback. The other is to preserve the joint clock/readout injection at actual bank changes, retaining clock-selected preparation and both later reset terms. The latest frustrated-spin source work supports the method of carrying actual residual directions, signed mixed terms and physical/background payments through a selected consumer. Its

28-spin or large-$N$ source, $N/\xi$ site-attempt clock and resolvent parameter remain distinct from this seven-spin gate and clock Laplace parameter. Reusing that method is an identifiable mathematical connection; assigning its numerical constants to Prime Event would change the source.[56]

SAM VOLUME III

# Jacobian Maps, Return and Retained History

Chapter XXVI

The Jacobian branch uses a polynomial map as a laboratory for the difference between returning endpoints and retained path history. The finite source results are preserved, together with corrections that prevent a chosen auxiliary grammar from becoming an unproved natural invariant. The proposed clean restart remains distinct from executed work.

The account below follows the chapter source as retained at the fixed 8 September 2026 SAMA revision. Engine names, installed interfaces and campaign status are historical source statements. They do not describe a freshly inspected October deployment. Later source amendments are distinguished where available; no later engine or campaign is inferred.

## XXVI.1 Overview

The Jacobian program asks how an exact polynomial map can return its target state while retaining a nontrivial history of sheets, braids, escape and projective completion. That is a mathematical and computational question, so JC1 and HF1–HF6TH belong in Volume III. Their earlier matter-volume-labelled files remain immutable provenance, but relocation changes placement and current interpretation rather than any executed result.

**Bounded status of the archived algebra.** The following formulas and fiber claims are source-reported archived algebra, not a globally verified theorem certified by this edition. Their printed coefficients lie in the rational numbers. If intended as a global polynomial map over the characteristic-zero complex field, an everywhere constant nonzero Jacobian together with distinct preimages of one target would amount to a counterexample to the Jacobian conjecture. This edition does not certify that global claim. The intended domain, source interpretation and independent mathematical adjudication remain unresolved here. The finite Exact Write endpoint/history comparison below is logically separate and does not require accepting a global counterexample.

The JC1 record reports the polynomial formulas

$$\begin{aligned} P(x,y,z) &= (1+xy)^3 z + y^2(1+xy)(4+3xy), \\ Q(x,y,z) &= y + 3x(1+xy)^2 z + 3xy^2(4+3xy), \\ R(x,y,z) &= 2x - 3x^2 y - x^3 z, \end{aligned}$$

and reports that their Jacobian determinant simplifies to

$$\det D(P,Q,R) = -2.$$

An exact elimination chart produces a cubic fiber polynomial, its resultant and discriminant. On the frozen slice $V = -16/27, R = 1$, one winding around $U = 0$ transposes two escaping sheets, while double winding returns the sheet permutation to the identity. The same winding programs compile through Complete Exact Write on one Theta18 axis. Four positive writes, four negative writes and eight positive writes all have zero compiled translation across all 262,144 addresses, yet their ordered W9 receipts retain four, four and eight events with different orientations and semantic hashes. JC1 passes 18/18; independent validation passes 9/9. Its exact classification is **The test result suggests strong contact with the concept.**

The portable result is narrower and stronger than an endpoint slogan: endpoint equality does not imply program-history equality. It does not by itself install JC1 monodromy, a quotient, a Newton field or a tensor/path-jet field as a matter payload.

HF1–HF6TH then form a valuable but historically scaffolded discovery chain: local reducers and a chosen `Z4^2` cell grammar; nonproper meridians and exact polynomial cells; finite versus projective sheet custody; path-dependent escape corrections; blinded path-jet discriminators; fresh-map transfer; Newton-polygon signatures; and a complete mixed-jet compiler. Every executed finding and correction remains in custody. The current clean restart does not assume that the HF2 torus, its generators, `H_SAM`, W9 or any carrier/matter meaning arises naturally from the polynomial map.

The restart therefore reverses the order of dependence: reconstruct the source algebra independently, derive invariants from actual fibers and the regular-value set, freeze algebraically generated loops, examine monodromy, properness and infinity, and only then reveal legacy HF observations. An optional SAM comparison requires an explicit compiler. A later matter application requires carrier compilation, reciprocal preservation, an identified matter consumer and a load-bearing controlled output change. None has executed. This future bridge is not standalone SAMA Volume IV, which is the RH program.

## XXVI.2 Placement, question and authority

### XXVI.2.1 Governing question

The clean governing question is:

> What global algebraic or topological invariant arises naturally from the exact JC1 polynomial map before an SLC rewrite grammar, W9 receipt, A-field-carrier role or matter interpretation is imported?

The reading order is

- exact polynomial source
- affine fibers and eliminant
- regular-value loops and monodromy
- nonproper and projective behavior
- exact endpoint-return/history distinction

- preserved HF clue deck
- unexecuted source-derived restart
- optional explicit SAM compiler
- optional four-join future matter-applicability test

#### XXVI.2.2 Current placement

H000211 relocates JC1/HF1–HF6TH from a former matter-volume interpretation into Volume III computation. The current rule is:

- executable result artifacts control their exact outputs;
- immutable history preserves the route, including superseded literal bridges and corrections;
- `volume_III/SAM_VOLUME_III_JACOBIAN_PROGRAM.md` controls current program routing; and
- live SLC authority remains frozen `SLCQ2-RZ`, whose delegated base installs no Jacobian or physical-matter result.

No historical file is deleted or retroactively reclassified. In particular, H000195/H000196 remain the record of a literal quotient/carrier bridge route, while H000197 and H000211 remove that bridge interpretation from current authority.

### XXVI.3 Exact JC1 source algebra

#### XXVI.3.1 Polynomial map and constant Jacobian

Let

$$F : \mathbb{A}^3 \to \mathbb{A}^3, \qquad F(x,y,z) = (P,Q,R)$$

with $P,Q,R$ as stated above. The derivative matrix is formed by differentiating each component with respect to ((x,y,z)). The archived record reports that polynomial expansion and factorization of this determinant cancel the variable-dependent terms and leave

$$\det DF = -2.$$

Within the archived statement, this determinant is nonzero at every affine source point and is read as absence of a local Jacobian singularity. The record then investigates escaping branches and properness. These global interpretations remain subject to the bounded-status qualification above; they are not newly certified by the finite history experiment.

#### XXVI.3.2 Elimination chart

JC1 uses

$$x = \frac{1}{m}, \qquad y = m(s-1), \qquad z = (5-3s-m)m^2.$$

Substitution into $P,Q,R$, followed by elimination of $m$, gives the exact resultant

$$\mathrm{Res}_m(U-Q, V-P) = s^3\Phi(U,V;s),$$

where

$$\Phi(U,V;s) = K(U,V)s^3 + (U^2 - 12V)s - 4V$$

and

$$K(U,V) = U^3 - U^2 - 18UV + 27V^2 + 16V.$$

The cubic discriminant factors as

$$\mathrm{Disc}_s\,\Phi = -4\big(U^3 - 18UV + 54V^2\big)^2 K(U,V).$$

Thus the two load-bearing target components are

$$D(U,V) = U^3 - 18UV + 54V^2$$

and $K(U,V)$. Their repeated and simple factors distinguish finite-root braid behavior from coefficient loss and projective escape.

#### XXVI.3.3 Exact calibration and loss fibers

In the archived record, the target $(P,Q,R) = (-1,1,-1)$ is assigned the eliminated $x$-fiber

$$(x-1)(3x+1)(3x+2),$$

with three exact sources

$$(1,-2,9), \quad (-1/3,4,27), \quad (-2/3,-1/2,-9/8).$$

The record reports that each maps to the same target and has Jacobian $-2$. This is precisely the source-reported claim whose global scope is not certified here.

At the coefficient-loss target

$$(U,V,R) = (0,-16/27,1),$$

the unique finite source is

$$(x,y,z) = (1/2,-8/3,16).$$

The eliminated $x$-fiber drops to $(2x-1)/2$. Two other cubic branches have moved to projective infinity rather than collided at an affine Jacobian zero.

### XXVI.4 Monodromy from the fiber

#### XXVI.4.1 Frozen slice

On $V = -16/27$,

$$K(U,-16/27) = \frac{U(3U^2 - 3U + 32)}{3}.$$

At $U = 0$, the cubic coefficient vanishes while

$$\Phi(0,-16/27;s) = \frac{64}{9}s + \frac{64}{27}.$$

One branch therefore tends to $s = -1/3$. Balancing the cubic and linear terms for the other two branches gives

$$s^2U \longrightarrow -\frac{2}{3}.$$

Those branches escape with magnitude proportional to $|U|^{-1/2}$.

#### XXVI.4.2 Loop continuation

Take a small circle

$$U(\theta) = \rho e^{i\theta}, \qquad V = -16/27, \qquad R = 1.$$

At each step, numerically continue the three roots of $\Phi$ by minimum matching to the preceding roots. A positive winding and a negative winding both return the finite branch and exchange the two escaping branches:

$$[0,1,2] \longmapsto [0,2,1].$$

The opposite loop orientation changes path direction but not the transposition. A double winding applies the transposition twice:

$$(1\ 2)^2 = e,$$

so the sheet permutation returns to `[0,1,2]`. The offset zero-winding control also returns identity. Maximum recorded fiber residuals stay below $5 \times 10^{-13}$ on the winding paths.

This chain separates target return, sheet permutation and path winding. Double winding can have identity permutation while still containing a nonempty ordered program.

### XXVI.5 Exact Write compilation and retained history

#### XXVI.5.1 Theta18 event map

Theta18 addresses form $\mathbb{Z}_4^9$ in packed two-bit coordinates. On axis $j$, one positive or negative event applies the exact signed $\mathbb{Z}_4$ step. For axis zero and initial address zero, the positive program visits

- 0 -> 1 -> 512 -> 513 -> 0

while the negative program visits

- 0 -> 513 -> 512 -> 1 -> 0.

Four equal signed steps are zero modulo four. Hence the programs

$$w_+ = (0,+1)^4, \qquad w_- = (0,-1)^4, \qquad w_{++} = (0,+1)^8$$

all compile to translation zero. Because the compiled action is a group translation, zero translation returns every one of the 262,144 input addresses, not only address zero. Sequential and compiled native execution agree on the full address set.

#### XXVI.5.2 Ordered receipt construction

Before endpoint collapse, retain for event $r$

$$(j_r, \varepsilon_r, a_{r-1}, a_r).$$

The JC1-to-W9 adapter assigns event $r$ to reciprocal channel $r$, with

$$L_r = \varepsilon_r, \qquad R_r = -\varepsilon_r.$$

The W9 common and directional components are

$$C_r = L_r + R_r = 0, \qquad A_r = L_r - R_r = 2\varepsilon_r.$$

Therefore the positive four-event receipt begins ((2,2,2,2)), the negative receipt begins ((-2,-2,-2,-2)), and the double receipt has eight positive entries. Their $L^1$ receipt norms are 8, 8 and 16, and their semantic hashes are distinct. Reciprocal reconstruction is

$$L_r = \frac{C_r + A_r}{2}, \qquad R_r = \frac{C_r - A_r}{2}.$$

The exact endpoint is the same; event count, sign, order and receipt remain different.

#### XXVI.5.3 JC1 result and boundary

The principal campaign passes 18/18 and independent validation passes 9/9. The result is classified:

**The test result suggests strong contact with the concept.**

The established portable statement is:

- endpoint compilation can erase which closed program occurred;
- an ordered event receipt can retain that distinction.

This directly joins the append-only reel of `SAMA-D000042`. It does not imply that a particular carrier, intersection or matter ledger consumes the full JC1 polynomial object.

### XXVI.6 Preserved HF1–HF6TH lineage

#### XXVI.6.1 HF1: reducer foundation

HF1 separates endpoint action in $\mathbb{Z}_4^9$ from an unwrapped signed lift in $\mathbb{Z}^9$. For an endpoint-returning word $w$,

$$\omega(w) = \widetilde{z}(w)/4 \in \mathbb{Z}^9.$$

It reduces only adjacent exact reciprocal writes with matching route, payload, account anchor and completion type. Four same-orientation writes retain a completed turn. Generic freeze passes 42/42, JC1 calibration 41/41 and independent reconstruction 54/54. A first 41/42 receipt is preserved: its validator mistook boundary prose for embedded map coefficients. The corrected validator checks actual coefficient markers. HF1's classification is **The test result suggests strong contact with the concept.**

### XXVI.6.2 HF2: chosen local `Z4^2` grammar

HF2 installs reciprocal bigons and signed commuting squares on a chosen (x,y) Z4^2 surface. Termination yields normal form

$$x^m y^n,$$

while endpoint compilation retains only $(m \bmod 4, n \bmod 4)$. The resulting history quotient is

$$H_{local} \cong \mathbb{Z}^2.$$

Among 87,381 words through length eight, 8,909 return the endpoint, 5,341 are locally null and 3,568 retain nonzero history. This falsifies the fixed target that every endpoint-returning word in that installed grammar is locally exact-null. The result does not establish that the chosen torus or its two generators are natural invariants of JC1.

### XXVI.6.3 HF3/HF4: nonproper obstruction and exact regular cell

HF3 follows the (K/x) meridian. Its radial filling reaches coefficient loss: one source stays finite while two escape with exponent $-1/2$. Positive, negative and double filling attempts remain NONPROPER_OBSTRUCTION; no exact cell is promoted. HF3 passes 28/28, separated promotion 24/24 and independent reconstruction 46/46, with strong contact at the obstruction-localization scope.

HF4 shows the obstruction is not a fixed-slice artifact by constructing the full nonproper hypersurface and its nonzero meridian linking. Separately, a (D/y) meridian has identity sheet permutation, pure-braid winding one and no escape. An exact Gaussian-rational source disk certifies boundary ((0,1)), so

$$\Lambda_{JC1}^{certified} = \langle (0,1) \rangle, \qquad \mathbb{Z}^2/\Lambda_{JC1}^{certified} \cong \mathbb{Z}.$$

Two implementation attempts are preserved: NumPy-boolean serialization and a closed-path final-sample tracking defect. Corrected execution passes 37/37; promotion and independent reconstructions pass their declared gates. The result suggests strong contact.

### XXVI.6.4 HF5 and HF5A: selected-sheet fill versus global sheet custody

HF5 constructs an exact source disk with $K$-winding one and certifies a regular finite sheet through the coefficient-loss center. Its first interpretation promoted ((1,0)), combined it with ((0,1)) and concluded a full lattice. H000195/H000196 preserve that literal bridge route and its exact calculations.

HF5A corrects the current interpretation: the finite-sheet disk does not fill all three typed sheets. Homogenizing

$$\Phi^h = KS^3 + BST^2 - 4VT^3$$

gives at projective infinity

$$\Phi^h|_{T=0} = KS^3.$$

At the loss center the local infinity equation is

$$\frac{64}{27} t^2 (t+3) = 0.$$

One finite root survives and a double root remains at infinity. The response $I_\infty(k,d) = k$ retains the (K/x) global-history class. Current preserved HF authority therefore returns to the rank-one certified lattice above. HF5A passes 26/26 with 22/22 independent reconstruction and suggests strong contact. Three pre-result field/dependency/alias faults remain preserved.

### XXVI.6.5 HF6 and the path-specific correction

HF6 emits anonymous finite-plus-projective receipts before attaching sector labels. It reconstructs:

- **`D/y.`** finite/infinity sheets: 3/0; infinity index: 0; permutation: identity; custody: GLOBALLY_DISCHARGED.
- **`K/x.`** finite/infinity sheets: 1/2; infinity index: 1; permutation: [0,2,1]; custody: MIXED.

Seven nonslice deformations preserve $K$-winding one, the transposition and the surviving class. Execution passes 19/19 and independent validation 24/24; the result suggests strong contact.

H000199 corrects one overgeneralized coefficient. The escape exponent remains $-1/2$, but the leading coefficient depends on the exact path jet. The radial route alone has $s^2U \to -2/3$; six other slopes and the HF5 disk have their own exact values. The correction passes 15/15 and 14/14 and does not change HF6's classification or quotient.

### XXVI.6.6 HF6T/HF6TF: blinded jet rule and transfer boundary

HF6T freezes the eliminant-conditioned rule

- a3(0)=0 and ord_path(a3)=1 and b1(0)!=0
- PROJECTIVE_CUSTODY_CANDIDATE
- otherwise
- FINITE_OR_UNRESOLVED

before reveal. It matches nine forward routes and nine reversals, reconstructs path-specific escape and preserves the failed pure-tensor partition. A first 18/19 run used the wrong determinant sign after swapping target coordinate order; correcting only that explicit sign gives 19/19. The result suggests strong contact for the JC1 path-jet discriminator, not a universal tensor law.

HF6TF freezes the rule unchanged on ten fresh maps. It returns four true positives, zero false positives, two false negatives and four true negatives. It is exact 7/7 on the admitted proper-cubic/simple-transverse domain, but it misses quadratic coefficient loss and simultaneous cubic/linear degeneration. Those misses remain evidence and motivate the next signature rather than a retroactive threshold change.

#### XXVI.6.7 HF6TG/HF6TH: Newton signature and mixed-jet compiler

For

$$\Phi_t(s) = \sum_j a_j(t) s^j, \qquad \nu_j = \operatorname{ord}_t a_j(t),$$

HF6TG takes the lower convex hull of $(j, \nu_j)$. Positive-slope edge width, slope and initial polynomial predict escaping-sheet count, exponent and regular cycle structure. On 15 fresh fibers, projective presence, sheet counts and coefficient winding close 15/15; 34/34 exponents and 10/10 regular cycle types reconstruct. A repeated edge remains `SINGULARITY_OBSTRUCTION`. The result suggests strong contact without installing a universal Newton law.

HF6TH then works in the triangular affine-path chart and freezes the complete mixed coefficient

$$\mathscr{T}_{rj} = \frac{1}{r!\,j!} D^{r+j} F^2(v^r, w^j).$$

For

$$G(x,y) = \sum_{p,j} c_{pj} x^p y^j,$$

exact Taylor expansion gives

$$\mathscr{T}_{rj} = \sum_{p \geq r} c_{pj} \binom{p}{r} u_0^{p-r}, \qquad \nu_j = \min\{r : \mathscr{T}_{rj} \neq 0\}.$$

The complete mixed jet therefore reconstructs the Newton signature in that chart. Known calibration closes 15/15; twelve fresh maps reconstruct all valuation tables, hulls, edge signatures and projective counts, with 40/40 escape exponents. This is a chart-specific exact compiler identity. Low-order compression, nontriangular maps, nonaffine paths and coordinate-invariant transfer remain open.

### XXVI.7 What the legacy lineage does and does not install

The HF lineage contains exact reducers, cells, obstructions, projective receipts, blind-transfer results and compiler identities. Its stages remain valid in their declared charts. The current program makes three distinctions:

1. the HF2 grammar is a constructed SLC object, not yet a source-derived JC1 invariant;
2. projective and mixed-jet findings do not constitute a universal tensor law;
3. historically matter-volume-labelled carrier supplements are preserved route artifacts, not current matter interfaces.

No current JC1-specific history, sheet, quotient, Newton or tensor field is consumed by AFC2–AFC4. No `q_A`, mass, SI energy or matter-generation relation is installed.

### XXVI.8 Clean restart: frozen method, no execution

#### XXVI.8.1 Premise firewall

JCR1 begins without the former matter-volume interpretation, A-field carrier, $q_A$, $A(r)$, W8/X1/W9, HF1 reducer roles, the HF2 torus, its generators, `H_SAM`, projective custody as information custody or tensor-to-matter interpretation. These remain available only after clean definitions are frozen.

#### XXVI.8.2 Stage sequence

- **JCR1-0.** Work: normalize exact source map, fields, coordinates, determinant, fibers, discriminant and assumptions; Required output or boundary: source normal form; no SLC encoding.
- **JCR1-1.** Work: independently reconstruct determinant, resultant, fibers, degree-loss set and homogenization; Required output or boundary: algebraic facts only; no history labels.
- **JCR1-2.** Work: derive candidates from regular-value complement, covering, monodromy, braid, homology, infinity divisors and cycles; Required output or boundary: typed invariant roster with domains and equivalences.
- **JCR1-3.** Work: freeze meridians, reversals, concatenations, null loops and map controls from algebraic components; Required output or boundary: clean loop campaign; failures retained.
- **JCR1-4.** Work: account for finite/infinity multiplicity, branch continuation, divisors and asymptotics; Required output or boundary: projective receipts with exact/numerical separation.
- **JCR1-5.** Work: ask the two-dimensional invariant question with all dimensions separated; Required output or boundary: no `Z4^2` substitution for polynomial geometry.
- **JCR1-6.** Work: reveal JC1/HF3–HF6TH observations one at a time; Required output or boundary: rediscovered/compatible/scaffold-dependent/conflicting/unresolved classifications.
- **JCR1-7.** Work: optionally compare stable invariant with SAM/SLC history; Required output or boundary: explicit compiler both ways or precise information loss.
- **JCR1-8.** Work: optionally test future matter applicability; Required output or boundary: four native bridge joins required.

#### XXVI.8.3 Candidate-invariant contract

Every candidate invariant must state its definition, domain, equivalence relation, composition law, base-point and coordinate dependence, loop-reversal and concatenation behavior, ordinary invertible-map value and JC1 value or unresolved status. A familiar shape is not enough to rename it `H_SAM`.

The legacy reveal occurs only after the clean source, definitions and first predictions are sealed. Agreement, disagreement, partial rediscovery and irrelevance are all retained. This prevents legacy success from becoming a hidden premise.

### XXVI.8.4 Current status

JCR1 is a draft for owner review and has not executed. It installs no new Volume III authority, invariant, tensor law, SLC training change, carrier payload or matter mapping. This chapter does not authorize its execution.

## XXVI.9 Future matter-bridge boundary

A future Jacobian object reaches a matter application only if one bridge-native campaign establishes all four joins:

1. an exact compiler from the source-derived invariant into a declared carrier payload;
2. exact preservation of that payload through reciprocal delivery;
3. an identified intersection or matter-ledger consumer that reads it; and
4. a load-bearing output change when the payload is altered under a controlled comparison.

The first two establish transport; the third establishes consumption; the fourth establishes consequence. A history hash that survives delivery without a consumer is not the completed bridge. A consumer that changes for an undeclared correlated field is not the controlled payload result.

No current JC1/HF field has completed this chain. Depth training, Q2 selection, W9 availability or A-field-carrier existence alone does not validate or promote a Jacobian-to-matter interpretation.

The source program called this a prospective `Volume IV` boundary. That exact wording remains in the immutable source and `SAMA-C000256-R001`. In this standalone atlas, Volume IV is the Riemann Hypothesis program. The matter bridge therefore remains unnumbered, unexecuted and outside the Volume IV dependency graph unless a later source-bound RH relation is separately established.

## XXVI.10 Established result and forward handoff

The established current result is the isolated JC1 algebra/monodromy and endpoint-history distinction, together with the preserved HF findings in their exact charts. The relocated program establishes where those results belong and which former interpretations are historical rather than current.

The next mathematical step is the unexecuted source-derived restart. Its first output is independent affine/projective reconstruction, not an SLC or matter bridge. Any later comparison must make the compiler and information loss explicit.

## XXVI.11 Sources and evidential scope

This chapter is assembled from the following fixed source manuscripts. Source-reported campaign outcomes retain their original scope; no new campaign was executed for this edition.

- Jacobian program and clean restart.

SAM VOLUME III

# Relational Composition and the Boundary of the Volume

Chapter XXVII

The closing chapter of the retained foundation changes evidential level. It preserves two source-originated proposals about composition and interaction, rather than presenting a newly validated universal law. Their role is to state the intended conceptual relation among the finite objects already developed. The volume can end with these questions explicit; they do not create an obligation to run another campaign.

The account below follows the chapter source as retained at the fixed 8 September 2026 SAMA revision. Engine names, installed interfaces and campaign status are historical source statements. They do not describe a freshly inspected October deployment. Later source amendments are distinguished where available; no later engine or campaign is inferred.

## XXVII.1 Orientation

SAM already contains three ingredients that are usually discussed at different scales: a nonzero accumulation field sourced by matter, an exact W8/X1/W9 write lifecycle with Theta carrier structure, and the accumulated effect of many stellar sources in galaxy halos. The Nonzero Amplification Principle identifies these as one recursive mechanism.

The central statement is:

> **Matter writes. The substrate carries. Nonzero writes accumulate.**

Matter does not have to occupy the route between two sources for information to cross that route. A source changes the substrate accumulation; the lifted substrate carries the source-conditioned write through matter-free space. The same composition that turns particle-scale writes into a body-scale presence can turn stellar writes into a galactic accumulation halo.

This document records an owner-originated conceptual bridge for continued mathematical development. It does not assign a test result classification.

## XXVII.2 Established inputs and new principle

### XXVII.2.1 Established inputs

The new bridge begins from established, separately typed SAM structures:

- the universal accumulation floor `A_0`;
- the spherical source accumulation lift `A(r)`;
- the closed-matter return to accumulation through source support and carrier compression;
- the W8/X1/W9 substrate-atom relation;
- Theta18 as a tensor carrier; and
- the Complete Exact Write computation surface.

These inputs retain their established definitions. The new concept does not replace `A_0`, redefine W9 as a separate object, turn Theta18 into matter, or replace the Exact Write with a continuum metaphor.

### XXVII.2.2 Owner-originated Nonzero Amplification Principle

> **Every nonzero constituent substrate write survives coherent composition, allowing particle-scale information to accumulate recursively into macroscopic and galactic accumulation structure.**

The principle adds a cross-scale interpretation to the established inputs:

1. a particle contributes a nonzero source-conditioned write;
2. many particle writes compose into the coherent write of a body;
3. the body-scale write is carried by matter-free substrate;
4. overlapping body writes establish local informational contact;
5. body writes compose again at larger scales; and
6. billions of stellar writes form the extended galactic accumulation halo.

## XXVII.3 The information-bearing substrate

Introduce provisional notation for the source-conditioned lift:

$$\Delta A_S(x) = A_{\text{total},S}(x) - A_0.$$

This notation distinguishes three states that must not be collapsed:

- **Matter present.** Meaning: A local material source participates in creating or changing the write..
- **Matter absent and `Delta A_S(x)>0`.** Meaning: The substrate remains source-conditioned and carries information without an intervening material messenger..
- **`Delta A_S(x)=0`, equivalently the source route has returned to `A=A_0`.** Meaning: The active source-conditioned connection is absent at that point..

The substrate is therefore not a passive geometric backdrop. Changes in `A` change the local write environment. In the owner-originated mechanism, those changes continually affect the W8/X1/W9 lifecycle and Theta carrier activity throughout the connected matter-free region.

Define the source-conditioned contact domain provisionally by

$$\Omega_S = \{x : \Delta A_S(x) > 0\}.$$

A route between two systems is in active source contact when it remains in the appropriate connected component of this lifted domain. A point at which the source-conditioned lift returns to zero supplies the proposed disconnection condition.

### XXVII.4 Exact composition rather than point-particle messaging

For a composite source S, write

$$\Delta A_S(x) = \mathscr{C}_S(\{\delta A_i(x) : i \in S\}),$$

where each constituent contribution is nonzero on its active route and mathcal C_S denotes the exact composition law.

The symbol mathcal C_S is deliberately not replaced by ordinary linear addition. SAM has an Exact Write, retained histories, typed carriers and directional computation. The correct composition law should be recovered or derived from that algebra so that cross terms, history and carrier structure remain intact.

The conceptual content is nevertheless exact enough to distinguish the mechanisms:

- an individual Earth particle does not send an isolated command to an individual Moon particle;
- all particle writes in Earth compose into an Earth-scale accumulated write;
- all particle writes in the Moon compose into a Moon-scale accumulated write; and
- their source-conditioned domains overlap in the matter-free substrate, where continuing changes in A, W8/X1/W9 and Theta provide local contact.

The Moon responds inside the Earth's accumulated substrate write while also contributing its own write to the same substrate.

### XXVII.5 One recursive bridge across three scales

#### XXVII.5.1 Quantum to macroscopic

The quantum bridge is the passage from exact local substrate transactions to a coherent composite source:

$$\{\text{particle Exact Writes}\} \xrightarrow{\mathscr{C}} \text{body-scale accumulated write.}$$

Object-level informational presence need not be introduced as an unrelated primitive if it is the coherent closure of the constituent writes.

#### XXVII.5.2 Earth to Moon

Earth and Moon participate in one shared lifted substrate. Their continuing internal particle activity changes their composite writes; those changes are carried through matter-free space by the substrate itself. The orbital response is therefore downstream of an information-bearing write process, not merely a description of the resulting path.

#### XXVII.5.3 Stars to galaxy

The galactic bridge applies the same composition recursively:

$$\{\text{stellar body writes}\} \xrightarrow{\mathscr{C}} \text{galaxy-scale accumulated halo.}$$

Each star contributes a nonzero effect at large distance. Billions of such contributions produce the extended galactic accumulation structure. The particle-to-body and star-to-galaxy steps are thus two applications of the same nonzero-preserving composition law.

### XXVII.6 Technical continuation

The next investigation begins with the exact algebra already present in SAM:

1. **Composition operator.** Identify mathcal C inside Complete Exact Write and SLC execution, including retained-history and directional terms.
2. **Nonzero preservation.** Determine the exact condition under which a constituent nonzero write survives composite closure.
3. **A-dependent lifecycle.** State the operator by which a local change in A changes W8/X1/W9 execution and Theta carrier activity.
4. **Contact domain.** Derive the connected A>A_0 route and the exact A=A_0 disconnection condition.
5. **Recursive closure.** Determine when a composite write is admitted as a source input to the same operation at the next scale.
6. **Existing SLC orbit.** Inspect whether the current particle-orbit emulator already contains part of this composition and transport law.
7. **Cross-scale comparison.** Express particle/body, Earth–Moon and star/galaxy examples through one shared receipt format.

This route is intended to discover the exact operator already implied by the combined SAM structures, rather than introduce separate quantum, orbital and galactic mechanisms.

### XXVII.7 Relational interaction and completed composition

### XXVII.8 Orientation

The first Nonzero Amplification document proposes that constituent substrate writes can compose recursively from particles to bodies and from stellar bodies to galaxies. The relational proposal adds the following intended correction: composition depends on how the constituents interact.

Two planets light-years apart and the same two planets closely orbiting have the same constituent inventory. They do not have the same reciprocal history, overlap, directional relation or completed substrate write. The nearby orbiting pair forms a coherent composite source; the distant pair approaches two separately carried source writes.

The central statement is:

**Nonzero amplification is relational. Interacting writes create completed information that is absent from an inventory-only sum.**

### XXVII.9 Inventory and interaction are differently typed

Let S be a composite system. Its source data includes at least:

- constituent writes {w_i};
- a typed active relation network R_S;
- retained reciprocal or directional history H_S;
- the local and route-dependent accumulation state; and
- the completed W8/X1/W9 and Theta carrier state.

The proposed completed composition is therefore written schematically as

$$\Delta A_S = \mathscr{C}_S(\{w_i\}, R_S, H_S).$$

An inventory-only control would instead evaluate

$$\Delta A_{\text{diag}} = \sum_i \delta A_i.$$

The difference

$$\mathscr{I}_S = \Delta A_S - \Delta A_{\text{diag}}$$

is a useful diagnostic representation of the interaction innovation. It does not define mathcal C_S as ordinary addition. The native composition must retain Exact Write types, histories, cross terms and completion.

### XXVII.10 Separation changes the completed write

For the same two constituents, consider two relational states.

#### XXVII.10.1 Distant state

The two source-conditioned writes are carried separately. Their active overlap and continuing reciprocal history are reduced, and a route returning to the universal floor supplies the proposed disconnection condition.

#### XXVII.10.2 Nearby orbiting state

Each source continually changes the accumulation environment of the other. Those changes alter the matter-free substrate between them, the local W8/X1/W9 lifecycle and Theta carrier/intersection activity. The pair therefore generates a continuing relational history and becomes a new composite source.

The distinction is not "two sources versus more sources." It is:

$$(w_1, w_2, R_{\text{distant}}, H_{\text{distant}}) \neq (w_1, w_2, R_{\text{orbit}}, H_{\text{orbit}}).$$

### XXVII.11 Many-body amplification

For many particles, the number of possible relations grows faster than the constituent count. Pair, cycle, face and higher closure information can therefore become load-bearing even when every individual particle write is small.

A schematic cluster display is

$$\Delta A_S \sim \sum_i A_i + \sum_{i<j} \mathscr{I}_{ij} + \sum_{i<j<k} \mathscr{I}_{ijk} + \cdots,$$

but the Exact Write composition owns the actual algebra. The display only shows what a spreadsheet-style constituent sum omits: interaction terms whose values depend on the completed relational state.

This gives Nonzero Amplification two inseparable aspects:

1. each constituent contributes a nonzero write; and
2. coherent interaction among those writes generates additional completed information.

### XXVII.12 ATOM3D consequence

ATOM3D has frequently represented candidate assemblies through counts, independent relation rows and separately scored receipt features. The current Li-6 action-frame frontier exposes the limit of that representation: complete W9P custody retains each relation but supplies no relation-to-frame identity, leaving 13,824 reciprocal frame assignments.

Relational Nonzero Amplification changes the unit evaluated by a successor. The unit should be the whole active relation network, with each relation read in the context of all co-active writes:

$$(e, f_e, R_S, H_S, \Delta A_S, \Theta_S) \longmapsto \mathscr{I}_e(S).$$

Here e is a relation, f_e its action frame and mathcal I_e(S) its context-dependent innovation inside the completed assembly. The same isolated edge placed in a different network need not produce the same innovation.

The direct successor is therefore an **interaction-aware composite-write receipt**, not another isolated edge selector. Candidate Li-6 assemblies can then be compared by exact completed closure while preserving the current bars against geometry fitting and opaque selection.

### XXVII.13 RH consequence

The corresponding RH distinction is between eventwise inventory and completed coupling. If W is assembled from event writes W_i, then

$$W^*W = \sum_i W_i^* W_i + \sum_{i \neq j} W_i^* W_j.$$

The off-diagonal products carry relational information. Current RH work already retains completed Gamma–prime–Robin coupling where eventwise positivity is insufficient, and its support extensions are expressed through transported carriers and entrant Schur innovations.

Relational Nonzero Amplification supplies a shared candidate origin for those cross terms: they are the mathematical receipt of coexisting source writes interacting inside one completed substrate history.

### XXVII.14 Technical continuation

1. Recover `mathcal C_S` from Complete Exact Write and current SLC receipts.
2. Compile whole-network interaction innovations without reducing relations to independent rows.
3. Compare identical inventories under distant, nearby, orbiting and relation-deleted configurations.
4. Determine which interaction data selects or reduces ATOM3D action frames.
5. Express the resulting operator as a completed Gram or Schur innovation.
6. Cross-walk that operator to the RH source-append architecture.

### XXVII.15 Sources and evidential scope

This chapter is assembled from the following fixed source manuscripts. Source-reported campaign outcomes retain their original scope; no new campaign was executed for this edition.

- Nonzero amplification quantum galactic bridge.
- Relational nonzero amplification interaction closure.

SAM VOLUME III

# Computational Interfaces Across the Three Volumes

Chapter XXVIII

The expanded computational record gives a concrete meaning to the handoff between the volumes. Volume I supplies substrate fields, propagation, clocks and physical records. Volume II supplies finite material sources, preparations, interactions and their receiving observables. Volume III supplies exact representations, executable transformations and retained results that can be passed to the next calculation. The interfaces are mathematical objects with declared maps, so a useful result can travel without changing what its symbols count or measure.[59, 60, 61]

## XXVIII.1 What crosses an interface

A sufficient computational handoff can be written schematically as

$$\mathscr{B} = (\mathscr{S}, \mu, \mathscr{O}, \mathscr{C}, \mathscr{R}, \mathscr{D}). \tag{XXVIII.1}$$

Here $\mathscr{S}$ identifies the source and preparation, $\mu$ its state law or metric, $\mathscr{O}$ the operator, $\mathscr{C}$ its clock convention, $\mathscr{R}$ the requested receiver and $\mathscr{D}$ the retained dependencies. This tuple is an editorial organization of the interfaces developed in the preceding chapters. The particular calculation supplies its actual entries; it is not an additional physical law.

For a joint spin table, $\mathscr{S}$ is the signed integer Hamiltonian with ordered ports, $\mu$ is the labelled configuration count and $\mathscr{R}$ might be a rational-field DOS or a crossing query. There need be no dynamical clock in that static request. For a Gibbs response, $\mu$ is the original Gibbs inner product, $\mathscr{O} = N(I-P)$ and the retained history scale is $N/\xi$ uniform site attempts. For a signed-Write trajectory, the source path produces calibrated elapsed intervals received by the full carrier dynamics. Chapters VI and XVIII give the corresponding implemented constructions.[43, 55, 56, 51]

This organization also explains how two calculations can share a native matrix operation while retaining different meanings. A Gram matrix may count states, measure a Gibbs residual, retain a completed arithmetic form or encode a receiver's observable directions. Its entries and metric establish the correspondence. Matching dimensions alone do not supply it.

## XXVIII.2 Volume I: source time, retained records and actual readouts

The signed-Write integral supplies an explicit source-to-clock map. It receives the actual phase state and path, retains orientation and waiting, and sends a certified duration interval into the unchanged receiver. Different source orders with equal endpoints then produce different complete histories. Native dependency preservation carries that distinction through fresh sessions and later readouts.[50, 51]

The common-receiver and directional reconstruction work develops the complementary direction: determine what history can be recovered from a declared observation set. A receiver with signed phase and axis occupancy resolves a larger class than either partial observation. A source construction and its inverse therefore meet through the actual measured or computed record, with preparation, direction and sampling specified by the task. The application chapters retain those complete observation contracts.

The prime-event construction supplies another direct clock interface. Its microscopic Ising gate accumulates all-spin and output-spin activity along retained source paths. The exact output quotient has a simple two-state generator in its own activity coordinate, while actual sampling occurs at source-dependent stopping times. Chapter XXV develops how the hidden state and those stopping times remain observable. That source clock, the integer candidate index and a physical seconds assignment retain separate roles.[62]

The resulting computational contribution to Volume I is specific: an identified source can produce a duration, ordered record or response function that a field or clock construction receives through a declared map. This lets accumulation, propagation and clock questions use the completed computational operands directly while keeping their own physical definitions.[59]

## XXVIII.3 Volume II: exact phase sources and complete material receivers

The phase-to-binary chart gives an immediate source correspondence. Its energy factor and aligned coordinate make the joint Ising table an executable phase-response object. Ordered boundaries retain which contact receives which sign. Conditional spectra and source constants remain available for thermal assembly; a ground-matched auxiliary interaction retains its own thermal factor.[44, 45, 60]

The later response work adds a second interface. Complete retained and hidden source functions generate Grams, Dirichlet forms and action residuals. A receiver carries the full signed mixed entries and the background terms that belong to that physical source. The latest edge calculation demonstrates a new source observable entering a sharper lower response through an explicit Schur extension. The source, clock and earlier maxima remain identifiable throughout.[56, 57]

The applied matter chapters show the same discipline in actual configurations: retain circulation ties, phase signs, incoming preparation and post-contact memory, then ask the declared readout question. Exact source selection, distinct-report selection and physical energy comparison are different operations. Each has its completed result and actual receiving interface. This gives Volume II reusable computational methods for source construction, history and response, with the material inventory preserved.[60]

### XXVIII.4 Volume III: several useful kinds of exact object

The volume now develops several related return types. A full integer DOS retains the number of configurations at each energy. A joint table adds magnetization and ordered boundary information. A layered ground certificate retains the minimum, its exact degeneracy, a witness and exhaustive accounting. A certified matrix operation returns enclosures and residual bounds. A retained method trace supplies its operation/dependency graph. A prime or factor certificate answers its specified arithmetic question. Their shared computational infrastructure makes them composable when a receiving map is supplied.[43, 47, 48]

This distinction is productive when choosing a method. A thermal response needs the energy distribution or a justified source integration. A ground-energy question can use a minimum-preserving reduction. An inverse-history question needs the observations and ambiguity classes left by the actual receiver. An arithmetic transition estimate needs its signed source and completed complementary terms. A source-clock calculation needs the original path and clock map. The complete object requested by the next step determines which retained result can serve it.

### XXVIII.5 The record carried into the next three-volume pass

The manuscript preserves the original twelve-chapter foundation and develops later completed work in separate chapters with an exact source register. Historical revisions remain attached to their results. Later installations, qualified methods and completed calculations have their own dated sources. Proposed successors and stopped or partial campaigns retain those states rather than being folded into completed totals.

That organization provides a usable starting point for the planned review of all three volumes. The next pass can follow a concrete chain: identify a receiving question, locate its existing source operands, determine the missing map or calculation, and select an applicable exact method. The present expansion supplies the completed work and its interfaces; the comparative selection of new research targets belongs to that subsequent review.

# Part 2

# The exact engine, GEN6 and the Duck, as installed on 25 October 2026

## Scope of this part

This volume is a source-bound account of the engine as it stands on 20 October 2026. Its purpose is to explain the machine, preserve the order in which it was built and the decisions that shaped it, state its capabilities and limits plainly, and show by example how the owner, the lead and the research peers work with it. A chapter's open items identify the boundary of the record; they are not conditions that must be met for the volume to be complete.

Four levels of statement are kept apart, as in Volume III. Installed facts (a version number, a file hash, a count of returns) are read from the repository and are exact. Procedures (how a run is prepared, what change control requires) are quoted from the owner's words or from the documents that record them. Readings of results (what a run showed) are the engineering-register summaries of the run reports, labelled as readings. Physical interpretation is not attempted here; it belongs to Volumes I–III and to the owner's decisions.

Three names recur. **Sean Brady** is the originator and director of the research; every change to the installed machine, every claim and every change of an idea's status is his. **The lead** (root) is the session that prepares a run, stays outside its window, and triages every outcome afterwards; since 17 October this has been Claude Code. **The peers** are research agents bound one each to a native controller during a run; in October they were GPT-6.1 Sol Ultra threads, then Codex sessions, then Claude-backed agents (Fable, then Opus by installed default). The native controllers, helpers and broker are CPU processes of the engine itself; they execute and keep receipts and author nothing.

SAM VOLUME III

# What the engine is for

Chapter I

The Substrate Accumulation Model asks for arithmetic that can be trusted after the fact. Its constructions are exact rational actions, histories of ordered events, logarithmic accounts, matrix exponentials and certified enclosures; its arguments move from a declared source to a readout through steps that each have to be reproducible, attributable and refusable. The engine exists to supply that kind of computation and nothing else. It does not interpret. It does not decide what is interesting. It computes what a declared job asks, in exact or certified-interval arithmetic, writes a receipt for every step, binds every step to the declared source, and refuses a request it cannot admit under its contracts.

Three properties follow from that purpose and recur through the volume.

**Exactness.** Scalars are integers or rational strings; where a quantity is irrational (a logarithm, an exponential, a square root), the engine returns a certified enclosure with exact lower and upper endpoints, at a declared working precision between 64 and 4096 bits. A comparison between two results is made on their enclosures; "identical" means identical endpoints, and "disjoint" means the enclosures do not overlap. Nothing in the engine rounds silently.

**Receipts and custody.** Every native operation returns a record with a content hash; every job's steps are appended to a session whose objects are content-addressed and whose root is committed atomically. A changed receipt fails custody. A run folder retains every submission, response, return, finding and ledger; a campaign folder in git retains a SHA-256 manifest of that bulk.

**Refusal.** Admission contracts are enforced before mathematics. A payload with an unknown field, a matrix past a limit, a coupling that must be positive but is zero, a construction whose source map differs from its task, a time outside a first-flight domain: each is a REFUSED outcome with its reason recorded, and a refusal is an outcome, not a failure of the machine. The research method (Chapter VI) builds on this: a peer may declare that a control is expected to be refused or to fail, and the engine labels the result as declared or against declaration.

The research question the engine serves was fixed by the owner in words that recur in the record: the Duck "doesn't answer questions, it gives ideas"; "you set the research target and just let it run, then we analyze the results"; and, after a run was framed as a test of the manuscript, "we are not 'testing' Volume I, we are trying to learn and this is not the way." The engine is a generator of exactly-computed material from which ideas are read; it is not a validator of the volumes.

SAM VOLUME III

# Lineage of the exact engine, August to October 2026

Chapter II

The engine's history is the SAM history: 2,748 immutable entries from 4 August to 20 October 2026, indexed chronologically. The lineage below follows the entries that changed what was installed.

**The Mersenne engine and SLC v1.0 (4–9 August).** The first SLC was a typed SAM/SLC branch of a public exact Mersenne-prime search (H000011, H000013). Through 5–9 August it ran the SLCMP production series (SLCMP02 through SLCMP3431, 795,316 public candidates by H000075 and the billion-exponent boundary by H000127), with an exact Lucas–Lehmer baseline, FFTW and AVX2 kernels and per-test custody. On 9 August the owner promoted SLCV1.0-H14F-T18 as the sole active revision (H000128) and the Exact Algebra of the Write became its native packed engine (H000132). The habit of exact arithmetic with full custody dates from this week.

**SLCV1.1 to SLCV2.2 and the depth library (10–14 August).** SLCV1.1 (H000142), SLCV1.2 with depth-9 training on 60,123,392 cases (H000143), SLCV1.3 with exact block depths 1–18 (H000207), then the graph-native state/history/custody successor SLCV2.0 (H000229), SLCV2.1 with depths to 27 (H000230) and a sparse fused D81 anchor as SLCV2.2 (H000231). On 14 August the owner established the co-authorship authority that the volumes still carry: Sean Brady as originator and conceptual director, OpenAI ChatGPT and Codex as AI research collaborators and co-authors (H000226). A correction of the same week (H000209) retracted an agent-manufactured scope and restored the installed revision: the pattern of recording an agent's drift and reverting it is as old as the engine.

**SLCV21R and the receipt Gram (15–17 August).** SLCV20R-D162 (H000233) and then SLCV21R with the AX1 successor program (H000243–H000253): exact 5,120-pair parity, the Starbreaker ratio interface, the event-local receipt Gram and its sign obstruction against the indefinite Weil targets, and the dense frustrated-spin N72 replay with bit-identical readback. SLCV21R was promoted sole current on 16 August (H000273) and later carried the direct Mersenne-prime Boolean through M127 and the adaptive cyclotomic routing through M521 (H000317–H000325). Volume III still names SLCV21R as the delegated arithmetic/application base.

**SLCQ2-RZ (23 August).** The frozen Q2 hybrid ordinal relative-quality engine was promoted sole current over the preserved SLCV21R base (H000551). Volume III's Chapter VII describes this: SLCQ2-RZ is the sole source-recorded default selection engine; it consumes the byte- identical base services and does not turn the earlier exact results into something else. From here the engine has two layers that it keeps today: a selector and control plane above, and an exact arithmetic runtime below.

**GEN2 and GEN3 (7–8 September).** The generation names begin here. SLC-GEN2-R4.1 (H001087) installed the Write foundation with 39 inherited operations as the sole global SLC. GEN3 began as a native history-learning campaign (H001070) and became the runtime that still executes everything: `gen3\_runtime.py`, "one learning and executing runtime" that owns source state, acquired relations, supported compositions, observation encounters, complete histories and exact logarithmic accounts. GEN3's contract for a positive source E defines the accounts that recur in every construction: delta = ln(E_after/E_before), U accumulates rises, D fall magnitudes, V = U + D, L = U  D = ln(E_final/E_initial), M the maximum excursion with every maximizing occurrence retained. Logs are exact positive-rational arguments with signs; arguments stay unfactored. Generation names GEN2 through GEN6 are operation-prefix lineages inside one engine, not folders.

**GEN4 and Furious Wildfowl (22 September).** GEN4 was a hardware-specific phase-one fork of GEN3 with a specialized domain, built on a rented pod (H001635). On the same day the owner named an exploratory process **Furious Wildfowl** and ran it for ten minutes (H001644): a research selector, a domain, a launcher whose Linux process name was `furiouswildfowl`. Its protocol (GEN4/furious_open1/PROTOCOL.md) is the seed of the Duck: generate equations, representations, methods and directions freely from the research web; then a brief second pass over every result recording one of four tags, possible follow-up, possible further reduction, no clear lead yet, needs inspection; then a run report with a per-result ledger. The tags are workflow tags, not scientific verdicts.

**GEN5: cooperating logarithmic kernels and live cooperation (8–9 October).** SLC-GEN5-R1, generation GEN5-COOPERATING-LOG1-20261008, installed twelve logarithmic opportunities as 28 mathematical operations and three Duck operations, 242 installed names, six functional kernels cooperating through one matrix algebra (H002493). The next day GEN5_LIVE_COOPERATION_V1 installed the single-writer durable discovery broker, four source-bound native process pools, revisioned causal tasks and a bridge to externally registered research agents (H002496). SLC-GEN5-R1 is still the sole global generation; everything since has been built beside it, not in place of it.

**GEN6: the one-principle core (11 October).** The GEN6 candidate series (H002571: paid original Write-to-flow, 350 qualified native calls) produced the private core `sam\_one\_principle` 0.10.0, carried byte-identical into the GEN6 Duck extension 1.0.0 on the same day the owner said "We are deploying GEN6 controllers not codex agents" (H002577): twelve local workers and six CPU helpers from one session, 132 directed greeting acknowledgments, 251 native operations on each ready menu. Core 0.10.1 and then 0.10.2 (source-clock integration, compiled operation access; author-

ity H002621) followed; 0.10.2 is the sealed core that every later version keeps unchanged.

**ironclad (17 October).** The owner carried the running machine out of the research repository into a dedicated private repository, re-qualified fresh, and named its first revision IRONCLAD-R1 with Duck 1.3.0 (H002709). Everything else became archive; nothing in ironclad points into it. From this day the engine has one installed identity (`INSTALLED.json`), a release tool that refuses to mint a revision when a sealed core module hash differs, a verifier, a qualification tool that refuses a record without a history row, and a change-control rule written by the owner. The 28 revisions since then (IRONCLAD-R1 to R28) are listed in Chapter V.

SAM VOLUME III

# The SLC today: SLC-GEN5-R1

Chapter III

The installed arithmetic engine is SLC-GEN5-R1 with its CE facade SLC-GEN5-CEV1-R1, selected through `CURRENT\_REVISION/REGISTRY.json` and `GENERATION.json`. The native runtime is `gen3\_runtime.py`; Python routes calculations and retains their authenticated records; the warm consumer is C++/GMP with exact FLINT and SymPy backends. The engine identities have not changed since 8 October; every ironclad release re-mints the selector hash chain from the tree and leaves the engine identity untouched.

## III.1 III.1 The session and its contract

Work enters through a `DomainSession`. The common payload is `\{"source\_binding":\\{...\},\"precision\_bits":\n,\"data":\\{...\}\}`. The source binding names the retained original source and reference; it is carried into every receipt and checked where accounts meet. Precision is 64 to 4096 bits. General matrices admit to dimension 2048; matrix logarithms to 512; specialized methods keep their own smaller ceilings (thermal compatibility algebra 128 and 16; finite protocols at most 128 states; polynomial factors at most 1024 of degree at most 128). A session commits immutable changed objects and one atomic root covering state, knowledge and readouts; a checkpoint authenticates the snapshot and complete closure; replication transfers only missing objects and requires a durable acknowledgement of the same root.

## III.2 III.2 The operation families used by the research

The capability menu lists about 250 operations. The research of October uses a small, stable subset, named here because the run reports cite them.

Two conventions matter for reading receipts. A word's `scale` factor may be a literal rational or a pointer to an earlier result, so an exact quantity computed in one step can scale a matrix in the next without leaving the engine. And a certified solve of a singular matrix returns COMPLETED with status INVERTIBILITY_UNRESOLVED, not FAILED; a control meant to break at a singularity must therefore be read by its status or declared REFUSED, a lesson recorded in Chapter IX.

## III.3 III.3 The mathematical library and the frozen releases

The engine carries a content-addressed mathematical library of 1,840 objects (`engines/SLC/mathematical\_library/objects`), checked by `./sam\verify`; the release tool refuses to mint when any object named by the index is missing or corrupt. Ten frozen SLC releases are read by live code through a pointer (`SLC/18\_SAM\_NATIVE\_QC/`). The domains (six, each with its `DOMAIN.json`) and the cooperation broker sit beside the engine. Runtime roots (`runs/`,

| family | operations | what they do |
|---|---|---|
| exact linear algebra | `GEN3\_COMPLEX\_PRODUCT`, `GEN3\_COMPLEX\_SOLVE`, `GEN3\_COMPLEX\_EXP`, `GEN3\_COMPLEX\_ADJOINT`, `GEN3\_COMPLEX\_RANK` | products, certified solves (status CERTIFIED_SOLVE, or INVERTIBILITY_UNRESOLVED on a singular matrix), matrix exponentials, adjoints and ranks on exact or interval matrices with declared bases |
| source words | `GEN3\_SOURCE\_OPERATOR\_WORD` | a small program over declared operators (`multiply`, `scale`, `add`, `subtract`, `transpose`, `identity`, `columns`) evaluated exactly, with per-column checks; payload exactly `\{name,\context,\source\_binding,\data\{operators,\program,\output\}\}`; matrices at most 64Œ64, at most 256 steps |
| logarithms and entropies | `GEN5\_LOG\_QUADRATIC\_FORM` (CERTIFIED_LOG), `GEN5\_LOG\_TRACE`, `GEN5\_ENTROPY` (CERTIFIED_ENTROPY, nats) | certified logarithmic readouts; the register ledgers of Chapter IX are built from these |
| modular dynamics | `GEN3\_MODULAR\_DYNAMICS\_GENERATOR`, `\_READOUT`, `\_MEMORY`, `GEN3\_MODULAR\_PROTOCOL\_COMPILE`, `\_APPLY`, `GEN3\_MODULAR\_PROTOCOL\_MEMORY\_SPLIT`, `GEN3\_MODULAR\_SPHERICAL\_HISTORY` | finite-state generators from declared energies, attempt rates and clocks; ordered observation protocols that propagate the full hidden state; projection memory kernels with lumpability diagnostics; the whole ordered spherical history with a horizon and a switching-work ledger |
| methods (recipes) | `GEN5\_METHOD\_ADMIT`, `GEN5\_METHOD\_COMPILE`, `GEN5\_METHOD\_EXECUTE` | a peer-authored method with its own checks, admitted once per controller, compiled against declared operands and executed with every check enforced |
| retrieval | `RESEARCH\_WEB\_SEARCH` | search of the run's immutable local index (a SQLite snapshot of the project's research web, never the public internet); findings carry query, source hash, excerpt and traversed path |

`state/`, `archive/`) are never committed; their contents are hashed into committed manifests.

SAM VOLUME III

# GEN6: the one-principle core and its adapter

Chapter IV

GEN6 is not a new arithmetic engine. It is a construction library, `sam\_one\_principle`, that compiles a declared SAM construction into a directed acyclic graph of native steps for the SLC to execute, and an adapter that executes that graph append-only in an authenticated session with no host mathematics. The library is the place where the volumes' constructions become jobs.

### IV.1 IV.1 The sealed core and its successors

Core 0.10.2 (authority H002621) is sealed: its directory `active\_core/sam\_one\_principle` is byte-identical to its freeze, its module hashes are in `CORE.json`, and the release tool refuses a revision in which any hash differs. The standing rule, written into the repository's instructions, is never to modify it; a change to the core is a new version in a new directory beside it. Two have been made:

- **0.11.0** (20 October, H002744, Duck 2.3.0 / IRONCLAD-R27): the Volume I pair. A `static\_profile` construction (a declared profile q(x) = 1 uš at two or more positions by rational rapidity u, both redshift readings, a light-path delay) and a derived `local\_metric\_clock` readout on the reservoir (candidate lapses q and q with the identity N_kinš N_coord = 0). The directory `core\_0\_11\_0/` is retained with its manifest.
- **0.11.1** (20 October, H002746, Duck 2.3.1 / IRONCLAD-R28): the feedback clock. A feedback's evolution time is the native end of its event history, taken by result pointer; a derived metric lapse N sets the coordinate time per source event to 1/N and scales the source block of the generator; the legacy chart is byte-identical to 0.10.2. Chapter IX recounts how this was specified, prototyped, tested and installed.

The library has 25 modules. Their first lines describe them: typed action/state composition over retained GEN5-native mathematics (`action\_state`); append-only source-bound native DAG execution without host mathematics (`adapter`); compilation of one rational common action to native exact coefficient and history calls (`common\_action`); the autonomous positive-action reservoir (`dynamical\_reservoir`); original J4 Write programs as native event and logarithmic histories (`events`); first-flight reciprocal source/field evolution from one positive action (`feedback`); the method of steps for the field's first return (`returning\_field`); the full 68/272 action with clocks and complete delayed histories (`full\_carrier`); joint field, history, resume and terminal constructions; paid events with complete native work accounting; portable typed-state custody; bound-state logarithmic reception; cold export and readback; the source clock; and the static profile. Mathematical operations belong to the session; the package's own docstring says it is "candidate orchestration".

### IV.2 IV.2 What "compile, do not compute" means

A construction module takes a parameter packet of exact rational strings (the `no\_floats` check refuses a float anywhere), validates it, and emits a job: a list of ordered steps, each a native operation with a payload whose inputs are either declared operators or pointers `\{"\$step":\id,\"path":\{[]...{]}\}` into earlier steps' results, each with explicit checks (`exact\_equal`, `truthy`, `native\_contract`, `interval\_contains`, `interval\_positive`), and a set of readouts with statuses. The adapter resolves pointers as results arrive, executes each step in the session, enforces each check, and writes `READOUTS.json`. The host never computes a number that reaches a readout: the common action's coefficients (_kin = /cš, K = /(2R),  = /(2c), M_eff = _kin R/6, = R/c, the ruler = L/c) are computed by a native word from the declared rationals and their reciprocal tokens, and the word's own residual columns certify that each reciprocal is exact.

Readout statuses are part of the contract. DERIVED_NATIVE must point directly at a native result; DERIVED_NATIVE_COMPOSITE may hold a structure of such pointers; SUPPLIED, ANALYTIC_CONTRACT, OPEN and INHERITED label what the construction declares rather than derives. A job whose readouts break these rules is refused before execution. The adapter's validation also requires every step to carry the common source binding and explicit check coverage.

### IV.3 IV.3 The constructions a run can use

Through the Duck a controller can apply: `gen6.events`, `gen6.coupled` (yielding a resumable state), `gen6.feedback`, `gen6.returning\_field`, `gen6.action\_flow` and `gen6.action\_continue` (typed action states on a store with parent checks), `gen6.paid\_event`, the reservoir (`gen6\_reservoir`), and since 0.11.0 `gen6.static\_profile`. Three carried constructions that bind the archived physical source inventory (`gen6.source\_clock`, `gen6.joint\_field`, `gen6.full\_carrier`) are installed hidden and shown only when a plan declares `show\_constructions`, with every operand checked against a committed source manifest; a mismatch is REFUSED with reason SOURCE_INVENTORY_MISMATCH. The owner's reason for hiding them is recorded: offering physical specifications to every run would let a controller apply them to a question that never declared them.

## IV.4 IV.4 The feedback clock, as an example of a construction's anatomy

The feedback construction (Volume I's reciprocal first flight: a source with metric W and port v coupled to one field coordinate a, with  = p,  = z + g a v,  = h/(2), h =  g vW(z  g a v)) is compiled as: a coefficient word (the common action, including the event-history times and `end`), a rank check on W, a certified log reference, a generator word G, an energy metric word B, a force row h, an energy certificate word whose residual GB + BG + hh/ must have rank 0, a scaled generator Gůt, a native exponential, the propagated state, the field force, an energy Gram and the complete energy account [E_source_initial, E_source_final, E_memory, E_outgoing], a reference energy and its reservoir account. Eighteen native calls; about two seconds on the T500. In 0.11.1 the factor t is either the supplied time (legacy chart) or the pointer to the coefficient word's `end` column (routed chart), and the source block of G is scaled by the declared lapse N. The energy certificate is what makes a change to the generator safe: the rank-0 check is evaluated natively on every run, and it held for every N tried.

SAM VOLUME III

# The Duck: name, lineage and the road to version 2

Chapter V

## V.1 V.1 From Furious Wildfowl to Duck (22 September – 1 October)

The process was named Furious Wildfowl on 22 September (H001644). On 27 September the owner named it "duck" for an RH continuation (H001847: "Sean Brady explicitly names Wildfowl 'duck' "; the engine and mode did not change). On 1 October it became the official name (H002089): keep the parent idea, elevate each selected result as a next-round target, add an organizer to rank findings; and the two-round cycle was fixed (H002090): a 300-second exploration window, an untimed brief review, a second 300-second window on the parent plus every selected subtarget, then documents. The protocol text insists on what the organizer may not do: ranking never deletes a promoted subtarget; review does not execute follow-ups; the parent stays fixed across rounds; a new cycle needs a new authorized run.

## V.2 V.2 Model peers, controllers, and the twelve-Duck recipe (9–10 October)

GEN5 live cooperation (H002496) made discovery delivery available within a round. On 10 October the owner authorized three groups of four actual controllers, each a persistent GPT-6.1 Sol peer accepting Ultra effort, bound to twelve installed Simple controllers with native workers, one broker and pool, root and three engineering assistants outside the roster (H002553, run `duck\_twelve\_isotope1`): 125 native tasks, 112 ok, a peer-chosen mechanical union of twenty second-round targets. The same day the owner authorized a rotating program of twelve-Duck broad discovery and four-Duck research across RH, Volume II, Volume I and closed spin (H002561), with the words "only use the frozen methodology". This recipe is the one the installed method restored on 19 October.

## V.3 V.3 Native controllers and the Codex era (11–17 October)

On 11 October the owner corrected course: "We are deploying GEN6 controllers not codex agents" (H002577). The GEN6 Duck extension deployed twelve native workers and six CPU helpers from one session with the private 0.10.0 core. On 13 October the owner asked Codex to choose a spin run's second round (H002603), and from 14 October campaign reports describe the constructions as "Codex research contributions" while crediting the native layer with execution. On 15 October the owner set the 6/3 topology, two concurrent sessions, two 180-second rounds (H002640). On 17 October the owner asked for the original division of responsibility to be restored, controller-owned methods and round-two selection with root doing brief review (H002694), and clarified "No researcher backends" (H002695); the native controllers, able only to pick catalogue operations, were recorded as NATIVE_SELECTION_UNDER_REPAIR. The owner's later reading of this period is recorded in the lead's memory: the 6/3 era produced fascinating results, then "the agent creeps in one step at a time"; Codex credited GEN6 with execution only.

## V.4 V.4 ironclad phases B to E (17–19 October)

The machine moved to ironclad on 17 October (IRONCLAD-R1, Duck 1.3.0). Phase C (gates 2–7, R2 to R10, Duck 1.4.0 to 1.11.0) built the native selector: a controller decision task kind and broker board; two modes, focused and research; the recipe route automated; GEN6 continuation and reservoir routes; 22 contract operations with a kind vocabulary; GEN6 constructions as a task kind; the carried hidden sources; topology as a per-run option (12/6 default; 6/3 profiles a and b that can run at once on disjoint CPU sets). Phase D gate 1 (R11, 19 October) added record enforcement: `.\sam\verify` warns when the live page does not cite the installed revision; `.\sam\qualify` refuses a record without a history row. Phase E (R12 to R14, Duck 1.12.0 to 1.14.0) made the controller loop native: each controller decides its own next calculation; root may be outside the window; round two was retired; controllers publish findings to one another.

## V.5 V.5 The peer recipe restored; Duck 2.0 (19 October)

On 19 October, after a record search found the 10 October recipe, the owner said "Restore it and test please" (H002729). Three revisions (R19 to R21, Duck 1.16.0 to 1.16.2) implemented peer-authored submissions bound to controllers with root outside the roster, `because` naming completed result keys, and PEER labels on returns. The restoration test used Claude-backed peers since the October GPT backends were not on the machine. The same evening the owner set one window of 900 seconds with no second round by default ("round two was from wildfowl, I am not sure it is needed"), asked that nothing be hidden from the menu, and said the last run's record "doesn't need to be the focus, it just needs to be there" (H002730). The lead moved the second-round discipline into the peers' briefing and documented the method-authoring route. Duck 2.0.0 (IRONCLAD-R22) promoted the method with its change control (`docs/DUCK\_2.0\_METHOD.md`).

## V.6 V.6 Versions 2.0.1 to 2.3.1 (20 October)

Each permanent change followed the owner's rule (Chapter VI.6): the phrase "Allow permanent changes" typed in the chat, a readback approved, a new version number, rollback to the previous tag verified in a worktree before install, a freeze of file hashes, a history entry, tag, commit and sync.

| version | revision | kind | what |
|---|---|---|---|
| 2.0.1 | R23 | PERMANENT | research peers on the Opus model by default, on two 900 s runs that matched or exceeded the Fable-peer run on every instrument number |
| 2.1.0 / 2.1.1 | R24 / R25 | TEMPORARY | result handles on every return for two FEOP runs; reverted to 2.0.1 after each; recovery tag kept |
| 2.2.0 | R26 | PERMANENT | five Duck-layer defects fixed in one release: pointer records carry `authored\_by`; research mode accepts an empty query list; memory-gate head-of-line bypass; second-return shape documented; `expected\_status` for declared controls |
| 2.3.0 | R27 | PERMANENT | core 0.11.0 beside the sealed 0.10.2: the Volume I pair; `common\_clock` on feedback refused with a message |
| 2.3.1 | R28 | PERMANENT | core 0.11.1: the feedback clock |

SAM VOLUME III

# The refined Duck method (2.3.1)

Chapter VI

The method as installed on 20 October, stated as procedure. It is the 10 October recipe, run natively, with the owner's two later rules (change control; conclusion of every run) and the lead's learning rule.

## VI.1 VI.1 Roles

**Owner.** Sets the research target in his words; approves every change to the installed machine; decides what is claimed and what an idea's status is. Nothing enters a volume without his word.

**Lead (root).** Writes the campaign: `TARGET.md` (the question and the material), `PEER\_BRIEFING.md` (one briefing shared by all peers), `CONSTRUCTIONS\_REFERENCE.md`, `RECIPE\_AUTHORING.md`, `TASK\_SHAPES.json`, the plan with its openings, and a provenance file hashing them. Prepares and deploys the run; spawns the peers once, each with the briefing path, the run path, its name and controller. Then stays outside the window: `root\_submissions: \"peers"` makes a root submission impossible, and the lead messages no peer. Reads the returns as they land and keeps a live capture in a scratchpad outside the campaign folder (a lesson from the first FEOP run, where a capture inside the folder was readable by peers). After finalization, triages every outcome, writes the run report, records the run in both repositories.

**Peers.** Research agents, one per controller (`question.peers\=\{NAME:\G6\_xx\}`), by installed default on the Opus model. A peer reads the briefing and the material, writes exact native tasks, and submits them for its own controller (`./sam\duck\submit\-\/-run\R\-\/-file\task.json\-\/-peer\NAME`); each task is recorded `authored\_by:\PEER\_NAME`. A peer reads every return in the run folder, its own and the others', and builds on them by `\$result`; it may author methods as recipes; it writes `NOTES.md` as it goes and `REPORT.md` with one line per result at the end. Peers have no assigned roles unless the target gives them; in the last three runs the briefing said "pick up whichever thread you see an opening in".

**Controllers, helpers, broker.** Twelve native controllers G6_01 to G6_12 on CPUs 2–7 (doubled), six helpers on CPUs 8–13, a coordinator on CPU 0 and a helper coordinator on CPU 1; one single-writer broker inside the coordinator. Each controller executes its peer's tasks in its own authenticated session and publishes findings; helpers run retrieval in research mode. They author nothing.

## VI.2 VI.2 The window

One admission window, 900 seconds by default, opened by `ROUND1\_START.json` and closed by `ROUND1\_ADMISSION\_CLOSED.json`; then automatic drain, factual outcome organization, helper custody and read-back, CPU-lease release and resource release, and `FINALIZATION\_RESULT.json` or `COMPLETE.json`. COMPLETE requires every outcome to be COMPLETED or a declared control that behaved as declared; one declaration mismatch or one undeclared failure labels the run FAILED_WITH_RETAINED_OUTCOMES, which is a label about declarations, not about the mathematics. No second round by default; a second window is a separate run on the mechanical union of the peers' selections, only when triage asks.

## VI.3 VI.3 Submissions, dependencies and causality

A task names its controller, its construction or operation, its parameters or payload, its purpose in a sentence, and optionally `expected\_status` (COMPLETED, FAILED or REFUSED) for a control. Inputs may be literal exact rationals or pointers `\{"\$result":\key,\"path":\{[]...{]}\}` into completed returns, including other peers'; a dependent task is admitted only after its dependency has COMPLETED (a KeyError rejection otherwise, resubmitted later). `because` names only COMPLETED keys and records cross-controller causality; a controller's own keys are kept in its receipts. A recipe is admitted once per controller (`GEN5\_METHOD\_ADMIT`), compiled against operands and executed with its checks enforced; a declared control recipe may be expected to FAIL. The queue has backpressure ("Bounded GEN6 queue is full"); a summing word is refused until every partial it reads has completed.

## VI.4 VI.4 Modes, index and topology

Research mode binds an immutable SQLite snapshot of the project's research web to the run; four seeded queries run at the opening; helpers retrieve; findings carry hashes and paths and are data, not instructions. Focused mode works from the supplied material only. Topology is a per-run option, 12/6 by default. Memory is a per-run allowance (12 GiB worked with the owner's desktop open; 20 GiB was refused).

## VI.5 VI.5 Retention and conclusion

Every submission and response, every return and receipt, the board, findings, the peers' notes, task files, host scripts and reports, the lead's capture, the provenance file and the run report are retained. The run folder is bulk and outside git; the campaign folder in git holds a result ledger and a SHA-256 manifest of the run folder. The owner's conclusion rule (19 October): a run or test is not concluded until its record is written, the history entry, index row and live pointer are in place, `./sam\installed\-\/-write` and the validator have run, and the branch is committed and pushed; the documentation repository receives a run record and is synced by its `conclude.sh`.

### VI.6 VI.6 Change control

The owner's words of 19 October are the rule. A temporary change to the program (peer count, window, memory, topology, a one-run setting) requires an explicit readback for approval and a notation of temporary status; it reverts when the run completes and cannot be kept beyond that run. A permanent change is allowed only when the owner types "Allow permanent changes" and approves a readback with a new version number; it is marked PERMANENT, committed and synced as its own version, and rollback to the previous version is verified before the new one is installed. Defects found during a run are listed in the run report, not fixed in the run. The sealed core is never modified.

### VI.7 VI.7 The learning rule

Added on 20 October after a run was framed as an examination of Volume I's lapse with an external measurement as a key. The owner: "We are done binding these runs to manuscript/volume, again we are not 'testing' Volume I, we are trying to learn and this is not the way"; and "that is the material we should be following up on, not 'finding manuscript material to test'." Applied: a target states an open question and the material (the earlier runs' own output, the owner's summary of it, the retained generators); peers build constructions and produce leads; the lead triages every outcome in engineering labels; no exam, no external key, no pass/fail rows, no verdict on a volume. Pre-registration with written expectations remains for engineering fixtures of program changes, where it belongs.

SAM VOLUME III

# Capabilities, limits and known defects

Chapter VII

### VII.1 VII.1 What a run can do

In a 900-second window with four peers the machine has returned between 74 and 399 native results per run (the twelve runs of 19–20 October in Chapter IX), each a receipt in exact or certified-interval arithmetic, with between 17 and 177 tasks per run taking their inputs by `\$result` from other results. Peers have authored and executed up to 12 recipes per run. Constructions compile in milliseconds and execute in about two seconds each; a 64Œ64 source word or a four-state protocol apply is of the same order. Fibering a residue system across several words and summing the partials by reference carried the FEOP bracket receipts far past the single-word ceiling within one window.

### VII.2 VII.2 Limits, as the peers met them

- A source word's matrices are at most 64Œ64 and its program at most 256 steps; `columns` needs at least two inputs; a `scale` factor is a literal or a pointer, never an expression. Fibering costs two passes through the queue per case and grows as pˆ(2k).
- A recipe has at most 32 parameters in 15 slots and 64 steps, no `add` step and no positivity predicate; returned steps need checks; an admitted method is controller-local.
- The feedback construction requires a positive coupling (a zero coupling is refused; the uncoupled source is a zero port), a time or a history inside the first-flight domain 0 t 2R/c, and at most 31 source directions.
- The 0.11.1 feedback clock takes one lapse for the whole history; a lapse that varies along a path cannot be expressed, and the ordered product must be built by hand (Chapter IX.6).
- A certified solve of a singular matrix is COMPLETED with INVERTIBILITY_UNRESOLVED; a rank of an exactly rank-deficient matrix may return RANK_UNRESOLVED. Controls meant to break there must be read by status.
- Research mode reads only the bound local index; when the index lacks the literature, the peers say so and the comparison is their reading.
- The peer harness sometimes hands back before finalization despite the briefing; it is recorded, not fixed.

### VII.3 VII.3 Known defects and their status

Of the six defects listed at the 2.0.0 promotion, five Ducklayer items were fixed in 2.2.0 and the `common\_clock` drop on feedback is refused with a message since 2.3.0. Open from that list: the protocol trace carries no per-stroke internal time; the recipe engine's inputs are scalar-only. Found since: the plan builder once rewrote a task's source binding without the copy inside the construction packet (refused at admission; fixed in the next builder); the 0.11.0 feedback lapse was a label with no dynamics (found by two peers independently, fixed in 0.11.1); the admission ledger keeps `because` for cross-controller links only (by design, documented). Every defect has a history entry or a run report line; none was fixed inside a run.

SAM VOLUME III

# Connections to Volumes I, II and III

Chapter VIII

## VIII.1 VIII.1 Volume I: substrate and accumulation

Volume I's accumulation a = 1  q and its clock readings are the common action's and the reservoir's readouts. The reservoir (dynamical_reservoir.py) is the autonomous positive action L = m eˆ š/2 +  š/2 with (0) = (0) = 0, whose all-time solution  = 2 log cosh(t), q = sechš(t), E_field = Eq, E_ = E(1 q) is evaluated natively at finite t with certified logs and solves; its 0.11.0 readout emits both candidate lapses, N_coord = q and N_kin = q, with their identity. The static profile (0.11.0) puts a declared q(x) at two or more positions with a light path, so that the two redshift readings and the delay are rows of one receipt. The runs of 19–20 October (Chapter IX.2 and IX.5) located Volume I's open question exactly: the catalogue carries both readings and nothing in it derives which is a resting clock's proper time; with the feedback clock of 0.11.1 a lapse acts on dynamics, and the identity found there (a lapse equals a rescale of the field inertia per source tick) says what construction would make the choice non-free: one that derives .

## VIII.2 VIII.2 Volume II: matter, spin sources and retained memory

Volume II's chapter on composition, thermal cancellation and intrinsic spin memory meets the engine in the modular dynamics family: finite generators from declared energies and clocks, protocols that propagate the full hidden state between ordered observations with no equilibrium reset, projection memory kernels with lumpability diagnostics, and the matched-present comparison p versus R C p that the spin_memory run implemented exactly on the declared 64-state source. The register ledgers of 20 October (Chapter IX.7 and IX.8) are built from GEN5_LOG_TRACE and GEN5_ENTROPY on native interval operands. What the volume's chapter would need from the engine is stated in the run reports and not yet built: a control Hamiltonian on the 64-state source, and the three-time tensor for SAM's own 54 preparations.

## VIII.3 VIII.3 Volume III: computation and reciprocal history

Volume III is the engine's own volume in the manuscript. Its Chapter II (SAM Language and exact program semantics) describes what Chapter III here calls the session contract: typed instruction tapes, exact state, deterministic receipts whose recomputed hash for a mutated object does not authorize it, predecessor custody as part of the chain. Its Chapter VII names SLCQ2-RZ as the sole source-recorded default selection engine over the delegated SLCV21R base, which is the two-layer structure of Chapter II and III here. Its Chapter IX, reciprocal history, is the native operation GEN3_MODULAR_SPHERICAL_HISTORY: ordered source increments at times t_i against a horizon , the whole free-field energy, drive, counterterm and switching work with a ledger whose residual must contain zero. The path-order results of 20 October (Chapter IX.6) are reciprocal history read in the clock constructions. Volume III's reader's contract, four levels of statement kept apart, is the contract this volume adopts.

SAM VOLUME III

# Collaborative work: nine runs and one release

Chapter IX

The examples below are drawn from the run reports of 19–20 October, after the peer method was restored. Each entry names the run, the peers, the numbers, what the constructions showed in the engineering register, what failed, and what the collaboration looked like. None is a claim.

## IX.1 IX.1 peer_restore_next (19 October, 22:00Z): the recipe works with Claude peers

Duck 1.16.2, four Claude-backed peers CLOCK, RULER, PROPAGATION, JOINT, one 900 s window, the method-authoring route documented for the first time. 133 returns (121 COMPLETED, 11 FAILED, 1 REFUSED); 110 submissions accepted, 4 rejected; 10 methods admitted, 30 compiled, 26 executed; every peer authored methods. CLOCK made a native division from a certified solve and authored the first ruler that consumes teeth (`ruler.teeth\_per\_tick`, `ruler.cadence\_identity` passing at 11 and failing by design at 10). The best idea of the three restoration runs was CLOCK's reading of JOINT's and RULER's operands by reference: the sechš and 1 xš laws of two earlier charts are endpoint readings of charts whose mean readings are the inverse pair tanh(x)/x and artanh(x)/x. RULER executed a method with three controllers' outputs in one chain and no declared metric. Failures: compiles binding the wrong reference, rerun; designed controls failing as designed; one refusal at the 2R/c chart. The owner set the Opus default two runs later on the strength of the instrument numbers.

## IX.2 IX.2 aq_identification and resting_clock (19 October, evening): locating Volume I's fork

Two 900 s runs on Duck 2.0.0 with Opus peers (TEMPORARY, read back, approved). aq_identification: 141 returns, 115 submissions accepted, 12 methods; the lead's own defect stated first in the report (two openings refused because the plan builder left a stale source map inside the construction packets; the peers rebuilt their operands within three minutes). Reading: the two accumulation laws that had seemed to disagree were the same function in different time coordinates. resting_clock: 108 returns, 85 submissions, all accepted, 9 methods, the reception construction used for the first time. All four peers reached the same structure independently: no construction in the catalogue decides between (q_e/q_r) and q_e/q_r because every exchange encloses both as separate rows of one construction; and three of them read from the source exactly what the catalogue lacked (`local\_metric\_clock` marked OPEN; the lapse forced to 1 in `feedback.py`; a two-position profile missing). That list became core 0.11.0.

## IX.3 IX.3 feop_bound_c02, its blind replication and its 2.0.1 control (20 October, 04:25Z onward)

The FEOP program under the reciprocal-history authority: exact local-factor recurrences and a remainder for a nested tuple family. Run 1 on Duck 2.1.1 (TEMPORARY result handles), peers LOCAL, CRT, BOUND, REVIEW: 74 returns, all COMPLETED; three peers converged on the reading that the candidate bound is the Selberg sieve with its standard remainder specialized to the family, with the main term a constant factor above the elementary count; the closed form D_g(L) = 1 (p1) _{hL} min(g,h) pˆ(2h) was found by LOCAL and CRT separately and certified on 41 and 64 rows; REVIEW withdrew its own "density against diameter" reading in its report. Then the owner asked whether the same run could be done without the knowledge it had gained: a blind replication in a work-tree at the pre-run commit with fresh peers, and a third run on the reverted 2.0.1 to make sure the temporary change had not outperformed the installed version. The three replicates gave a recurrence table; the exact remainder's leading constant c 1.44 against the candidate's 324 came from runs 2 and 3.

## IX.4 IX.4 feop_sidebyside and feop_bracket_fiber (20 October, 06:06Z and 06:49Z): past the 64-row limit without a program change

Side by side (Duck 2.0.1, no temporary change consumed): 210 returns, 198 COMPLETED, 12 FAILED first attempts that broke the source-word payload contract and were resubmitted; 177 of 201 peer task files took inputs by `\$result`; 693 sign checks passed in authored gate recipes; raw- controls reported 1 while completing. The bracket's steps were verified natively where the word allowed and their hypotheses located by controls (p Q breaks both; odd separation breaks one; a flat tuple breaks the other); the report's own scope limit said plainly that at p = 11 or k 2 nothing had a receipt. The owner then said "Try the recipe route for the matrix limit first": the fiber run (298 returns, 281 COMPLETED, 17 FAILED of which eight were RECIPE's declared controls) fibered the residue system across several words and summed the partials by reference, giving native receipts far past the ceiling within one window and answering the engine-release question in the negative. Its cost, two queue passes per case and growth as pˆ(2k), is in the report. The payload-contract and word-grammar failures of these two runs were written into the authoring guide for every later run.

## IX.5 IX.5 voli_lapse (20 October, 08:08Z): the run that changed the method

Duck 2.3.0 / core 0.11.0, peers READINGS, LAPSE, CONTROL, REVIEW: 148 returns, 142 COMPLETED, 4 REFUSED as declared, 1 FAILED as declared, 1 COMPLETED against declaration (a singular solve the engine reports as completed-unresolved). The constructions showed the two

readings' coefficients (1/(2q_e) with remainder 3aš/8; the coordinate ratio exactly (a_e  a_r)/q_e), the two charts meeting in q and in a = tanhš within 10 in both readings, the text's floor A = 1/(12) bracketed natively, and a map-independent ratio of light-path excess to redshift (1 for the coordinate reading, tending to 2 for the kinetic). Two peers found independently that the 0.11.0 feedback lapse was inert. The lead had framed the target as an examination against Gravity Probe A with pre-registered rows; during the window the owner retired that framing for all runs (Chapter VI.7). The run report records the readings and no verdict.

### IX.6 IX.6 explore_voli_threads_2 (20 October, 22:22Z): exploration from the owner's material

Duck 2.3.1 / core 0.11.1, peers ONE to FOUR without assigned roles, the owner's own summary of the earlier runs as the material. 285 returns, 273 COMPLETED; six undeclared refusals from a zero coupling (a documentation gap, fixed) and one recipe missing its source binding (its revision passed). ONE found, and verified entry by entry in the rational generator before exponentiation in five instances, that a feedback at lapse N over N source ticks equals the legacy chart at time N with  Nů: the q versus q fork is a choice of field inertia per tick, and nothing present derives . THREE built a Pythagorean profile so that both readings were exact and showed, with ONE, that ordered products over segments with different lapses are certified disjoint from their reverse (the 0.11.1 clock collapses a history to its end). TWO and FOUR priced the spin work's register: the history gain was 1/838 of the erasure floor of the one decision bit it needed. ONE's admitted recipe asserting order independence FAILED as declared, which is the engine saying what the peers said.

### IX.7 IX.7 spin_work_1 (20 October, 22:46Z): the gain follows the ceiling once the family is widened

Same installation, 399 returns, all COMPLETED, a clean COMPLETE. All four peers took the same open question from four sides. ONE: a current-only policy with the field strength tuned per present reading beats the earlier history policy by 0.039, 170 times its gain; the earlier advantage was a weak-benchmark effect. FOUR: a field on the hidden spin chosen from the visible record gives a history gain 42 times the earlier one, 96 % of it at dwell 4, converting 0.42–0.43 of the hidden information at separations 1 and 3/2 and peaking between 1 and 2 as the information ceiling does. THREE: a single quench with both fields converts 0.45; staircases of two and four stages convert 0.65 and 0.79, with the fraction bracketed by signs. TWO: the dwell route survives only below separation 1/2 and moves opposite to the ceiling; the hidden-field route follows it. Cross-controller agreement to 1e-13 on shared rows; THREE corrected its own first benchmark in the window and said so; ONE's two-field staircase did not finish natively and its total is host arithmetic, stated as such. The register still dominates the account.

### IX.8 IX.8 The feedback clock release (20 October): specification by the owner, prototype in the sandbox, fixture before install

The owner, after withdrawing a rollback: "I need a precise definition connecting lapse, event count, elapsed time and the feedback evolution time. A nonzero-history test demonstrating the intended effect of changing lapse. Simply replacing zero events with one is not enough: the propagator uses a separately supplied time. Side-by-side regression checks against 0.10.2 for unchanged feedback and reservoir inputs." The lead wrote the definition with its two conventions labelled, prototyped the compiler change in a sandbox copy of the core, executed the test matrix natively offline through the core's own adapter (N = 1, 4/5, 16/25 at the same end: pairwise disjoint; N = 1 routed through one and three intervals: identical to 0.10.2; the uncoupled source at N = 4/5: identical to 0.10.2 at the rescaled time; the energy identity exactly zero), and ran the regression (feedback seed: 16/16 readouts and 18/18 steps identical between 0.10.2 and 0.11.0; reservoir: 14/14 shared readouts). The owner adopted the conventions ("proceed with your recommendations"), the lead prepared core 0.11.1 on a branch with tests and a pre-registered fixture, verified rollback, and installed only after "Allow permanent changes, readback approved." The fixture then ran as a Duck run and every pre-registered relation held on the Duck returns. The same evening a peer found, in the new construction, the identity of IX.6 that says what the construction still cannot decide.

### IX.9 IX.9 What the collaboration looks like from inside a window

The pattern across these runs: a peer writes its expectation before a scan, submits, reads the return, and either builds on it by reference or reads the failure and resubmits; peers read one another's notes and returns and cite them; a peer that finds its own error says so in its report and the lead quotes it; the engine's refusals and declared-control labels are the arbiter of whether a control did what its author said. The lead never submits and never steers. Independence, in the sense the repository's rules require, is none: the peers are Claude instances reading the same code, and the run reports say so in every case.

SAM VOLUME III

# Lessons of method

Chapter X

Recorded as the lead learned them, in the order they were learned, each with the run that taught it.

1. **Ideas are not instructions; the Duck gives ideas.** Target in, lead outside, controllers and peers generate, brief all-outcome triage after; no decoys, no steering, no expected-outcome lists for research runs (owner, 19 October).
2. **Pre-register fixtures, not research conclusions.** Written expectations and hashes belong to the engineering fixture of a program change, where a run must come out as written; for a research run they become a key, and a key turns the Duck into a validator (voli_lapse).
3. **Runs learn from the material the runs produce.** The owner's summary of what earlier runs found is a better target than any passage of a volume; the volumes are material to read, not hypotheses to grade (20 October).
4. **Give both policies the same family.** A history-versus-current-only comparison is only a comparison when both have the same control family: strength, stages, hidden field. The 0.00023 gain of the spin work did not (spin_work_1).
5. **Blind replication is cheap and worth it.** A work-tree at the pre-run commit with fresh peers reproduces a run without the knowledge it gained; a third run on the reverted installed version shows whether a temporary change mattered (FEOP runs 2 and 3).
6. **Documentation is capability.** Nothing was hidden by a flag; the effective unlock of the method-authoring route was writing it down, and every payload-contract failure of one run became a line in the next run's authoring guide (peer_restore_next, sidebyside, fiber).
7. **Keep the lead's capture out of the peers' reach** until the close (FEOP run 1).
8. **Controls should report a break by sign or status, not by a declared FAILED**, because the engine labels singular solves and ranks as unresolved rather than failed (voli_lapse).
9. **Fix defects in releases, not in runs; never touch the sealed core; verify rollback before install; freeze; tag.** Every permanent change of 20 October followed this, and the one bulk-in-git slip of the day was taken out of the index and reported.
10. **Conclude every run and test the same way, before anything else**: record, history entry, index, live pointer, installed manifest, validator, commit, push, and the documentation repository's record (owner, 19 October).

SAM VOLUME III

# Status, limits and what is not decided

Chapter XI

**Installed on 20 October 2026, 23:00Z.** Duck 2.3.1, IRONCLAD-R28, core 0.11.1 beside 0.11.0 and the sealed 0.10.2; SLC-GEN5-R1; 156 tests passing; `./sam\verify` PASS with no warnings; 2,748 history entries; rollback target `duck-2.3.0` verified. One campaign is staged and not deployed; nothing is pending that the owner has not been told.

**What this volume does not claim.** That any engine result is a physical result. That the peers' readings are verified: within-run agreement is between Claude instances reading the same code, and the repository's independence rule counts none of it. That the engine decides the q versus q question, the spin work's closed-cycle account, or anything about the original 54 spin preparations; the run reports say what construction each would need. That the method is finished: the per-event ordered lapse, a construction deriving , a control Hamiltonian on the 64-state source, finite-dwell staircases and register reuse over repeated cycles are logged as open ideas awaiting the owner's word.

**Caveats on this text.** Dates and counts are read from the history index and run reports as of the writing; the engine-lineage chapter compresses roughly a thousand entries from August and September into a page and names only those that changed what was installed. The descriptions of the GPT-peer and Codex eras rest on the history entries and the lead's verified lineage notes, not on the lead's own observation. The owner may renumber or retitle this volume; it is a working manuscript, authored in Markdown with no PDF source, and its change log begins with this edition.

Directed by Sean Brady. Executed with Claude Code.

SAM VOLUME III

# Continuation to the 25 October 2026 state

Chapter XII

Sean Brady, 2026-10-25: "a modernized vol ii and vol iii updated with the latest spin/feop work, potentially add a 'part 2' specifically for SCL work and its implementation. The most recent version Claude has produced of Duck has been the most successful and consistent to date. A starting manuscript has been made for the SLC work already." This manuscript is that starting text; from this date it is Volume III, Part 2. The sections below continue Chapters VII, IX, X and XI to the installed state of 25 October 2026 in the same register: installed facts read from the repository, procedures quoted, readings labelled as readings, no physical interpretation. Directed by Sean Brady. Executed with Claude Code.

## XII.1 Chapter IX continued — the runs and releases of 21–25 October 2026

Twenty-one further runs and three releases since the edition of 20 October. Duck runs are 1500 s windows with four Opus peers and the lead outside unless stated; RH entries are domain experiments in the RH DomainSession (host enumeration, native transcendentals and solves) with one Duck verification window.

Releases between runs: **Duck 2.3.3** (IRONCLAD-R30; `expected\_output\_status`, `expected\_reason`, `supersedes`: a control can name the engine status it expects, the reason it expects, and a repaired attempt can supersede its own failed first attempt), **Duck 2.3.4** (IRONCLAD-R31; `expected\_failed\_check`: a control names the recipe step whose check must fail, and a generic failure phrase no longer passes as a declaration), **Core 0.11.3 / Duck 2.3.5** (above). Each followed the ğ13 procedure: readback document, branch, tests, pre-registered fixture, rollback verified, "Allow permanent changes" typed by Sean Brady, install, freeze, tag.

## XII.2 Chapter VII addendum — limits and defects found 21–25 October

- **The drain has no deadline.** After a window closes the Duck drains the running tasks with no time limit. In volii_li6_1 one exact 36Œ36 rank with entries near 10 held the drain for 49 minutes until the lead stopped the run (`./sam\duck\stop`), which labels the run STOPPED_INCOMPLETE, queued tasks DEFERRED and the running task INTERRUPTED. Logged as a Duck defect (ideas file); the process remedy proposed to the owner is three briefing lines: a size rule for exact tasks (depth 3, operands 10źš, else modular ranks or the host), a late-window rule (nothing heavy in the last three minutes), and the lead's drain clock (stop five minutes after close, recorded as an operator stop). Awaiting Sean Brady's agreement; the code is untouched.
- **Source words.** `GEN3\_SOURCE\_OPERATOR\_WORD` takes exactly the fields name, source_binding, context, data (operators, program, output) and refuses any other ("Supply exactly the source operation fields"); the output must carry the declared basis row count ("Word output basis differs"). Products of rational matrices are certified enclosures of width about 2źš, not exact; identities are read by `interval\_contains`. Ranks of rational matrices are exact integers.
- **Recipes.** Matrices inside a recipe are limited to 512 entries; larger objects go as single operations.
- **Peers' host tools.** sympy's `.rank()` on a 36-dimensional span with entries near 10źź did not finish in thirty minutes; exact Fraction elimination finishes in seconds. `pkill\-f\\textless{}script\textgreater{}` matches the shell that issues it; use `pgrep` and kill by pid. Both are process notes, not engine defects.
- **Lead slips, recorded as such.** Framing-table numbers typed rather than computed (voli_ceiling_5; the rule now: compute every framing number and host-check every framing identity before hashing); a briefing that claimed width-zero products (volii_li6_1; corrected after the offline precheck).
- **Closed since the edition.** The 2.3.3 and 2.3.4 fields close Chapter VII.2's note that controls must be read by status: a control now names the status, the reason or the failing check it expects, and the ledger counts the rest against the run.

## XII.3 Chapter X continued — lessons 11–18

11. **The verification form** (adopted 21 October, spin_verify_1): everything given, separate code, predictions first. A result stands when two implementations written from the rules alone agree to the enclosure width and the lead's pre-registered predictions are scored, failures included, in the report.
12. **Weights first** (rh_transition_3 to the window addenda): the receiver's weights decide the sign of a change before the arithmetic does. The first line of any chart is what it weights; a prediction is a prediction of a receiver's reading.
13. **Compute every framing number** (voli_ceiling_5). A number typed into a framing table is a number the peers will inherit; host-check every identity before the preregistration hash.
14. **A control's declared check must be reachable** before any step that could raise (voli_ceiling_3 and after); a control that breaks earlier is right for the wrong reason, and 2.3.4 now labels it AGAINST_DECLARATION.
15. **Supersedes is for first attempts**, never for a control, and must name the latest failed attempt (2.3.3; the ceilings).
16. **Precheck openings offline** through the core's own

| Date | Run or release | Record | Machine | Returns / status | What it showed (engineering register) |
|---|---|---|---|---|---|
| 21 Oct | spin_verify_1 | H002760 | Duck 2.3.4 | verification form adopted | five spin results verified to the enclosure width by three implementations; S2d's reversal stands; one account corrected |
| 21 Oct | feop_verify_1 | H002761 | Duck 2.3.4 | two code bases, predictions first | transfer operator re-derived four times, verified at p = 11…23 and joint moduli; hypotheses mapped; reconciles the two forms |
| 21 Oct | volii_matter_1 | H002762 | Duck 2.3.4 | one control against declaration (basis error) | `coupled` construction's first research use; the two recipients of a lapse separated natively |
| 22 Oct | volii_matter_2 | H002763 | Duck 2.3.4, staircase rev 2 | — | lapse orders invisible under the clock recipient, order-sensitive under the energy recipient |
| 22 Oct | volii_matter_3 | H002764 | Duck 2.3.4 | 250 returns, 0 late, 0 undeclared | field-plus-lapse total optimised natively per chain; THREE's front recipe |
| 22 Oct | volii_matter_4 | H002765 | Duck 2.3.4, front library | 235 / 0 / 0 | order 1's total lead 1.83–1.90e-4 on every family; front cross-checked to 1e-40 |
| 22 Oct | volii_matter_5 | H002766 | Duck 2.3.4, staircase rev 3 | 247 / 0 / 0 | every chain's continuous optimum certified a local maximum; the lead's mechanism fails host and native |
| 22 Oct | volii_matter_6 | H002767 | Duck 2.3.4 | 207 / 0 / 0 | the switched share at N = 1/4 above N = 1/2, witnessed; Sean Brady's four conventions as rules |
| 22 Oct | rh_transition_1 | H002768 | RH DomainSession | 1262 calls, 1185 checks | the original source's transition to 1367; two host implementations agree to 8e-80; published target to every digit |
| 23 Oct | rh_transition_2 (window) + native ledger + cloud replica | H002769 | Duck 2.3.4; GCE c3d-highcpu-180 | 218 returns, 0 late; 1673 calls, 881 checks | native exterior ledger PASS; four separate implementations agree; the replica proved exact by `./sam\verify` (now STOPPED) |
| 23 Oct | rh_transition_3 | H002770 | T500 + cloud | 5067 calls, 880 checks, identical enclosures | adaptive receiver reproduced; its rise structural; the sign belongs to the receiver's weights; four of five predictions failed |
| 23 Oct | rh_transition_4 | H002771 | T500, unattended | three PASSes | account 1367 1371 both receivers; the two-cut combiner form; storage covers the crossing by 1 + /m exactly |
| 24 Oct | rh_sweep_1371_1389 | H002772 | T500, 9.47 h unattended | fourteen PASSes, 44,066 calls | seven accounts to 1389, ledger seamless; sharp 8.66e-4 per account; flip at 1383 1385 as declared |
| 24 Oct | rh_sweep addenda | H002773–H002775 | host + native checks | — | the change is dP/dY Œ Y; the sign is the admission rule's choice; the adaptive sign a 2 % residual decided by the weights; standing prediction 1389 1391: sharp 8.6305e-4, adaptive +6.449e-6 |
| 24 Oct | voli_ceiling_1 | H002776 | Duck 2.3.4, 900 s | FAILED_WITH_RETAINED_OUTCOMES (one undeclared), 44 COMPLETED | four candidates for the 1/ road; the ceiling a floor beside the reservoir |
| 24 Oct | voli_ceiling_2 | H002777 | Duck 2.3.4, 900 s | FAILED_WITH_RETAINED_OUTCOMES 93 COMPLETED | the floor sight as a staircase; smallest lawful non-road ledger has four hidden stages |
| 24 Oct | voli_ceiling_3 | H002778 | Duck 2.3.4, 900 s | eleven methods | the regression gate as three native faces; lawfulness global, invisible to any finite jet |
| 24 Oct | **Core 0.11.3 / Duck 2.3.5 / IRONCLAD-R32** | H002779 | PERMANENT | 182 tests OK; verify PASS; rollback duck-2.3.4 verified; tag duck-2.3.5 | declared reservoir floor and the `line\_of\_sight` construction (derivation OPEN) |
| 24 Oct | voli_ceiling_4 | H002780 | Duck 2.3.5, 900 s | COMPLETE, 143 returns, 0 undeclared, 0 late | the two constructions are one object on coshš = eˆ{ds}; the delay is the accumulation gap |

adapter before deploying; a refused opening in the window costs the peers minutes and the lead a slip (volii_li6_1's word payload, found offline).

17. **Admit on your own controller; keep a running task count; read the board before duplicating.** The briefing lines that cut the duplicated submissions of the matter runs to zero.
18. **Size exact tasks for the window** (proposed after volii_li6_1, awaiting the owner): depth, operand size and timing rules in the briefing rather than a change to the drain.

### XII.4 Chapter XI — status at 25 October 2026

**Installed.** Duck 2.3.5, IRONCLAD-R32, core 0.11.3 beside 0.11.2, 0.11.1, 0.11.0 and the sealed 0.10.2 (`active\_core/`, never modified); SLC-GEN5-R1; 182 tests OK at the install; `./sam\verify` PASS; history entries through H002783 (2,373 index rows); rollback target `duck-2.3.4` verified; research peers on the Opus model by default. The Google Cloud replica ironclad-c3d-20261022 is STOPPED; session stores are archived to Google Drive by SHA manifest and removed locally. Nothing is deployed; nothing is pending that the owner has not been told, except his decision on the three process lines above.

**What changed in the method since the edition.** Controls declare what they expect in three ways and a peer can supersede its own failed attempt; the verification form is standard for reproductions; the recipe libraries (staircase, front, identities, gate faces) are admitted at the start of a run and consumed by `\$result`; runs are prepared from the owner's words and the volumes' own chapters as material, never as keys; the lead triages every outcome after the close and writes its own slips first.

**What this part does not claim** is unchanged from the edition of 20 October: no engine result is a physical result; within-run agreement between Claude instances is not independence; the engine decides none of the volumes' open joins (the lapse, , the floor, the count, the connector action, the common energy map, the all-scale bound), and every run report names the construction each would need.

Directed by Sean Brady. Executed with Claude Code.

## Record of sources (Part 2)

History entries (ironclad `SAM\_HISTORY/entries/`): H000011, H000013, H000075, H000127, H000128, H000132, H000142, H000143, H000207, H000209, H000226, H000229, H000230, H000231, H000233, H000243–H000253, H000273, H000317–H000325, H000551, H001070, H001087, H001635, H001644, H001847, H002086, H002089, H002090, H002493, H002496, H002553, H002561, H002571, H002577, H002603, H002621, H002640, H002694, H002695, H002709–H002718, H002729, H002730, H002734, H002736, H002743–H002748.

Installed documents (ironclad): `CLAUDE.md` ğğ1–14; `docs/DUCK\_2.0\_METHOD.md`; `docs/DUCK\_2.2\_RELEASE\_PLAN.md`; `docs/CORE\_0\_11\_PLAN.md`; `docs/CORE\_0\_11\_1\_PLAN.md`; `docs/ARCHITECTURE\_MAP.md`; `docs/PHASE\_B\_PLAN.md` to `PHASE\_E\_PLAN.md`; `SAM\_PROJECT/gen6\_duck\_t500/README.md`, `RESEARCH\_MODES.md`, `INTEGRATION\_SUCCESSOR.md`; `CURRENT\_REVISION/engines/SLC/README.md`, `CAPABILITIES.md`, `GEN5.md`; `qualification/RELEASES.json`.

Campaign run reports (ironclad `campaigns/`): peer_restore_20261019 (NEXT_RUN_REPORT.md), aq_identification_20261019, resting_clock_20261019, feop_bound_c02_20261020 (RUN_REPORT.md, rerun/, run3/), feop_sidebyside_20261020, feop_bracket_fiber_20261020, voli_lapse_20261020, explore_voli_threads_20261020, spin_work_20261020; the retained Codex campaign spin_memory_20261019; sandbox branch sandbox/feedback-lapse-20261020 (FEEDBACK_LAPSE_DEFINITION.md, RESULTS.md).

Archive (SAM_Research_Project, read-only): GEN4/furious_open1/PROTOCOL.md; GEN4/duck_process1/RUN_CYCLE.md; GEN4/duck_twelve_isotope1.

Documentation repository: `runs/20261020\_*/RECORD.md`; `decisions/DECISIONS.md`; `ideas/IDEAS.md`.

## Reproducibility and source register

The original twelve chapters retain their chapter-level links to the fixed SAMA source revision. Their complete supplied source manuscripts and provenance are included under `source_material/historical/volume_iii` in the companion package. The following numbered references support the new chapters. Each entry identifies an exact repository file copied into the source package; `SOURCE_REGISTER.json` records its byte count and SHA-256 identity. Reused aliases map to one entry for the same source file. These files supply the mathematical exposition and reported results; source reports identify the canonical larger native result archives and datasets.

**Chapter I source manuscripts.** N100 computational grammar; H14F grammar-to-work factorization. Both are fixed at the original SAMA source revision. Their chapter source note is collected here to avoid an isolated short page.

## Repository sources for the expansion


[1] Sean Brady with OpenAI ChatGPT and Codex. *Starbreaker J4 simulation on SLC-GEN2-R4*. SAM Research Project. Source: `SAM_REVIEW/campaigns/STARBREAKER_GEN2_R4_J4_SIMULATION1/RESULT.md`. Snapshot read 19 October 2026.

[2] Sean Brady with OpenAI ChatGPT and Codex. *Common reception identifies every declared native history*. SAM Research Project. Source: `SAM_REVIEW/campaigns/VOLUME_I_A3D41_COMMON_RECEIVER3/RESULT.md`. Snapshot read 19 October 2026.

[3] Sean Brady with OpenAI ChatGPT and Codex. *Why the two channels admit a local inverse*. SAM Research Project. Source: `SAM_REVIEW/campaigns/VOLUME_I_A3D41_COMMON_RECEIVER3/DERIVATION.md`. Snapshot read 19 October 2026.

[4] Sean Brady with OpenAI ChatGPT and Codex. *SLC GEN 2 information-bridge series: completed results*. SAM Research Project. Source: `SAM_REVIEW/campaigns/SLC_GEN2_INFORMATION_BRIDGE_TESTS1/RESULT.md`. Snapshot read 19 October 2026.

[5] Sean Brady with OpenAI ChatGPT and Codex. *First shared dynamics installation and SB/A3D41 pilots*. SAM Research Project. Source: `GEN4/dynamics_install1/REPORT.md`. Snapshot read 19 October 2026.

[6] Sean Brady with OpenAI ChatGPT and Codex. *Shared modular capability and declared finite-state dynamics contracts*. SAM Research Project. Source: `GEN4/dynamics_install1/GUIDE.md`. Snapshot read 19 October 2026.

[7] Sean Brady with OpenAI ChatGPT and Codex. *SB dynamics extension through N5000*. SAM Research Project. Source: `GEN4/sb_dynamics_extension1/REPORT.md`. Snapshot read 19 October 2026.

[8] Sean Brady with OpenAI ChatGPT and Codex. *Li-6 construction grammar, retained circulation and hidden support*. SAM Research Project. Source: `SAM_REVIEW/campaigns/A3D41_LI6_GRAMMAR_ASSEMBLY1/README.md`. Snapshot read 19 October 2026.

[9] Sean Brady with OpenAI ChatGPT and Codex. *GEN2 separates A3D41 tied Li-6 constructions*. SAM Research Project. Source: `SAM_REVIEW/campaigns/A3D41_GEN2_SIGNED_READOUT1/RESULT.md`. Snapshot read 19 October 2026.

[10] Sean Brady with OpenAI ChatGPT and Codex. *GEN3 native distinct-observation selection*. SAM Research Project. Source: `SAM_REVIEW/campaigns/GEN3_ATOM3D_DISTINCT_SELECTOR1/RESULT.md`. Snapshot read 19 October 2026.

[11] Sean Brady with OpenAI ChatGPT and Codex. *A3D41 memory-preserving parity qualification*. SAM Research Project. Source: `GEN4/a3d_memory1/REPORT.md`. Snapshot read 19 October 2026.

[12] Sean Brady with OpenAI ChatGPT and Codex. *Exact spin-reversal propagation with measurement memory*. SAM Research Project. Source: `GEN4/a3d_memory1/DERIVATION.md`. Snapshot read 19 October 2026.

[13] Sean Brady with OpenAI ChatGPT and Codex. *GEN3 tau semantic and training-method learning pilot*. SAM Research Project. Source: `SAM_REVIEW/campaigns/TAU_GEN3_SEMANTIC_LEARNING1/RESULT.md`. Snapshot read 19 October 2026.

[14] Sean Brady with OpenAI ChatGPT and Codex. *Current tau reconstruction application and storage/status authority*. SAM Research Project. Source: `SAM_LIVE/09_TAU_RECO_CURRENT.md`. Snapshot read 19 October 2026.

[15] Sean Brady with OpenAI ChatGPT and Codex. *Exact Type II reduction and completed directed source production*. SAM Research Project. Source: `GEN4/rh_typeii_prefix1/REPORT.md`. Snapshot read 19 October 2026.

[16] Sean Brady with OpenAI ChatGPT and Codex. *Exact signed interval-prefix evaluator with strict moving integer boundaries*. SAM Research Project. Source: `GEN4/rh_typeii_prefix1/DERIVATION.md`. Snapshot read 19 October 2026.

[17] Sean Brady with OpenAI ChatGPT and Codex. *Common-grid Type II covariance and explicit all-real geometric payments*. SAM Research Project. Source: `GEN4/rh_typeii_block_transport1/REPORT.md`. Snapshot read 19 October 2026.

[18] Sean Brady with OpenAI ChatGPT and Codex. *Near-critical all-q payments and precise remaining source target*. SAM Research Project. Source: `GEN4/rh_subpower_gen4_review1/REPORT.md`. Snapshot read 19 October 2026.

[19] Sean Brady with OpenAI ChatGPT and Codex. *Qualified ordered Gram forest, complete source cuts and measured arithmetic scaling*. SAM Research Project. Source: `GEN4/rh_gram_tree_pod1/REPORT.md`. Snapshot read 19 October 2026.

[20] Sean Brady with OpenAI ChatGPT and Codex. *Ordered source aggregate, full prime cocycle, logarithmic credit and cross-block task*. SAM Research Project. Source: `GEN4/rh_gram_tree_pod1/METHOD.md`. Snapshot read 19 October 2026.

[21] Sean Brady with OpenAI ChatGPT and Codex. *Original complete RH crossing, actual signed payment and selected shell derivative*. SAM Research Project. Source: GEN4/rh_source_action_residual1_20261015/REPORT.md. Snapshot read 19 October 2026.

[22] Sean Brady with OpenAI ChatGPT and Codex. *Source-zero quotient, complete exterior square and explicit crossing storage*. SAM Research Project. Source: GEN4/rh_source_action_residual1_20261015/DERIVATION.md. Snapshot read 19 October 2026.

[23] Sean Brady with OpenAI ChatGPT and Codex. *Completed compensated-reference, cycle-allocation and finite boundary-price recipients*. SAM Research Project. Source: GEN4/duck_rh_a0_cycle_price3/REPORT.md. Snapshot read 19 October 2026.

[24] Sean Brady with OpenAI ChatGPT and Codex. *Installed exact certificate validation, known-evidence reuse and bulk elimination*. SAM Research Project. Source: SAM_REVIEW/campaigns/MP_GEN2_R4_DOMAIN1/RESULT.md. Snapshot read 19 October 2026.

[25] Sean Brady with OpenAI ChatGPT and Codex. *MP source-to-native mapping, canonical proof reuse and certified-F history*. SAM Research Project. Source: SAM_REVIEW/campaigns/MP_GEN2_R4_DOMAIN1/CAMPAIGN_MAP.md. Snapshot read 19 October 2026.

[26] Sean Brady with OpenAI ChatGPT and Codex. *B300 completed cost-learning loop, quiet-pod regression and owner-paused handoff*. SAM Research Project. Source: SAM_REVIEW/campaigns/MP_B300_TRAIN1/TOMORROW_HANDOFF.md. Snapshot read 19 October 2026.

[27] Sean Brady with OpenAI ChatGPT and Codex. *MP52 finite exact native trial and recoverable KEEP_LOOKING frontier*. SAM Research Project. Source: SAM_REVIEW/campaigns/MP52_GEN2_R4_TRIAL1/RESULT.md. Snapshot read 19 October 2026.

[28] Sean Brady with OpenAI ChatGPT and Codex. *Completed BULK_FIRST farm learning, matched later-row outcomes and installed adoption*. SAM Research Project. Source: SAM_REVIEW/campaigns/MP_GEN2_R4_FARM_TRAINING1/RESULT.md. Snapshot read 19 October 2026.

[29] Sean Brady with OpenAI ChatGPT and Codex. *Exact action definitions and chronological empirical-reward map for CSV training*. SAM Research Project. Source: SAM_REVIEW/campaigns/MP_R4_CSV_TRAINING1/README.md. Snapshot read 19 October 2026.

[30] Sean Brady with OpenAI ChatGPT and Codex. *Completed farm methods and final deep-CSV equal-terminal lane comparison*. SAM Research Project. Source: GEN4/mp_transition1/FARM_REUSE.md. Snapshot read 19 October 2026.

[31] Sean Brady with OpenAI ChatGPT and Codex. *Exact inverse root support, recursive child reduction and resident GPU history readout*. SAM Research Project. Source: SAM_REVIEW/campaigns/MP_B300_TRAIN1/FARM_AND_INVERSE_RESULT.md. Snapshot read 19 October 2026.

[32] Sean Brady with OpenAI ChatGPT and Codex. *Qualified local support/power-plan prototype with native certificate admission*. SAM Research Project. Source: SAM_REVIEW/campaigns/GEN3_PRIME95_RESEARCH1/README.md. Snapshot read 19 October 2026.

[33] Sean Brady with OpenAI ChatGPT and Codex. *Exact 4620-wheel sparse survivor iterator with fresh matched confirmation.* SAM Research Project. Source: GEN4/mp_sparse1/REPORT.md. Snapshot read 19 October 2026.

[34] Sean Brady with OpenAI ChatGPT and Codex. *Held-out GPU factorer configuration-learning result*. SAM Research Project. Source: GEN4/mp_gpu_training2/RESULT.md. Snapshot read 19 October 2026.

[35] Sean Brady with OpenAI ChatGPT and Codex. *Exact modular sieve-offset transition and retained near-tied total-runtime comparison.* SAM Research Project. Source: GEN4/mp_transition1/RESULT.md. Snapshot read 19 October 2026.

[36] Sean Brady with OpenAI ChatGPT and Codex. *Current Volume III Jacobian program and clean-restart/bridge boundaries*. SAM Research Project. Source: volume_III/SAM_VOLUME_III_JACOBIAN_PROGRAM.md. Snapshot read 19 October 2026.

[37] Sean Brady with OpenAI ChatGPT and Codex. *Jacobian original-result, validation, compiler and routing source manifest*. SAM Research Project. Source: volume_III/JACOBIAN_SOURCE_MANIFEST.csv. Snapshot read 19 October 2026.

[38] Sean Brady with OpenAI ChatGPT and Codex. *Two-coordinate RH determinant, exact source transport and finite signed certificates*. SAM Research Project. Source: GEN4/rh_determinant_wildfowl1/REPORT.md. Snapshot read 19 October 2026.

[39] Sean Brady with OpenAI ChatGPT and Codex. *Whole auxiliary complementary inverse and original full signed source kernels*. SAM Research Project. Source: GEN4/duck_rh_complementary_resolvent1/RUN_REPORT.md. Snapshot read 19 October 2026.

[40] Sean Brady with OpenAI ChatGPT and Codex. *Original kernel/pivot signed Schur recipient and retained source requirements*. SAM Research Project. Source: GEN4/duck_rh_complementary_resolvent1/FULL_SOURCE_RECIPIENT.md. Snapshot read 19 October 2026.

[41] Sean Brady with OpenAI ChatGPT and Codex. *Exact source-conditioned finite energy-gap certificates across prime bases*. SAM Research Project. Source: GEN4/rh_gap_wildfowl1/REPORT.md. Snapshot read 19 October 2026.

[42] Sean Brady with OpenAI ChatGPT and Codex. *Signed rough-product support, two-coordinate Gram and endpoint-corrected source memory*. SAM Research Project. Source: GEN4/rh_support_memory1/REPORT.md. Snapshot read 19 October 2026.

[43] Sean Brady with OpenAI ChatGPT and Codex. *Shared exact packet and joint spin capability contracts*. SAM Research Project. Source: CURRENT_REVISION/engines/SLC/SPIN_JOINT.md. Snapshot read 19 October 2026.

[44] Sean Brady with OpenAI ChatGPT and Codex. *Completed packet/joint capability installation and actual application adoption*. SAM Research Project. Source: GEN4/spin_joint_install1/RESULT.md. Snapshot read 19 October 2026.

[45] Sean Brady with OpenAI ChatGPT and Codex. *Exact source decomposition, observable-dependent boundary equality and C4 phase map*. SAM Research Project. Source: GEN4/spin_joint_install1/ANALYSIS.md. Snapshot read 19 October 2026.

[46] Sean Brady with OpenAI ChatGPT and Codex. *Joint energy/magnetization/boundary readout and exact rational crossing example*. SAM Research Project. Source: GEN4/frustrated_spin_joint_readout1/RESULT.md. Snapshot read 19 October 2026.

[47] Sean Brady with OpenAI ChatGPT and Codex. *Exact graph solving, joint readout, finite contraction and layered-ground contracts*. SAM Research Project. Source: CURRENT_REVISION/engines/SLC/SPIN_TOOLS.md. Snapshot read 19 October 2026.

[48] Sean Brady with OpenAI ChatGPT and Codex. *Latest layered dense-source exact ground method and recorded Masquerade use H002708*. SAM Research Project. Source: GEN4/spin_layered_install1_20261017/REPORT.md. Snapshot read 19 October 2026.

[49] Sean Brady with OpenAI ChatGPT and Codex. *Layered residual frontier dynamic program, exact counts and witness implementation*. SAM Research Project. Source: CURRENT_REVISION/engines/SLC/gen3/spin_layered.py. Snapshot read 19 October 2026.

[50] Sean Brady with OpenAI ChatGPT and Codex. *Complete five-case signed-source integration successor and independent reproduction*. SAM Research Project. Source: GEN4/gen6_spin_clock_successor1_20261014/REPORT.md. Snapshot read 19 October 2026.

[51] Sean Brady with OpenAI ChatGPT and Codex. *Original signed source paths, two-coordinate quadrature and complete receiver composition*. SAM Research Project. Source: GEN4/gen6_spin_clock_successor1_20261014/DERIVATION.md. Snapshot read 19 October 2026.

[52] Sean Brady with OpenAI ChatGPT and Codex. *Independent reconstruction of original source paths and full carrier trajectories*. SAM Research Project. Source: GEN4/gen6_spin_clock_successor1_20261014/INDEPENDENT_DERIVATION.md. Snapshot read 19 October 2026.

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